Introduction to Analysis II

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Abstract

Lecture note of Introduction to Analysis II.

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Chapter 1

Several Variable Differential Calculus

Lecture 1

In this chapter, we want to approximate non-linear functions by linear maps. If we consider 24 Feb.

$$f : \mathbb{R}^n \to \mathbb{R}^m \quad f(\underbrace{x_1, x_2, \ldots, x_n}_{x}) = (f_1(x), f_2(x), \ldots, f_m(x)),$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ for all i. Now given a real-valued function $F : \mathbb{R}^n \to \mathbb{R}$ which is differentiable at a point x_0. We know that near $x_0 \in \mathbb{R}^n$ we can approximate $F(x)$ in the following way:

$$F(x) \approx F(x_0) + \nabla F(x_0) \cdot (x - x_0)$$

where

$$\nabla F(x_0) = \left(\frac{\partial F(x_0)}{\partial x_1}, \frac{\partial F(x_0)}{\partial x_2}, \ldots, \frac{\partial F(x_0)}{\partial x_n}\right) \in \mathbb{R}^n \text{ with } x_0 = (x_1, x_2, \ldots, x_n)$$

and thus

$$\begin{aligned}\nabla F(x_0) \cdot (x - x_0) &= \left\langle \frac{\partial F(x_0)}{\partial x_1}, \frac{\partial F(x_0)}{\partial x_2}, \ldots, \frac{\partial F(x_0)}{\partial x_n} \right\rangle \cdot \langle x_1, x_2, \ldots, x_n \rangle \\ &= \sum_{i=1}^{n} \frac{\partial F(x_0)}{\partial x_i} x_i.\end{aligned}$$

Hence,

$$f(x) = \begin{pmatrix} f_1(x_0) \\ f_2(x_0) \\ \vdots \\ f_n(x_0) \end{pmatrix} \approx \begin{pmatrix} f_1(x_0) + \nabla f_1(x)(x - x_0) \\ f_2(x_0) + \nabla f_2(x)(x - x_0) \\ \vdots \\ f_n(x_0) + \nabla f_n(x)(x - x_0) \end{pmatrix},$$

which gives

$$f(x) - f(x_0) \approx \begin{pmatrix} \nabla f_1(x_0)(x - x_0) \\ \nabla f_2(x_0)(x - x_0) \\ \vdots \\ \nabla f_n(x_0)(x - x_0) \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \\ \nabla f_2(x_0) \\ \vdots \\ \nabla f_n(x_0) \end{pmatrix} \cdot \underbrace{(x - x_0)}_{\text{column vector}}.$$

1.1 Linear Transformation

Definition 1.1.1 (Row vectors). Let $n \geq 1$ be an integer. We refer to elements of $\mathbb{R}^n$ an n-dimensional row vectors. A typical row vector is $x = (x_1, x_2, \ldots, x_n)$ which we abbreviate as $(x_i)_{1 \leq i \leq m}$. The components $x_1, x_2, \ldots, x_n$ are real numbers. If x and y are two row vectors in $\mathbb{R}^n$, we can define vector sum by

$$x + y = (x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n).$$

If $c \in \mathbb{R}$ is any real number, we define scalar multiplications by

$$cx = c(x_1, x_2, \ldots, x_n) = (cx_1, cx_2, \ldots, cx_n).$$

Remark 1.1.1.

(1) $-x := (-1) \cdot x = (-x_1, -x_2, \ldots, -x_n)$.

(2) zero vector is denoted by 0, i.e. $(0, 0, \ldots, 0)$.

Lemma 1.1.1 ($\mathbb{R}^n$ is a vector space). Let x, y, z be vectors in $\mathbb{R}^n$, and let $c, d \in \mathbb{R}$. Then the following properties hold:

(a) $x + y = y + x$.

(b) $(x + y) + z = x + (y + z)$.

(c) $x + 0 = 0 + x = x$.

(d) $x + (-x) = (-x) + x = 0$.

(e) $(c \cdot d)x = c \cdot (dx)$.

(f) $c(x + y) = cx + cy$.

(g) $(c + d)x = cx + dx$.

(h) $1x = x$.

Definition 1.1.2. Let $x = (x_1, x_2, \ldots, x_n)$ be row vector. Its transpose is the n-dimensional column vector

$$x^T = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Definition 1.1.3. The standard basis of $\mathbb{R}^n$ consists of $e_1, e_2, \ldots, e_n$, where e_j has 1 in the j-th position and 0 elsewhere:

$$e_j = (0, \ldots, 0, \underbrace{1}_{j\text{-th}}, 0, \ldots, 0).$$

Every row vector

$$x = (x_1, x_2, \ldots, x_n) = \sum_{j=1}^{n} x_j e_j.$$

Similarly,

$$x^T = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^{n} x_j e_j^T.$$

Definition 1.1.4 (Linear transformation)**.** A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is any function from one Euclidean space to another that satisfies the following two properties:

(a) Additivity: For $x, y \in \mathbb{R}^n$, $T(x + y) = T(x) + T(y)$.

(b) Homogenity: For $x \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$, $T(cx) = cT(x)$.

Remark 1.1.2. This definition is equivalent to the following:

$$T(c_1v_1 + \cdots + c_kv_k) = c_1T(v_1) + \cdots + c_kT(v_k)$$

where $v_1, \ldots, v_k \in \mathbb{R}^n$ and $c_1, c_2, \ldots, c_n \in \mathbb{R}$.

Definition 1.1.5. Let $m, n \geq 1$ be integers. An $m \times n$ ordered matrix is an ordered rectangular array of real numbers

$$A = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

consisting of m rows and n columns, where

(a) The entry a_{ij} denote the number in the i-th row and j-th column.

(b) We denote the set of all $m \times n$ matrices by $\mathbb{R}^{m \times n}$.

(c) In particular, a row vector is a $1 \times n$ matrix, a column vector is a $n \times 1$ vector.

Definition 1.1.6 (Matrix multiplication)**.** Given an $m \times n$ matrix $A = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$ and an $n \times p$ matrix $B = (b_{jk})_{\substack{1 \leq j \leq n \\ 1 \leq k \leq p}}$, we define AB to be the $m \times p$ matrix $(c_{ik})_{\substack{1 \leq i \leq m \\ 1 \leq k \leq p}}$ where

$$c_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}.$$

Definition 1.1.7 (Matrix-vector multiplication)**.** Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ and $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ be a column vector. We define

$$Ax = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{pmatrix}$$

Remark 1.1.3. In our class, we just treat $\mathbb{R}^n, \mathbb{R}^m$ as column vector spaces, and $L_A(x) = Ax$ is a $m \times 1$ column vector.

Theorem 1.1.1. Let A be a $m \times n$ matrix, then $L_A(x) = Ax$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Proof. ■ DIY

Proposition 1.1.1. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. For each $j = 1, 2, \ldots, n$, let e_j denote the j-th standard basis vector in $\mathbb{R}^n$ and write $T(e_j) = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$. Define the matrix $A = (a_{ij})$, then $T(x) = Ax$.

Proof. Let $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$. We can write $x = \sum_{j=1}^n x_j e_j$, then we know

$$T(x) = T\left(\sum_{j=1}^n x_j e_j\right) = \sum_{j=1}^n x_j T(e_j) = \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = Ax.$$

■

Lemma 1.1.2. Let A be a $m \times n$ matrix and let B be a $n \times p$ matrix. Then $L_A \circ L_B = L_(AB)$.

Proof. It suffices to show that $(L_A \circ L_B)(x) = L_{AB}(x)$ for $x \in \mathbb{R}^p$, and the rest is easy. ■

As previously seen. $f : E \to \mathbb{R}$ where E is a subset of $\mathbb{R}$, then

$$f'(x_0) = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{f(x) - f(x_0)}{x - x_0}.$$

Suppose now $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, we can't define

$$f'(x) = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{f(x) - f(x_0)}{x - x_0}$$

since the denominator and the numerator are vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$.

1.2 Derivatives in Several Variable Calculus

Lemma 1.2.1. Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$ be a function and let $L \in \mathbb{R}$ and x_0 be a limit point of E. Then the following two statements are equivalent:

(a) f is differentiable at x_0 and $f'(x_0) = L$.

(b) $\lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{|f(x) - f(x_0) - L(x - x_0)|}{|x - x_0|} = 0$.

Proof. Note that

$$\frac{f(x) - f(x_0)}{x - x_0} = L + \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \text{ if } x \neq x_0,$$

so we have

$$\frac{f(x) - f(x_0)}{x - x_0} - L = \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \text{ if } x \neq x_0,$$

and thus

$$0 = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \left| \frac{f(x) - f(x_0)}{x - x_0} - L \right| = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \left| \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \right|.$$

■

Lecture 2

Definition 1.2.1 (Differentiability)**.** Let E be a subset of $\mathbb{R}^n$, let $f : E \to \mathbb{R}^m$ be a function, and let x_0 be a limit point of E. Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. We say f is differentiable at x_0 with derivative L if 26 Feb.

$$\lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0,$$

or equivalently, given $\varepsilon > 0$, $\exists \delta > 0$ s.t. for all $x \in E$ satisfying $0 < \|x - x_0\| < \delta$, we have

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} < \varepsilon.$$

Remark 1.2.1. $\|f(x) - f(x_0) - L(x - x_0)\|$ is the length of a vector in $\mathbb{R}^m$ and $\|x - x_0\|$ is the length of a vector in $\mathbb{R}^n$.

Remark 1.2.2. x_0 is a limit point of $E \subseteq \mathbb{R}^n$ if for any $r > 0$, $B(x_0, r) \cap (E \setminus \{x_0\}) \neq \varnothing$. That is, for every $r > 0$, $\exists x \in E$ s.t. $0 < \|x - x_0\| < r$.

Example 1.2.1. Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ has

$$f(x) = \begin{cases} 0, & \text{if } x \in E = \{(x_1, x_2, \dots, x_n) \mid x_n \geq 0\}; \\ x, & \text{if } x \in \mathbb{R}^n \setminus E \end{cases}$$

then f is differentiable on E and $f'(x) = 0$ and f is not differentiable on $\mathbb{R}^n$ at 0. This is because on E, f is a constant function, so

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0$$

for $x_0 \in E, x \in E \setminus \{x_0\}$. On $\mathbb{R}^n$ and $x_0 = 0$, then for $x \in \mathbb{R}^n \setminus \{x_0\} = \mathbb{R}^n \setminus \{0\}$, then if we approach to 0 along E, then it gives

$$\lim_{\substack{x \to x_0 \\ x \in E}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0,$$

but if we approach 0 along $x_n < 0$, then it becomes

$$\lim_{\substack{x \to x_0 \\ x \in \mathbb{R}^n \setminus E}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = \lim_{\substack{x \to x_0 \\ x \in \mathbb{R}^n \setminus E}} \frac{\|x - 0 - L(x - x_0)\|}{\|x\|} = 1$$

for any linear map L since any linear map is differentiable and thus continuous. Thus, f is not differentiable at 0 on $\mathbb{R}^n$.

Remark 1.2.3. Recall that x_0 is said to be an interior point of $E \subseteq \mathbb{R}^n$ if there exists $r > 0$ s.t.

$$B(x_0, r) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\} \subseteq E.$$

If x_0 is an interior point of E, then f is differentiable at x_0 is equivalent to

$$\lim_{\substack{h \to 0 \\ x_0 + h \in E}} \frac{\|f(x_0 + h) - f(x_0) - L(h)\|}{\|h\|} = 0$$

since if $\|h\| < r$, then $x_0 + h \in B(x_0, r)$, which gives $x_0 + h \in E$, and thus $f(x_0 + h)$ is well-defined.

Remark 1.2.4. Here $\|\cdot\|$ denoted the standard Euclidean norm on $\mathbb{R}^n$ (and on $\mathbb{R}^m$):

$$\|(x_1, x_2, \ldots, x_n)\| = \left(x_1^2 + x_2^2 + \cdots + x_n^2\right)^{\frac{1}{2}}.$$

Example 1.2.2. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$f(x, y) = \left(x^2, y^2\right),$$

let $x_0 = (1, 2)$, and let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map

$$L(x, y) = (2x, 4y).$$

We claim that f is differentiable at x_0 with derivative L.

Proof. By definition, f is differentiable at x_0 with derivative L if and only if

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L(h)\|}{\|h\|} = 0,$$

where $h = (a, b)$. Thus, we know the above equation becomes

$$\frac{\|f(1 + a, 2 + b) - f(1, 2) - L(a, b)\|}{\|(a, b)\|} = \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}},$$

and since

$$0 \le \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}} \le \sqrt{a^2 + b^2},$$

so by the Squeeze Theorem, we have

$$\lim_{(a,b) \to (0,0)} \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}} = 0,$$

and we're done. ⊛

Lemma 1.2.2 (Uniqueness of derivative)**.** Let E be a subset of $\mathbb{R}^n$, and let $f : E \to \mathbb{R}^n$ be a function, and let $x_0 \in E$ be an interior point of E. Suppose $L_1, L_2 : \mathbb{R}^n \to \mathbb{R}^m$ are linear transformations s.t. f is differentiable at x_0 with derivative L_1 and also differentiable at x_0 with derivative L_2. Then $L_1 = L_2$.

Proof. Since x_0 is an interior point. By Remark 1.2.3, we have

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L_1(h)\|}{\|h\|} = 0 \text{ and } \lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L_2(h)\|}{\|h\|} = 0.$$

Thus,

$$\begin{aligned} 0 \le \frac{\|L_1(h) - L_2(h)\|}{\|h\|} &= \frac{\|L_1(h) - f(x_0 + h) + f(x_0) + f(x_0 + h) - f(x_0) - L_2(h)\|}{\|h\|} \\ &\le \frac{\|L_1(h) - f(x_0 + h) + f(x_0)\|}{\|h\|} + \frac{\|f(x_0 + h) + f(x_0) - L_2(h)\|}{\|h\|}. \end{aligned}$$

Thus, by squeeze theorem, we know

$$\lim_{h \to 0} \frac{\|L_1(h) - L_2(h)\|}{\|h\|} = 0.$$

Now take $v \in \mathbb{R}^n \setminus \{0\}$, and let $h = tv$, so

$$\lim_{t \to 0} \frac{\|L_1(tv) - L_2(tv)\|}{\|tv\|} = 0.$$

That is,

$$\lim_{t \to 0} \frac{|t| \, \|L_1(v) - L_2(v)\|}{|t|\|v\|} = 0.$$

Hence, we have

$$\lim_{t \to 0} \frac{\|L_1(v) - L_2(v)\|}{\|v\|} = 0,$$

and since $v \neq 0$, so we know $L_1(v) = L_2(v)$, and since v can be arbitrary non-zero vector in $\mathbb{R}^n \setminus \{0\}$, so $L_1 = L_2$. $(L_1(0) = L_2(0) = 0)$ ■

Remark 1.2.5. Because of Lemma 1.2.2, the derivative of f at interior points x_0 is unique, and thus we may safely write it as $f'(x_0)$ or $Df(x_0)$. Thus $f'(x_0)$ is the unique linear transformation $f'(x_0) : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0 \; x \neq x_0} \frac{\|f(x) - (f(x_0) + f'(x_0)(x - x_0))\|}{\|x - x_0\|} = 0.$$

Informally, this condition means that near x_0, the function f can be approximated by its linearization:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

This approximation is sometimes referred to as Newton's approximation in higher dimension.

Remark 1.2.6. A useful consequence of Lemma 1.2.2 is the following: If two functions f and g satisfy $f(x) = g(x)$ for all $x \in E$, and both are differentiable at an interior point x_0, then their derivatives coincide at x_0, i.e. $f'(x_0) = g'(x_0)$. This will be important in later arguments, where one extends functions to larger domains or modifies them on sets of measure zero.

Remark 1.2.7. As we have emphasized, Lemma 1.2.2 guarantees the uniqueness of the derivative only at interior points of the domain E. If x_0 is instead a boundary point of E, the derivative may fail to be uniquely determined.

1.3 Partial and Directional Derivatives

We now begin to relate the concept of differentiability to the more classical notions of partial and directional derivatives. Directional derivatives describe how f changes when we move from a point x_0 in a fixed direction v.

Definition 1.3.1 (Directional derivative)**.** Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$ be a function, let x_0 be an interior point of E and let $v \in \mathbb{R}^n$. If the limit

$$\lim_{\substack{t \to 0, t > 0 \\ x_0 + tv \in E}} \frac{f(x_0 + tv) - f(x_0)}{t} = \lim_{\substack{t \to 0^+ \\ x_0 + tv \in E}} \frac{f(x_0 + tv) - f(x_0)}{t}$$

exists, then we say f is differentiable in the direction v at x_0. This limit is called the directional derivative of f at x_0 in the direction v, and we denote it by

$$D_v f(x_0) \coloneqq \lim_{t \to 0^+} \frac{f(x_0 + tv) - f(x_0)}{t} \in \mathbb{R}^m.$$

Remark 1.3.1. Under this definition,

$$D_{-v}f(x_0) = \lim_{t\to 0^+} \frac{f(x_0 + t(-v)) - f(x_0)}{t} = \lim_{t\to 0^+} \frac{f(x_0 - tv) - f(x_0)}{t}.$$

In usual definition, one define

$$D_v f(x_0) = \lim_{t\to 0} \frac{f(x_0 + tv) - f(x_0)}{t}.$$

This usual definition implies that

$$D_v f(x_0) = -D_{-v} f(x_0)$$

since

$$\begin{aligned} D_v f(x_0) &= \lim_{t\to 0^+} \frac{f(x_0 + tv) - f(x_0)}{t} = \lim_{t\to 0^-} \frac{f(x_0 + tv) - f(x_0)}{t} \\ &= \lim_{s\to 0^+} = \frac{f(x_0 + (-s)v) - f(x_0)}{(-s)} = -\lim_{s\to 0^+} \frac{f(x_0 - sv) - f(x_0)}{s} = -(D_{-v} f(x_0)). \end{aligned}$$

Remark 1.3.2. This definition should be compared with the definition of differentiability. Here we divide by the scalar t rather than by a vector, so the expression always makes sense algebraically. The directional derivative $D_v f(x_0)$ is a vector in $\mathbb{R}^m$.

Remark 1.3.3. It is sometimes possible to define directional derivatives at boundary points of E, provided that the vector v points inward toward the domain. However, we shall restrict attention to interior points in what follows.

Lemma 1.3.1. Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, let $x_0 \in \operatorname{Int}(E)$ and $v \in \mathbb{R}^n$. If f is differentiable at x_0, then f is differentiable in the direction v at x_0, and

$$D_v f(x_0) = f'(x_0)(v).$$

Proof. Since $x_0 \in \operatorname{Int}(E)$ and f is differentiable at x_0, so

$$\lim_{h\to 0} \frac{\|f(x_0 + h) - f(x_0) - f'(x_0)(h)\|}{\|h\|} = 0.$$

Note that for $t \neq 0$, we have

$$\frac{f(x_0 + tv) - f(x_0)}{t} = \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t} + f'(x_0)(v),$$

so

$$\frac{f(x_0 + tv) - f(x_0)}{t} - f'(x_0)(v) = \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t}.$$

Now we show that

$$\lim_{t\to 0^+} \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t} = 0.$$

Recall that f is differentiable at x_0 and $x_0 \in \operatorname{Int}(E)$, so

$$\lim_{h\to 0} \frac{\|f(x_0 + h) - f(x_0) - f'(x_0)(h)\|}{\|h\|} = 0.$$

Let $h = tv$ when $v \neq 0$, then

$$\lim_{t\to 0^+} \frac{\|f(x_0 + tv) - f(x_0) - f'(x_0)(tv)\|}{\|tv\|} = 0 \Rightarrow \lim_{t\to 0^+} \frac{\|f(x_0 + tv) - f(x_0) - f'(x_0)(tv)\|}{|t|} = 0,$$

which is equivalent to

$$\lim_{t\to 0^+}\frac{f(x_0+tv)-f(x_0)-f'(x_0)(tv)}{t}=0,$$

(since the numerator should tend to 0 so the limit will exist), so we know

$$\lim_{t\to 0^+}\frac{f(x_0+tv)-f(x_0)}{t}-f'(x_0)(v)=0,$$

i.e.

$$\lim_{t\to 0^+}\frac{f(x_0+tv)-f(x_0)}{t}=f'(x_0)(v).$$

■

Remark 1.3.4. One consequence of this lemma is that total differentiability implies directional differentiability. However, the converse is not true: the existence of all directional derivatives does not guarantee differentiability.

Definition 1.3.2 (Partial Derivative). Let $E\subseteq\mathbb{R}^n$, let $f:E\to\mathbb{R}^m$, and let $x_0\in\mathrm{Int}(E)$, and let $1\le j\le n$. The partial derivative of f with respect to the variable x_j at x_0, denoted $\frac{\partial f}{\partial x_j}(x_0)$, is defined by

$$\frac{\partial f}{\partial x_j}(x_0):=\lim_{t\to 0,t\neq 0}\frac{f(x_0+te_j)-f(x_0)}{t}$$

provided the limit exists.

Remark 1.3.5. If $f=\begin{pmatrix}f_1\\ \vdots\\ f_m\end{pmatrix}$, then differentiation is componentwise:

$$\frac{\partial f}{\partial x_j}(x_0)=\begin{pmatrix}\frac{\partial f_1}{\partial x_j}(x_0)\\ \vdots\\ \frac{\partial f_m}{\partial x_j}(x_0)\end{pmatrix}.$$

We say that f is continuously differentiable if all partial derivatives exist and are continuous.

Remark 1.3.6. If f is differentiable at x_0, then

$$\frac{\partial f(x_0)}{\partial x_j}=D_{e_j}f(x_0)=f'(x_0)(e_j).$$

Remark 1.3.7. If f is differentiable at x_0, then

$$D_vf(x_0)=f'(x_0)(v)=f'(x_0)\left(\sum_{j=1}^n v_je_j\right)=\sum_{j=1}^n v_jf'(x_0)(e_j)=\sum_{j=1}^n v_j\frac{\partial f(x_0)}{\partial x_j}.$$

Theorem 1.3.1. Let $E\subseteq\mathbb{R}^n$, let $f:E\to\mathbb{R}^m$, $F\subseteq E$, and let $x_0\in\mathrm{Int}(F)$ (the partial derivative exists on a neighborhood of x_0). If all partial derivative $\frac{\partial f}{\partial x_j}$ exists on F and are continuous at x_0, then f is differentiable at x_0, and

$$f'(x_0)(v)=\sum_{j=1}^n v_j\frac{\partial f}{\partial x_j}(x_0)$$

for all $v \in \mathbb{R}^n$.

Proof. Define a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ by

$$L(v) := \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

We prove that f is differentiable at x_0 with derivative L. It suffices to show that for any $\varepsilon > 0$, $\exists \delta > 0$ s.t.

$$\|f(x) - f(x_0) - L(x - x_0)\| < \varepsilon \|x - x_0\| \text{ whenever } x \in F \text{ and } \|x - x_0\| < \delta$$

Since $x_0 \in \mathrm{Int}(F)$, $\exists r > 0$ s.t. $B(x_0, r) \subseteq F$. Because each partial derivative $\frac{\partial f_i}{\partial x_j}$ is continuous at x_0, for every pair (i, j) there exists $\delta_{ij} > 0$ s.t.

$$\left| \frac{\partial f_i}{\partial x_j}(x) - \frac{\partial f_i}{\partial x_j}(x_0) \right| < \frac{\varepsilon}{nm} \text{ whenever } \|x - x_0\| < \delta_{ij}.$$

Let $\delta := \min\{r, \delta_{ij} : 1 \le i \le m, 1 \le j \le n\}$. Now fix $x \in F$ with $\|x - x_0\| < \delta$ and write

$$x - x_0 = \sum_{j=1}^{n} v_j e_j, \text{ so } x = x_0 + \sum_{j=1}^{n} v_j e_j.$$

Then

$$\|x - x_0\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad |v_j| \le \|x - x_0\|,$$

Write $f = (f_1, f_2, \ldots, f_m)$. To estimate $f(x) - f(x_0)$, we vary the coordinates one at a time. Consider

$$\begin{aligned} x^{(0)} &= x_0, \\ x^{(j)} &= x_0 + \sum_{k=1}^{j} v_k e_k \text{ for } j = 1, 2, \ldots, n. \end{aligned}$$

Thus, $x^{(n)} = x$ and

$$f(x) - f(x_0) = \sum_{j=1}^{n} \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) \right) = \sum_{j=1}^{n} \sum_{i=1}^{m} \left(f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) \right) e_i.$$

Fix j and consider the j-th increment. For each component f_i, define

$$\varphi_i^j(t) = f_i\left(x^{(j-1)} + t v_j e_j\right), \quad 0 \le t \le 1.$$

Then

$$f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) = \varphi_i^j(1) - \varphi_i^j(0).$$

Since f_i has continuous partial derivatives, φ_i^j is differentiable. By MVT, there exists $t_j \in (0, 1)$ s.t.

$$\varphi_i^j(1) - \varphi_i^j(0) = \varphi_i^{j\prime}(t_j).$$

By the chain rule (See Lemma 1.3.2),

$$\varphi_i^{j\prime}(t) = \nabla f_i\left(x^{(j-1)} + t v_j e_j\right) \cdot (v_j e_j) = \frac{\partial f_i}{\partial x_j}\left(x^{(j-1)} + t v_j e_j\right) v_j.$$

Hence,

$$f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) = \frac{\partial f_i}{\partial x_j}(\xi_j)\, v_j,$$

where
$$\xi_j = x^{(j-1)} + t_j v_j e_j.$$
Subtracting $\frac{\partial f_i}{\partial x_j}(x_0)v_j$, we obtain
$$\left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| = \left| \frac{\partial f_i}{\partial x_j}(\xi_j) - \frac{\partial f_i}{\partial x_j}(x_0) \right| |v_j|.$$
Because $\|\xi_j - x_0\| < \delta$, continuity of partial derivatives yields,
$$\left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| < \frac{\varepsilon}{nm}|v_j|.$$
Summing over $i = 1, 2, \ldots, m$ and using $\|u\| \le \sum_i |u_i|$, we conclude
$$\begin{aligned}
&\left\| f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right\| = \left\| \sum_{i=1}^{m} \left(f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right) e_i \right\| \\
&\le \sum_{i=1}^{m} \left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| \le m\frac{\varepsilon}{nm}|v_j| = \frac{\varepsilon}{n}|v_j| \le \frac{\varepsilon}{n}\|x - x_0\|.
\end{aligned}$$
Finally, summing over $j = 1, 2, \ldots, n$ and applying the triangle inequality,
$$\begin{aligned}
&\left\| f(x) - f(x_0) - \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(x_0)v_j \right\| = \left\| \sum_{j=1}^{n} \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right) \right\| \\
&\le \sum_{j=1}^{n} \left\| \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right) \right\| \le \sum_{j=1}^{n} \frac{\varepsilon}{n}\|x - x_0\| = \varepsilon\|x - x_0\|.
\end{aligned}$$
Since $L(x - x_0) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0)$, so we have
$$\|f(x) - f(x_0) - L(x - x_0)\| \le \varepsilon\|x - x_0\|,$$
and we're done. ■

Remark 1.3.8. From Theorem 1.3.1 and Lemma 1.3.1 we conclude the following important fact:

If the partial derivatives of a function $f : E \to \mathbb{R}^m$ exist and are continuous on a set $F \subseteq E$, then at every interior point x_0 of F all directional derivatives exist, and they are given by
$$D_{(v_1, v_2, \ldots, v_n)} f(x_0) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

- The scalar-valued case. If $f : E \to \mathbb{R}$ is real-valued, we define the gradient of f at x_0 to be the row vector
$$\nabla f(x_0) := \left(\frac{\partial f}{\partial x_1}(x_0), \ldots, \frac{\partial f}{\partial x_n}(x_0) \right).$$
Whenever x_0 lies in the interior of a region where the partial derivatives exist and are continuous, the directional derivative takes the familiar form
$$D_v f(x_0) = \nabla f(x_0) \cdot v.$$
- The vector-valued case. Now let $f : E \to \mathbb{R}^m$ be vector-valued, say
$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}.$$

If x_0 lies in the interior of a region where all partial derivatives exist and are continuous, then

$$f'(x_0)(v_1, v_2, \ldots, v_n) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

Writing this componentwise,

$$f'(x_0)(v_1, \ldots, v_n) = \sum_{j=1}^{n} v_j \begin{pmatrix} \frac{\partial f_1}{\partial x_j}(x_0) \\ \vdots \\ \frac{\partial f_m}{\partial x_j}(x_0) \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{n} v_j \frac{\partial f_1}{\partial x_j}(x_0) \\ \vdots \\ \sum_{j=1}^{n} v_j \frac{\partial f_m}{\partial x_j}(x_0) \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \cdot v \\ \vdots \\ \nabla f_m(x_0) \cdot v \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \\ \vdots \\ \nabla f_m(x_0) \end{pmatrix} v$$

- The derivative matrix (Jacobian matrix). We therefore define the derivative matrix (or Jacobian matrix) of f at x_0 by

$$Df(x_0) \coloneqq \left(\frac{\partial f_i}{\partial x_j}(x_0) \right)_{\substack{1 \le i \le m \\ 1 \le j \le n}}.$$

Explicitly,

$$Df(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix}.$$

With this notation, the derivative acts by matrix multiplication:

$$D_v f(x_0) = f'(x_0)v = Df(x_0)v.$$

Lecture 3

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Lemma 1.3.2. Let $E \subseteq \mathbb{R}^n$ and let $f = (f_1, f_2, \ldots, f_m) : E \to \mathbb{R}^m$. Fix $1 \le i \le m$ and $1 \le j \le n$. Let $z \in E$, $v \in \mathbb{R}$ and define $\phi_i^j(t) = f_i(z + tve_j)$ for $0 \le t \le 1$. Assume that for some $t_0 \in (0, 1)$, $\exists \rho > 0$ s.t.

$$z + (t_0 + s)ve_j \in E \quad \text{for all } |s| < \rho,$$

and that the partial derivative $\frac{\partial f_i}{\partial x_j}$ exists at $p = z + t_0 v e_j$. Then, ϕ_i^j is differentiable at t_0 and

$$\left(\phi_i^j\right)'(t_0) = \frac{\partial f_i(p)}{\partial x_j}(v).$$

Proof. Note that

$$\begin{aligned} \left(\phi_i^j\right)'(t_0) &= \lim_{h \to 0} \frac{\phi_i^j(t_0 + h) - \phi_i^j(t_0)}{h} = \lim_{h \to 0} \frac{f_i(z + (t_0 + h)ve_j) - f_i(z + t_0 ve_j)}{h} \\ &= \lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{h}. \end{aligned}$$

Now if $v = 0$, then this limit is equal to 0. Now if $v \neq 0$, then

$$\lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{h} = \lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{vh} \cdot v = \lim_{s \to 0} \frac{f_i(p + se_j) - f_i(p)}{s} \cdot v = \frac{\partial f_i(p)}{\partial x} \cdot v.$$

∎

Example 1.3.1. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (x^2 + xy, y^2)$, then

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x}(x^2 + xy) & \frac{\partial}{\partial y}(x^2 + xy) \\ \frac{\partial}{\partial x}(y^2) & \frac{\partial}{\partial y}(y^2) \end{pmatrix} = \begin{pmatrix} 2x + y & x \\ 0 & 2y \end{pmatrix}.$$

Thus,

$$D_{(v,w)}f(x,y) = \begin{pmatrix} 2x+y & x \\ 0 & 2y \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} (2x+y)v + xw \\ 2yw \end{pmatrix}.$$

1.4 Chain Rule

If $f : X \to Y$ and $g : Y \to Z$, then their composition $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$. We want to show that

$$D(g \circ f)(x) = (Dg(f(x))) \cdot (Df(x))$$

under suitable conditions.

Definition 1.4.1. Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be square matrix. The trace of A, denoted $\mathrm{Tr}(A)$, is defined by $\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{in}$. Let $M = (m_{ij})$ be an $r \times s$ matrix. We may regard M as an element of $\mathbb{R}^{r\times s}$. We can define its Frobenius norm by

$$\|M\|_F = \left(\sum_{i=1}^{r}\sum_{j=1}^{s} m_{ij}^2\right)^{\frac{1}{2}},$$

which can be written as $\|M\|_F = (\mathrm{Tr}\,(MM^t))^{\frac{1}{2}}$.

Theorem 1.4.1. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. Define the Frobenius norm by

$$\|M\|_F = \sqrt{\mathrm{Tr}\,(MM^t)} = \left(\sum_{i=1}^{r}\sum_{j=1}^{s} m_{ij}^2\right)^{\frac{1}{2}}$$

with $M \in \mathbb{R}^{r\times s}$. Then,

$$\|AB\|_F \le \|A\|_F\|B\|_F.$$

In particular, for any column vector $v \in \mathbb{R}^n$, we have

$$\|Av\| \le \|A\|_F\|v\|,$$

and the linear map $x \mapsto Ax$ is continuous on $\mathbb{R}^n$.

Proof. Note that

$$(AB)_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk},$$

so

$$|(AB)_{ik}| = \left|\sum_{j=1}^{n} a_{ij}b_{jk}\right| \le \left(\sum_{j=1}^{n} a_{ij}^2\right)^{\frac{1}{2}} \left(\sum_{j=1}^{n} b_{jk}^2\right)^{\frac{1}{2}}.$$

Hence, we have

$$\|AB\|_F^2 = \sum_i\sum_k (AB)_{ik}^2 \le \sum_i\sum_k \left(\sum_{j=1}^{n} a_{ij}^2\right)\left(\sum_{j=1}^{n} b_{jk}^2\right) = \|A\|_F^2\|B\|_F^2.$$

For the second part, since we have

$$\|Av\|_F \le \|A\|_F\|v\|_F,$$

and we know for column vector, Frobenius norm coincides with the Euclidean norm, so we're done.

Finally, for $x, x_0 \in \mathbb{R}^n$, we know $Ax - Ax_0 = A(x - x_0)$, and thus

$$0 \le \|Ax - Ax_0\| = \|A(x - x_0)\| \le \|A\|_F\|x - x_0\|,$$

and if $x \to x_0$, we know

$$\|A\|_F \|x - x_0\| \to 0$$

since $\|A\|_F$ is a constant, and thus $\|A(x - x_0)\|$ also converges to 0, which shows the map $x \mapsto Ax$ is linear. ■

Theorem 1.4.2. Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, and let x_0 be an interior point of E. If f is differentiable at x_0, then f is continuous at x_0.

Proof. If f is differentiable at x_0, then there exists $L : \mathbb{R}^n \to \mathbb{R}^m$ s.t.

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

Thus,

$$\lim_{x \to x_0} \|f(x) - f(x_0) - L(x - x_0)\| = \lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} \lim_{x \to x_0} \|x - x_0\| = 0.$$

Hence, we know

$$\begin{aligned} \|f(x) - f(x_0)\| &= \|f(x) - f(x_0) - L(x - x_0) + L(x - x_0)\| \\ &\leq \|f(x) - f(x_0) - L(x - x_0)\| + \|L(x - x_0)\| \end{aligned}$$

and the former converges to 0 as $x \to x_0$, while the latter also converges to 0 as $x \to x_0$ since L is continuous. ■

Theorem 1.4.3 (Chain Rule)**.** Let $E \subseteq \mathbb{R}^n$, let $F \subseteq \mathbb{R}^m$, and let $f : E \to F$ and $g : F \to \mathbb{R}^p$. Let $x_0 \in \mathrm{Int}(E)$. Suppose f is differentiable at x_0, that $f(x_0)$ lies in the interior of F, and that g is differentiable at $f(x_0)$. Then, $g \circ f : E \to \mathbb{R}^p$ is differentiable at x_0, and

$$(g \circ f)'(x_0) = g'(f(x_0)) \circ f'(x_0)$$

Proof. Set $y_0 = f(x_0)$ and denote the derivatives by $A = f'(x_0) : \mathbb{R}^n \to \mathbb{R}^m$, $B : g'(f(x_0)) = g'(y_0) : \mathbb{R}^m \to \mathbb{R}^p$. Since g is differentiable at y_0, define $\eta : \mathbb{R}^m \to \mathbb{R}^p$ by

$$\eta(k) = \begin{cases} \dfrac{g(y_0 + k) - g(y_0) - B(k)}{\|k\|}, & \text{if } k \neq 0; \\ 0, & \text{if } k = 0. \end{cases}$$

Note that $\lim_{k \to 0} \eta(k) = 0$ since g is differentiable at y_0, so η is continuous at 0. Moreover, we have

$$g(y_0 + k) - g(y_0) - Bk = \eta(k)\|k\| \quad \text{true for all } k.$$

Since $x_0 \in \mathrm{Int}(E)$, $x_0 + h \in E$ for small enough $\|h\|$. Note that

$$\begin{aligned} (g \circ f)(x_0 + h) - g(f(x_0)) &= g(f(x_0 + h)) - g(f(x_0)) = g(f(x_0) + (f(x_0 + h) - f(x_0))) - g(f(x_0)) \\ &= B(f(x_0 + h) - f(x_0)) + \eta(f(x_0 + h) - f(x_0)) \cdot \|f(x_0 + h) - f(x_0)\|, \end{aligned}$$

so we have

$$(g \circ f)(x_0 + h) - g(f(x_0)) - BAh = B(f(x_0 + h) - f(x_0) - Ah) + \eta(f(x_0 + h) - f(x_0)) \cdot \|f(x_0 + h) - f(x_0)\|.$$

Now since

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|} = 0,$$

so

$$\frac{\|B(f(x_0 + h) - f(x_0) - Ah)\|}{\|h\|} \leq \frac{\|B\|_F \|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|}$$

implies

$$\lim_{h\to 0}\frac{\|B(f(x_0+h)-f(x_0)-Ah)\|}{\|h\|}=0.$$

Now we claim that

$$\lim_{h\to 0}\frac{\eta\,(f(x_0+h)-f(x_0))\cdot\|f(x_0+h)-f(x_0)\|}{\|h\|}=0.$$

Note that

$$\frac{\eta\,(f(x_0+h)-f(x_0))\cdot\|f(x_0+h)-f(x_0)\|}{\|h\|}=\eta\,(f(x_0+h)-f(x_0))\,\frac{\|f(x_0+h)-f(x_0)\|}{\|h\|}.$$

Now since η is continuous at 0, so

$$\lim_{h\to 0}\eta(f(x_0+h)-f(x_0))=\eta(0)=0.$$

Also,

$$\begin{aligned}\frac{\|f(x_0+h)-f(x_0)\|}{\|h\|}&=\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)+f'(x_0)(h)\|}{\|h\|}\\&\leq\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)\|+\|f'(x_0)(h)\|}{\|h\|}\\&\leq\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)\|}{\|h\|}+\frac{\|h\|\|f'(x_0)\|_F}{\|h\|}\\&\to\|A\|_F\text{ as }h\to 0.\end{aligned}$$

Thus,

$$\lim_{h\to 0}\frac{\|(g\circ f)(x_0+h)-g(f(x_0))-BAh\|}{\|h\|}=0.$$

■

Corollary 1.4.1. We may express the chain rule in matrix form.

$$D(g\circ f)(x_0)=Dg(f(x_0))Df(x_0).$$

Remark 1.4.1. We prove the chain rule of linear transformations in the last theorem, and earlier we have found that in fact the linear transformation $f'(x_0)$ is the Jacobian matrix $Df(x_0)$, so we can write the matrix form of the chain rule, i.e. the chain rule on Jacobian matrix.

Example 1.4.1. Let $f,g:\mathbb{R}^n\to\mathbb{R}$ be differentiable functions. Recall that

$$\nabla f=\left(\frac{\partial f}{\partial x_1},\frac{\partial f}{\partial x_2},\dots,\frac{\partial f}{\partial x_n}\right),\quad \nabla g=\left(\frac{\partial g}{\partial x_1},\frac{\partial g}{\partial x_2},\dots,\frac{\partial g}{\partial x_n}\right),$$

and

$$\nabla f=\sum_{i=1}^{n}\frac{\partial f}{\partial x_i}e_i,$$

then we have

$$\nabla(fg)=\sum_{i=1}^{n}\frac{\partial}{\partial x_i}(fg)e_i=\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}g+f\frac{\partial g}{\partial x_i}\right)e_i=g\nabla f+f\nabla g$$

by ordinary product rule(前微後不微加上前不微後微).

Example 1.4.2. Consider $f : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ defined by

$$f(A) = A^t A,$$

then $Df(A) = ?$

Proof. We use two methods to solve this problem.

Method 1: Direct Expansion (Using the Definition) To find the derivative of f at A, we apply a small matrix perturbation $H \in \mathbb{R}^{n\times n}$ and expand $f(A+H)$:

$$f(A+H) = (A+H)^t(A+H) = \left(A^t + H^t\right)(A+H) = A^tA + A^tH + H^tA + H^tH.$$

Hence, we have

$$f(A+H) - f(A) = \left(A^tH + H^tA\right) + H^tH.$$

Hence, we may choose $L : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ by

$$L(H) = A^tH + H^tA.$$

Thus, we have

$$\lim_{H\to 0} \frac{\|f(A+H) - f(A) - L(H)\|}{\|H\|} = \lim_{H\to 0} \frac{\|H^tH\|}{\|H\|} \leq \lim_{H\to 0} \frac{\|H^t\|\,\|H\|}{\|H\|} = \lim_{H\to 0} \left\|H^t\right\| = 0.$$

Method 2: Decomposition and the Chain Rule We can deconstruct f into

$$A \overset{\Phi}{\longmapsto} (A^t, A) \overset{\Psi}{\longmapsto} A^tA$$

Note that

$$\Phi(A+h) - \Phi(A) - \left(H^t, H\right) = 0,$$

so

$$\lim_{\|H\|\to 0} \frac{\|\Phi(A+H) - \Phi(A) - (H^t, H)\|}{\|H\|} = 0.$$

Thus, $\Phi(A)'(H) = (H^t, H)$. Also, we can check $\Psi'(B,C)(K,L) = KC + BL$, i.e.

$$\lim_{(K,L)\to(0,0)} \frac{\|\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL)\|}{\|(K,L)\|} = 0.$$

Note that

$$\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL) = KL.$$

and

$$\|(K,L)\| = \sqrt{\|K\|^2 + \|L\|^2} \geq \frac{\|K\| + \|L\|}{\sqrt{2}}.$$

Thus,

$$\begin{aligned} \frac{\|\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL)\|}{\|(K,L)\|} &\leq \sqrt{2}\frac{\|KL\|}{\|K\| + \|L\|} \leq \sqrt{2}\frac{\|K\|\|L\|}{\|K\| + \|L\|} \\ &\leq \sqrt{2}\frac{\|K\|\|L\|}{\|L\|} = \sqrt{2}\|K\|. \end{aligned}$$

This shows

$$0 \leq \lim_{(K,L)\to(0,0)} \frac{\|\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL)\|}{\|(K,L)\|} \leq \lim_{(K,L)\to(0,0)} \sqrt{2}\|K\| = 0.$$

By chain rule, we know

$$f'(A)(H) = \Psi'(\Phi(A))\,(\Phi'(A)(H)) = H^t A + A^t H.$$

⊛

Lecture 4

If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then $T'(x) = T$ for every x. This is because for every x 5 March.

$$\lim_{h\to 0} \frac{\|T(x+h) - T(x) - T(h)\|}{\|h\|} = 0.$$

Hence, for any differentiable $f : E \to \mathbb{R}^n$, the composition Tf is differentiable and

$$(Tf)'(x_0) = T'(f(x_0)) \circ f'(x_0) = T \circ f'(x_0) = T(f'(x_0)).$$

This generalizes the familiar one-variable rule $(cf)' = cf'$.

Another useful special case of the chain rule is the following. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable and $x_j : \mathbb{R} \to \mathbb{R}$ are differentiable functions for $j = 1, 2, \ldots, n$, then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x_1(t), x_2(t), \ldots, x_n(t)) = \sum_{j=1}^{n} x_j'(t) \frac{\partial f}{\partial x_j}(x_1(t), x_2(t), \ldots, x_n(t))$$

since we can suppose $f = (f_1, f_2, \ldots, f_m)$ and $g(t) = (x_1(t), x_2(t), \ldots, x_n(t))^t$ and thus

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} f(x_1(t), x_2(t), \ldots, x_n(t)) &= \frac{\mathrm{d}}{\mathrm{d}t} f \circ g(t) = f'(g(t)) \circ g'(t) \\
&= \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(g(t)) & \frac{\partial f_1}{\partial x_2}(g(t)) & \cdots & \frac{\partial f_1}{\partial x_n}(g(t)) \\ & \vdots & & \\ \frac{\partial f_m}{\partial x_1}(g(t)) & \frac{\partial f_m}{\partial x_2}(g(t)) & \ldots & \frac{\partial f_m}{\partial x_n}(g(t)) \end{pmatrix} \begin{pmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{pmatrix} \\
&= \begin{pmatrix} \sum_{i=1}^n \frac{\partial f_1}{\partial x_i}(g(t)) x_i'(t) \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m}{\partial x_i}(g(t)) x_i'(t) \end{pmatrix} = \sum_{i=1}^{n} x_i'(t) \begin{pmatrix} \frac{\partial f_1}{\partial x_i}(g(t)) \\ \vdots \\ \frac{\partial f_m}{\partial x_i}(g(t)) \end{pmatrix} \\
&= \sum_{i=1}^{n} x_i'(t) \frac{\partial f}{\partial x_i}(g(t)) = \sum_{i=1}^{n} x_i'(t) \frac{\partial f}{\partial x_i}(x_1(t), x_2(t), \ldots, x_n(t)).
\end{aligned}$$

The Mean Value Theorem for functions $f : \mathbb{R} \to \mathbb{R}$ asserts that

$$f(y) - f(x) = f'(z)(y - x)$$

where $z \in (x, y)$. We begin by extending this statement to real-valued functions defined on open subsets of $\mathbb{R}^n$.

Theorem 1.4.4 (Mean Value Theorem for Real-Valued Functions)**.** Let V be open in $\mathbb{R}^n$ and suppose that $f : V \to \mathbb{R}$ is differentiable on V. Given $a, x \in V$, write $L(x; a)$ for the (closed) line segment joining a to x, namely

$$L(x; a) \coloneqq \{a + t(x - a) : 0 \leq t \leq 1\}.$$

Assume that $L(x;a) \subseteq V$. Then there exists $c \in L(x;a)$ such that

$$f(x) - f(a) = \nabla f(c) \cdot (x-a).$$

Proof. Suppose $g(t) := a + t(x-a)$ for $t \in \mathbb{R}$. Then, $g : \mathbb{R} \to \mathbb{R}^n$ is differentiable and

$$Dg(t) = x - a \quad \forall t \in \mathbb{R}.$$

Since V is open and the compact segment $L(x;a) = \{g(t) : t \in [0,1]\}$ is contained in V, there exists $\delta > 0$ s.t.

$$g(t) \in V \quad \text{for all } t \in I_\delta = (-\delta, 1+\delta).$$

Hence, $f \circ g : I_\delta \to \mathbb{R}$ is well-defined and differentiable. By the chain rule, for each $t \in I_\delta$ we have

$$D(f \circ g)(t) = Df(g(t))D(g(t)) = Df(g(t))(x-a).$$

Because f is real-valued, the differential $Df(y)$ may be identified with the gradient by

$$Df(y)(h) = \nabla f(y) \cdot h, y \in V, h \in \mathbb{R}^n.$$

Therefore the above equation becomes

$$(f \circ g)'(t) = \nabla f(g(t)) \cdot (x-a), t \in I_\delta.$$

Now apply the one-dimensional MVT to $f \circ g$ on $[0,1]$. There exists $t_0 \in (0,1)$ such that

$$(f \circ g)(1) - (f \circ g)(0) = (f \circ g)(0)'(t_0).$$

Since $(f \circ g)(1) = f(x)$ and $(f \circ g)(0) = f(a)$, we obtain

$$f(x) - f(a) = \nabla f(g(t_0)) \cdot (x-a).$$

We can set $c := g(t_0) \in L(x;a)$, and we're done. ∎

The conclusion of Theorem 1.4.4 relies essentially on the fact that f is real-valued. The following remark shows that no analogous statement can hold in general for vector-valued functions, even when the domain is one-dimensional.

Remark 1.4.2. The function $f(t) = (\cos t, \sin t)$ is differentiable on $\mathbb{R}$ and satisfies $f(w\pi) = f(0)$, but there is no $c \in \mathbb{R}$ s.t. $Df(c) = (0,0)$.

Now we want to extend the MVT above to $f : \mathbb{R}^n \to \mathbb{R}^m$ with $m > 1$, one can recover a valid statement by testing against linear functionals on $\mathbb{R}^m$. Equivalently, we take the dot product with an arbitrary vector $a \in \mathbb{R}^m$ and apply the real-valued mean value theorem to the scalar function $t \mapsto a \cdot f(x+tu)$. This yields the following useful generalization.

In the following statement, we let

$$L(x;y) = \{tx + (1-t)y : 0 \leq t \leq 1\}.$$

Theorem 1.4.5 (Mean-Value Theorem)**.** Let S be an open subset of $\mathbb{R}^n$ and assume that $f : S \to \mathbb{R}^m$ is differentiable at each point of S. Let x and y be two points in S s.t. $L(x;y) \subseteq S$. Then for every vector $a \in \mathbb{R}^m$ there is a point $z \in L(x;y)$ s.t.

$$a \cdot \{f(y) - f(x)\} = a \cdot \{f'(z)(y-x)\}.$$

Proof. Let $u = y - x$. Since S is open and $L(x;y) \subseteq S$, there is a $\delta > 0$ s.t.

$$x + tu \in S \text{ for all real } t \in (-\delta, 1+\delta).$$

Let a be a fixed vector in $\mathbb{R}^m$ and define the real-valued function F on $(-\delta, 1+\delta)$ by

$$F(t) = a \cdot f(x+tu).$$

Then F is differentiable on $(-\delta, 1+\delta)$ since $F = T \circ f$ where $T(x) = a \cdot x$ is a linear map, and note that

$$F'(t) = a \cdot \{f'(x+tu)(u)\}.$$

By the usual MVT for real-valued functions, we have

$$F(1) - F(0) = F'(\theta) \text{ for some } \theta \text{ with } 0 < \theta < 1.$$

Now

$$F'(\theta) = a \cdot \{f'(x+\theta u)(u)\} = a \cdot \{f'(z)(y-x)\},$$

where $z = x + \theta u \in L(x;y)$. But

$$F(1) - F(0) = a \cdot \{f(y) - f(x)\},$$

so we're done. ∎

Remark 1.4.3. If S is convex, then $L(x;y) \subseteq S$ for all $x, y \in S$, so Theorem 1.4.5 holds for all x and y ion S.

1.5 Double Derivatives and Clairaut's theorem

Definition 1.5.1. Let $E \subseteq \mathbb{R}^n$ be open and let $f : E \to \mathbb{R}^m$. We say f is twice continuously differentiable on E if

(1) f is continuously differentiable on E, i.e. $\frac{\partial f}{\partial x_i}$ exists and are continuous on E

(2) for each $j = 1, 2, \ldots, n$, the partial derivative $\frac{\partial f}{\partial x_j} : E \to \mathbb{R}^m$ is continuously differentiable on E.

Theorem 1.5.1 (Clairaut's theorem)**.** Let E be an open set in $\mathbb{R}^n$, and let $f : E \to \mathbb{R}^m$ be twice continuously differentiable on E. Then, for all $1 \leq i, j \leq n$ and all $x_0 \in E$

$$\frac{\partial^2 f(x_0)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x_0)}{\partial x_j \partial x_i}.$$

Proof. Since $f = (f_1, f_2, \ldots, f_m)$ where $f_i : E \to \mathbb{R}$, we may just assume $f : E \to \mathbb{R}$. We may also assume $x_0 = 0$. The $i = j$ case is trivial. Given any $\varepsilon > 0$, $\exists \delta_0 > 0$ s.t.

$$\left| \frac{\partial^2 f(x)}{\partial x_i \partial x_j} - \frac{\partial^2 f(0)}{\partial x_i \partial x_j} \right| < \varepsilon \text{ and } \left| \frac{\partial^2 f(x)}{\partial x_j \partial x_i} - \frac{\partial^2 f(0)}{\partial x_j \partial x_i} \right| < \varepsilon$$

whenever $\|x\| \leq 2\delta_0$ since the second order derivative is continuous at 0. Fix $\delta \in (0, \delta_0)$. Consider the square in the (x_i, x_j)-plane with vertices $0, \delta e_i, \delta e_j, \delta e_i + \delta e_j$ and define the second finite difference

$$X \coloneqq f(\delta e_i + \delta e_j) - f(\delta e_i) - f(\delta e_j) + f(0).$$

Define

$$\phi(t) \coloneqq f(te_i + \delta e_j) - f(te_i), \quad t \in [0, \delta].$$

Then, $X = \phi(\delta) - \phi(0)$. By MVT, $\exists c_i \in (0, \delta)$ s.t.

$$X = \delta \phi'(c_i) = \delta \left[\frac{\partial f}{\partial x_i}(c_i e_i + \delta e_j) - \frac{\partial f}{\partial x_i}(c_i e_i) \right].$$

Now consider

$$g(s) \coloneqq \frac{\partial f}{\partial x_i}(c_i e_i + s e_j), \quad s \in [0, \delta].$$

Then
$$g(\delta) - g(0) = \frac{\partial f}{\partial x_i}(c_i e_i + \delta e_j) - \frac{\partial f}{\partial x_i}(c_i e_i).$$
Applying MVT one more time, then we know there exists $c_j \in (0, \delta)$ s.t.
$$g(\delta) - g(0) = g'(c_j)\delta = \delta \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i e_i + c_j e_j).$$
Substituting into the expression for X gives
$$X = \delta^2 \frac{\partial^2 f}{\partial x_j x_i}(\xi), \quad \xi = c_i e_i + c_j e_j.$$
Since $\|\xi\| = \sqrt{c_i^2 + c_j^2} < \delta\sqrt{2} < 2\delta_0$, continuity implies
$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_j \partial x_i}(0) \right| \leq \varepsilon.$$
If we write X as
$$X = f(\delta e_i + \delta e_j) - f(\delta e_j) - f(\delta e_i) + f(0)$$
and define
$$\psi(s) := f(\delta e_i + s e_j) - f(s e_j), \quad s \in [0, \delta].$$
Then, we can use similar method to show
$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right| \leq \varepsilon.$$
Thus, by triangle inequality, we know
$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(0) - \frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right| \leq 2\varepsilon$$
for any $\varepsilon > 0$, and we're done. ■

Lecture 5

1.6 The Contraction Mapping Theorem

10 March.

Definition 1.6.1. Let (X, d) be a metric, and let $f : X \to X$ be a map. We say f is a contraction if
$$d(f(x), f(y)) \leq d(x, y) \quad \text{for all } x, y \in X.$$
We say f is a strict contraction if there exists a constant $0 < c < 1$ s.t.
$$d(f(x), f(y)) \leq c d(x, y) \quad \text{for all } x, y \in X.$$
We call c a contraction constant of f.

Example 1.6.1. Suppose $f : \mathbb{R} \to \mathbb{R}$,

(a) if $f(x) = x + 1$, then
$$f(x) - f(y) = (x + 1) - (y + 1) = x - y \Rightarrow |f(x) - f(y)| = |x - y|,$$
so f is a contraction but not a strict contraction.

(b) if $f(x) = \frac{x}{2}$, then
$$|f(x) - f(y)| = \frac{|x-y|}{2},$$
so f is a strict contraction.

Example 1.6.2. Suppose $f : [0,1] \to [0,1]$ and $f(x) = x - x^2$, then $f'(x) = 1 - 2x$, so
$$|f'(x)| \leq 1 \quad \text{on } [0,1].$$
By MVT,
$$f(x) - f(y) = f'(c)(x-y) \text{ for all } x, y \in [0,1] \text{ for some } c \in (x,y).$$
Now since $|f'(c)| \leq 1$, so we know f is a contraction. But f is not a strict contraction. Suppose it is a strict contraction, then
$$|f(x)| = |f(x) - f(0)| \leq c|x| \text{ for some } 0 < c < 1 \text{ for all } x \in [0,1].$$
However,
$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = f'(0) = 1,$$
which is impossible.

Definition 1.6.2. Let $f : X \to X$ be a map and let $x \in X$. We say that x is a fixed point of f if $f(x) = x$.

Remark 1.6.1. Not every contraction has a fixed point.

Remark 1.6.2. Suppose $X \subseteq \mathbb{R}$. The existence of a fixed point x s.t. $f(x) = x$ means that the graph of $y = f(x)$ intersect with $y = x$.

Definition 1.6.3. A metric space (X,d) is said to be complete if every Cauchy sequence in X converges to a point of X.

Example 1.6.3. In $\mathbb{R}^n$, $\mathbb{R}^n$ is complete with the standard metric
$$d(x,y) = \|x - y\| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \text{ with } x = (x_1, x_2, \ldots, x_n),\ y = (y_1, y_2, \ldots, y_n).$$

Remark 1.6.3. Recall that

- Any closed subset of a complete metric space is also a complete metric space.
- Any closed set in $\mathbb{R}^n$ is complete. In particular, a closed ball
$$\overline{B(x_0, r)} = \{y : \|y - x_0\| \leq r\}$$
is complete.

Theorem 1.6.1 (Contraction Mapping Theorem)**.** Let (X,d) be a metric space and $f : X \to X$ be a strict contractor. Then f have at most one fixed point. Moreover, if we also assume X is non-empty and complete, then f has exactly one fixed point.

Proof. Suppose x and y are fixed points of a strict contractor f. Then, $f(x) = x$ and $f(y) = y$, and $\exists 0 < c < 1$ s.t.

$$d(x, y) = d(f(x), f(y)) \le cd(x, y),$$

so $0 \le (1 - c)d(x, y) \le 0$, which shows $x = y$.

Now we assume X is non-empty and complete. Choose any point $x_0 \in X$. Consider the sequence $x_1 = f(x_0), x_2 = f(x_1), \ldots$. For $n \ge 1$, we have

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \le cd(x_n, x_{n-1}) \le \cdots \le c^n d(x_1, x_0).$$

This shows $\{x_n\}_{n=0}^{\infty}$ is a Cauchy sequence. Now since X is complete, so $\lim_{n\to\infty} x_n = x^*$ for some $x^* \in X$. Now we claim $f(x^*) = x^*$. By triangle inequality and $f(x_{n-1}) = x_n$,

$$\begin{aligned} d(x^*, f(x^*)) &\le d(x^*, f(x_n)) + d(f(x_n), f(x^*)) \le d(x^*, x_{n+1}) + cd(x_n, x^*) \\ &\le d(x^*, x_{n+1}) + d(x_n, x^*) \le d(x_{n+1}, x_n) \to 0. \end{aligned}$$

■

Lemma 1.6.1. Let $B(0, r)$ be an open ball in $\mathbb{R}^n$, and let $g : B(0, r) \to \mathbb{R}^n$ be a map with $g(0) = 0$ and

$$\|g(x) - g(y)\| \le \frac{1}{2}\|x - y\| \text{ for all } x, y \in B(0, r).$$

Then, the function $f : B(0, r) \to \mathbb{R}^n$ defined by $f(x) = x + g(x)$ is one to one and the image of $f(B(0, r))$ contains the ball $B\left(0, \frac{r}{2}\right)$.

Proof. First, we show that f is one to one. Suppose $f(x) = f(y)$, then

$$x + g(x) = y + g(y) \Leftrightarrow x - y = g(y) - g(x) \Leftrightarrow \|x - y\| = \|g(x) - g(y)\| \le \frac{1}{2}\|x - y\|,$$

so $x = y$.

Now we show $B\left(0, \frac{r}{2}\right) \subseteq f(B(0, r))$. It suffices to show that for any $y \in B\left(0, \frac{r}{2}\right)$, $\exists x \in B(0, r)$ s.t. $f(x) = y$, or equivalently, $x = y - g(x)$. Define $F(x) = y - g(x)$, then we want to show that for any $y \in B\left(0, \frac{r}{2}\right)$, $\exists x \in B(0, r)$ s.t. $F(x) = x$. Now $y \in B\left(0, \frac{r}{2}\right)$, then $\exists \varepsilon > 0$ s.t. $2\|y\| \le r - \varepsilon$.

Claim 1.6.1. $F : \overline{B(0, r - \varepsilon)} \to \overline{B(0, r - \varepsilon)}$ is a strict contraction.

Proof. We know

$$\|F(x)\| \le \|y\| + \|g(x)\| \le \frac{r - \varepsilon}{2} + \|g(x)\|.$$

Since $g(0) = 0$, so

$$\|g(x)\| = \|g(x) - g(0)\| \le \frac{1}{2}\|x - 0\| \le \frac{r - \varepsilon}{2}.$$

Thus,

$$\|F(x)\| \le \frac{r - \varepsilon}{2} + \frac{r - \varepsilon}{2} = r - \varepsilon.$$

Thus, $F(x) \in \overline{B(0, r - \varepsilon)}$. Also,

$$\|F(x_1) - F(x_2)\| = \|g(x_1) - g(x_2)\| \le \frac{1}{2}\|x_1 - x_2\|.$$

Thus, F is a strict contraction. ⊛

Now since closed ball is non-empty and complete, so by Theorem 1.6.1, F has a fixed point. ■

Lecture 6

12 March.

1.7 Inverse Function Theorem

Recall that if $f : \mathbb{R} \to \mathbb{R}$ is invertible, i.e. $f^{-1} : \mathbb{R} \to \mathbb{R}$ exists s.t. $f \circ f^{-1}(x) = f^{-1} \circ f(x) = x$ for all $x \in \mathbb{R}$, and $f'(x_0) \neq 0$ for some $x_0 \in \mathbb{R}$, then f^{-1} is differentiable at $f(x_0)$ and

$$\left(f^{-1}\right)'(f(x_0)) = \frac{1}{f'(x_0)},$$

which can be proved by chain rule.

In fact, one can say something even when f' is not invertible, as long as we know that f is continuously differentiable. If $f'(x_0) \neq 0$, then $f'(x_0)$ must be either strictly positive or strictly negative (since we assume f' to be continuous by Theorem 1.4.2), which implies $f'(x)$ is strictly positive for x near x_0, or strictly negative for x near x_0, and thus one-to-one in such small interval. In either case, f will become invertible if we restrict the domain and codomain of f to be sufficiently close to x_0 and to $f(x_0)$, respectively.

Now if $f : \mathbb{R}^n \to \mathbb{R}^n$ and $f \in C^1$, i.e. f is continuously differentiable. Suppose

$$f'(x_0) = Df(x_0) = \left[\frac{\partial f_i(x_0)}{\partial x_j}\right]$$

is invertible. Then, $\exists$ a neighborhood U of x_0 and V of $f(x_0)$ s.t. $f : U \to V$ is bijective and its inverse $f^{-1} : V \to U$ is continuous and differentiable at $f(x_0)$ with

$$\left(f^{-1}\right)'(f(x_0)) = \left[f'(x_0)\right]^{-1}.$$

This is the Inverse Function Theorem, and we will prove it later.

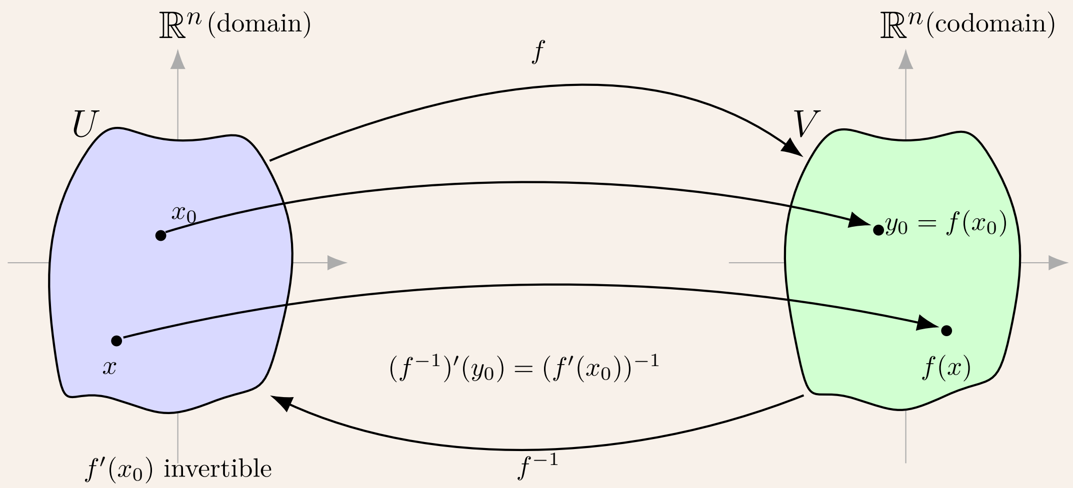
First, we prepare a lemma.

Lemma 1.7.1. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear transformation . Then $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is also linear.

Proof. ■ DIY

Theorem 1.7.1 (The Inverse Function Theorem)**.** Let E be an open subset of $\mathbb{R}^n$ and let $f : E \to \mathbb{R}^n$ be a function which is continuously differentiable on E. Suppose $x_0 \in E$ is the point s.t. the linear map $f'(x_0) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible, then there exists an open set $U \subseteq E$ s.t. $x_0 \in U$ and an open set $V \subseteq \mathbb{R}^n$ s.t. $f(x_0) \in V$ so that $f : U \to V$ is a bijection. In particular, there is an inverse map $f^{-1} : V \to U$. Furthermore, this inverse map is differentiable at $f(x_0)$, and

$$\left(f^{-1}\right)'(f(x_0)) = \left(f'(x_0)\right)^{-1}.$$



Proof. Suppose such f^{-1} exists, i.e. $\exists U, V$ s.t. $f : U \to V$ and $f^{-1} : V \to U$ are bijective where $x_0 \in U$ and $f(x_0) \in V$. Since $f'(x_0)$ is invertible and $f^{-1}(f(x_0))$ exists, we know

$$f^{-1}(f(x)) = x \text{ for } x \in U \text{ and } \left(f^{-1}\right)'(f(x_0)) \circ (f'(x_0)) = Id \text{ by chain rule.}$$

Now we have to show such U, V exists. Let $A = f'(x_0)$, then

$$\lim_{x\to 0} \frac{\|f(x_0 + x) - f(x) - A(x)\|}{\|x\|} = 0.$$

Note that A^{-1} exists and

$$\begin{aligned} A^{-1}\left(f(x_0 + x) - f(x_0) - Ax\right) &= A^{-1}(f(x_0 + x) - f(x)) - A^{-1}Ax \\ &= A^{-1}\left(f(x_0 + x) - f(x_0)\right) - x. \end{aligned}$$

Note that

$$\lim_{x\to 0} \frac{\left\|A^{-1}(f(x_0 + x) - f(x_0)) - x\right\|}{\|x\|} = 0$$

since

$$\begin{aligned} 0 \leq \lim_{x\to 0} \frac{\left\|A^{-1}(f(x_0 + x) - f(x_0)) - x\right\|}{\|x\|} &= \lim_{x\to 0} \frac{\left\|A^{-1}\left(f(x_0 + x) - f(x) - A(x)\right)\right\|}{\|x\|} \\ \leq \lim_{x\to 0} \frac{\|A^{-1}\| \, \|f(x_0 + x) - f(x) - A(x)\|}{\|x\|} &= 0. \end{aligned}$$

Let $g(x) = A^{-1}(f(x_0 + x) - f(x_0)) - x$. Thus,

$$\lim_{x\to 0} \frac{\|g(x)\|}{\|x\|} = 0.$$

Note that $g(0) = 0$. Thus,

$$\lim_{x\to 0} \frac{\|g(x) - g(0) - 0\|}{\|x\|} = 0,$$

which shows g is differentiable at 0 with derivative $g'(0) = 0$. Now let's analyze $g(x)$. Since f is continuously differentiable, g is also continuously differentiable. Since E is open and $x_0 \in E$, there exists $r_0 > 0$ s.t. $B(x_0, r_0) \subseteq E$. Hence, g is defined on $B(0, r_0)$. (For $x \in B(0, r_0)$, $\|x\| < r_0$, so $\|(x + x_0) - x_0\| < r_0$, i.e. $x + x_0 \in B(x_0, r_0) \subseteq E$, so $g(x)$ is well-defined.)

Thus, for all i, j, $\exists r > 0$ s.t.

$$\left|\frac{\partial g_j}{\partial x_i}\right| \leq \frac{1}{2\sqrt{n^5}} \text{ for } x \in B(0, r)$$

since $\frac{\partial g_j(0)}{\partial x_i} = 0$ (since $g'(0) = 0$ and thus g's Jacobian matrix at 0 is 0 matrix) and $\frac{\partial g_j}{\partial x_i}$ is continuous, and hence

$$\left|\frac{\partial g_j(x)}{\partial x_i} - \frac{\partial g_j(0)}{\partial x_i}\right| \leq \frac{1}{2\sqrt{n^5}} \text{ if } \|x\| < r \text{ for some small enough } r.$$

By this, we can choose r small enough so that g is defined on $B(0, r)$ and

$$\left\|\frac{\partial g(x)}{\partial x_j}\right\| = \left(\sum_{i=1}^{n} \left(\frac{\partial g_i(x)}{\partial x_j}\right)^2\right)^{\frac{1}{2}} \leq \frac{1}{2n^2} \text{ for all } x \in B(0, r), \quad j = 1, 2, \ldots$$

Now if $x, y \in B(0,r)$, then $x + t(y-x) \in B(0,r)$ if $0 \le t \le 1$. Also,

$$g(y) - g(x) = \int_0^1 \left(\frac{\mathrm{d}}{\mathrm{d}t} \left[g(x + t(y-x)) \right] \right) \mathrm{d}t = \int_0^1 \left[Dg(x + t(y-x)) \circ (y-x) \right] \mathrm{d}t$$
$$= \int_0^1 \left(D_{(y-x)} g(x + t(y-x)) \right) \mathrm{d}t.$$

Now we estimate $D_v g(x)$ for $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$.

$$\|D_v g(x)\| = \left\| \sum_{j=1}^n v_j \frac{\partial g(x)}{\partial x_j} \right\| \le \sum_{j=1}^n \|v_j\| \left\| \frac{\partial g(x)}{\partial x_j} \right\| \le \sum_{j=1}^n \|v\| \frac{1}{2n^2} = \frac{\|v\|}{2n} \le \frac{\|v\|}{2}.$$

Thus,

$$\left\| D_{(y-x)} g(x + t(y-x)) \right\| \le \frac{\|x-y\|}{2}.$$

Hence, we have

$$\|g(y) - g(x)\| = \left\| \int_0^1 \left(D_{(y-x)} g(x + t(y-x)) \right) \mathrm{d}t \right\| \le \frac{\|x-y\|}{2}.$$

Recall $A^{-1}(f(x_0 + x) - f(x)) = g(x) + x$ and $g(0) = 0$, and we have shown

$$\|g(x) - g(y)\| \le \frac{\|x-y\|}{2} \text{ for } x, y \in B(0,r).$$

Let $\Phi(x) = x + g(x)$. By Lemma 1.6.1, $\Phi : B(0,r) \to \mathbb{R}^n$ is one to one and $B\left(0, \frac{r}{2}\right) \subseteq \Phi\left(B(0,r)\right)$. Since Φ is one to one on $B(0,r)$. Let $W = \Phi^{-1}\left(B\left(0, \frac{r}{2}\right)\right)$, then $\Phi : W \to B\left(0, \frac{r}{2}\right)$ is one to one and onto. Now we know Φ is continuous (since g is composition of differentiable and thus continuous map, so g is continuous, and thus Φ is continuous), so W is open (preimage of open set is open for continuous map). Also, $0 \in W$ since $\Phi(0) = 0 \in B\left(0, \frac{r}{2}\right)$. Now we claim $\Phi^{-1} : B\left(0, \frac{r}{2}\right) \to W$ is continuous. Note that for all $x, y \in W$

$$\|\Phi(x) - \Phi(y)\| = \|x - y + g(x) - g(y)\| \ge \|x-y\| - \|g(x) - g(y)\| \ge \|x-y\| - \frac{1}{2}\|x-y\| = \frac{\|x-y\|}{2}.$$

Let $x = \Phi^{-1}(u)$ and $y = \Phi^{-1}(v)$ for $u, v \in B\left(0, \frac{r}{2}\right)$, then

$$\left\| \Phi\left(\Phi^{-1}(u)\right) - \Phi\left(\Phi^{-1}(v)\right) \right\| \ge \frac{1}{2} \|\Phi^{-1}(u) - \Phi^{-1}(v)\|.$$

Thus,

$$\|u - v\| \ge \frac{1}{2} \left\| \Phi^{-1}(u) - \Phi^{-1}(v) \right\|.$$

This gives

$$\left\| \Phi^{-1}(u) - \Phi^{-1}(v) \right\| \le 2\|u - v\|,$$

and implies Φ^{-1} is continuous. Thus, $\Phi : W \to B\left(0, \frac{r}{2}\right)$ is a homomorphism, i.e. Φ is continuous and Φ^{-1} is also continuous. Since $\Phi(0) = 0$ and Φ is one-to-one, $\Phi^{-1}(0) = 0$. Also, note that

$$\lim_{z \to 0} \frac{\left\| \Phi^{-1}(z) - \Phi^{-1}(0) - z \right\|}{\|z\|} = \lim_{z \to 0} \frac{\left\| \Phi^{-1}(z) - z \right\|}{\|z\|}.$$

Note that for $z \neq 0$, if we let $\Phi^{-1}(z) = x$, then $z = \Phi(x) = x + g(x)$, and $x \neq 0$, so

$$\frac{\left\| \Phi^{-1}(z) - z \right\|}{\|z\|} = \frac{\|x - (x + g(x))\|}{\|z\|} = \frac{\|g(x)\|}{\|z\|} = \frac{\|g(x)\|}{\|x\|} \cdot \frac{\|x\|}{\|z\|}.$$

Hence,

$$\lim_{z\to 0}\frac{\left\|\Phi^{-1}(z)-z\right\|}{\|z\|}=\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}\cdot\frac{\|x\|}{\|z\|}=\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}\cdot\frac{\left\|\Phi^{-1}(z)\right\|}{\|z\|}$$

Now if $z\to 0$, then since Φ^{-1} is continuous, so $x=\Phi^{-1}(z)\to\Phi^{-1}(0)=0$. Also, when $z\to 0$, we know

$$\left\|\Phi^{-1}(z)-\Phi^{-1}(0)\right\|\leq 2\|z-0\|\Rightarrow\frac{\left\|\Phi^{-1}(z)\right\|}{\|z\|}\leq 2,$$

and since

$$\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}=\lim_{x\to 0}\frac{\|g(x)\|}{\|x\|}=0,$$

so by Squeeze theorem,

$$\lim_{z\to 0}\frac{\left\|\Phi^{-1}(z)-z\right\|}{\|z\|}=0.$$

This shows $\left(\Phi^{-1}\right)'(0)=I$. Recall that

$$\Phi(x)=x+g(x)=A^{-1}(f(x+x_0)-f(x_0))$$

and $\Phi:W\to B\left(0,\frac{r}{2}\right)$ is a homomorpism. Now set

$$U=x_0+W,\quad V=f(x_0)+A\left(B\left(0,\frac{r}{2}\right)\right).$$

Since W is open and A is a linear isomorphism, both U and V are open subsets of $\mathbb{R}^n$. Moreover, $x_0\in U$ and $f(x_0)\in V$. We claim that f maps U bijectively to V. Let $u=x+x_0\in U$, where $x\in W$, then since $\Phi(W)=B\left(0,\frac{r}{2}\right)$, we have $\Phi(x)\in B\left(0,\frac{r}{2}\right)$, and thus

$$f(u)=f(x+x_0)=f(x_0)+A\Phi(x)\in f(x_0)+A\left(B\left(0,\frac{r}{2}\right)\right)=V.$$

This shows $f(U)\subseteq V$. Conversely, let $y\in V$, then

$$A^{-1}(y-f(x_0))\in B\left(0,\frac{r}{2}\right).$$

Because $\Phi:W\to B\left(0,\frac{r}{2}\right)$ is surjective, there exists $x\in W$ s.t.

$$\Phi(x)=A^{-1}(y-f(x_0)).$$

For this x we have

$$f(x+x_0)=f(x_0)+A\Phi(x)=y.$$

This shows $V\subseteq f(U)$ and thus $f(U)=V$, i.e. $f:U\to V$ is surjective. Since Φ is one to one, so the x above is unique. It follows that f is one to one on U. Thus, $f:U\to V$ is bijective.

Now for $y\in V$ we know $y=f(x+x_0)$ for some $x\in W$, so $x+x_0=f^{-1}(y)$. Recall

$$\Phi(x)=A^{-1}(f(x+x_0)-f(x_0))=A^{-1}(y-f(x_0)).$$

Hence,

$$x=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right)\Rightarrow f^{-1}(y)-x_0=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right).$$

Thus,

$$f^{-1}(y)=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right)+x_0.$$

This shows

$$\left(f^{-1}\right)'(f(x_0))=\left(\Phi^{-1}\right)'\left(A^{-1}(f(x_0)-f(x_0))\right)(A^{-1})'(0)=(\Phi^{-1})'(0)\circ A^{-1}=IA^{-1}=(f'(x_0))^{-1}$$

since $x_0\in U$ so $f(x_0)\in V$ and every linear transformation is differentiable at every point where $\left(A^{-1}\right)(x)=A^{-1}$ for all x. Thus, we're done. ■

Remark 1.7.1. Since $\det f'(x_0) \neq 0$ and det is continuous, there exists a neighborhood B of x_0 such that $f'(x)$ is invertible for all $x \in B$.

Applying the inverse function theorem at each point $x \in B$, we obtain neighborhoods U_x such that f^{-1} is differentiable on $f(U_x)$ and

$$(f^{-1})'(f(x)) = (f'(x))^{-1}.$$

Restrict to a sufficiently small neighborhood $U' \subset B$ of x_0 and set $V' = f(U')$. Then f^{-1} is differentiable on V'.

Moreover, for $y \in V'$,

$$(f^{-1})'(y) = (f'(f^{-1}(y)))^{-1}.$$

Since f^{-1} is continuous, f' is continuous, and matrix inversion is continuous on the set of invertible matrices, it follows that $(f^{-1})'$ is continuous on V'.

Hence, f^{-1} is continuously differentiable on V'.

Lecture 7

Remark 1.7.2. The requirement of $f'(x_0)$ is invertible is necessary to get the differentiability of f^{-1} at x_0. 17 March.

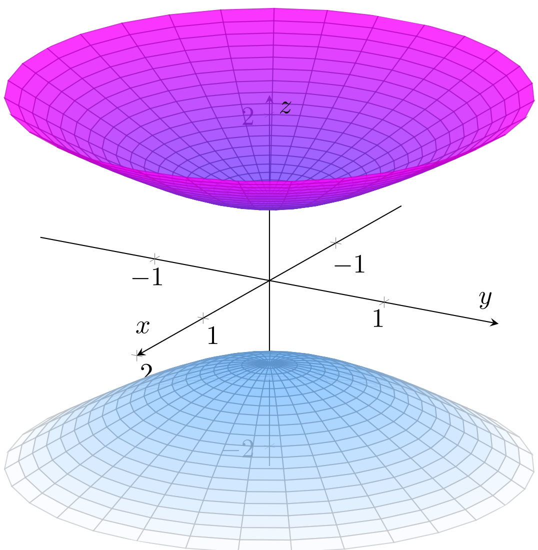
1.8 The Implicit Function Theorem

In $\mathbb{R}^2$, $S = \{(x, y) \mid x^2 + y^2 = 1\}$ is a circle. However, S is not the graph of one function. S can be expressed as the union of the graph of four functions:

$$\begin{cases} y = \sqrt{1 - x^2} \text{ for } -1 < x < 1 \\ y = -\sqrt{1 - x^2} \text{ for } -1 < x < 1 \\ x = \sqrt{1 - y^2} \text{ for } -1 < y < 1 \\ x = -\sqrt{1 - y^2} \text{ for } -1 < y < 1. \end{cases}$$

For a $\mathbb{R}^3$ case, consider $S = \{(x, y, z) \mid x^2 + y^2 - z^2 + 1 = 0\}$, then S is the union of 2 graph:

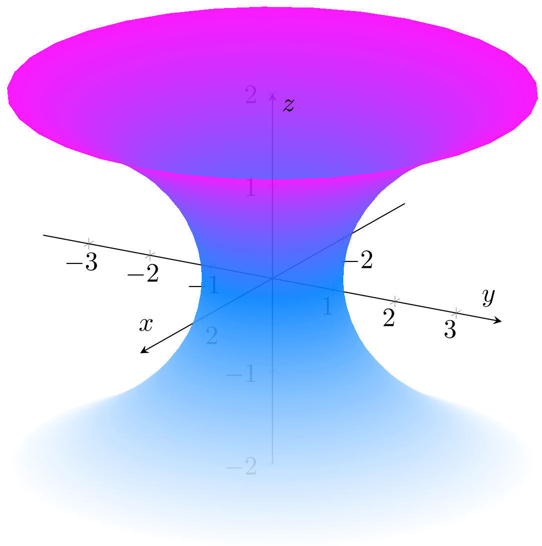
$$\begin{cases} z = \sqrt{x^2 + y^2 + 1} \text{ for } z > 0 \\ z = -\sqrt{x^2 + y^2 + 1} \text{ for } z < 0. \end{cases}$$



For another example, consider $S = \left\{(x,y,z) \mid x^2+y^2 = \cosh^2(z)\right\}$, where

$$\cosh(z) = \frac{e^z+e^{-z}}{2}.$$

Then S is symmetric with respect to z.



Theorem 1.8.1 (Implicit Function Theorem). Let E be an open subset of $\mathbb{R}^n$, let $f : E \to \mathbb{R}$ be continuously differentiable, and let $y = (y_1, y_2, \dots, y_n) \in E$ s.t. $f(y) = 0$ and $\frac{\partial f(y)}{\partial x_n} \neq 0$. Then, $\exists$ an open subset $U \subseteq \mathbb{R}^{n-1}$ containing $(y_1, y_2, \dots, y_{n-1})$, an open subset $V \subseteq E$ containing y, and a function $g : U \to \mathbb{R}$ s.t.

$$g(y_1, y_2, \dots, y_{n-1}) = y_n$$

and

$$\{x = (x_1, \dots, x_n) \in V : f(x) = 0\} = \{(x_1, \dots, x_{n-1}, g(x_1, \dots, x_{n-1})) : (x_1, \dots, x_{n-1}) \in U\}.$$

In other words, the set $\{x \in V : f(x) = 0\}$ is a graph of a function over U. Moreover, g is differentiable at $(y_1, \dots, y_{n-1})$ and we have

$$\frac{\partial g}{\partial x_j}(y_1, y_2, \dots, y_{n-1}) = -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)}$$

for all $1 \leq j \leq n-1$.

Proof. Define $F : E \to \mathbb{R}^n$ by

$$(x_1, x_2, \dots, x_n) \to (x_1, x_2, \dots, x_{n-1}, f(x_1, x_2, \dots, x_n)).$$

Since $f \in C^1(E)$, so $F \in C^1(E)$. Thus,

$$DF(x) = \begin{bmatrix} \nabla x_1 \\ \nabla x_2 \\ \vdots \\ \nabla x_{n-1} \\ \nabla f \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ & & \ddots & \\ 0 & \cdots & 1 & 0 \\ \frac{\partial f}{\partial x_1}(x) & \cdots & \frac{\partial f}{\partial x_{n-1}}(x) & \frac{\partial f}{\partial x_n}(x) \end{bmatrix},$$

which gives

$$\det(DF(x)) = \frac{\partial f(x)}{\partial x_n}.$$

Hence,

$$\det(DF(y)) = \frac{\partial f(y)}{\partial x_n} \neq 0,$$

so $F'(y)$ is invertible. Now we can apply inverse function theorem, so $\exists$ an open set $V \subseteq E$ and $W \subseteq \mathbb{R}^m$ with $y \in V$ and $F(y) = (y_1, \ldots, y_{n-1}, 0) \in W$ s.t. $F : V \to W$ is a C^1 diffeomorphism. Write the inverse in coordinate

$$F^{-1}(\underbrace{u_1, u_2, \ldots, u_n}_{u}) = (h_1(u), h_2(u), \ldots, h_n(u)).$$

Since $F(F^{-1}(u)) = u$ for $u \in W$, so

$$(h_1(u), \ldots h_{n-1}(u), f(h_1(u), \ldots, h_n(u))) = F(h_1(u), h_2(u), \ldots, h_n(u)) = (u_1, u_2, \ldots, u_n).$$

Hence,

$$\begin{aligned} h_1(u) &= u_1 \\ h_2(u) &= u_2 \\ &\vdots \\ h_{n-1}(u) &= u_{n-1} \\ f(h_1(u), \ldots, h_n(u)) &= f(u_1, u_2, \ldots, u_{n-1}, h_n(u)) = u_n. \end{aligned}$$

Thus, for $u_1, \ldots, u_n \in W$,

$$F^{-1}(u_1, \ldots, u_n) = (u_1, \ldots, u_{n-1}, h_n(u)).$$

By this, we can plug $u_n = 0$ into it and get

$$F^{-1}(u_1, \ldots, u_{n-1}, 0) = (u_1, \ldots, u_{n-1}, h_n(u_1, \ldots, u_{n-1}, 0)),$$

where

$$0 = u_n = f(u_1, \ldots, u_{n-1}, h_n(u_1, \ldots, u_{n-1}, 0)).$$

Now we want to restrict $u_n = 0$ and define g. Set

$$U = \left\{x' = (x_1, \ldots, x_{n-1}) \in \mathbb{R}^{n-1}, (x', 0) \in W\right\}.$$

Then U is open and contains $(y_1, \ldots, y_{n-1})$ since

$$F(y_1, \ldots, y_n) = (y_1, \ldots, y_{n-1}, f(y_1, \ldots, y_n)) = (y_1, \ldots, y_{n-1}, 0) \in W.$$

Define $g : U \to \mathbb{R}$ by

$$g(x') = g(x_1, \ldots, x_{n-1}) = h_n(x_1, \ldots, x_{n-1}, 0).$$

Since $f(y_1, \ldots, y_n) = 0$, we know $F(y) = (y_1, \ldots, y_n, 0)$. Hence,

$$(y_1, \ldots, y_n) = y = F^{-1}(F(y)) = F^{-1}(y_1, \ldots, y_n, 0) = (y_1, \ldots, y_{n-1}, h_n(y_1, \ldots, y_{n-1}, 0)).$$

Thus,

$$y_n = h_n(y_1, \ldots, y_{n-1}, 0) = g(\underbrace{y_1, \ldots, y_{n-1}}_{y'}) \Rightarrow g(y') = y_n.$$

Moreover, g is differentiable since h_n is differentiable.

Now we want to show the zero set is the graph of g.

Claim 1.8.1. $\{x \in V : f(x) = 0\} = \{(x', g(x')) : x' \in U\}$.

Proof. Let $A = \{x \in V : f(x) = 0\}$ and $B = \{(x', g(x')) : x' \in U\}$. We first show $A \subseteq B$. Let $x = (x', x_n) \in V$ and $f(x', x_n) = 0$. Then

$$F(x) = (x', f(x', x_n)) = (x', 0) \in W.$$

Since $x' \in U$,

$$x = F^{-1}(F(x)) = F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x')).$$

Hence, $A \subseteq B$.

Now we show $B \subseteq A$. Suppose $x' \in U$, then

$$x = (x', g(x')) = (x', h_n(x', 0)).$$

Thus,

$$F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x')).$$

Thus,

$$(x', 0) = F(F^{-1}(x', 0)) = F(x', g(x')) = (x', f(x', g(x'))).$$

Thus,

$$f(x', g(x')) = 0,$$

which shows $x \in A$. ⊛

Now for $(x_1, x_2, \ldots, x_{n-1}) \in U$, we know

$$f(x_1, x_2, \ldots, x_{n-1}, g(x_1, x_2, \ldots, x_{n-1})) = 0.$$

Fix $j \in [n-1]$, and set $y' = (y_1, \ldots, y_{n-1})$. Consider

$$\psi(t) \coloneqq f(y' + te_j, g(y' + te_j)),$$

defined for t near 0. Note that $\psi(t) \equiv 0$ for all t sufficiently close to 0. Hence, $\psi'(0) = 0$. Since f is differentiable at $y = (y', g(y'))$ and g is differentiable at y', we may apply the chain rule to compute $\psi'(0)$. Suppose $\psi(t) = f(\gamma(t))$, where $\gamma : \mathbb{R} \to \mathbb{R}^n$ is defined by

$$t \mapsto (y' + te_j, g(y' + te_j)).$$

Now since

$$\psi'(t) = f'(\gamma(t)) \circ \gamma'(t),$$

and f is a function from a subset of $\mathbb{R}^n$ to $\mathbb{R}$, so

$$f'(\gamma(t)) = \nabla f(\gamma(t)) = \left(\frac{\partial f(\gamma(t))}{\partial x_1}, \frac{\partial f(\gamma(t))}{\partial x_2}, \ldots, \frac{\partial f(\gamma(t))}{\partial x_n}\right),$$

which gives

$$f'(\gamma(0)) = \left(\frac{\partial f(y)}{\partial x_1}, \frac{\partial f(y)}{\partial x_2}, \ldots, \frac{\partial f(y)}{\partial x_n}\right).$$

Also,

$$\gamma'(t) = \left(0, \ldots, 1, \ldots, 0, \frac{\mathrm{d}g(y' + te_j)}{\mathrm{d}t}\right) \Rightarrow \gamma'(0) = \left(0, \ldots, 1, \ldots, 0, \frac{\mathrm{d}g(y')}{\mathrm{d}t}\right),$$

and note that

$$\frac{\mathrm{d}g(y' + te_j)}{\mathrm{d}t} = \nabla g(y' + te_j) \circ e_j \Rightarrow \frac{\mathrm{d}g(y')}{\mathrm{d}t} = \nabla g(y') \circ e_j = \frac{\partial g(y')}{\partial x_j}.$$

Hence,

$$\begin{aligned}\psi'(0) &= \left(\frac{\partial f(y)}{\partial x_1}, \frac{\partial f(y)}{\partial x_2}, \dots, \frac{\partial f(y)}{\partial x_n}\right) \circ \left(0, \dots, 1, \dots, 0, \frac{\partial g(y')}{\partial t}\right) \\ &= \frac{\partial f}{\partial x_j}(y) + \frac{\partial f}{\partial x_n}(y)\frac{\partial g}{\partial x_j}(y'),\end{aligned}$$

and thus

$$0 = \psi'(0) = \frac{\partial f}{\partial x_j}(y) + \frac{\partial f}{\partial x_n}(y)\frac{\partial g}{\partial x_j}(y').$$

Now since $\frac{\partial f}{\partial x_n}(y) \neq 0$, we have

$$\frac{\partial g}{\partial x_j}(y') = -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)}.$$

■

Example 1.8.1. Let $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid xy + yz + xz = -1\right\}$, which we rewrite as

$$\left\{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) = 0\right\},$$

where $f : \mathbb{R}^3 \to \mathbb{R}$ is the function

$$f(x, y, z) = xy + yz + xz + 1.$$

Clearly f is continuously differentiable, and $\frac{\partial f}{\partial z} = y + x$. Thus, for any $(x_0, y_0, z_0) \in S$ sith $y_0 + x_0 \neq 0$, one can write this surface (near (x_0, y_0, z_0)) as a graph of the form

$$\{(x, y, g(x, y)) : (x, y) \in U\}$$

for some open set U containing (x_0, y_0), and some function g which is differentiable at (x_0, y_0). Indeed one can implicitly obtain

$$\frac{\partial g}{\partial x}(x_0, y_0) = -\frac{y_0 + z_0}{y_0 + x_0}, \quad \text{and} \quad \frac{\partial g}{\partial y}(x_0, y_0) = -\frac{x_0 + z_0}{y_0 + x_0}.$$

Chapter 2

Lebesgue measure

Lecture 8

To overcome the limitations of the Riemann integral, particularly its inability to handle pointwise limits of functions, this chapter constructs a more robust theory of integration starting with the concept of measure. 19 March.

2.1 The goal: Lebsgue measure

Our goal is to define a concept of measurable set, which will be a special kind of subset of $\mathbb{R}^n$. For every such measurable set $\Omega \subseteq \mathbb{R}^n$, we'll define the Lebesgue measure $m(\Omega)$ to be a number in $[0, \infty]$.

Definition 2.1.1 (Measurable set). The concept of measurable set will obey the following properties:

(i) **(Borel property)** Every open set in $\mathbb{R}^n$ is measurable, as is every closed set.

(ii) **(Complementarity)** If Ω is measurable, then $\mathbb{R}^n \setminus \Omega$ is also measurable.

(iii) **(Boolean algebra propery)** If $(\Omega_j)_{j\in J}$ is any finite collection of measurable sets, then $\bigcup_{j\in J} \Omega_j$ and $\bigcap_{j\in J} \Omega_j$ are also measurable.

(iv) **(σ-algebra property)** If $(\Omega_j)_{j\in J}$ are any countable collection of measurable sets (so J is countable), then $\bigcup_{j\in J} \Omega_j$ and $\bigcap_{j\in J} \Omega_j$ are also measurable.

Note that some of these properties are redundant; for instance, (iv) will imply (iii), and once one knows all open sets are measurable, (ii) will imply that all closed sets are measurable also. The properties (i)-(iv) will ensure that virtually every set one cares about is measurable; though there do exist non-measurable sets.

Definition 2.1.2 (Lebsegue measure). To every measurable set Ω, we associate the Lebsegue measure $m(\Omega)$ of Ω, which will obey the following properties:

(v) **(Empty set)** The empty set $\varnothing$ has measure $m(\varnothing) = 0$.

(vi) **(Positivity)** We have $0 \leq m(\Omega) \leq +\infty$ for every measurable set Ω.

(vii) **(Monotocity)** If $A \subseteq B$, and A and B are both measurable, then $m(A) \leq m(B)$.

(viii) **(Finite sub-additivity)** If $(A_j)_{j\in J}$ are a finite collection of measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) \leq \sum_{j\in J} m(A_j).$$

(ix) **(Finite additivity)** If $(A_j)_{j\in J}$ are a finite collection of *disjoint* measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m(A_j).$$

(x) **(Countable sub-additivity)** If $(A_j)_{j\in J}$ are a countable collection of measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) \leq \sum_{j\in J} m(A_j).$$

(xi) **(Countable additivity)** If $(A_j)_{j\in J}$ are a countable collection of *disjoint* measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m(A_j).$$

(xii) **(Normalization)** The unit cube

$$[0,1]^n = \{(x_1,\ldots,x_n)\in\mathbb{R}^n : 0\leq x_j\leq 1 \text{ for all } 1\leq j\leq n\}$$

has measure $m([0,1]^n)=1$.

(xiii) **(Translation invariance)** If Ω is a measurable set, and $x\in\mathbb{R}^n$, then $x+\Omega := \{x+y : y\in\Omega\}$ is also measurable, and $m(x+\Omega)=m(\Omega)$.

Again, many of these properties are redundant; for instance, the countable additivity property can be used to deduce the finite additivity property, which in turn can be used to derive monotonicity (when combined with the positivity property). One can also obtain the sub-additivity properties from the additivity ones. Note that $m(\Omega)$ can be $+\infty$, and so in particular some of the sums in the above properties may also equal $+\infty$; in this chapter we adopt the convention that an infinite sum $\sum_{j\in J} a_j$ of non-negative quantities a_j is equal to $+\infty$ if the sum is not absolutely convergent. (Since everything is non0negative we will never have to deal with indeterminate forms such as $(-\infty)+(+\infty)$.)

Our goal for this chapter can then be stated:

Theorem 2.1.1 (Existence of Lebsegue measure). There exists a concept of a measurable set and a way to assign a number $m(\Omega)$ to every measurable set $\Omega\subseteq\mathbb{R}^n$, which obeys all of the properties (i) to (xiii).

Lecture 9

2.2 First attempt: Outer Measure

24 March.

Definition 2.2.1. An open box B in $\mathbb{R}^n$ is any set of the form

$$B=\prod_{i=1}^n (a_i,b_i) := \{(x_1,\ldots,x_n)\in\mathbb{R}^n : x_i\in(a_i,b_i) \text{ for all } 1\leq i\leq n\},$$

where $b_i\geq a_i$ are real numbers. We define the volume $\mathrm{vol}(B)$ of this box to be the number

$$\mathrm{vol}(B) := \prod_{i=1}^n (b_i-a_i).$$

Definition 2.2.2. Let $\Omega\subseteq\mathbb{R}^n$ be a subset of $\mathbb{R}^n$. We say a collection $(B_j)_{j\in J}$ of boxes cover Ω iff $\Omega\subseteq\bigcup_{j\in J} B_j$.

Remark 2.2.1. An index set J is countable if it is finite or $\exists$ a one-to-one map $f : J \to \mathbb{N}$ where $\mathbb{N} = \{1, 2, \dots\}$ is the set of natural numbers.

Remark 2.2.2. The set of rational numbers is countable. We usually denote it by

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid q \neq 0, p, q \in \mathbb{Z} \right\}.$$

Definition 2.2.3 (Outer measure). If Ω is a set in $\mathbb{R}^n$, we define the outer measure $m^*(\Omega)$ of Ω to be the quantity

$$m^*(\Omega) := \inf \left\{ \sum_{j \in J} \mathrm{vol}(B_j) : (B_j)_{j \in J} \text{ is a collection of boxes covers } \Omega \text{ and } J \text{ at most countable.} \right\}.$$

Note that $m^*(\Omega)$ may be ∞. Also, since $\mathrm{vol}(B_j) \geq 0$, so $m^*(\Omega) \geq 0$.

Lemma 2.2.1. The space $\mathbb{R}^n$ can be covered by countably many translates of the open cube $\left(-\frac{1}{2}, \frac{1}{2}\right)^n$.

Proof. For each $m \in \mathbb{Z}^n$, define

$$Q_m := \frac{m}{2} + \left(-\frac{1}{2}, \frac{1}{2}\right)^n.$$

We claim $\mathbb{R}^n = \bigcup_{m \in \mathbb{Z}^n} Q_m$. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$. For each i, set $m_i := \lfloor 2x_i \rfloor \in \mathbb{Z}$, then

$$\frac{m_i}{2} \leq x_i < \frac{1}{2} + \frac{m_i}{2} \Rightarrow x \in \frac{m}{2} + \left(-\frac{1}{2}, \frac{1}{2}\right)^n.$$

Note that $\mathbb{Z}^n$ is countable since $\mathbb{Z}$ is countable, and thus we're done. ■

Lemma 2.2.2 (Properties of outer measure). Outer measure has the following 6 properties:

(v) **(Empty set)** The empty set $\varnothing$ has outer measure 0, i.e. $m^*(\varnothing) = 0$.

(vi) **(Positivity)** We have $0 \leq m^*(\Omega) \leq \infty$ for all measurable set Ω.

(vii) **(Monotocity)** If $A \subseteq B \subseteq \mathbb{R}^n$, then $m^*(A) \leq m^*(B)$.

(viii) **(Finite sub-additivity)** If $(A_j)_{j \in J}$ are finite collection of subsets of $\mathbb{R}^n$, then

$$m^*\left(\bigcup_{j \in J} A_j \right) \leq \sum_{j \in J} m^*(A_j).$$

(x) **(Countable sub-additivity)** If $(A_j)_{j \in J}$ are countable collection of subsets of $\mathbb{R}^n$, then

$$m^*\left(\bigcup_{j \in J} A_j \right) \leq \sum_{j \in J} m^*(A_j).$$

(xiii) **(Translation invariance)** If Ω is a subset of $\mathbb{R}^n$, and $x \in \mathbb{R}^n$, then $m^*(x + \Omega) = m^*(\Omega)$.

proof of (v). From the definition we know $m^*(\Omega) \geq 0$ for all sets $\Omega \subseteq \mathbb{R}^n$. Hence, $m^*(\varnothing) \geq 0$. Now given $\varepsilon > 0$, we know $\left(0, \varepsilon^{\frac{1}{n}}\right)^n$ is an open box that covers $\varnothing$. Thus,

$$m^*(\varnothing) \leq \mathrm{vol}\left(\left(0, \varepsilon^{\frac{1}{n}}\right)^n \right) = \varepsilon.$$

Thus, $m^*(\Omega) = 0$. ■

proof of (vi). It follows from the definition of m^*. ■

proof of (vii). If $m^*(B) = \infty$, then this is trivial. Now suppose $m^*(B) < \infty$. Recall

$$m^*(B) = \inf \left\{ \sum_{j\in J} \mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are open boxes cover } B \right\}.$$

Also, for any countable collection of boxes covering B, say $(B_j)_{j\in J}$, it also covers A, so

$$\left\{ \sum_{j\in J} \mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are open boxes cover } B \right\} \subseteq \left\{ \sum_{j\in J} \mathrm{vol}(A_j) : (A_j)_{j\in J} \text{ are open boxes cover } A \right\},$$

which shows

$$m^*(A) \leq m^*(B).$$

■

proof of (x). If one of $m^*(A_j) = \infty$, then

$$\infty = m^*(A_j) \leq m^* \left(\bigcup_{j\in J} A_j \right),$$

so we're done. Now we assume $m^*(A_j) < \infty$ for all $j \in J$. Given $\varepsilon > 0$, $\exists$ open boxes $B_{j,k}$ s.t. $A_j \subseteq \bigcup_{k=1}^{\infty} B_{j,k}$ and

$$\sum_{k=1}^{\infty} \mathrm{vol}(B_{j,k}) \leq m^*(A_j) + \frac{\varepsilon}{2^j}$$

since the definition of m^* is infimum. Now since J is countable, so we can re-label it to $\mathbb{N}$. Hence,

$$\bigcup_{j=1}^{\infty} A_j \subseteq \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{\infty} B_{j,k},$$

and note that R.H.S. is the union of a collection of boxes, so $\bigcup_{j=1}^{\infty} A_j$ is covered by them. This shows

$$m^* \left(\bigcup_{j=1}^{\infty} A_j \right) \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \mathrm{vol}(B_{j,k}) \leq \sum_{j=1}^{\infty} m^*(A_j) + \frac{\varepsilon}{2^j} = \sum_{j=1}^{\infty} m^*(A_j) + \varepsilon.$$

Now since this is true for all $\varepsilon > 0$, so

$$m^* \left(\bigcup_{j=1}^{\infty} A_j \right) \leq \sum_{j=1}^{\infty} m^*(A_j).$$

■

Proposition 2.2.1 (Outer measure of closed box)**.** For any closed box

$$B = \prod_{i=1}^{n} [a_i, b_i] = \{(x_1, x_2, \ldots, x_n) \mid a_i \leq x_i \leq b_i \text{ for } 1 \leq i \leq n\},$$

we have

$$m^*(B) = \prod_{i=1}^{n} (b_i - a_i).$$

Proof. Write

$$\mathrm{vol}(B) = \prod_{i=1}^{n}(b_i - a_i).$$

We first show that

$$m^*(B) \leq \mathrm{vol}(B).$$

Let $\varepsilon > 0$, then the box

$$U_\varepsilon \coloneqq \prod_{i=1}^{n}(a_i - \varepsilon, b_i + \varepsilon)$$

contains B. Thus,

$$m^*(B) \leq \mathrm{vol}(U_\varepsilon) = \prod_{i=1}^{n}(b_i - a_i + 2\varepsilon).$$

Take $\varepsilon \to 0$, then we can show

$$m^*(B) \leq \mathrm{vol}(B).$$

Now we show

$$m^*(B) \geq \prod_{i=1}^{n}(b_i - a_i).$$

Claim 2.2.1. For any countable family of open boxes $\{B_j\}_{j\in J}$ covering B. We have

$$\mathrm{vol}(B) \leq \sum_{j\in J}\mathrm{vol}(B_j).$$

Proof. Since B is compact, $\exists$ a finite subcover of $(B_j)_{j\in J}$

$$(U_j)_{j=1}^{N} \text{ s.t. } B \subseteq \bigcup_{j=1}^{N} U_j.$$

Thus,

$$\sum_{j=1}^{N}\mathrm{vol}(U_j) \leq \sum_{j\in J}\mathrm{vol}(B_j).$$

Choose a closed box Q s.t. $B \subseteq Q$ and $U_k \subseteq Q$ for $1 \leq k \leq N$. Define functions on Q by

$$f(x) = \begin{cases} 1, & \text{if } x \in B; \\ 0, & \text{if } x \in Q \setminus B. \end{cases} \qquad f_k(x) = \begin{cases} 1, & \text{if } x \in U_k; \\ 0, & \text{if } x \in Q \setminus U_k. \end{cases}$$

Thus,

$$\int_Q f(x)\,\mathrm{d}x = \mathrm{vol}(B) \text{ and } \int_Q f_k(x)\,\mathrm{d}x = \mathrm{vol}(U_k).$$

Note that this is integrable since $B, (U_k)_{k=1}^{N}$ are all boxes and Q is also a box. Now since

$$f(x) \leq \sum_{j=1}^{k} f_j(x) \text{ on } Q.$$

(If $x \in B$, then $x \in U_j$ for some j.) Thus,

$$\mathrm{vol}(B) = \int_Q f(x)\,\mathrm{d}x \leq \int_Q \sum_{j=1}^{k} f_j(x)\,\mathrm{d}x = \sum_{j=1}^{k}\int_Q f_j(x)\,\mathrm{d}x = \sum_{j=1}^{k}\mathrm{vol}(U_j) \leq \sum_{j\in J}\mathrm{vol}(B_j).$$

⊛

This implies vol(B) is a lower bound of

$$\left\{ \sum_{j\in J} \mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are boxes covering } B \right\},$$

so by the definition of m^*,

$$m^*(B) \geq \mathrm{vol}(B).$$

■

Lecture 10

26 March.

Corollary 2.2.1. For any open box

$$B = \prod_{i=1}^{n}(a_i, b_i) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid a_i < x_i < b_i \text{ for } 1 \leq i \leq n\},$$

we have $m^*(B) = \prod_{i=1}^{n}(b_i - a_i)$.

Proof. First since B cover itself, we have $m^*(B) \leq \mathrm{vol}(B) = \prod_{i=1}^{n}(b_i - a_i)$. Choose $\varepsilon_i < \frac{b_i - a_i}{2}$. Then,

$$[a_i + \varepsilon_i, b_i - \varepsilon_i] \subseteq (a_i, b_i).$$

Hence,

$$\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i] \subseteq \prod_{i=1}^{n}(a_i, b_i),$$

which shows

$$m^*\left(\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i]\right) \leq m^*(B),$$

and by previous proposition, we have

$$\prod_{i=1}^{n}(b_i - a_i - 2\varepsilon_i) = m^*\left(\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i]\right) \leq m^*(B).$$

Now pick all $\varepsilon_i \to 0$, then we have $m^*(B) \geq \mathrm{vol}(B)$. ■

Example 2.2.1. Let $r > 0$, then $(-r, r) \subseteq \mathbb{R}$, and thus

$$2r = m^*((-r, r)) \leq m^*(\mathbb{R})$$

for all r, so $m^*(\mathbb{R}) = \infty$. Similarly, $m^*(\mathbb{R}^n) = \infty$.

Example 2.2.2.

$$\mathbb{Q} = \text{ the set of rational numbers } = \bigcup_{q\in\mathbb{Q}} \{q\},$$

and $\mathbb{Q}$ is countable, so

$$m^*(\mathbb{Q}) \leq \sum_{q\in\mathbb{Q}} m^*(\{q\}) = 0.$$

In particular, $\exists$ an countable family of open boxes (B_j) s.t. $\mathbb{Q} \subseteq \bigcup_{j\in J} B_j$ and

$$\sum_{j\in J} \mathrm{vol}(B_j) \leq m^*(\mathbb{Q}) + \varepsilon = \varepsilon.$$

Example 2.2.3. Since $\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q}$, we have

$$m^*(\mathbb{R}) \leq m^*(\mathbb{R} \setminus \mathbb{Q}) + m^*(\mathbb{Q}),$$

and since $m^*(\mathbb{R}) = \infty$ and $m^*(\mathbb{Q}) = 0$, we know $m^*(\mathbb{R} \setminus \mathbb{Q}) = \infty$.

Claim 2.2.2. $m^*([0,1] \setminus \mathbb{Q}) = 1$.

Proof. Since

$$[0,1] = ([0,1] \setminus \mathbb{Q}) \cup ([0,1] \cap \mathbb{Q}),$$

we have

$$1 = m^*([0,1]) \leq m^*([0,1] \setminus \mathbb{Q}) + m^*([0,1] \cap \mathbb{Q}) = m^*([0,1] \setminus \mathbb{Q})$$

since

$$[0,1] \cap \mathbb{Q} \subseteq \mathbb{Q} \text{ and } 0 \leq m^*([0,1] \cap \mathbb{Q}) \leq m^*(\mathbb{Q}) = 0.$$

Now since

$$[0,1] \setminus \mathbb{Q} \subseteq [0,1],$$

so

$$m^*([0,1] \setminus \mathbb{Q}) \leq m^*([0,1]) = 1.$$

Thus,

$$m^*([0,1] \setminus \mathbb{Q}) = 1.$$

⊛

Claim 2.2.3. Let

$$L = \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^2,$$

then $m^*(L) = 0$.

Proof. Let $L_n = \{(x,0) \mid -n \leq x \leq n\} \subseteq \mathbb{R}^2$, then

$$L_n \subseteq [-n, n] \times [-\varepsilon, \varepsilon],$$

so

$$0 \leq m^*(L_n) \leq (2n) \cdot (2\varepsilon).$$

Let $\varepsilon \to 0$, then we have

$$m^*(L_n) = 0.$$

Now since

$$L = \bigcup_{n=1}^{\infty} L_n,$$

so

$$m^*(L) \leq \sum_{n=1}^{\infty} m^*(L_n) = 0,$$

which shows $m^*(L) = 0$. ⊛

2.3 Outer measure is not additive

We only know that

$$m^*\left(\bigcup_{j \in J} A_j\right) \leq \sum_{j \in J} m^*(A_j)$$

if J is countable. To construct a measure, we require that if $A_i \cap A_j = \varnothing$ for all $i, j \in J$ and $i \neq j$,

$$m^*\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m^*(A_j).$$

Proposition 2.3.1 (Failure of countable additivity)**.** There exists a countable collection $(A_j)_{j\in J}$ of disjoint subsets of $\mathbb{R}$ s.t.

$$m^*\left(\bigcup_{j\in J} A_j\right) \neq \sum_{j\in J} m^*(A_j).$$

Proof. We shall construct such a family from the coset of $\mathbb{Q}$ in $\mathbb{R}$.

- Step 1. Coset of $\mathbb{Q}$. For $x \in \mathbb{R}$, let

  $$x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}.$$

  This is called a coset of $\mathbb{Q}$. Two properties of cosets:

  (1) $(x + \mathbb{Q}) \cap (y + \mathbb{Q}) \neq \varnothing$ implies $x + \mathbb{Q} = y + \mathbb{Q}$.

  (2) Every coset meets the interval $[0, 1]$. (We can pick $-x \leq q \leq 1 - x$ so that $0 \leq q + x \leq 1$)

- Step 2. Choose one representative from each coset. Let $\mathbb{R}/\mathbb{Q}$ denote the collection of all cosets. For each coset $A \in \mathbb{R}/\mathbb{Q}$. By (2) in Step 1, there exists $x_A \in [0, 1]$ s.t. $x_A \in A \cap [0, 1]$(By axiom of choice), we can choose one and define

  $$E = \{x_A : A \in \mathbb{R}/\mathbb{Q}, x_A \in [0, 1]\} \subseteq [0, 1].$$

- Step 3. Rational translates of E. Consider the set

  $$X := \bigcup_{q\in\mathbb{Q}\cap[-1,1]} (q + E).$$

  Since $E \subseteq [0, 1]$ and $-1 \leq q \leq 1$, we have $X \subseteq [-1, 2]$. Now we claim $[0, 1] \subseteq X$. Let y be any number in $[0, 1]$. Let $y \in [0, 1]$, then y is in some coset of $\mathbb{R}/\mathbb{Q}$. Thus, $\exists$ cosets A and $x_A \in E \subseteq [0, 1]$ s.t. $y \in A = x_A + \mathbb{Q} = y + \mathbb{Q}$. Let $y = x_A + q$ for some $q \in \mathbb{Q}$. Thus, $q = y - x_A \in [-1, 1]$. Thus, $q \in \mathbb{Q} \cap [-1, 1]$. Hence,

  $$y = q + x_A \in q + E \subseteq X.$$

  Thus,

  $$[0, 1] \subseteq X \subseteq [-1, 2].$$

- Step 4. the sets $q + E$ are pairwise disjoint. We now show that the family

  $$\{q + E\}_{q\in\mathbb{Q}\cap[-1,1]}$$

  is pairwise disjoint. Suppose, for contradiction, there are distinct $q, q' \in \mathbb{Q} \cap [-1, 1]$ we had

  $$(q + E) \cap (q' + E) \neq \varnothing.$$

  Then, there exists cosets $A, A' \in \mathbb{R}/\mathbb{Q}$ s.t.

  $$q + x_A = q' + x_{A'}.$$

Hence,

$$x_{A'} - x_A = q - q' \in \mathbb{Q}.$$

Now since $x_A \in A = x_A + \mathbb{Q}$ and $x_{A'} \in A' = x_{A'} + \mathbb{Q}$, so $A = A'$. But E contains exactly one representative from each coset, which means $x_A = x_{A'}$. Thus, $q = q'$, which contradicts to the assumption, so the sets $q + E$ are pairwise disjoint.

- Step 5. Comparing the two sides. Now since $[0, 1] \subseteq X \subseteq [-1, 2]$,

$$1 = m^*([0, 1]) \leq m^*(X) \leq m^*([-1, 2]) = 3.$$

On the other hand,

$$m^*(q + E) = m^*(E) \text{ for every } q \in \mathbb{Q} \cap [-1, 1].$$

Thus,

$$\sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q + E) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E).$$

Note that

$$\sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E) = \begin{cases} 0, & \text{if } m^*(E) = 0; \\ \infty, & \text{if } m^*(E) > 0. \end{cases}$$

In particular, it cannot lie in the interval $[1, 3]$. But $m^*(X) \in [1, 3]$. Hence,

$$m^*(X) \neq \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q + E).$$

Since $\mathbb{Q} \cap [-1, 1]$ is countably infinite, so we're done.

■

Proposition 2.3.2. There exists a finite collection $(A_j)_{j \in J}$ of disjoint subsets of $\mathbb{R}$ s.t.

$$m^* \left(\bigcup_{j \in J} A_j \right) \neq \sum_{j \in J} m^*(A_j).$$

Proof. Suppose m^* was finitely additive, and let E and X to be same as previous proposition, then

$$m^*(X) \leq \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q + E) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E).$$

Since we know that $1 \leq m^*(X) \leq 3$, we thus have $m^*(E) \neq 0$, or otherwise we will have $m^*(X) \leq 0$, but we have $m^*(X) \in [1, 3]$.

Since $m^*(E) > 0$, so $m^*(E) > \frac{1}{n}$ for some $n \in \mathbb{N}$. Now let J be a finite subset of $\mathbb{Q} \cap [-1, 1]$ of cardinality $3n$. If m^* were finitely additive, then we would have

$$m^* \left(\bigcup_{j \in J} (q + E) \right) = \sum_{q \in J} m^*(q + E) = \sum_{q \in J} m^*(E) > 3n \cdot \frac{1}{n} = 3.$$

But we know $\bigcup_{j \in J} (q + E)$ is a subset of X, so it has outer measure at most 3, so we have a contradiction. ■

Lecture 11

31 March.

2.4 Measurable sets

In this chapter, our goal is to construct a measure. Note that we constructed outer measure in previous lecture, but countbale additivity fails on it.

Definition 2.4.1 (Lebesgue measurability)**.** Let $E \subseteq \mathbb{R}^n$. We say E is Lebesgue measurable, or measurable, iff we have the identity

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E) \text{ for all } A \subseteq \mathbb{R}^n.$$

Notation. If E is measurable, then we define $m(E) = m^*(E)$.

Remark 2.4.1. We know $A = (A \cap E) \cup (A \setminus E)$, so

$$m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E).$$

Thus, E is measurable iff

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

Note that $\mathbb{R}^n$ is measurable since $A \cap \mathbb{R}^n = A$ and $A \setminus \mathbb{R}^n = \varnothing$. Also, $\varnothing$ is measurable since $A \cap \varnothing = \varnothing$ and $A \setminus \varnothing = A$.

Exercise 2.4.1. If A is an open box in $\mathbb{R}^n$ and $E = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_n > 0\}$, then

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

Proof. We first give a claim:

Claim 2.4.1. If A is an open interval in $\mathbb{R}$, then

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

Proof. Suppose $A = (a_i, b_i)$ s.t. $a_i \leq b_i$, then

- Case 1: $0 \leq a_i \leq b_i$, then $A \cap (0, \infty) = A$ and $A \setminus (0, \infty) = \varnothing$, so this is true.
- Case 2: $a_i \leq 0 \leq b_i$. Similar.
- Case 3: $a_i \leq b_i \leq 0$. Similar.

$\circledast$

Suppose $A = \prod_{i=1}^n I_i$. Note that

$$m^*(A) = \text{vol}(A) = \prod_{i=1}^n \text{vol}(I_i) = \prod_{i=1}^n m_1^*(I_i).$$

Also,

$$A \cap E = \left(\prod_{i=1}^{n-1} I_i\right) \times (I_n \cap (0, \infty)) \quad A \setminus E = \left(\prod_{i=1}^{n-1} I_i\right) \times (I_n \setminus (0, \infty)).$$

Thus, $A \cap E$ and $A \setminus E$ are both open boxes. By the claim, we have

$$m_1^*(I_n) = m_1^*(I_n \cap (0, \infty)) + m_1^*(I_n \setminus (0, \infty)).$$

Thus,

$$\begin{aligned} m^*(A) &= \prod_{i=1}^{n} m_1^*(I_i) = \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n) = \prod_{i=1}^{n-1} m_1^*(I_i) \times (m_1^*(I_n \cap (0,\infty)) + m_1^*(I_n \setminus (0,\infty))) \\ &= \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n \cap (0,\infty)) + \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n \setminus (0,\infty)) \\ &= m^*(A \cap E) + m^*(A \setminus E). \end{aligned}$$

■

Lemma 2.4.1 (Half-spaces are measurable)**.** The half space $\{(x_1, \dots, x_n) \in \mathbb{R}^n : x_n > 0\}$ is measurable.

Proof. Let $E = \{x \in \mathbb{R}^n \mid x_n > 0\}$. Then

$$m^*(A) \le m^*(A \cap E) + m^*(A \setminus E)$$

is true for all $A \subseteq \mathbb{R}^n$, so it suffices to show that

$$m^*(A) \ge m^*(A \cap E) + m^*(A \setminus E) \text{ for all } A \subseteq \mathbb{R}^n.$$

If $m^*(A) = \infty$, then this is obviously true. Now we assume $m^*(A) < \infty$. Given $\varepsilon > 0$, $\exists$ a countable collection of open boxes (B_k) s.t. $A \subseteq \bigcup_{k=1}^{\infty} B_k$ and

$$\sum_{k=1}^{\infty} \mathrm{vol}(B_k) \le m^*(A) + \varepsilon.$$

We know that

$$m^*(B_k) = m^*(B_k \cap E) + m^*(B_k \setminus E) \text{ by Exercise 2.4.1}.$$

Thus,

$$\mathrm{vol}(B_k) = m^*(B_k \cap E) + m^*(B_k \setminus E).$$

This gives

$$\begin{aligned} \sum_{k=1}^{\infty} \mathrm{vol}(B_k) = \sum_{k=1}^{\infty} m^*(B_k \cap E) + \sum_{k=1}^{\infty} m^*(B_k \setminus E) &\ge m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \cap E\right) + m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \setminus E\right) \\ &\ge m^*(A \cap E) + m^*(A \setminus E). \end{aligned}$$

Hence,

$$m^*(A) + \varepsilon \ge \sum_{k=1}^{\infty} \mathrm{vol}(B_k) \ge m^*(A \cap E) + m^*(A \setminus E),$$

and since this is true for all $\varepsilon > 0$, so we're done. ■

Remark 2.4.2. A similar argument will also shows that any half-space of the form

$$\{(x_1, \dots, x_n) \in \mathbb{R}^n : x_j > 0\} \text{ or } \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_j < 0\}$$

for all $1 \le j \le n$ is measurable.

Lemma 2.4.2 (Properties of measurable sets)**.**

(a) If E is measurable, then $\mathbb{R}^n \setminus E$ is also measurable.

(b) If E is measurable, and $x \in \mathbb{R}^n$, then $x + E$ is also measurable.

(c) If E_1 and E_2 are measurable, then $E_1 \cap E_2$ and $E_1 \cup E_2$ are measurable.

(d) If $E_1, \dots, E_N$ are measurable, then $\bigcup_{j=1}^N E_j$ and $\bigcap_{j=1}^N E_j$ are measurable.

(e) Every open box and close box is measurable.

(f) Any set E of outer measure zero is measurable.

proof of (a). Since E is measurable, $m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$ for any $A \subseteq \mathbb{R}^n$. Also,

$$A \cap (\mathbb{R}^n \setminus E) = A \setminus E \text{ and } A \setminus (\mathbb{R} \setminus E) = A \cap E.$$

Thus,

$$m^*(A) = m^*(A \setminus (\mathbb{R}^n \setminus E)) + m^*(A \cap (\mathbb{R}^n \setminus E))$$

for all $A \subseteq \mathbb{R}^n$, which means $\mathbb{R}^n \setminus E$ is measurable. ■

proof of (b). Suppose E is meausrable, then $m^*(B) = m^*(B \cap E) + m^*(B \setminus E)$ for any $B \subseteq \mathbb{R}^n$. To show $x + E$ is measurable, if we let $A \subseteq \mathbb{R}^n$ arbitrarily, we have

$$m^*(A - x) = m^*((A - x) \cap E) + m^*((A - x) \setminus E).$$

Now since $m^*(A) = m^*(A - x)$, and

$$(A - x) \cap E = (A \cap (E + x)) - x \text{ and } (A - x) \setminus E = (A \setminus (E + x)) - x,$$

we have

$$\begin{aligned} m^*(A) &= m^*(A - x) = m^*((A - x) \cap E) + m^*((A - x) \setminus E) \\ &= m^*((A \cap (E + x)) - x) + m^*((A \setminus (E + x)) - x) = m^*(A \cap (E + x)) + m^*(A \setminus (E + x)). \end{aligned}$$

Thus, $E + x$ is measurable. ■

proof of (c). It sufficies to show that $E_1 \cup E_2$ is measurable. Then, by Demorgan's law,

$$E_1 \cap E_2 = \mathbb{R}^n \setminus ((\mathbb{R}^n \setminus E_1) \cup (\mathbb{R}^n \setminus E_2)),$$

so we can show $E_1 \cap E_2$ is measurable. Now we want to show

$$m^*(A) = m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

Since E_1 is measurable,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1).$$

Next, since E_2 is measurable,

$$m^*(A \setminus E_1) = m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Hence,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1) = m^*(A \cap E_1) + m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Notice that $(A \setminus E_1) \cap E_2 = A \cap (E_2 \setminus E_1)$ and $(A \setminus E_1) \setminus E_2 = A \setminus (E_1 \cup E_2)$. Thus,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap (E_2 \setminus E_1)) + m^*(A \setminus (E_1 \cup E_2)).$$

Note that

$$(A \cap E_1) \cup (A \cap (E_2 \setminus E_1)) = A \cap (E_1 \cup E_2),$$

so

$$m^*(A \cap (E_1 \cup E_2)) \leq m^*(A \cap E_1) + m^*(A \cap (E_2 \setminus E_1)).$$

This gives

$$m^*(A) \geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

On the other hand, we have $A = (A \cap (E_1 \cup E_2)) \cup (A \setminus (E_1 \cup E_2))$, so we have

$$m^*(A) = m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

■

proof of (d). This can be proved by induction on N with (c). ■

proof of (e). Note that every open box is the intersection of many half-spaces of the form

$$\{x : x_j > a_j\} \text{ or } \{x : x_j < b_j\}.$$

Thus, by Lemma 2.4.1 and (d), we know every open box is measurable. Closed boxes are complement of union of open half spaces and thus measurable. ■

proof of (f). Suppose $m^*(E) = 0$. For any $A \subseteq \mathbb{R}^n$, we have $A \cap E \subseteq E$, so

$$m^*(A \cap E) \leq m^*(E) = 0.$$

Hence,

$$m^*(A \cap E) + m^*(A \setminus E) = m^*(A \setminus E) \leq m^*(A).$$

Since

$$m^*(A \cap E) + m^*(A \setminus E) \geq m^*(A)$$

is trivial, so we're done. ■

Lemma 2.4.3 (Finite additivity). If $(E_j)_{j\in J}$ are a finite collection of disjoint measurable sets, then for any set A (not necessarily measurable), we have

$$m^*\left(A \cap \left(\bigcup_{j\in J} E_j\right)\right) = \sum_{j\in J} m^*(A \cap E_j).$$

Furthermore, we have

$$m\left(\bigcup_{j\in J} E_j\right) = \sum_{j\in J} m(E_j).$$

Proof. It sufficies to show the case of two sets: If E and F are disjoint measurable sets, then for any $A \subseteq \mathbb{R}^n$,

$$m^*(A \cap (E \cup F)) = m^*(A \cap E) + m^*(A \cap F).$$

Assume the statement is true for k pairwise disjoint measurable sets, and let $E_1, E_2, \ldots, E_k, E_{k+1}$ be pairwise disjoint measurable sets. Applying the two sets case,

$$m^*(A \cap (\bigcup_{i=1}^{k} E_i \cup E_{k+1})) = m^*(A \cap \bigcup_{i=1}^{k} E_i) + m^*(A \cap E_{k+1}).$$

Now the induction hypothesis gives

$$m^*(A \cap \bigcup_{i=1}^{k} E_i) = \sum_{i=1}^{k} m^*(A \cap E_i),$$

so

$$m^*(A \cap \left(\bigcup_{i=1}^{k+1} E_i\right)) = \sum_{i=1}^{k+1} m^*(A \cap E_i).$$

Now we show the two set cases. Suppose E and F are measurable and disjoint, then $U := E \cup F$ is measurable, and since E is measurable,

$$m^*(A \cap U) = m^*((A \cap U) \cap E) + m^*((A \cap U) \setminus E).$$

Note that

$$A \cap U \cap E = A \cap (E \cup F) \cap E = A \cap E \text{ and } (A \cap U) \setminus E = (A \cap (E \cup F)) \setminus E = A \cap F$$

since $E \cap F = \varnothing$. Hence,

$$m^*(A \cap U) = m^*(A \cap E) + m^*(A \cap F),$$

and we're done.

Now if we have pairwise disjoint $E_1, E_2, \ldots, E_k$, we can pick $A = \mathbb{R}^n$, and thus

$$m(\bigcup_{j \in J} E_j) = \sum_{j \in J} m(E_j).$$

■

Corollary 2.4.1. If $A \subseteq B$ and A, B ar both measurable, then $B \setminus A$ is measurable and

$$m(B \setminus A) + m(A) = m(B).$$

Proof. Since

$$B \setminus A = B \cap (\mathbb{R}^n \setminus A),$$

and A, B are measurable, so we can deduce $B \setminus A$ is measurable. Also, since $A \cap (B \setminus A) = \varnothing$, so by Lemma 2.4.3, we know

$$m(B) = m(A) + m(B \setminus A).$$

■

Lemma 2.4.4 (Countable additivity). Let $(E_j)_{j \in J}$ be a countable collection of pairwise disjoint measurable subsets of $\mathbb{R}^n$, then

$$E := \bigcup_{j \in J} E_j$$

is measurable, and

$$m(E) = \sum_{j \in J} m(E_j).$$

Proof. Since J is countable, we may enumerate it as $\{j_1, j_2, \ldots\}$. For each $N \in \mathbb{N}$, define $F_N = \bigcup_{k=1}^N E_{j_k}$. Thus, F_N is measurable. By previous result,

$$m(F_N) = \sum_{k=1}^N m(E_{j_k}).$$

Let $E = \bigcup_{k=1}^\infty E_{j_k}$, we have $E = \bigcup_{N=1}^\infty F_N$ and $F_1 \subseteq F_2 \subseteq \ldots$.

We first show E is measurable. It sufficies to show that

$$m^*(A \cap E) + m^*(A \setminus E) \leq m^*(A) \text{ for all } A \subseteq \mathbb{R}^n.$$

Since F_N is measurable and $F_N \subseteq E$, we have

$$m^*(A) = m^*(A \cap F_N) + m^*(A \setminus F_N)$$

and

$$A \setminus E \subseteq A \setminus F_N \Rightarrow m^*(A \setminus F_N) \geq m^*(A \setminus E).$$

Also,

$$m^*(A \cap F_N) = \sum_{k=1}^N m^*(A \cap E_{j_k}).$$

Hence,

$$m^*(A) \geq \sum_{k=1}^{N} m^*(A \cap E_{j_k}) + m^*(A \setminus E) \text{ true for all } N.$$

Thus,

$$m^*(A) \geq \sum_{k=1}^{\infty} m^*(A \cap E_{j_k}) + m^*(A \setminus E).$$

Since

$$\bigcup_{k=1}^{\infty} (A \cap E_{j_k}) = A \cap \left(\bigcup_{k=1}^{\infty} E_{j_k} \right) = A \cap E,$$

we have

$$m^*(A \cap E) \leq \sum_{k=1}^{\infty} m^*(A \cap E_{j_k}),$$

so

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

This shows E is measurable.

Now we show

$$m(E) = \sum_{k=1}^{\infty} m(E_{j_k}).$$

First, since $E = \bigcup_{i=1}^{\infty} E_{j_k}$, so

$$m(E) \leq \sum_{k=1}^{\infty} m(E_{j_k}).$$

Now since $F_N \subseteq E$ for all N, so

$$\sum_{k=1}^{N} m(E_{j_k}) = m(F_N) \leq m(E),$$

so

$$m(E) = \sum_{k=1}^{\infty} m(E_{j_k}).$$

■

Lemma 2.4.5 (*σ-algebra property*)**.** If $(\Omega_j)_{j \in J}$ is a countable collection of measurable sets, then $\bigcup_{j \in J} \Omega_j$ and $\bigcap_{j \in J} \Omega_j$ are both measurable.

Proof. Write $J = \{j_1, j_2, \dots\}$ and define $E_1 = \Omega_{j_1}$ and

$$E_N = \Omega_N \setminus \left(\bigcup_{k=1}^{N-1} \Omega_{j_k} \right) \text{ for all } N \geq 2.$$

Hence, E_i is measurable for all $i \in \mathbb{N}$ and disjoint by construction. Note that

$$\bigcup_{k=1}^{\infty} E_k = \bigcup_{k=1}^{\infty} \Omega_{j_k}.$$

By this, $\bigcup_{k=1}^{\infty} \Omega_{j_k}$ is measurable since we have shown that the union of countably infinitely many pairwise disjoint measurable sets is measurable.

Now since

$$\bigcap_{k=1}^{\infty} \Omega_{j_k} = \mathbb{R}^n \setminus \bigcup_{k=1}^{\infty} (\mathbb{R}^n \setminus \Omega_{j_k}),$$

we know $\bigcap_{k=1}^{\infty} \Omega_{j_k}$ is measurable.

■

Lemma 2.4.6. Every open set can be written as a countable or finite union of open boxes.

Proof. We say an open box is rational if $B = \prod_{i=1}^{n}(a_i, b_i)$ where $a_i, b_i \in \mathbb{Q}$. The set of rational open boxes is countable (since this set is isomorphic to a subset of $\mathbb{Q}^{2n}$).

We make the following claim: given any open ball $B(x, r)$, $\exists$ a rational box B with $x \in B \subseteq B(x, r)$. To show this, let $x = (x_1, \ldots, x_n)$ by any point in $\mathbb{R}^n$. Let a_i, b_i be rational numbers s.t.

$$x_i - \frac{r}{n} < a_i < x_i < b_i < x_i + \frac{r}{n},$$

then we show that $B \subseteq B(x, r)$. Let $y \in B$, then

$$x_i - \frac{r}{n} < a_i < y_i < b_i < x_i + \frac{r}{n},$$

so

$$|y_i - x_i| < \frac{r}{n}.$$

Hence,

$$\|y - x\| = \sqrt{\sum_{i=1}^{n}(y_i - x_i)^2} \leq \sum_{i=1}^{n}|y_i - x_i| < \frac{r}{n} \cdot n = r.$$

Hence, $y \in B(x, r)$. Thus, we're done.

Now let E be an open set in $\mathbb{R}^n$. For any $x \in E$, $\exists r_x > 0$ s.t. $B(x, r_x) \subseteq E$. Thus, $E = \bigcup_{x \in E} B(x, r_x)$. Also, for any $B(x, r_x)$, there exists a rational box B_x s.t. $x \in B_x \subseteq B(x, r_x)$. Thus,

$$\bigcup_{x \in E} B_x = E.$$

■

Lemma 2.4.7. Every open set, and every closed set, is Lebesgue measurable.

Proof. By previous lemma, every open set can be written as the union of countable or finite rational open boxes, and every rational open box is measurable, so every open set is measurable. Since every closed box is $\mathbb{R}^n \setminus E$ for some open E, so it is also measurable. ■

Lecture 12

2.5 Measurable Functions

7 April.

Definition 2.5.1 (Measurable functions)**.** Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Then, f is measurable iff $f^{-1}(V)$ is measurable for every open set $V \subseteq \mathbb{R}^m$, where

$$f^{-1}(V) = \{x \in \Omega \mid f(x) \in V\}$$

is the pre-image(or inverse image) of V under f.

Remark 2.5.1. In measure theory, this type of function is called Borel measurable.

Definition 2.5.2. Given a set X. A collection of subsets $\sigma(X)$ of X is called a σ-algebra if

(1) $X \in \sigma(X)$.

(2) $E \in \sigma(X)$ implies $X \setminus E \in \sigma(X)$.

(3) $E_i \in \sigma(X)$ for all $i \in \mathbb{N}$ implies $\bigcup_{i=1}^{\infty} E_i \in \sigma(X)$.

Remark 2.5.2. Since

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \left(\bigcup_{i=1}^{\infty} (X \setminus E_i) \right),$$

so $E_i \in \sigma(X)$ for all $i \in \mathbb{N}$ implies $\bigcap_{i=1}^{\infty} E_i \in \sigma(X)$.

Definition 2.5.3. In $\mathbb{R}^n$, a set is called Borel measurable if it is in the smallest σ-algebra that contains all open sets.

Lemma 2.5.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and $f : \Omega \to \mathbb{R}^m$ be continuous, then f is also measurable.

Proof. Let V be any open subset of $\mathbb{R}^m$, then since f is continuous, so $f^{-1}(V)$ is open, and since all open sets are measurable, we know $f^{-1}(V)$ is measurable, and thus f is measurable. ∎

Lemma 2.5.2. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Then f is measurable if and only if $f^{-1}(B)$ is measurable for every open box B.

Proof.

($\Rightarrow$) If f is measurable, then since every open box B is open, $f^{-1}(B)$ is measurable.

($\Leftarrow$) Given any open set V, $\exists$ a countable union of open boxes s.t. $V = \bigcup_{\ell=1}^{\infty} B_\ell$ by Lemma 2.4.6, and

$$f^{-1}(V) = f^{-1}\left(\bigcup_{\ell=1}^{\infty} B_\ell \right) = \bigcup_{\ell=1}^{\infty} f^{-1}(B_\ell),$$

and since $f^{-1}(B_\ell)$ is measurable for all $\ell \in \mathbb{N}$, and thus $\bigcup_{\ell=1}^{\infty} f^{-1}(B_\ell)$ is measurable by Lemma 2.4.5.

∎

Corollary 2.5.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Suppose $f = (f_1, f_2, \ldots, f_m)$, where $f_j : \Omega \to \mathbb{R}$ is the j-th coordinate of f. Then f is measurable if and only if f_j is measurable for all $1 \le j \le m$.

Proof.

($\Rightarrow$) Fix $j \in \{1, 2, \ldots, m\}$. Let $I \subseteq \mathbb{R}$ be an open interval. Consider the open set $B_j = \mathbb{R} \times \cdots \times I \times \cdots \times \mathbb{R}$, where I is in the j-th coordinate, then for $x \in \Omega$, we know

$$x \in f^{-1}(B_j) \Leftrightarrow f(x) \in B_j \Leftrightarrow f_j(x) \in I \Leftrightarrow x \in f_j^{-1}(I).$$

Hence, $f^{-1}(B_j) = f_j^{-1}(I)$. Since f is measurable, $f^{-1}(B_j)$ is measurable since B_j is an open box. Hence, $f_j^{-1}(I)$ is measurable, and since this is true for all open box I of $\mathbb{R}$, so by Lemma 2.5.2, f_j is measurable for all $1 \le j \le m$.

($\Leftarrow$) Let $B = I_1 \times I_2 \times \cdots \times I_m$ be an open box in $\mathbb{R}^m$. Then,

$$\begin{aligned} x \in f^{-1}(B) &\Leftrightarrow f(x) = (f_1(x), f_2(x), \ldots, f_m(x)) \in B = I_1 \times I_2 \times \cdots \times I_m \\ &\Leftrightarrow f_j(x) \in I_j \text{ for all } 1 \leq j \leq m \Leftrightarrow x \in f_j^{-1}(I_j) \text{ for all } 1 \leq j \leq m \\ &\Leftrightarrow x \in \bigcap_{j=1}^{m} f_j^{-1}(I_j). \end{aligned}$$

Hence, $f^{-1}(B) = \bigcap_{j=1}^m f_j^{-1}(I_j)$. Since for all $1 \leq j \leq m$, f_j is measurable and I_j is an open box in $\mathbb{R}$, so $f^{-1}(I_j)$ is measurable and thus $\bigcap_{j=1}^m f_j^{-1}(I_j)$ is measurable, so $f^{-1}(B)$ is measurable, and thus f is measurable.

■

Lemma 2.5.3. Let Ω be a measurable subset of $\mathbb{R}^n$, and let W be an open subset of $\mathbb{R}^m$. If $f : \Omega \to W$ is measurable and $g : W \to \mathbb{R}^p$ is continuous, then $g \circ f : \Omega \to \mathbb{R}^p$ is measurable.

Proof. Let V be an open set in $\mathbb{R}^p$, then $(g \circ f)^{-1}(V) = f^{-1}\left(g^{-1}(V)\right)$. Since g is continuous and V is open, $g^{-1}(V)$ is open in W, so $g^{-1}(V) = U \cap W$, where U is open in $\mathbb{R}^m$. Since W is also open in $\mathbb{R}^m$, $W \cap U$ is also open in $\mathbb{R}^m$, so $g^{-1}(V)$ is open in $\mathbb{R}^m$. Now since f is measurable, $f^{-1}\left(g^{-1}(V)\right)$ is measurable, i.e. $(g \circ f)^{-1}(V)$ is measurable. ■

Corollary 2.5.2. Let Ω be a measurable subset of $\mathbb{R}^n$. If $f : \Omega \to \mathbb{R}$ is measurable, then so is $|f|, \max\{f, 0\}$, and $\min\{f, 0\}$.

Proof. Since $g_1(x) \coloneqq |x|, g_2(x) \coloneqq \max\{x, 0\}$, and $g_3(x) \coloneqq \min\{x, 0\}$ are all continuous, so by Lemma 2.5.3, $g_1 \circ f, g_2 \circ f, g_3 \circ f$ are all measurable.

Remark 2.5.3.
$$\max\{x, 0\} = \frac{x + |x|}{2}, \quad \min\{x, 0\} = \frac{x - |x|}{2}.$$

■

Corollary 2.5.3. Let Ω be a measurable subset of $\mathbb{R}^n$. If $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ are measurable, then so are $f + g, f - g, f \cdot g, \max\{f, g\}$, and $\min\{f, g\}$. If $g(x) \neq 0$ for all $x \in \Omega$, then f/g is also measurable.

Proof. Consider the map $F : \Omega \to \mathbb{R}^2$ defined by $F(x) = (f(x), g(x))$, then by Corollary 2.5.1, F is also measurable. Define $h_+(u, v) = u + v$, where $h_+ : \mathbb{R}^2 \to \mathbb{R}$ is continuous. Then $h_+ \circ F$ is measurable, where $(h_+ \circ F)(x) = f(x) + g(x)$. Similarly, we can define

$$\begin{aligned} h_*(u, v) &= uv, \\ h_{\max}(u, v) &= \max(u, v) = \frac{u + v + |u - v|}{2}, \\ h_{\min}(u, v) &= \min(u, v) = \frac{u + v - |u - v|}{2}, \end{aligned}$$

and they are all continuous, so $h_* \circ F, h_{\max} \circ F, h_{\min} \circ F$ are all measurable. As for the division, it is similar. ■

Lemma 2.5.4. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}$ be a function, then f is measurable iff $f^{-1}((a, \infty))$ is measurable for any $a \in \mathbb{R}$.

Proof.

($\Rightarrow$) Since (a, ∞) is open, so $f^{-1}((a, \infty))$ is measurable.

($\Leftarrow$) Let I be an open interval in $\mathbb{R}$ with $I=(\alpha,\beta)$ where $\alpha<\beta$, then it suffices to show $f^{-1}(I)$ is measurable by Lemma 2.5.2. Note that $(\alpha,\beta)=(\alpha,\infty)\cap(\mathbb{R}\setminus[\beta,\infty))$, and

$$[\beta,\infty)=\bigcap_{m=1}^{\infty}\left(\beta-\frac{1}{m},\infty\right).$$

Therefore,

$$f^{-1}([\beta,\infty))=f^{-1}\left(\bigcap_{m=1}^{\infty}\left(\beta-\frac{1}{m},\infty\right)\right)=\bigcap_{m=1}^{\infty}f^{-1}\left(\left(\beta-\frac{1}{m},\infty\right)\right),$$

which is measurable since $f^{-1}(a,\infty)$ is measurable for all $a\in\mathbb{R}$. Hence,

$$f^{-1}(\alpha,\beta)=f^{-1}(\alpha,\infty)\cap\left(\Omega\setminus f^{-1}(\beta,\infty)\right)$$

is measurable.

■

Definition 2.5.4 (Measurable functions in the extended reals). Let Ω be a measurable subset of $\mathbb{R}^n$. Set

$$\mathbb{R}^*:=\mathbb{R}\cup\{+\infty\}\cup\{-\infty\}.$$

A function $f:\Omega\to\mathbb{R}^*$ is said to be measurable if $f^{-1}((a,\infty])$ is measurable for all $a\in\mathbb{R}$, where

$$f^{-1}((a,\infty])=f^{-1}((a,\infty))\cup f^{-1}(\{\infty\}).$$

Lecture 13

9 April.

Example 2.5.1. Suppose $f(x)=\frac{1}{x^2}$ is defined on $\mathbb{R}$, then we can extend it to $f:\mathbb{R}\to\mathbb{R}^*$ so that

$$f(x)=\begin{cases}\dfrac{1}{x^2}, & \text{if } x\neq 0;\\ \infty, & \text{if } x=0.\end{cases}$$

Then f is measurable. Note that if we define f by $f:\Omega\to\mathbb{R}$, then $f^{-1}((a,\infty])=f^{-1}((a,\infty))$ for all $a\in\mathbb{R}$.

Lemma 2.5.5. Let $\Omega\subseteq\mathbb{R}^n$ be measurable. For each positive integer n, let $f_n:\Omega\to\mathbb{R}^*$ be a measurable function. Then, $\sup_{n\in\mathbb{N}}f_n, \inf_{n\in\mathbb{N}}f_n, \limsup_{n\to\infty}f_n, \liminf_{n\to\infty}f_n$ are measurable. In particular, if $\lim_{n\to\infty}f_n(x)=f(x)$ for $f:\Omega\to\mathbb{R}^*$, then f is also measurable.

Proof. Let $g(x)=\sup_{n\in\mathbb{N}}f_n(x)$, then g is measurable iff $g^{-1}((a,\infty])$ is measurable for any $a\in\mathbb{R}$. Also,

$$\begin{aligned}x\in g^{-1}((a,\infty])\Leftrightarrow g(x)>a\Leftrightarrow \sup_{n\in\mathbb{N}}f_n(x)>a\Leftrightarrow \exists k\in\mathbb{N}\text{ s.t. } f_k(x)>a\\ \Leftrightarrow \exists k\in\mathbb{N}\text{ s.t. } x\in f_k^{-1}((a,\infty])\Leftrightarrow x\in\bigcup_{n=1}^{\infty}f_n^{-1}((a,\infty]).\end{aligned}$$

Hence, $g^{-1}((a,\infty])=\bigcup_{n=1}^{\infty}f_n^{-1}((a,\infty])$, so it sufficies to show that $\bigcup_{n=1}^{\infty}f_n^{-1}((a,\infty])$ is measurable. However, since for all $n\in\mathbb{N}$, f_n is measurable, so $f_n^{-1}((a,\infty])$ is measurable, and thus we're done.

Let $h(x)=\inf_{n\in\mathbb{N}}f_n(x)$, then h is measurable iff $h^{-1}((a,\infty])$ is measurable for all $a\in\mathbb{R}$. Now since

$$x \in h^{-1}((a,\infty]) \Leftrightarrow h(x) \in (a,\infty] \Leftrightarrow \inf_{n\in\mathbb{N}} f_n(x) \in (a,\infty] \Leftrightarrow \inf_{n\in\mathbb{N}} f_n(x) > a \Leftrightarrow \forall n \in \mathbb{N},\ f_n(x) > a$$
$$\Leftrightarrow \forall n \in \mathbb{N},\ f_n(x) \in (a,\infty] \Leftrightarrow \forall n \in \mathbb{N},\ x \in f_n^{-1}((a,\infty]) \Leftrightarrow x \in \bigcap_{n=1}^{\infty} f_n^{-1}((a,\infty]).$$

Hence, $h^{-1}((a,\infty]) = \bigcap_{n=1}^{\infty} f_n^{-1}((a,\infty])$. However, since for all $n \in \mathbb{N}$, f_n is measurable, so $f_n^{-1}((a,\infty])$ is measurable, and thus we're done.

Now we define $g_N(x) = \sup_{n\geq N} f_n(x)$, then we know g_N is measurable for all $N \in \mathbb{N}$ by what we've proved. Now since

$$\limsup_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sup_{k\geq n} f_k(x) = \inf_{N\geq 1} \sup_{n\geq N} f_n(x) = \inf_{N\geq 1} g_N(x),$$

so by what we've proved, we know $\inf_{N\geq 1} g_N(x)$ is measurable, and we're done.

Similar argument can be used to prove $\liminf_{n\to\infty} f_n(x)$ is measurable.

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of real-valued functions on a set Ω, and suppose that

$$f_n(x) \to f(x) \quad \text{for every } x \in \Omega.$$

We will show that for every $x \in \Omega$,

$$\limsup_{n\to\infty} f_n(x) = f(x) = \liminf_{n\to\infty} f_n(x).$$

Fix $x \in \Omega$. Then $\{f_n(x)\}$ is a sequence of real numbers converging to $f(x)$.

Recall the definitions:

$$\limsup_{n\to\infty} f_n(x) = \lim_{n\to\infty} \left(\sup_{k\geq n} f_k(x)\right), \quad \liminf_{n\to\infty} f_n(x) = \lim_{n\to\infty} \left(\inf_{k\geq n} f_k(x)\right).$$

We first show that

$$\limsup_{n\to\infty} f_n(x) \leq f(x).$$

Let $\varepsilon > 0$. Since $f_n(x) \to f(x)$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \varepsilon,$$

which implies

$$f_n(x) \leq f(x) + \varepsilon.$$

Hence, for all $n \geq N$,

$$\sup_{k\geq n} f_k(x) \leq f(x) + \varepsilon.$$

Taking the limit as $n \to \infty$, we obtain

$$\limsup_{n\to\infty} f_n(x) \leq f(x) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\limsup_{n\to\infty} f_n(x) \leq f(x).$$

Next, we show that

$$\liminf_{n\to\infty} f_n(x) \geq f(x).$$

Again let $\varepsilon > 0$. Since $f_n(x) \to f(x)$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \varepsilon,$$

which implies

$$f_n(x) \geq f(x) - \varepsilon.$$

Hence, for all $n \geq N$,
$$\inf_{k \geq n} f_k(x) \geq f(x) - \varepsilon.$$
Taking the limit as $n \to \infty$, we obtain
$$\liminf_{n\to\infty} f_n(x) \geq f(x) - \varepsilon.$$
Since $\varepsilon > 0$ is arbitrary,
$$\liminf_{n\to\infty} f_n(x) \geq f(x).$$

Combining the two inequalities, we have
$$\limsup_{n\to\infty} f_n(x) \leq f(x) \leq \liminf_{n\to\infty} f_n(x).$$
But in general,
$$\liminf_{n\to\infty} f_n(x) \leq \limsup_{n\to\infty} f_n(x),$$
so all three quantities are equal:
$$\limsup_{n\to\infty} f_n(x) = f(x) = \liminf_{n\to\infty} f_n(x).$$

Since $x \in \Omega$ was arbitrary, the result holds for all x, hence
$$f = \limsup_{n\to\infty} f_n = \liminf_{n\to\infty} f_n.$$

This shows that if $f_n \to f$ pointwise, then f is measurable. ■

Chapter 3

Lebesgue Integration

What is Lebesgue integration?

Lebesgue integration is a way of defining the integral of a function by grouping together points that have *similar output values*. Instead of cutting the *x-axis* into many thin intervals, as in Riemann integration, Lebesgue integration studies the sets of points where the function lies in certain value ranges and measures the size of those sets.

Informally, the two viewpoints are:

- **Riemann integration:** slice the domain into vertical strips and add "width × height."
- **Lebesgue integration:** slice the range into horizontal bands and add "value × measure of the set having that value."

This change of perspective makes Lebesgue integration much more flexible. It can integrate many functions that are badly behaved from the Riemann point of view, and it works especially well in probability, Fourier analysis, and modern analysis.

The main idea

Suppose a function f takes the value approximately y on a set of points whose length is approximately m. Then its contribution to the integral is approximately

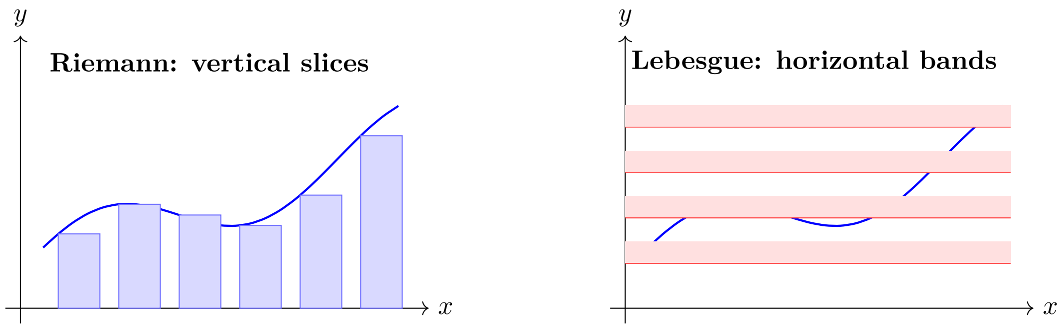
$$y \cdot m.$$

Adding these contributions over all output levels leads to the Lebesgue integral.

For nonnegative functions, one can think of the construction as approximating f by step functions built from *horizontal* layers. In this way, the Lebesgue integral measures area by organizing the graph according to function values rather than according to input intervals.

How is it different from Riemann integration?

The essential difference is the direction of the partition.



- In **Riemann integration**, the interval on the x-axis is divided into many small subintervals. One samples the function in each interval and sums rectangle areas.
- In **Lebesgue integration**, the y-axis is divided into small bands. For each band, one looks at all x such that $f(x)$ falls inside that band, then measures how large that set is.

So Riemann organizes the area by *where the input lies*, while Lebesgue organizes it by *what values the function takes*.

Why Lebesgue integration is more powerful

Lebesgue integration handles limits of functions much better. For example, if a sequence of functions converges in a reasonable way, the Lebesgue integral often allows us to interchange limit and integral. This is one of the main reasons it became the standard notion of integration in advanced mathematics.

It also integrates some functions that are not Riemann integrable. The classical example is the Dirichlet function on $[0, 1]$:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

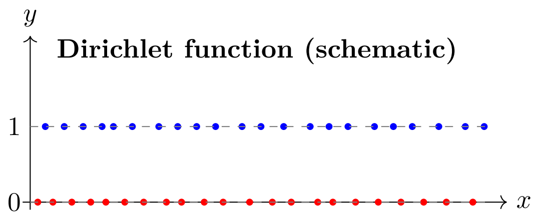
This function is discontinuous at every point, so it is not Riemann integrable. However, the rational numbers in $[0, 1]$ have Lebesgue measure 0, so from the Lebesgue point of view the function is equal to 0 almost everywhere. Therefore,

$$\int_0^1 f(x)\, dx = 0$$

in the Lebesgue sense.

A visual example

The next picture shows the Dirichlet function schematically. The dense blue points near $y = 1$ represent rational inputs, and the dense red points near $y = 0$ represent irrational inputs. Since the rational set has measure zero, the upper layer contributes no area in the Lebesgue sense.



Summary

- **Riemann integration** partitions the domain and adds areas of vertical rectangles.
- **Lebesgue integration** partitions the range and measures the size of the sets where the function takes certain values.
- Lebesgue integration is better suited to complicated functions and to taking limits.
- Every Riemann integrable function is Lebesgue integrable, but not every Lebesgue integrable function is Riemann integrable.

In short, Lebesgue integration is a more robust and more powerful notion of area, built to handle modern analysis.

3.1 Simple Functions

Definition 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let $f : \Omega \to \mathbb{R}$ be a measurable function. We say f is a simple function if the image of $f(\Omega)$ is finite, i.e. $\exists c_1, \ldots, c_N$ s.t. for any $c \in \Omega$, $f(x) = c_j$ for some $1 \leq j \leq N$.

Example 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let E be a measurable subset of Ω. Define the characteristic function $\chi_E : \Omega \to \mathbb{R}$ by

$$\chi_E(x) = \begin{cases} 1, & \text{if } x \in E; \\ 0, & \text{if } x \in \Omega \setminus E. \end{cases}$$

Then, we claim that χ_E is measurable. Note that for any open interval I, we have

$$\chi_E^{-1}(I) = \begin{cases} \varnothing, & \text{if } 0 \notin I, 1 \notin I; \\ E, & \text{if } 0 \notin I, 1 \in I; \\ \Omega \setminus E, & \text{if } 0 \in I, 1 \notin I; \\ \Omega, & \text{if } 0 \in I, 1 \in I. \end{cases}$$

Thus, χ_E is measurable.

Lemma 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ be simple function. Then, $f + g$ is also simple and for any $c \in \mathbb{R}$, the function cf is also simple.

Proof. Suppose $f(\Omega) = \{a_1, \ldots, a_N\}$ and $g(\Omega) = \{b_1, \ldots, b_M\}$, then $(f+g)(\Omega) = \{a_i + b_j\}_{\substack{1 \leq i \leq N \\ 1 \leq j \leq M}}$, which is finite, so $f + g$ is simple.

Also, $(cf)(\Omega) = \{ca_1, \ldots, ca_N\}$, which is also finite, so cf is simple. ■

Lemma 3.1.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ be a simple function. Then, there exists finitely many distinct real numbers $c_1, \ldots, c_N$ and finitely many pairwise disjoint measurable sets $E_1, \ldots, E_N \subseteq \Omega$ s.t. $f = \sum_{j=1}^{N} c_j \chi_{E_j}$.

Proof. Suppose $f(\Omega) = \{c_1, \ldots, c_N\}$, where $c_i \neq c_j$ for all $i \neq j$. Define $E_j = f^{-1}(\{c_j\})$ for each $1 \leq j \leq N$, then $(E_j)_{j=1}^{N}$ are pairwise disjoint. Since $f(x) = c_\ell$ for some $1 \leq \ell \leq N$ for all $x \in \Omega$, $\Omega = \bigcup_{j=1}^{N} E_j$. Note that

$$E_j = f^{-1}(\{c_j\}) = \bigcap_{k=1}^{\infty} f^{-1}\left(\left(c_j - \frac{1}{k}, c_j + \frac{1}{k}\right)\right),$$

then since f is measurable, we know E_j is measurable. Now for $x \in \Omega$, we know $x \in E_k$ for some $1 \leq k \leq N$. Hence, $f(x) = c_k$, and thus

$$\sum_{j=1}^{N} c_j \chi_{E_j}(x) = c_k \chi_{E_k}(x) = c_k = f(x).$$

Thus, $f = \sum_{j=1}^{N} c_j \chi_{E_j}$. ■

Remark 3.1.1. This shows f is simple implies f is measurable since we can show $f^{-1}((a, \infty))$ is measurable for all $a \in \mathbb{R}$ with knowing that in fact $f^{-1}((a, \infty))$ is some intersection and union of E_i's.

Lemma 3.1.3. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to [0, \infty]$ be a measurable function. Then there exists a sequence of simple functions $\{f_n\}_{n=1}^{\infty}$ with $f_n : \Omega \to \mathbb{R}$ s.t. each $f_n \geq 0$, and the sequence is increasing:

$$0 \leq f_1(x) \leq f_2(x) \leq \dots \quad \text{for all } x \in \Omega,$$

and $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in \Omega$.

Proof. For each integer $n \geq 1$, define

$$E_{n,k} \coloneqq \left\{ x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}, \quad k = 0, 1, \dots, n2^n - 1,$$

and $F_n \coloneqq \{x \in \Omega : f(x) \geq n\}$. Then define

$$f_n(x) \coloneqq \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \chi_{E_{n,k}}(x) + n\chi_{F_n}(x).$$

Thus, f_n is obtained from f by truncating at height n and then rounding down to the nearest multiple of 2^{-n}. Note that If $f(x) < n$, then $\exists 0 \leq k \leq n2^n - 1$ s.t.

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n},$$

so $f_n(x) = \frac{k}{2^n}$, and if $f(x) \geq n$, then $f_n(x) = n$.

Now we try to write down the explicit formula for f_n. Fix $x \in \Omega$, then write $a \coloneqq f(x) \in [0, \infty]$.

Claim 3.1.1.

$$f_n(x) = \min\left(n, \frac{\lfloor 2^n a \rfloor}{2^n}\right) = \min\left(n, \frac{\lfloor 2^n f(x) \rfloor}{2^n}\right).$$

Proof. If $a = f(x) \geq n$, then $f_n(x) = n$, and $n \leq \frac{\lfloor 2^n f(x) \rfloor}{2^n}$, so this is true. If $a = f(x) < n$, then $\exists 0 \leq k \leq n2^n - 1$ s.t.

$$\frac{k}{2^n} \leq a < \frac{k+1}{2^n} \Leftrightarrow k \leq 2^n a < k+1,$$

so $k = \lfloor 2^n a \rfloor$. Since $f(x) \in E_{n,k}$, we have

$$f_n(x) = \frac{k}{2^n} = \frac{\lfloor 2^n a \rfloor}{2^n}.$$

Thus, in this case, $f_n(x) = \min\left(n, \frac{\lfloor 2^n a \rfloor}{2^n}\right)$.

⊛

By this claim, we know $=0 \leq f_n(x) \leq f(x)$ for all n.

Now we show $f_n(x) \leq f_{n+1}(x)$.

- Case 1: $n \leq f(x)$, then $f_n(x) = n$, and $f_{n+1}(x) = \min\left(n+1, \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}}\right) \geq n$, so we're done.
- Case 2: $f(x) < n$, then

$$f_n(x) = \min\left(n, \frac{\lfloor 2^n f(x) \rfloor}{2^n}\right) = \frac{\lfloor 2^n f(x) \rfloor}{2^n}.$$

Thus, $\exists$ unique $m = \lfloor 2^n f(x) \rfloor \leq 2^n f(x) < m+1$, and

$$f_{n+1}(x) = \min\left(n+1, \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}}\right),$$

and note that
$$2m \le 2^{n+1} f(x) < 2(m+1),$$
so
$$f_n(x) = \frac{\lfloor 2^n f(x) \rfloor}{2^n} = \frac{m}{2^n} = \frac{2m}{2^{n+1}} \le \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}} = f_{n+1}(x),$$
where the last equality holds since $f(x) < n < n+1$.

Now we prove $\lim_{n\to\infty} f_n(x) = f(x)$.

- Case 1: $f(x) = \infty$, then $f_n(x) = n$, so $\lim_{n\to\infty} f_n(x) = x$.
- Case 2: $f(x) < \infty$, then $\exists N \in \mathbb{N}$ s.t. $f(x) < N$. If $n \ge N$, then
$$f_n(x) = \frac{k}{2^n} \text{ where } \frac{k}{2^n} \le f(x) < \frac{k+1}{2^n}.$$
Thus, $f_n(x) \le f(x) \le f_n(x) + \frac{1}{2^n}$, i.e.
$$|f(x) - f_n(x)| < \frac{1}{2^n}.$$
Thus, $\lim_{n\to\infty} f_n(x) = f(x)$.

■

Lecture 14

21 April.

Definition 3.1.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to [0,\infty)$ be a simple function which is non-negative, thus f is measurable and the image $f(\Omega)$ is finite and contained in $[0,\infty)$. Thus, we can define the Lebesgue integral $\int_\Omega f$ of f on Ω by
$$\int_\Omega f = \sum_{\lambda \in f(\Omega), \lambda > 0} \lambda m\left(\{x \in \Omega : f(x) = \lambda\}\right) = \sum_{\lambda \in f(\Omega)} \lambda m\left(\{x \in \Omega : f(x) = \lambda\}\right).$$
We'll also sometimes write $\int_\Omega f$ as $\int_\Omega f \,\mathrm{d}m$.

Remark 3.1.2. By definition, we know simple function can be written as
$$f = \sum_{j=1}^N c_j \chi_{E_j},$$
where $\{c_1, \ldots, c_N\}$ are distinct and $\{E_j\}_{j=1}^N$ are disjoint, measurable subsets of Ω defined by $E_j = f^{-1}(\{c_j\})$, and thus
$$\int_\Omega f = \sum_{j=1}^N c_j m(E_j).$$

Example 3.1.2. Let $f : \mathbb{R} \to \mathbb{R}$ be the function which is equal to 3 on $[1,2]$ and equal to 4 on $(2,4)$ and is zero elsewhere. Then,
$$\int_\mathbb{R} f = 3m([1,2]) + 4m([2,4]) = 11.$$

Now we show the simple integral of a simple function can equal $+\infty$. (The reason why we restrict this integral to non-negative functions is to avoid ever encountering the indefinite form $+\infty + (-\infty)$.)

Example 3.1.3. Suppose $g : \mathbb{R} \to \mathbb{R}$ is the function which is 1 on $[0, \infty)$ and is zero on $(-\infty, 0)$, then $g = 1 \cdot \chi_{[0.\infty)}$ and

$$\int_{\mathbb{R}} g = 1 \cdot m([0, \infty)) = \infty.$$

Lemma 3.1.4. Let $\Omega \subseteq \mathbb{R}^n$ be measurable and let $E_1, E_2, \ldots, E_N$ be pairwise disjoint measurable subset of Ω. Let $c_1, \ldots, c_N$ be non-negative real numbers (not required to be distinct). Then,

$$\int_{\Omega} \sum_{j=1}^{N} c_j \chi_{E_j} = \sum_{j=1}^{N} c_j m(E_j).$$

Proof. Define $f := \sum_{j=1}^{N} c_j \chi_{E_j}$, then f is simple by definition. Let $d_1, \ldots, d_L$ be the distinct positive numbers that occurs among $c_1, \ldots, c_N$. Let

$$F_\ell := \bigcup_{\{j : c_j = d_\ell\}} E_j.$$

Then since each E_j is measurable, so F_ℓ is measurable. Note that $F_1, F_2, \ldots, F_\ell$ are pairwise disjoint since $E_1, \ldots, E_N$ are pairwise disjoint. Also, note that

$$f = \sum_{\ell=1}^{L} d_\ell \chi_{F_\ell}.$$

Now since $d_1, \ldots, d_L$ are distinct and $F_1, F_2, \ldots, F_L$ are pairwise disjoint, by definition we have

$$\int_{\Omega} f = \sum_{\ell=1}^{L} d_\ell m(F_\ell) = \sum_{\ell=1}^{L} d_\ell \sum_{j : c_j = d_\ell} m(E_j) = \sum_{\ell=1}^{L} \sum_{j : c_j = d_\ell} c_j m(E_j) = \sum_{j=1}^{N} c_j m(E_j).$$

■

Proposition 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to [0, \infty)$ and $g : \Omega \to [0, \infty)$ be non-negative simple functions.

(a) We have $0 \leq \int_{\Omega} \leq \infty$. Furthermore, we have $\int_{\Omega} f = 0$ iff $m(\{x \in \Omega : f(x) \neq 0\}) = 0$.

(b) $\int_{\Omega} f + g = \int_{\Omega} f + \int_{\Omega} g$.

(c) For any $c > 0$, we have $\int_{\Omega} cf = c \int_{\Omega} f$.

(d) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_{\Omega} f \leq \int_{\Omega} g$.

Remark 3.1.3. We make a very convenient notational convention: if a property $P(x)$ holds for all points in Ω, except for a set of measure zero, then we say that P holds for almost every point in Ω. Thus (a) asserts that $\int_{\omega} f = 0$ if and only if f is zero for almost every point in Ω.

Proof. Since

$$f = \sum_{\lambda \in f(\Omega) \setminus \{0\}} \lambda \cdot \chi_{\{x \in \Omega : f(x) = \lambda\}},$$

we can write $f = \sum_{j=1}^{N} c_j \chi_{E_j}$, where $E_1, \ldots, E_N$ are pairwise disjoint, and each of c_j is positive. Similarly, we write $g = \sum_{j=1}^{M} d_j \chi_{F_j}$, where $F_1, \ldots, F_M$ are pairwise disjoint and $d_1, \ldots, d_M$ are positive.

(a) Since $\int_{\Omega} f = \sum_{j=1}^{N} c_j m(E_j)$, so it is clear that $0 \leq \int_{\Omega} f \leq \infty$. If f is zero almost everywhere, then all of E_j must have measure 0 (because $E_j \subseteq \{x \in \Omega : f(x) \neq 0\}$), and thus $\int_{\Omega} f = 0$.

Conversely, if $\int_\Omega f = 0$, then $\sum_{j=1}^N c_j m(E_j) = 0$, which can only happen when all of $m(E_j)$ are 0, and hence f is zero almost everywhere.

(b) Set

$$E_0 = \Omega \setminus \bigcup_{j=1}^N E_j, \quad c_0 = 0,$$

then $E_0, E_1, \ldots, E_N$ are pairwise disjoint, and their union is Ω, and $f = \sum_{j=0}^N c_j \chi_{E_j}$. Similarly, we can define

$$F_0 = \Omega \setminus \bigcup_{k=1}^M F_k, \quad d_0 = 0,$$

and $g = \sum_{k=0}^M d_k \chi_{F_k}$, where $F_0, F_1, \ldots, F_M$ are pairwise disjoint and their union is Ω. Now we know

$$\{E_j \cap F_k : 0 \le j \le N, 0 \le k \le M\}$$

forms a partition of Ω. Now since

$$\chi_{E_j} = \sum_{k=0}^M \chi_{E_j \cap F_k}, \quad \chi_{F_k} = \sum_{j=0}^N \chi_{E_j \cap F_k},$$

we have

$$f = \sum_{j=0}^N \sum_{k=0}^M c_j \chi_{E_j \cap F_k}, \quad g = \sum_{j=0}^N \sum_{k=0}^M d_k \chi_{E_j \cap F_k}.$$

(Note that we need to define E_0, F_0 to write this thing.) Hence,

$$f + g = \sum_{j=0}^N \sum_{k=0}^M (c_j + d_k) \chi_{E_j \cap F_k},$$

which shows

$$\int_\Omega (f+g) = \sum_{j=0}^N \sum_{k=0}^M (c_j + d_k) m(E_j \cap F_k) = \int_\Omega f + \int_\Omega g$$

by previous lemma ($c_j + d_k$ may repeat, so we have to use the lemma) since

$$\int_\Omega f = \sum_{j=0}^N c_j m(E_j) = \sum_{j=0}^N \sum_{k=0}^M c_j m(E_j \cap F_k),$$

and similarly we have

$$\int_\Omega g = \sum_{k=0}^M d_k m(F_k) = \sum_{k=0}^M \sum_{j=0}^N d_k m(E_j \cap F_k).$$

(c) Since $cf = \sum_{j=1}^N (cc_j) \chi_{E_j}$, we have

$$\int_\Omega cf = \sum_{j=1}^N cc_j m(E_j) = c \int_\Omega f.$$

(d) Write $h := g - f$, then h is simple and non-negative and $g = h + f$. Hence, by (b), $\int_\Omega g = \int_\Omega h + \int_\Omega f$, but by (a) $\int_\Omega h \ge 0$, so we're done.

■

3.2 Integration of non-negative measurable functions

We now pass from integrating non-negative simple functions to integrating general non- negative measurable functions. In this section we allow our measurable functions to take the value $+\infty$.

Definition 3.2.1. Let $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ be functions, we say that f majorizes g, or equivalently that g minorizes f, if

$$f(x) \geq g(x) \quad \forall x \in \Omega.$$

We sometimes use the phrase "f dominates g"instead of "f majorizes g"

Definition 3.2.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and $f : \Omega \to [0, \infty]$ be measurable and non-negative. We define the Lebesgue integral $\int_\Omega f$ of f on Ω by

$$\begin{aligned} \int_\Omega f &:= \sup \left\{ \int_\Omega s : s \text{ is simple and non-negative, and } s \text{ minorizes } f \right\} \\ &= \sup \left\{ \int_\Omega s : s \text{ is simple and non-negative, and } s(x) \leq f(x) \quad \forall x \in \Omega \right\}. \end{aligned}$$

Remark 3.2.1. If $\Omega' \subseteq \Omega$ is a measurable subset of Ω, then

$$\int_{\Omega'} f = \int_{\Omega'} f|_{\Omega'}.$$

Proposition 3.2.1. Let Ω be a measurable set, and let $f : \Omega \to [0, \infty]$ and $g : \Omega \to [0, \infty]$ be non-negative measurable functions.

(a) We have $0 \leq \int_\Omega f \leq \infty$. Furthermore, $\int_\Omega f = 0$ iff $f(x) = 0$ for almost every $x \in \Omega$.

(b) For any positive c, $\int_\Omega cf = c \int_\Omega f$.

(c) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_\Omega f \leq \int_\Omega g$.

(d) If $f(x) = g(x)$ for almost every $x \in \Omega$, then $\int_\Omega f = \int_\Omega g$.

(e) If $\Omega' \subseteq \Omega$ is measurable, then
$$\int_{\Omega'} f = \int_\Omega f\chi_{\Omega'} \leq \int_\Omega f.$$

Proof.

(a) $0 \leq \int_\Omega f \leq \infty$ since zero function is simple and non-negative, and minorizes f for all f. Now we show
$$\int_\Omega f = 0 \Leftrightarrow f(x) = 0 \text{ almost everywhere in } \Omega.$$
If $f \equiv 0$ almost everywhere in Ω, then for any simple non-negative function s minorizes f, we have $s(x) = 0$ whenever $f(x) = 0$. This shows $s \equiv 0$ almost everywhere in Ω. Hence, $\int_\Omega s = 0$ by definition. This implies $\int_\Omega f = 0$.

Now if $\int_\Omega f = 0$. For each $k \in \mathbb{N}$, define
$$A_k := \left\{ x \in \Omega : f(x) \geq \frac{1}{k} \right\}.$$
Since f is measurable, so $A_k = f^{-1}\left(\left(\frac{1}{k}, \infty\right)\right)$ is measurable for all $k \in \mathbb{N}$. Now define

$s_k := \frac{1}{k}\chi_{A_k}$. Then, s_k is a non-negative simple function. Note that $s_k \leq f$ on Ω. Thus,

$$\int_\Omega s_k \leq \int_\Omega f = 0$$

by definition, which shows $\int_\Omega s_k = 0$. Note that

$$0 = \int_\Omega s_k = \frac{1}{k} m(A_k) \Rightarrow m(A_k) = 0.$$

This is true for all $k \in \mathbb{N}$, and since

$$\{x \in \Omega : f(x) > 0\} = \bigcup_{k=1}^{\infty} A_k,$$

we have

$$m(\{x \in \Omega : f(x) > 0\}) \leq \sum_{k=1}^{\infty} m(A_k) = 0,$$

and we're done.

(b) We first show $\int_\Omega cf \geq c\int_\Omega f$. Let s be any simple non-negative function minorizing f, then cs minorizes cf. Note that

$$\int_\Omega cs = c\int_\Omega s,$$

Hence,

$$\int_\Omega cf \geq \int_\Omega cs = c\int_\Omega s.$$

Since this is true for all simple non-negative s minorizing f, so

$$\int_\Omega cf \geq c \sup\left\{\int_\Omega s : s \text{ simple and non-negative and minorizes } f\right\} = c\int_\Omega f.$$

For the reverse inequality, let t be any simple non-negative function s.t. $t \leq cf$, then since $c > 0$, we have $\frac{1}{c}t$ simple, non-negative, and $\frac{1}{c}t \leq f$. Hence,

$$\int_\Omega t = c\int_\Omega \frac{1}{c}t \leq c\int_\Omega f.$$

Since this is true for all simple, non-negative t s.t. $t \leq cf$, so we can take supremum over all such t and gets

$$\int_\Omega cf \leq c\int_\Omega f.$$

(c) Let s be any simple, non-negative function s.t. $s \leq f$, then $s \leq g$. Thus,

$$\sup\left\{\int_\Omega s : s \text{ simple, non-negative, and } s \leq f\right\} \leq \sup\left\{\int_\Omega s : s \text{ simple, non-negative, and } s \leq g\right\},$$

i.e. $\int_\Omega f \leq \int_\Omega g$.

(d) Let $N = \{x \in \Omega : f(x) \neq g(x)\}$, then $m(N) = 0$ and N is measurable, and we have $f = g$ on $\Omega \setminus N$. We first prove that $\int_\Omega f \leq \int_\Omega g$. Let s be any non-negative simple function s.t. $s \leq f$ on Ω, then define $t := s\chi_{\Omega\setminus N}$. Note that t is simple and non-negative. Moreover, if $x \in \Omega \setminus N$, then

$$t(x) = s(x) \leq f(x) = g(x),$$

while if $x \in N$, then $t(x) = 0 \le g(x)$. Thus, $t \le g$ on Ω. Since s and t only differ on N, so $s = t$ almost everywhere in Ω. In other words, $s - t \equiv 0$ almost everywhere in Ω. This shows $\int_\Omega s - t = 0$, i.e. $\int_\Omega s = \int_\Omega t$. Note that we have

$$\int_\Omega s = \int_\Omega t \le \int_\Omega g$$

by definition of $\int_\Omega g$, and this is true for any simple, non-negative $s \le f$, so taking supremum we have $\int_\Omega f \le \int_\Omega g$. Now we can interchanging the role of f and g, and we can get $\int_\Omega g \le \int_\Omega f$.

(e) We first show that $\int_{\Omega'} f = \int_\Omega f\chi_{\Omega'}$. First note that $\int_{\Omega'} f = \int_{\Omega'} f|_{\Omega'}$. We then first show

$$\int_{\Omega'} f \le \int_\Omega f\chi_{\Omega'}.$$

Let s be any simple non-negative function on Ω' s.t. $s \le f|_{\Omega'}$. Extend s to a function $\widetilde{s}$ on Ω by

$$\widetilde{s}(x) := \begin{cases} s(x), & \text{if } x \in \Omega'; \\ 0, & \text{if } \Omega \setminus \Omega'. \end{cases}$$

Then, $\widetilde{s}$ is simple and non-negative on Ω, and we have $\widetilde{s} \le f\chi_{\Omega'}$. Moreover, we have

$$\int_\Omega \widetilde{s} = \int_{\Omega'} s$$

by definition. Hence, every simple, non-negative s with $s \le f|_{\Omega'}$ produces a simple, non-negative $\widetilde{s}$ with $\widetilde{s} \le f\chi_{\Omega'}$ and $s, \widetilde{s}$ has same Lebesgue integral. Hence, by taking supremum,

$$\int_{\Omega'} f|_{\Omega'} \le \int_\Omega f\chi_{\Omega'} \Rightarrow \int_{\Omega'} f \le \int_\Omega f\chi_{\Omega'}.$$

Now we show $\int_\Omega f\chi_{\Omega'} \le \int_{\Omega'} f$. Let t be any simple, non-negative function on Ω s.t. $t \le f\chi_{\Omega'}$, then $t(x) = 0$ for every $x \notin \Omega'$. Also, we have $t|_{\Omega'} \le f|_{\Omega'}$, and $t|_{\Omega'}$ is a simple, non-negative function on Ω'. Hence,

$$\int_\Omega t = \int_{\Omega'} t|_{\Omega'} \le \int_{\Omega'} f|_{\Omega'} = \int_{\Omega'} f.$$

Since this is true for all simple, non-negative t s.t. $t \le f\chi_{\Omega'}$, so by taking supremum, we have

$$\int_\Omega f\chi_{\Omega'} \le \int_{\Omega'} f.$$

Now we have shown $\int_{\Omega'} f = \int_\Omega f\chi_{\Omega'}$, and since $f\chi_{\Omega'} \le f$ on Ω, so

$$\int_\Omega f\chi_{\Omega'} \le \int_\Omega f.$$

■

Lecture 15

Remark 3.2.2. (d) of Proposition 3.2.1 shows that one can modify the values of a function on any measure zero set (e.g. every rational number), and not affect its integral at all. 23 April.

Remark 3.2.3. We havn't shown that $\int_\Omega (f+g) = \int_\Omega f + \int_\Omega g$. From definition, it is easy to show $\int_\Omega (f+g) \ge \int_\Omega f + \int_\Omega g$, but proving equality requires more work and will be done later.

As we have seen in previous chapters, we cannot always interchange an integral with a limit (or with limit-like concepts such as supremum). However, for the Lebesgue integral it is possible to do so when the functions are increasing:

Theorem 3.2.1 (Lebesgue monotone convergence theorem). Let Ω be a measurable subset of $\mathbb{R}^n$, and let $(f_n)_{n=1}^{\infty}$ be a sequence of non-negative measurable functions from Ω to $[0,\infty]$ which are increasing in the sense that

$$0 \le f_1(x) \le f_2(x) \le \dots \quad \forall x \in \Omega.$$

(Note that we are assuming $f_n(x)$ is increasing with respect to n; this is different notion from $f_n(x)$ increasing with respect to x.) Then we have

$$0 \le \int_\Omega f_1 \le \int_\Omega f_2 \le \int_\Omega f_3 \le \dots$$

and

$$\int_\Omega \sup_n f_n = \sup_n \int_\Omega f_n.$$

Proof. Since $f_n(x) \le f_{n+1}(x)$ for all $x \in \Omega$, then by (c) of Proposition 3.2.1, we have

$$\int_\Omega f_n \le \int_\Omega f_{n+1} \quad \forall n \ge 1.$$

In particular,

$$0 \le \int_\Omega f_1 \le \int_\Omega f_2 \le \int_\Omega f_3 \le \dots.$$

It remains to prove that

$$\int_\Omega \sup_n f_n = \sup_n \int_\Omega f_n.$$

Set $f \coloneqq \sup_n f_n$, then by Lemma 2.5.5, f is also measurable and takes values in $[0,\infty]$. We first show

$$\int_\Omega f \ge \sup_n \int_\Omega f_n.$$

Since $f_n \le f$ pointwise for every n, (c) of Proposition 3.2.1 yields

$$\int_\Omega f_n \le \int_\Omega f \quad \forall n.$$

Taking supremum over n, then we're done.

Now we show

$$\int_\Omega f \le \sup_n \int_\Omega f_n.$$

Since

$$\int_\Omega f = \sup \left\{ \int_\Omega s : s \text{ is simple, non-negative, and } s \le f \right\},$$

it suffices to show that for every simple, non-negative function s satisfying $s \le f$, one has

$$\int_\Omega s \le \sup_n \int_\Omega f_n.$$

Thus, we fix a non-negative simple function s with

$$s(x) \le f(x) = \sup_n f_n(x) \quad \forall x \in \Omega.$$

We will prove that $\int_\Omega s \le \sup_n \int_\Omega f_n$.

Now let $0 < \varepsilon < 1$. For each $n \ge 1$, define

$$E_n \coloneqq \{x \in \Omega : f_n(x) \ge (1-\varepsilon)s(x)\}.$$

Now we have f_n and s are measurable, we know that each set E_n is measurable since

$$E_n = (f_n - (1-\varepsilon)s)^{-1}([0,\infty]),$$

and since

$$A_{n,k} := (f_n - (1-\varepsilon)s)^{-1}((-\frac{1}{k}, \infty])$$

is measurable by definition for all $k \in \mathbb{N}$, we have $E_n = \bigcup_{k=1}^{\infty} A_{n,k}$ measurable.

Claim 3.2.1. $E_1 \subseteq E_2 \subseteq E_3 \subseteq \dots$ and $\bigcup_{n=1}^{\infty} E_n = \Omega$.

Proof. We first show $E_n \subseteq E_{n+1}$. This follows from $f_n \leq f_{n+1}$, so if $x \in E_n$, i.e. $f_n(x) \geq (1-\varepsilon)s(x)$, then

$$f_{n+1}(x) \geq f_n(x) \geq (1-\varepsilon)s(x) \Rightarrow x \in E_{n+1}.$$

To prove that $\bigcup_{n=1}^{\infty} E_n = \Omega$, fix $x \in \Omega$. Since

$$s(x) \leq f(x) = \sup_n f_n(x),$$

there exists some index $N(x)$ s.t. $f_{N(x)}(x) \geq s(x) \geq (1-\varepsilon)s(x)$. Thus, $x \in E_{N(x)}$, and we're done. ⊛

Since s is a non-negative simple function, we know

$$s = \sum_{j=1}^{N} c_j \chi_{F_j}$$

for some $c_1, \dots, c_N \geq 0$ and $F_1, \dots, F_N$ are pairwise disjoint measurable subsets of Ω. For each n,

$$\chi_{F_j}\chi_{E_n} = \chi_{F_j \cap E_n},$$

and hence

$$s\chi_{E_n} = \sum_{j=1}^{N} c_j \chi_{F_j}\chi_{E_n} = \sum_{j=1}^{N} c_j \chi_{F_j \cap E_n}.$$

Therefore,

$$\int_{E_n} s = \int_{\Omega} s\chi_{E_n} = \int_{\Omega} \sum_{j=1}^{N} c_j \chi_{F_j \cap E_n} = \sum_{j=1}^{N} c_j m(F_j \cap E_n).$$

Now since $E_n \uparrow \Omega$ by the claim, so for each j,

$$F_j \cap E_1 \subseteq F_j \cap E_2 \subseteq \dots \quad \text{and} \quad \bigcup_{n=1}^{\infty} (F_j \cap E_n) = F_j.$$

Since each $F_j \cap E_n$ is measurable, continuity from belowe of Lebesgure measure gives

$$m(F_j \cap E_n) \uparrow m(F_j) \quad \text{as } n \to \infty.$$

Thus,

$$\int_{E_n} s = \sum_{j=1}^{N} c_j m(F_j \cap E_n) \to \sum_{j=1}^{N} c_j m(F_j) = \int_{\Omega} s \text{ as } n \to \infty.$$

Note that since $E_1 \subseteq E_2 \subseteq \dots$, we have

$$\int_{E_1} s \leq \int_{E_2} s \leq \int_{E_3} s \leq \dots$$

and $\lim_{n\to\infty} \int_{E_n} s = \int_{\Omega} s$, so

$$\sup_n \int_{E_n} s = \int_{\Omega} s.$$

Now by definition of E_n, we have

$$f_n(x) \geq (1-\varepsilon)s(x) \quad \text{for every } x \in E_n.$$

Hence, by (c), (e) of Proposition 3.2.1,

$$\int_\Omega f_n \geq \int_{E_n} f_n \geq \int_{E_n} (1-\varepsilon)s = (1-\varepsilon)\int_{E_n} s.$$

Taking supremum over n,

$$\sup_n \int_\Omega f_n \geq (1-\varepsilon) \sup_n \int_{E_n} s = (1-\varepsilon)\int_\Omega s.$$

Since this holds for all $0 < \varepsilon < 1$, letting $\varepsilon \to 0$ gives

$$\sup_n \int_\Omega f_n \geq \int_\Omega s.$$

Hence, we're done. ■

Lemma 3.2.1 (Interchange of addition and integration)**.** Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f, g : \Omega \to [0,\infty]$ be measurable functions. Then,

$$\int_\Omega (f+g) = \int_\Omega f + \int_\Omega g.$$

Proof. By Lemma 3.1.3, there exist increasing sequence of non-negative simple functions

$$s_1 \leq s_2 \leq \cdots \leq f, \quad t_1 \leq t_2 \leq \cdots \leq g$$

s.t. $s_n(x) \uparrow f(x)$ and $t_n(x) \uparrow g(x)$ for every $x \in \Omega$. For each n, $s_n + t_n$ is again a non-negative simple function. Moreover, since both (s_n) and (t_n) are increasing, so is $(s_n + t_n)$. For each $x \in \Omega$, we have

$$s_n(x) + t_n(x) \uparrow f(x) + g(x),$$

and hence $s_n + t_n \uparrow f + g$ pointwise on Ω. Applying the monotone convergence theorem to the three increasing sequecnes

$$s_n \uparrow f, \quad t_n \uparrow g, \quad s_n + t_n \uparrow f + g,$$

we obtain

$$\int_\Omega f = \sup_n \int_\Omega s_n, \quad \int_\Omega g = \sup_n \int_\Omega t_n, \quad \int_\Omega (f+g) = \sup_n \int_\Omega (s_n + t_n).$$

Note that

$$\int_\Omega (s_n + t_n) = \int_\Omega s_n + \int_\Omega t_n \quad \text{for every } n$$

since s_n, t_n are non-negative simple functions. Thus,

$$\int_\Omega (f+g) = \sup_n \int_\Omega (s_n + t_n) = \sup_n \left(\int_\Omega s_n + \int_\Omega t_n \right)$$

Now let $a_n := \int_\Omega s_n$ and $b_n := \int_\Omega t_n$, then (a_n) and (b_n) are increasing seuqences in $[0,\infty]$. Consequently,

$$a_n + b_n \uparrow \left(\sup_n a_n\right) + \left(\sup_n b_n\right),$$

Thus, $\sup_n (a_n + b_n) = \sup_n a_n + \sup_n b_n$. It follows that

$$\int_\Omega (f+g) = \sup_n \left(\int_\Omega s_n + \int_\Omega t_n \right) = \sup_n \int_\Omega s_n + \sup_n \int_\Omega t_n = \int_\Omega f + \int_\Omega g.$$

■

Lecture 16

Once we can interchange an integral with a sum of two functions, we can handle fininte sums by induction. More suprisingly, we can handle infinite sums of non-negative functions as well: 28 April.

Corollary 3.2.1. If $\Omega \subseteq \mathbb{R}^n$ is measurable and $g_1, g_2, \ldots$ are a sequence of non-negative measurable functions from Ω to $[0, \infty]$, then

$$\int_\Omega \sum_{n=1}^\infty g_n = \sum_{n=1}^\infty \int_\Omega g_n.$$

Proof. For each integer $N \geq 1$, define $G_N := \sum_{n=1}^N g_n$. Since g_n is measurable and non-negative, it follows that each G_N is also measurable and non-negative. Moreover, the sequence (G_N) is increasing pointwise, because

$$G_{N+1}(x) = G_N(x) + g_{N+1}(x) \geq G_N(x) \quad \forall x \in \Omega.$$

Hence, we may apply the Lebesgue monotone convergence theorem to the sequence (G_N). We obtain

$$\int_\Omega \sup_N G_N = \sup_N \int_\Omega G_N.$$

Now since for each $x \in \Omega$, $G_1(x) \leq G_2(x) \leq \ldots$, so $\sup_N G_N(x) = \lim_{N\to\infty} G_N(x) = \sum_{n=1}^\infty g_n(x)$. This gives

$$\int_\Omega \sup_N G_N = \int_\Omega \sum_{n=1}^\infty g_n.$$

Now since for each $N \in \mathbb{N}$,

$$\int_\Omega G_N = \int_\Omega \sum_{n=1}^N g_n = \sum_{n=1}^N \int_\Omega g_n.$$

Hence,

$$\sup_N \int_\Omega G_N = \sup_N \sum_{n=1}^N \int_\Omega g_n.$$

Note that the sequence $\left(\sum_{n=1}^N \int_\Omega g_n\right)_{N=1}^\infty$ is increasing, so

$$\sup_N \sum_{n=1}^N \int_\Omega g_n = \sum_{n=1}^\infty \int_\Omega g_n.$$

Thus,

$$\int_\Omega \sum_{n=1}^\infty g_n = \int_\Omega \sup_N G_N = \sup_N \int_\Omega G_N = \sum_{n=1}^\infty \int_\Omega g_n.$$

■

Remark 3.2.4. We do not need to assume anything about convergence of above sums; it may well happen that both sides equal $+\infty$. However, we do need to assume non-negativity.

One might similarly ask whether we can interchange limits and integrals; i.e. whether

$$\int_\Omega \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int_\Omega f_n$$

holds in general. Unfortunately this is not true: For $n = 1, 2, 3, \ldots$ let $f_n : \mathbb{R} \to \mathbb{R}$ be $f_n = \chi_{[n,n+1)}$. Then, $f_n(x) \to 0$ for every x, but $\int_\mathbb{R} f_n = 1$ for every n since f_n is a simple function, and thus

$$\lim_{n\to\infty} \int_\mathbb{R} f_n = 1 \neq 0.$$

However, the following lemma of Fatou shows that the reverse cannot happen: the limiting function cannot have larger integral than the (liminf of the) original integrals.

Lemma 3.2.2 (Fatou's lemma)**.** Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f_1, f_2, \cdots : \Omega \to [0, \infty]$ be a sequence of non-negative measurable functions. Then

$$\int_\Omega \liminf_{n\to\infty} f_n \le \liminf_{n\to\infty} \int_\Omega f_n.$$

Proof. For each $N \in \mathbb{N}$, define

$$g_N(x) := \inf_{n \ge N} f_n(x) \quad \forall x \in \Omega.$$

Then g_N is non-negative and measurable. Moreover, $g_1(x) \le g_2(x) \le \dots$ for all $x \in \Omega$. By the definition of liminf,

$$\liminf_{n\to\infty} f_n(x) = \sup_{N\ge 1} g_N(x).$$

Since $(g_N(x))$ is increasing, this means that $g_N(x) \uparrow \liminf_{n\to\infty} f_n(x)$ for all $x \in \Omega$. Note that we can apply Lebesgue monotone convergence theorem since (g_N) is increasing:

$$\int_\Omega \liminf_{n\to\infty} f_n(x) = \int_\Omega \sup_{N\ge 1} g_N = \sup_{N\ge 1} \int_\Omega g_N.$$

On the other hand, for every N and every $n \ge N$, we have $g_N \le f_n$. Hence,

$$\int_\Omega g_N \le \int_\Omega f_n \quad \forall n \ge N.$$

Taking infinimum over all $n \ge N$, we get $\int_\Omega g_N \le \inf_{n\ge N} \int_\Omega f_n$. Now take the supremum over N to obtain

$$\sup_{N\ge 1} \int_\Omega g_N \le \sup_{N\ge 1} \inf_{n\ge N} \int_\Omega f_n.$$

Combining previous result, we have

$$\int_\Omega \liminf_{n\to\infty} f_n(x) = \sup_{N\ge 1} \int_\Omega g_N \le \sup_{N\ge 1} \inf_{n\ge N} \int_\Omega f_n = \liminf_{n\to\infty} f_n.$$

■

Note that we are allowing our functions to take the value $+\infty$ at some points. It is even possible for a function to take the value $+\infty$ but still have a finite integral; for instance, if E is a set of measure zero, and $f : \Omega \to \mathbb{R}$ is equal to $+\infty$ on E but equal 0 everywhere else, then $\int_\Omega f = 0$ by (a) of Proposition 3.2.1. However, if the integral is finite, the function must be finite almost everywhere:

Lemma 3.2.3. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to [0, \infty]$ be a non-negative measurable function s.t. $\int_\Omega f$ is finite. Then f is finite almost everywhere, i.e. the set $\{x \in \Omega : f(x) = +\infty\}$ has measure zero.

Proof. For each integer $k \ge 1$, define $A_k := \{x \in \Omega : f(x) \ge k\}$. Since f is measurable and $[k, \infty]$ is a Borel subset of $[0, \infty]$, each set A_k is measurable.

On the set A_k, we have $f(x) \ge k$, while outside A_k the function $k\chi_{A_k}$ vanishes. Thus, for every $x \in \Omega$, $k\chi_{A_k}(x) \le f(x)$. Because both $k\chi_{A_k}$ and f are non-negative measurable functions, the monotonicity property gives $\int_\Omega k\chi_{A_k} \le \int_\Omega f$. Now

$$\int_\Omega k\chi_{A_k} = k \int_\Omega \chi_{A_k} = km(A_k),$$

so we obtain $km(A_k) \le \int_\Omega f$. Since $\int_\Omega f < \infty$, it follows that $m(A_k) \le \frac{1}{k} \int_\Omega f$, and thus

$$m(A_k) \to 0 \quad \text{as } k \to \infty.$$

Note that

$$\{x \in \Omega : f(x) = +\infty\} = \bigcap_{k=1}^{\infty} A_k.$$

Also, $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$ by definition, and by continuity of measure (HW5 P2),

$$m\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} m(A_k).$$

Since we have shown $\lim_{k\to\infty} m(A_k) = 0$, so we have

$$m\left(\{x \in \Omega : f(x) = +\infty\}\right) = m\left(\bigcap_{k=1}^{\infty} A_k\right) = 0.$$

■

Lemma 3.2.4 (Borel-Cantelli lemma)**.** Let $\Omega_1, \Omega_2, \dots$ be measurable subsets of $\mathbb{R}^n$ such that

$$\sum_{n=1}^{\infty} m(\Omega_n) < \infty.$$

Then the set

$$\{x \in \mathbb{R}^n : x \in \Omega_k \text{ for infinitely many } k\}$$

has measure zero. Equivalently, for almost every $x \in \mathbb{R}^n$, there exists $N = N(x)$ s.t. $x \notin \Omega_k$ for all $k \geq N$.

Proof. Define

$$G(x) \coloneqq \sum_{n=1}^{\infty} \chi_{\Omega_n}(x), \quad \forall x \in \mathbb{R}^n,$$

where the value $+\infty$ is allowed.

For each n, χ_{Ω_n} is measurable and non-negative. Hence, the partial sums

$$G_N(x) \coloneqq \sum_{n=1}^{N} \chi_{\Omega_n}(x)$$

are non-negative measurable functions. Since $G_N(x) \uparrow G(x)$ for every $x \in \mathbb{R}^n$. It follows that $G = \sup_N G_N$ is also measurable and non-negative. By Corollary 3.2.1, we have

$$\int_{\mathbb{R}^n} G = \int_{\mathbb{R}^n} \sum_{n=1}^{\infty} \chi_{\Omega_n} = \sum_{n=1}^{\infty} \int_{\mathbb{R}^n} \chi_{\Omega_n}.$$

Since $\int_{\mathbb{R}^n} \chi_{\Omega_n} = m(\Omega_n)$, this becomes

$$\int_{\mathbb{R}^n} G = \sum_{n=1}^{\infty} m(\Omega_n) < \infty.$$

By Lemma 3.2.3,

$$m\left(\{x \in \mathbb{R}^n : G(x) = +\infty\}\right) = 0.$$

Note that

$$G(x) = +\infty \Leftrightarrow \sum_{n=1}^{\infty} \chi_{\Omega_n}(x) = \infty \Leftrightarrow x \in \Omega_k \text{ for infinitely many } k,$$

so

$$\{x \in \mathbb{R}^n : G(x) = +\infty\} = \{x \in \mathbb{R}^n : x \in \Omega_k \text{ for infinitely many } k\},$$

and we're done. ■

3.3 Integration of Absolutely Integrable Functions

We have now completed the theory of the Lebesgue integral for non-negative functions. We now consider how to integrate functions which can be both positive and negative. However, we wish to avoid the indefinite expression $+\infty + (-\infty)$, so we restrict our attention to a subclass of measurable functions: the absolutely integrable functions.

Definition 3.3.1. Let Ω be a measurable subset of $\mathbb{R}^n$. A measurable function $f : \Omega \to \mathbb{R}^*$ is said to be absolutely integrable if the integral $\int_\Omega |f|$ is finite.

Of course, $|f|$ is always non-negative, so this definition makes sense even if f changes sign. Absolutely integrable functions are also known as $L^1(\Omega)$ functions.

If $f : \Omega \to \mathbb{R}^*$ is a function, we define the positive part $f^+ : \Omega \to [0, \infty]$ and the negative part $f^- : \Omega \to [0, \infty]$ by

$$f^+ := \max(f, 0), \quad f^- := -\min(f, 0).$$

Equivalently, for each $x \in \Omega$,

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}.$$

From Corollary 2.5.3, which can be extended to $\mathbb{R}^*$-valued functions without difficulty, we know that f^+, f^- are measurable. Clearly f^+ and f^- are non-negative.

We also have the pointwise identities

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Definition 3.3.2 (Lebesgue integral)**.** Let $f : \Omega \to \mathbb{R}^*$ be an absolutely integrable function. We define the Lebesgue integral $\int_\Omega f$ of f to be the quantity

$$\int_\Omega f := \int_\Omega f^+ - \int_\Omega f^-.$$

Remark 3.3.1. Note that since f is absolutely integrable, $\int_\Omega f^+$ and $\int_\Omega f^-$ are less than or equal to $\int_\Omega |f|$ and hence are finite. Thus, $\int_\Omega f$ is always finite; we are never encountering the indeterminate form $+\infty - (+\infty)$.

Remark 3.3.2. We have the useful triangle inequality

$$\left|\int_\Omega f\right| \leq \int_\Omega |f|.$$

This is because

$$\left|\int_\Omega f\right| = \left|\int_\Omega f^+ - \int_\Omega f^-\right| \leq \left|\int_\Omega f^+\right| + \left|\int_\Omega f^-\right| = \int_\Omega f^+ + \int_\Omega f^- = \int_\Omega f^+ + f^- = \int_\Omega |f|.$$

Proposition 3.3.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f, g : \Omega \to \mathbb{R}^*$ be absolutely integrable functions. Then the following statements hold.

(a) $\forall c \in \mathbb{R}$, the function cf is absolutely integrable, and $\int_\Omega cf = c\int_\Omega f$.

(b) The function $f + g$ is absolutely integrable, and $\int_\Omega (f + g) = \int_\Omega f + \int_\Omega g$.

(c) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_\Omega f \leq \int_\Omega g$.

(d) If $f(x) = g(x)$ for almost every $x \in \Omega$, then $\int_\Omega f = \int_\Omega g$.

Proof.

(a) We first show cf is absolutely integrable. Since $|cf| = |c||f|$, we have $\int_\Omega |cf| = |c| \int_\Omega |f|$ since f is absolutely integrable and thus $\int_\Omega |f| < \infty$.

Now it remains to compute its integral. We consider two cases.
Case 1: $c \geq 0$. In this case, $(cf)^+ = cf^+$ and $(cf)^- = cf^-$. Thus,

$$\begin{aligned}\int_\Omega cf &= \int_\Omega (cf)^+ - \int_\Omega (cf)^- = \int_\Omega cf^+ - \int_\Omega cf^- = c\int_\Omega f^+ - c\int_\Omega f^- \\ &= c\left(\int_\Omega f^+ - \int_\Omega f^-\right) = c\int_\Omega f.\end{aligned}$$

Case 2: $c < 0$. Now we have $(cf)^+ = -cf^-$ and $(cf)^- = -cf^+$. Hence,

$$\begin{aligned}\int_\Omega cf &= \int_\Omega (cf)^+ - \int_\Omega (cf)^- = \int_\Omega (-c)f^- - \int_\Omega (-c)f^+ = (-c)\int_\Omega f^- - (-c)\int_\Omega f^+ \\ &= c\int_\Omega f^+ - c\int_\Omega f^- = c\left(\int_\Omega f^+ - \int_\Omega f^-\right) = c\int_\Omega f.\end{aligned}$$

(b) We first show that $f + g$ is absolutely integrable. Note that

$$\int_\Omega |f+g| \leq \int_\Omega |f| + |g| = \int_\Omega |f| + \int_\Omega |g| < \infty$$

by the properties of Lebesgue integral on non-negative measurbale functions, and hence we're done.

Now since

$$f + g = (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-),$$

where $f^+ + g^+$ and $f^- + g^-$ are non-negative measurable functions. Hence,

$$\begin{aligned}\int_\Omega (f+g) &= \int_\Omega (f^+ + g^+) - (f^- + g^-) \\ &= \int_\Omega (f^+ + g^+) - \int_\Omega (f^- + g^-) = \left(\int_\Omega f^+ + \int_\Omega g^+\right) - \left(\int_\Omega f^- + \int_\Omega g^-\right) \\ &= \left(\int_\Omega f^+ - \int_\Omega f^-\right) + \left(\int_\Omega g^+ - \int_\Omega g^-\right) = \int_\Omega f + \int_\Omega g.\end{aligned}$$

(c) Assume that $f(x) \leq g(x)$ for all $x \in \Omega$, then $g - f$ is non-negative everywhere on Ω. By (a),(b), we know $g - f = g + (-f)$ is absolutely integrable. Also,

$$\int_\Omega g = \int_\Omega (f + (g - f)) = \int_\Omega f + \int_\Omega (g - f).$$

Since $g - f$ is non-negative and measurable, $\int_\Omega (g - f) \geq 0$, and thus $\int_\Omega g \geq \int_\Omega f$, and we're done.

(d) Assume that $f(x) = g(x)$ for almost every $x \in \Omega$. Define $h := f - g$. Then $h = 0$ almost everywhere. Also, since f, g are absolutely integrable, (a) and (b) shows that h is absolutely integrable. Because $h = 0$ almost everywhere, we also have $|h| = 0$ almost everywhere. This shows $\int_\Omega |h| = 0$ by properties of Lebesgue integral for non-negative measurable functions. Now apply the traingle inequality:

$$\left|\int_\Omega h\right| \leq \int_\Omega |h| = 0,$$

which shows $\int_\Omega h = 0$. This shows

$$0 = \int_\Omega h = \int_\Omega f - g = \int_\Omega f - \int_\Omega g,$$

and thus $\int_\Omega f = \int_\Omega g$.

■

Lecture 17

As discussed in the previous section, limits and integrals cannot in general be interchanged. That is, even if $f_n \to f$ pointwise on Ω, it may happen that 30 April.

$$\lim_{n\to\infty} \int_\Omega f_n \neq \int_\Omega \lim_{n\to\infty} f_n.$$

Thus, in order to justify passing a limit through an integral, one needs an additional hypothesis. The Lebesgue dominated convergence theorem provides one of the most useful such criteria: if the sequence is dominated by a single absolutely integrable function, then the interchange of limit and integral is valid.

Theorem 3.3.1 (Lebesgue dominated convergence theorem). Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f_1, f_2, \ldots$ be measurable functions from Ω to $\mathbb{R}^*$ such that $f_n(x)$ converges pointwise for every $x \in \Omega$. Suppose moreover that there exists an absolutely integrable function $F : \Omega \to [0, \infty]$ such that $|f_n(x)| \leq F(x)$ for all $x \in \Omega$ and all $n \geq 1$. Then, if we write $f(x) := \lim_{n\to\infty} f_n(x)$, we have

$$\int_\Omega f = \lim_{n\to\infty} \int_\Omega f_n.$$

Proof. Since F is non-negative, measurable, and absolutely integrable, we knoe $F < \infty$ almost everywhere by Lemma 3.2.3. In other words, $E := \{x \in \Omega : F(x) = +\infty\}$ has measure zero. If we replace Ω by $\Omega \setminus E$, then none of the relevant integrals changes, because deleting a measure zero set does not affect the integral of an absolutely integrable function. Hence, without loss of generality, we may assume from now on that $F(x) < \infty$ for every $x \in \Omega$. Since $|f_n(x)| \leq F(x)$ for all n, this also implies that each $f_n(x)$ is finite for every $x \in \Omega$.

Because each f_n is measurable and $f_n \to f$ pointwise, f is measurable by Lemma 2.5.5. Next, since $|f_n(x)| \leq F(x)$ for all n and all $x \in \Omega$, we may pass to the limit in n and obtain

$$|f(x)| \leq F(x) \quad \text{for all } x \in \Omega.$$

Therefore,

$$\int_\Omega |f| \leq \int_\Omega F < \infty,$$

so f is absolutely integrable. The same argument can shows that f_n is absolutely integrable, because

$$\int_\Omega |f_n| \leq \int_\Omega F < \infty.$$

Thus, all of the integrals appearing below are finite and well-defined.

For each n and each $x \in \Omega$, we have $|f_n(x)| \leq F(x)$. Hence,

$$-F(x) \leq f_n(x) \leq F(x).$$

It follows that $F(x) + f_n(x) \geq 0$ for all $x \in \Omega$. So the function $F + f_n$ are non-negative and measurable. Moreover, since $f_n(x) \to f(x)$, we also have

$$F(x) + f_n(x) \to F(x) + f(x) \quad \text{pointwise on } \Omega.$$

Fatou's lemma therefore gives

$$\int_\Omega (F+f) = \int_\Omega \liminf_{n\to\infty}(F+f_n) \le \liminf_{n\to\infty} \int_\Omega (F+f_n).$$

Using additivity of the integral for absolutely integrable functions, this becomes

$$\int_\Omega F + \int_\Omega f \le \liminf_{n\to\infty}\left(\int_\Omega F + \int_\Omega f_n\right).$$

Since $\int_\Omega F$ is a fixed finite number, we may subtract it from both sides and obtain

$$\int_\Omega f \le \liminf_{n\to\infty} \int_\Omega f_n.$$

Similarly, since $|f_n(x)| \le F(x)$, we have $F(x) - f_n(x) \ge 0$ for all $x \in \Omega$. Thus, $F - f_n$ is non-negative and measurable for all n. Since $f_n(x) \to f(x)$ pointwise, we also have

$$F(x) - f_n(x) \to F(x) - f(x) \quad \text{pointwise on } \Omega.$$

Applying Fatou's lemma once more, we obtain

$$\int_\Omega (F-f) = \int_\Omega \liminf_{n\to\infty}(F-f_n) \le \liminf_{n\to\infty} \int_\Omega (F-f_n).$$

By additivity,

$$\int_\Omega F - \int_\Omega f \le \liminf_{n\to\infty}\left(\int_\Omega F - \int_\Omega f_n\right) = \int_\Omega F - \limsup_{n\to\infty} \int_\Omega f_n.$$

This gives

$$\int_\Omega f \ge \limsup_{n\to\infty} \int_\Omega f_n.$$

Now we have

$$\limsup_{n\to\infty} \int_\Omega f_n \le \int_\Omega f \le \liminf_{n\to\infty} \int_\Omega f_n.$$

But for real sequence (a_n) we always has

$$\liminf_{n\to\infty} a_n \le \limsup_{n\to\infty} a_n.$$

Therefore, the only possibility is that

$$\liminf_{n\to\infty} \int_\Omega f_n = \limsup_{n\to\infty} \int_\Omega f_n = \int_\Omega f.$$

Hence, the sequence $(\int_\Omega f_n)$ converges, and its limit is $\int_\Omega f$ by Theorem B.0.1. That is,

$$\lim_{n\to\infty} \int_\Omega f_n = \int_\Omega f.$$

This proves the dominated convergence theorem. ■

Finally, we record a lemma which is not particularly interesting in itself, but we will have some useful consequences later in these notes.

Definition 3.3.3 (Upper and lower Lebesgue integral)**.** Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}$ (not necessarily measurable). We define the upper Lebesgue integral $\overline{\int_\Omega} f$ to be

$$\overline{\int_\Omega} f := \inf\left\{\int_\Omega g : g \text{ is an absolutely integrable function from } \Omega \text{ to } \mathbb{R} \text{ that majorizes } f\right\},$$

and the lower Lebesgue integral $\underline{\int_\Omega} f$ to be

$$\underline{\int_\Omega} f := \sup\left\{\int_\Omega g : g \text{ is an absolutely integrable function from } \Omega \text{ to } \mathbb{R} \text{ that minorizes } f\right\}.$$

We first record two elementary consequences of this definition.

It is easy to see that

$$\underline{\int_\Omega} f \le \overline{\int_\Omega} f.$$

Next, suppose f is absolutely integrable. Then f is both an absolutely integrable minorant and an absolutely integrable majorant of itself. Therefore,

$$\underline{\int_\Omega} f = \sup\left\{\int_\Omega u : u \text{ is absolutely intergable and } u \le f\right\} \ge \int_\Omega f.$$

On the other hand, if $u \le f$ for some absolutely integrable u, then

$$\int_\Omega u \le \int_\Omega f.$$

Taking the supremum over all absolutely integrable $u \le f$, we obtain

$$\underline{\int_\Omega} f \le \int_\Omega f.$$

Hence, $\underline{\int_\Omega} f = \int_\Omega f$.

Similarly, since f is an absolutely integrable majorant of itself,

$$\overline{\int_\Omega} f = \inf\left\{\int_\Omega v : v \text{ is absolutely integrable and } f \le v\right\} \le \int_\Omega f.$$

Conversely, if $f \le v$ for some absolutely integrbale v, then

$$\int_\Omega f \le \int_\Omega v.$$

Taking the infimum over all absolutely integrbale $v \ge f$, we get $\int_\Omega f \le \overline{\int_\Omega} f$. Thus, $\overline{\int_\Omega} f = \int_\Omega f$. Consequently, whenever f is absolutely integrable,

$$\underline{\int_\Omega} f = \int_\Omega f = \overline{\int_\Omega} f.$$

The converse is also true:

Lemma 3.3.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}$ be a function, not assumed a priori to be measurable. Suppose that there exists $A \in \mathbb{R}$ s.t.

$$\overline{\int_\Omega} f = \underline{\int_\Omega} f = A.$$

Then f is absolutely integrable, and

$$\int_\Omega f = \overline{\int_\Omega} f = \underline{\int_\Omega} f = A.$$

Proof. We will show that f agree almost everywhere with an absolutely integrable function, and hence f itself is absolutely integrable with integral A.

By definition,

$$\overline{\int_\Omega} f = \inf\left\{\int_\Omega g : g \text{ is absolutely integrable and } g \ge f\right\}.$$

Since this infimum is equal to A, for every $n \ge 1$, there exists an absolutely integrable function

$f_n^+ : \Omega \to \mathbb{R}$ s.t. f_n^+ majorizes f and

$$\int_\Omega f_n^+ \le A + \frac{1}{n}.$$

Similarly, since

$$A = \underline{\int_\Omega} f = \sup\left\{\int_\Omega g : g \text{ is absolutely integrable and } g \le f\right\},$$

for every integer $n \ge 1$ there exists an absolutely integrable function $f_n^- : \Omega \to \mathbb{R}$ s.t. f_n^- minorizes f and

$$\int_\Omega f_n^- \ge A - \frac{1}{n}.$$

Thus, for every n we have $f_n^- \le f \le f_n^+$ on Ω.

Now set

$$F^+(x) := \inf_{n\ge1} f_n^+(x), \quad F^-(x) := \sup_{n\ge1} f_n^-(x), \quad x \in \Omega.$$

Since each f_n^+ and f_n^- is measurable, we know F^+ and F^- are measurable by Lemma 2.5.5. Since $f(x) \le f_n^+(x)$ for all n, we have

$$f(x) \le \inf_{n\ge1} f_n^+(x) = F^+(x).$$

Similarly, we have

$$F^-(x) = \sup_{n\ge1} f_n^-(x) \le f(x).$$

Hence,

$$F^-(x) \le f(x) \le F^+(x) \quad \forall x \in \Omega.$$

We first consider F^+. Since $F^+ \le f_1^+$ pointwise and f_1^+ is absolutely integrable, we would like to conclude that F^+ is also absolutely integrable. For this we also need a lower bound by an absolutely integrable functoin.

Because $f_n^- \le f \le F^+$ for every n, in particular we have

$$f_1^- \le F^+ \le f_1^+.$$

Hence,

$$|F^+| \le \max\left\{|f_1^-|, |f_1^+|\right\} \le |f_1^-| + |f_1^+|.$$

Since f_1^- and f_1^+ are absolutely integrable, so is $|f_1^-| + |f_1^+|$. By comparison, F^+ is absolutely integrable.

The same argument applies to F^-, by $f_1^- \le F^- \le f_1^+$, we have

$$|F^-| \le |f_1^-| + |f_1^+|$$

and thus F^- is also absolutely integrable.

Now since $F^+ \le f_n^+$ for all n and they are both absolutely integrable,

$$\int_\Omega F^+ \le \int_\Omega f_n^+ \le A + \frac{1}{n} \quad \forall n \ge 1.$$

Letting $n \to \infty$, we obtain

$$\int_\Omega F^+ \le A.$$

Similarly, we have

$$A - \frac{1}{n} \le \int_\Omega f_n^- \le \int_\Omega F^-.$$

Letting $n \to \infty$, we have

$$A \le \int_\Omega F^-.$$

On the other hand, since $F^- \leq F^+$ pointwise, we have

$$\int_\Omega F^- \leq \int_\Omega F^+.$$

Combining the three inequalities,

$$A \leq \int_\Omega F^- \leq \int_\Omega F^+ \leq A.$$

Thus,

$$\int_\Omega F^- = \int_\Omega F^+ = A.$$

Now since

$$\int_\Omega (F^+ - F^-) = \int_\Omega F^+ - \int_\Omega F^- = A - A = 0,$$

we know $F^+ - F^-$ is zero almost everywhere since $F^+ - F^-$ is a non-negative measurable function. Hence, $F^+ = F^-$ almost everywhere on Ω.

Recall that

$$F^-(x) \leq f(x) \leq F^+(x) \quad \forall x \in \Omega.$$

At every point where $F^+(x) = F^-(x)$, we have

$$f(x) = F^+(x) = F^-(x).$$

Since $F^+ = F^-$ almost everywhere, it follows that $f = F^+$ almost everywhere. Now F^+ is measurable and absolutely integrable. A function which agrees almost everywhere with a measurable function is measurable by Theorem B.0.2, so f is measurable. Also, $|f| = |F^+|$ almost everywhere, and therefore

$$\int_\Omega |f| = \int_\Omega |F^+| < \infty.$$

Thus, f is absolutely integrable.

Finally, since $f = F^+$ almost everywhere, we have

$$\int_\Omega f = \int_\Omega F^+ = A.$$

Therefore,

$$\int_\Omega f = \overline{\int_\Omega} f = \underline{\int_\Omega} f = A.$$

■

3.4 Comparison with the Riemann Integral

We have spent a lot of effort constructing the Lebesgue integral, but have not yet addressed the question of how to actually compute any Lebesgue integrals, and whether Lebesgue integration is any dierent from the Riemann integral (say for integrals in one dimension). We now show that the Lebesgue integral is a generalization of the Riemann integral. To clarify the following discussion, we shall temporarily distinguish the Riemann integral from the Lebesgue integral by writing the Riemann integral as $R. \int_I f$. Recall that, in the definition of the Riemann integral on a bounded interval $I \subseteq \mathbb{R}$, one starts with a bounded function $f : I \to \mathbb{R}$. The boundedness of both the interval I and the function f will be important. Our objective in this section is to prove that every Riemann integrable function on a bounded interval is Lebesgue integrable, indeed absolutely integrable, and that the two integrals agree.

Proposition 3.4.1. Let $I \subseteq \mathbb{R}$ be a bounded interval, and let $f : I \to \mathbb{R}$ be a Riemann integrable function. Then f is also absolutely integrable, and

$$\int_I f = R.\int_I f.$$

Proof. Write $A \coloneqq R.\int_I f$. Since f is Riemann integrable, f's upper integral and lower integral agrees, and thus for every $\varepsilon > 0$, there exists a partition $\mathcal{P}$ of I into finitely many subintervals J s.t.

$$A - \varepsilon \leq \sum_{J \in \mathcal{P}} |J| \inf_{x \in J} f(x) \leq A \leq \sum_{J \in \mathcal{P}} |J| \sup_{x \in J} f(x) \leq A + \varepsilon,$$

where $|J|$ represents the length of J. Note that $m(J) = |J|$ since J is a one-dimensional box.

Define simple functions

$$f_\varepsilon^-(x) = \sum_{J \in \mathcal{P}} \left(\inf_{t \in J} f(t) \right) \chi_J(x), \quad f_\varepsilon^+(x) = \sum_{J \in \mathcal{P}} \left(\sup_{t \in J} f(t) \right) \chi_J(x).$$

By construction, we have $f_\varepsilon^- \leq f \leq f_\varepsilon^+$ pointwise on I. Note that $f_\varepsilon^-, f_\varepsilon^+$ are simple, and thus measurable. Moreover, since I is bounded, so it has finite measure, so f_ε^- and f_ε^+ are absolutely integrable. Also,

$$\int_I f_\varepsilon^- = \sum_{J \in \mathcal{P}} \inf_{t \in J} f(t) \cdot |J|, \quad \int_I f_\varepsilon^+ = \sum_{J \in \mathcal{P}} \sup_{t \in J} f(t) \cdot |J|.$$

Therefore,

$$A - \varepsilon \leq \int_I f_\varepsilon^- \leq A \leq \int_I f_\varepsilon^+ \leq A + \varepsilon.$$

Since f_ε^+ majorizes f, and f_ε^+ is absolutley integrable, by definition we have $\overline{\int_I} f \leq \int_I f_\varepsilon^+$. Similarly, since f_ε^- minorizes f and is absolutely integrable, we have $\underline{\int_I} f \geq \int_I f_\varepsilon^-$. Hence,

$$A - \varepsilon \leq \underline{\int_I} f \leq \overline{\int_I} f \leq A + \varepsilon \quad \forall \varepsilon > 0.$$

This shows that $\underline{\int_I} f = \overline{\int_I} f = A$. By the previous lemma, f is absolutely integrable and $\int_I f = A$. Thus,

$$\int_I f = R.\int_I f.$$

■

8.5 Fubini's Theorem

In one dimension, we have already seen how the Lebesgue integral is related to the Riemann integral. We now turn to the corresponding question in higher dimensions. To keep the discussion simple, we shall focus on two-dimensional integrals, although the same ideas extend naturally to higher dimensions. We shall study integrals of the form

$$\int_{\mathbb{R}^2} f.$$

Once integration over $\mathbb{R}^2$ is understood, integration over a measurable subset $\Omega \subseteq \mathbb{R}^2$ follows by writing

$$\int_\Omega f = \int_{\mathbb{R}^2} f \chi_\Omega.$$

Let $f(x, y)$ be a function of two variables. There are, in principle, three natural ways to integrate f over $\mathbb{R}^2$. First, we may regard f as a function on $\mathbb{R}^2$ and take its two-dimensional Lebesgue integral:

$$\int_{\mathbb{R}^2} f(x, y) dx\, dy.$$

Second, we may fix x, integrate first with respect to y, and then integrate the resulting function of x:

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx.$$

Third, we may reverse the order of integration: fix y, integrate first with respect to x, and then integrate the resulting function of y:

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)dx \right) dy.$$

A fundamental question is whether these three quantities are equal. The answer is provided by Fubini's theorem and Tonelli's theorem. Roughly speaking, Tonelli's theorem applies to non-negative measurable functions, while Fubini's theorem applies to absolutely integrable functions. Together, these theorems allow us to compute many higher-dimensional integrals by reducing them to successive one-dimensional integrals. Fortunately, if the function f is absolutely integrable on $\mathbb{R}^2$, then all three integrals are equal:

Theorem 3.4.1 (Fubini's theorem)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be an absolutely integrable function. Then there exist absolutely integrable functions $F : \mathbb{R} \to \mathbb{R}$ and $G : \mathbb{R} \to \mathbb{R}$ such that for almost every x, $f(x,y)$ is absolutely integrable in y with

$$F(x) = \int_{\mathbb{R}} f(x,y)dy,$$

and for almost every y, $f(x,y)$ is absolutely integrable in x with

$$G(y) = \int_{\mathbb{R}} f(x,y)dx.$$

Finally, we have

$$\int_{\mathbb{R}} F(x)dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G(y)dy.$$

Proof. The proof of Fubini's theorem is somewhat delicate, mainly because the one-dimensional integrals $\int_{\mathbb{R}} f(x,y)dy$ and $\int_{\mathbb{R}} f(x,y)dx$ need not exist for every x and y. We shall prove the theorem by a sequence of reductions, keeping track of where the "almost everywhere" statements enter.

Step 1: Reduction to non-negative functions. Suppose first that the theorem has been proved for non-negative integrable functions. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be absolutely integrable. Then

$$f = f^+ - f^-,$$

with $f^+, f^- \geq 0$, and $\int_{\mathbb{R}^2} f^+ < \infty$, $\int_{\mathbb{R}^2} f^- < \infty$. Thus f^+ and f^- are both non-negative integrable functions on $\mathbb{R}^2$. By the non-negative case, there exist absolutely integrable functions $F^+, F^- : \mathbb{R} \to \mathbb{R}$ such that, for almost every x, $F^+(x) = \int_{\mathbb{R}} f^+(x,y)dy$, $F^-(x) = \int_{\mathbb{R}} f^-(x,y)dy$. Moreover, $\int_{\mathbb{R}} F^+(x)dx = \int_{\mathbb{R}^2} f^+$, $\int_{\mathbb{R}} F^-(x)dx = \int_{\mathbb{R}^2} f^-$. For almost every x, both one-dimensional integrals above are finite. Hence, for almost every x,

$$\int_{\mathbb{R}} |f(x,y)|dy = \int_{\mathbb{R}} f^+(x,y)dy + \int_{\mathbb{R}} f^-(x,y)dy < \infty.$$

Therefore $y \mapsto f(x,y)$ is absolutely integrable for almost every x, and for such x,

$$\int_{\mathbb{R}} f(x,y)dy = \int_{\mathbb{R}} f^+(x,y)dy - \int_{\mathbb{R}} f^-(x,y)dy.$$

Define $F(x) := F^+(x) - F^-(x)$. Then $F \in L^1(\mathbb{R})$, and for almost every x, $F(x) = \int_{\mathbb{R}} f(x,y)dy$. Also,

$$\int_{\mathbb{R}} F(x)dx = \int_{\mathbb{R}} F^+(x)dx - \int_{\mathbb{R}} F^-(x)dx = \int_{\mathbb{R}^2} f^+ - \int_{\mathbb{R}^2} f^- = \int_{\mathbb{R}^2} f.$$

The same argument, with x and y interchanged, gives the corresponding function $G \in L^1(\mathbb{R})$. Hence it suffices to prove the theorem for non-negative integrable functions. From now on, assume that $f \geq 0$.

Step 2: Reduction to bounded support. We next reduce to the case where f is supported in a bounded box. For each $N \in \mathbb{N}$ define $f_N := f\chi_{[-N,N]\times[-N,N]}$. Then $0 \leq f_1 \leq f_2 \leq \cdots \leq f$ and

$f_N(x,y) \to f(x,y)$ for every $(x,y) \in \mathbb{R}^2$. By the monotone convergence theorem,

$$\int_{\mathbb{R}^2} f_N \longrightarrow \int_{\mathbb{R}^2} f.$$

Assume the theorem has been proved for non-negative integrable functions supported in bounded boxes. Then, for each N,

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_N(x,y)dy \right) dx = \int_{\mathbb{R}^2} f_N.$$

For each fixed x, the functions $f_N(x,\cdot)$ increase pointwise to $f(x,\cdot)$. Hence, by the one-dimensional monotone convergence theorem, $\int_{\mathbb{R}} f_N(x,y)dy \longrightarrow \int_{\mathbb{R}} f(x,y)dy$. Applying monotone convergence once more in the x-variable, we get

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)dy \right) dx = \lim_{N\to\infty} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_N(x,y)dy \right) dx = \lim_{N\to\infty} \int_{\mathbb{R}^2} f_N = \int_{\mathbb{R}^2} f.$$

The same argument works with x and y interchanged. Thus it suffices to prove the theorem for non-negative functions supported in a bounded box. Henceforth assume that

$$f \geq 0 \text{ and } \operatorname{supp}(f) \subseteq [-N,N] \times [-N,N].$$

Step 3: Reduction to simple functions and then to characteristic functions. By Lemma 3.1.3, there exists an increasing sequence of non-negative simple functions s_k such that $0 \leq s_1 \leq s_2 \leq \cdots \leq f$ and $s_k(x,y) \to f(x,y)$ for every $(x,y) \in [-N,N] \times [-N,N]$. We may also assume that each s_k is supported in $[-N,N] \times [-N,N]$. Assume the theorem has been proved for non-negative simple functions supported in $[-N,N] \times [-N,N]$. Then

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} s_k(x,y)dy \right) dx = \int_{\mathbb{R}^2} s_k.$$

Letting $k \to \infty$ and using monotone convergence in the y-variable, then in the x-variable, and finally in $\mathbb{R}^2$, gives

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)dy \right) dx = \int_{\mathbb{R}^2} f.$$

Again the same argument applies with x and y interchanged. Thus it remains to prove the theorem for non-negative simple functions supported in $[-N,N] \times [-N,N]$. Since every such simple function is a finite linear combination of characteristic functions of measurable sets, it is enough, by finite linearity, to prove the result for $f = \chi_E$, where $E \subseteq [-N,N] \times [-N,N]$ is measurable.

Step 4: The key identity for characteristic functions. Let $Q := [-N,N] \times [-N,N]$ and let $E \subseteq Q$ be measurable. We shall prove

$$\int_{[-N,N]} \left(\int_{[-N,N]} \chi_E(x,y)dy \right) dx = m(E), \tag{3.1}$$

where the inner integral exists for almost every x. For each fixed $x \in [-N,N]$ define

$$U(x) := \overline{\int_{[-N,N]}} \chi_E(x,y)dy, \quad L(x) := \underline{\int_{[-N,N]}} \chi_E(x,y)dy.$$

Thus $U(x)$ and $L(x)$ are respectively the upper and lower one-dimensional Lebesgue integrals of the slice function $y \mapsto \chi_E(x,y)$. We shall first prove the upper estimate

$$\overline{\int_{[-N,N]}} U(x)dx \leq m(E). \tag{3.2}$$

Assuming Equation 3.2 for the moment, we show how it implies Equation 3.1. Apply Equation 3.2 to the measurable set $Q \setminus E$. Since

$$m(Q \setminus E) = m(Q) - m(E) = 4N^2 - m(E),$$

we get

$$\overline{\int}_{[-N,N]} \left(\overline{\int}_{[-N,N]} \chi_{Q\setminus E}(x,y)dy \right) dx \le 4N^2 - m(E).$$

But on Q, $\chi_{Q\setminus E}(x,y) = 1 - \chi_E(x,y)$. Therefore, for each fixed x,

$$\overline{\int}_{[-N,N]} \chi_{Q\setminus E}(x,y)dy = \overline{\int}_{[-N,N]} (1 - \chi_E(x,y))dy.$$

By the elementary relation between upper and lower integrals,

$$\overline{\int}_{[-N,N]} (1 - \chi_E(x,y))dy = 2N - \underline{\int}_{[-N,N]} \chi_E(x,y)dy = 2N - L(x).$$

Hence

$$\overline{\int}_{[-N,N]} (2N - L(x))dx \le 4N^2 - m(E).$$

Using again the relation between upper and lower integrals, now in the x-variable, we have

$$\overline{\int}_{[-N,N]} (2N - L(x))dx = 4N^2 - \underline{\int}_{[-N,N]} L(x)dx.$$

Thus

$$4N^2 - \underline{\int}_{[-N,N]} L(x)dx \le 4N^2 - m(E),$$

and therefore

$$\underline{\int}_{[-N,N]} L(x)dx \ge m(E).$$

Since $L(x) \le U(x)$ for every x, we also have

$$\underline{\int}_{[-N,N]} U(x)dx \ge \underline{\int}_{[-N,N]} L(x)dx \ge m(E).$$

Combining this with Equation 3.2, we obtain

$$\underline{\int}_{[-N,N]} U(x)dx = \overline{\int}_{[-N,N]} U(x)dx = m(E).$$

By Lemma 3.3.1, U is absolutely integrable on $[-N,N]$, and

$$\int_{[-N,N]} U(x)dx = m(E).$$

Similarly, since $L \le U$, we have

$$\overline{\int}_{[-N,N]} L(x)dx \le \overline{\int}_{[-N,N]} U(x)dx \le m(E),$$

while we already proved $\underline{\int}_{[-N,N]} L(x)dx \ge m(E)$. Therefore

$$\underline{\int}_{[-N,N]} L(x)dx = \overline{\int}_{[-N,N]} L(x)dx = m(E).$$

Again by Lemma 3.3.1, L is absolutely integrable on $[-N, N]$, and $\int_{[-N,N]} L(x)dx = m(E)$. Now $0 \leq L \leq U$. Hence $U - L \geq 0$ and

$$\int_{[-N,N]} (U(x) - L(x))dx = \int_{[-N,N]} U(x)dx - \int_{[-N,N]} L(x)dx = m(E) - m(E) = 0$$

since we have shown U, L are absolutely integrable. Since $U - L$ is non-negative and measurable, it follows that $U(x) = L(x)$ for almost every $x \in [-N, N]$. For every such x, Lemma 3.3.1 applied to the one-dimensional function $y \mapsto \chi_E(x, y)$ shows that this slice is Lebesgue integrable on $[-N, N]$, and

$$\int_{[-N,N]} \chi_E(x, y)dy = U(x) = L(x).$$

Consequently,

$$\int_{[-N,N]} \left(\int_{[-N,N]} \chi_E(x, y)dy \right) dx = \int_{[-N,N]} U(x)dx = m(E).$$

This proves Equation 3.1, provided that the estimate Equation 3.2 is known. The same argument, with x and y interchanged, proves $\int_{[-N,N]} \left(\int_{[-N,N]} \chi_E(x, y)dx \right) dy = m(E)$. It remains only to prove the upper estimate Equation 3.2.

Step 5: Proof of the estimate Equation 3.2. Let $\varepsilon > 0$. Since E is measurable, $m(E) = m^*(E)$. By the definition of outer measure, there exists an at most countable collection of boxes $(B_j)_{j \in J}$ such that

$$E \subseteq \bigcup_{j \in J} B_j \text{ and } \sum_{j \in J} m(B_j) \leq m(E) + \varepsilon.$$

Write each box as $B_j = I_j \times J_j$ where $I_j, J_j \subseteq \mathbb{R}$ are intervals. For a single box $B_j = I_j \times J_j$, we have $m(B_j) = |I_j||J_j|$. Also,

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_{B_j}(x, y)dy \right) dx = |I_j||J_j| = m(B_j).$$

Since $E \subseteq Q = [-N, N] \times [-N, N]$ and since only the values on $[-N, N] \times [-N, N]$ are relevant for the estimate, we may write

$$\int_{[-N,N]} \left(\int_{[-N,N]} \chi_{B_j}(x, y)dy \right) dx \leq m(B_j).$$

The inequality is enough for our purposes; equality would hold if B_j were already contained in Q. Now define

$$H(x, y) := \sum_{j \in J} \chi_{B_j}(x, y).$$

This is a non-negative measurable function, possibly taking the value $+\infty$. Since $E \subseteq \bigcup_{j \in J} B_j$, we have $\chi_E(x, y) \leq H(x, y)$ for all $(x, y) \in Q$. Therefore, for each fixed $x \in [-N, N]$,

$$\overline{\int_{[-N,N]}} \chi_E(x, y)dy \leq \int_{[-N,N]} H(x, y)dy$$

In other words,

$$U(x) \leq \int_{[-N,N]} H(x, y)dy.$$

Thus the function $x \mapsto \int_{[-N,N]} H(x, y)dy$ is an admissible majorant for U. Hence, by the definition of the upper Lebesgue integral,

$$\int_{[-N,N]} U(x)dx \leq \int_{[-N,N]} \left(\int_{[-N,N]} H(x, y)dy \right) dx.$$

Since $H = \sum_{j\in J} \chi_{B_j}$ is non-negative, Corollary 8.2.11 allows us to interchange the countable sum and the integral. Therefore

$$\begin{aligned}\int_{[-N,N]} \left(\int_{[-N,N]} H(x,y)dy\right) dx &= \int_{[-N,N]} \left(\int_{[-N,N]} \sum_{j\in J} \chi_{B_j}(x,y)dy\right) dx \\ &= \sum_{j\in J} \int_{[-N,N]} \left(\int_{[-N,N]} \chi_{B_j}(x,y)dy\right) dx \\ &\leq \sum_{j\in J} m(B_j) \leq m(E) + \varepsilon\end{aligned}$$

Hence

$$\overline{\int_{[-N,N]}} U(x)dx \leq m(E) + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we obtain $\overline{\int_{[-N,N]}} U(x)dx \leq m(E)$. This proves Equation 3.2.

Step 6: Conclusion. We have proved Fubini's theorem for characteristic functions χ_E where $E \subseteq [-N,N] \times [-N,N]$ is measurable. By finite linearity, the theorem holds for non-negative simple functions supported in $[-N,N] \times [-N,N]$. By approximation from below and the monotone convergence theorem, it holds for all non-negative measurable functions supported in bounded boxes. By applying the result to $f_N = f\chi_{[-N,N]\times[-N,N]}$ and then letting $N \to \infty$, it holds for all non-negative integrable functions on $\mathbb{R}^2$. Finally, by Step 1, the theorem follows for every absolutely integrable function $f : \mathbb{R}^2 \to \mathbb{R}$. Thus there exist absolutely integrable functions $F, G : \mathbb{R} \to \mathbb{R}$ such that, for almost every x, $F(x) = \int_{\mathbb{R}} f(x,y)dy$, and, for almost every y, $G(y) = \int_{\mathbb{R}} f(x,y)dx$. Moreover,

$$\int_{\mathbb{R}} F(x)dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G(y)dy.$$

This completes the proof. ■

Remark 3.4.1. Very roughly speaking, Fubini's theorem says that

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)dy\right) dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)dx\right) dy.$$

This allows us to compute two-dimensional integrals by splitting them into two one-dimensional integrals. The reason why we do not write Fubini's theorem this way, though, is that it is possible that the integral $\int_{\mathbb{R}} f(x,y)dy$ does not actually exist for every x, and similarly $\int_{\mathbb{R}} f(x,y)dx$ does not exist for every y; Fubini's theorem only asserts that these integrals only exist for almost every x and y. For instance, if $f(x,y)$ is the function which equals 1 when $y > 0$ and $x = 0$, equals -1 when $y < 0$ and $x = 0$ and is zero otherwise, then f is absolutely integrable on $\mathbb{R}^2$ and $\int_{\mathbb{R}^2} f = 0$ (since f equals zero almost everywhere in $\mathbb{R}^2$), but $\int_{\mathbb{R}} f(x,y)dy$ is not absolutely integrable when $x = 0$ (though it is absolutely integrable for every other x).

Theorem 3.4.2 (Tonelli's Theorem)**.** Let $E \subset \mathbb{R}^r$ and $F \subset \mathbb{R}^s$ be measurable sets, and let $f : E \times F \to \mathbb{R}$ be a nonnegative measurable function. Then:

1. For almost every $x \in E$, the function $f(x,\cdot)$ is measurable on F.

2. For almost every $y \in F$, the function $f(\cdot,y)$ is measurable on E.

3. The function
$$g(x) = \int_F f(x,y)\,dy$$
is measurable on E.

4. The function
$$h(y) = \int_E f(x,y)\,dx$$
is measurable on F.

5. The following equality holds:
$$\int_{E\times F} f(x,y)\,dx\,dy = \int_E \left(\int_F f(x,y)\,dy\right) dx = \int_F \left(\int_E f(x,y)\,dx\right) dy.$$

Proof. This theorem is a consequence of Fubini's Theorem and the Monotone Convergence Theorem. Define an increasing sequence of bounded nonnegative integrable functions $\{f_n\}_{n=1}^{\infty}$ as follows. For each $n \in \mathbb{N}$, let
$$C_n = [-n,n]^{r+s}.$$
Then define
$$f_n(x,y) = \begin{cases} \min\{f(x,y), n\}, & (x,y) \in C_n, \\ 0, & (x,y) \notin C_n. \end{cases}$$
Clearly, $f_n \uparrow f$ on $E \times F$. Note that f_n is measurable and absolutely integrable for each n.

By Fubini's Theorem, for almost every $x \in \mathbb{R}^r$, the function $y \mapsto f_n(x,y)$ is absolutely integrable on $\mathbb{R}^s$, and
$$g_n(x) = \int_{\mathbb{R}^s} f_n(x,y)\,dy$$
is absolutely integrable on $\mathbb{R}^r$.

Since $f(x,\cdot)$ is nonnegative, the integral
$$g(x) = \int_F f(x,y)\,dy$$
exists, possibly equal to $+\infty$. Moreover, for almost every $x \in E$, we have
$$f_n(x,\cdot) \uparrow f(x,\cdot),$$
so by the Monotone Convergence Theorem,
$$g_n(x) \uparrow g(x) = \int_F f(x,y)\,dy.$$
Hence g is measurable on E.

Applying the Monotone Convergence Theorem again,
$$\int_E g(x)\,dx = \lim_{n\to\infty} \int_E g_n(x)\,dx.$$
Using Fubini's Theorem for each f_n,
$$\int_E g_n(x)\,dx = \int_E \left(\int_F f_n(x,y)\,dy\right) dx = \int_{E\times F} f_n(x,y)\,dx\,dy.$$
Since $f_n \uparrow f$, another application of the Monotone Convergence Theorem gives
$$\lim_{n\to\infty} \int_{E\times F} f_n(x,y)\,dx\,dy = \int_{E\times F} f(x,y)\,dx\,dy.$$
Therefore,
$$\int_E \left(\int_F f(x,y)\,dy\right) dx = \int_{E\times F} f(x,y)\,dx\,dy.$$

Interchanging the roles of x and y yields

$$\int_F \left(\int_E f(x,y)\,dx \right) dy = \int_{E\times F} f(x,y)\,dx\,dy.$$

This proves the theorem. ■

Proposition 3.4.2. Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be arbitrary sets. Then

$$m^*(A \times B) = m^*(A)m^*(B),$$

where m^* denotes Lebesgue outer measure in the relevant Euclidean space.

Proof. We prove the two inequalities separately.

Step 1: Upper bound. We first show that

$$m^*(A \times B) \le m^*(A)m^*(B).$$

Fix $\epsilon > 0$. By the definition of outer measure, there exist countable collections of open boxes $\{Q_i\}_{i=1}^\infty \subset \mathbb{R}^n$, $\{R_j\}_{j=1}^\infty \subset \mathbb{R}^m$ such that $A \subset \bigcup_{i=1}^\infty Q_i$, $B \subset \bigcup_{j=1}^\infty R_j$, and

$$\sum_{i=1}^\infty \mathrm{vol}(Q_i) < m^*(A) + \epsilon, \quad \sum_{j=1}^\infty \mathrm{vol}(R_j) < m^*(B) + \epsilon.$$

Then $A \times B \subset \bigcup_{i,j=1}^\infty (Q_i \times R_j)$. Each $Q_i \times R_j$ is an open box in $\mathbb{R}^{n+m}$, and its volume is $\mathrm{vol}(Q_i \times R_j) = \mathrm{vol}(Q_i)\mathrm{vol}(R_j)$. Therefore,

$$\begin{aligned} m^*(A \times B) &\le \sum_{i,j=1}^\infty \mathrm{vol}(Q_i \times R_j) \\ &= \sum_{i,j=1}^\infty \mathrm{vol}(Q_i)\mathrm{vol}(R_j) \\ &= \left(\sum_{i=1}^\infty \mathrm{vol}(Q_i) \right) \left(\sum_{j=1}^\infty \mathrm{vol}(R_j) \right) \\ &< (m^*(A) + \epsilon)(m^*(B) + \epsilon). \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, we conclude that $m^*(A \times B) \le m^*(A)m^*(B)$.

Step 2: Lower bound. We now show that $m^*(A \times B) \ge m^*(A)m^*(B)$. Let $G \subset \mathbb{R}^{n+m}$ be any open set such that $A \times B \subset G$. For each $y \in \mathbb{R}^m$, define the section $G_y := \{x \in \mathbb{R}^n : (x,y) \in G\}$. Since G is open, each section G_y is an open subset of $\mathbb{R}^n$ hence Lebesgue measurable. Now let $y \in B$. If $x \in A$, then $(x,y) \in A \times B \subset G$, so $x \in G_y$. Thus $A \subset G_y$ for every $y \in B$. By monotonicity of outer measure, $m(G_y) \ge m^*(A)$ for every $y \in B$. Consider the function $f(y) := m(G_y)$, $y \in \mathbb{R}^m$. Since G is open, it is measurable, and by Tonelli's theorem for the indicator function χ_G,

$$\begin{aligned} m(G) = \int_{\mathbb{R}^{n+m}} \chi_G &= \int_{\mathbb{R}^m} \left(\int_{\mathbb{R}^n} \chi_G(x,y)\,\mathrm{d}x \right) \mathrm{d}y \\ &= \int_{\mathbb{R}^m} \left(\int_{\mathbb{R}^n} \chi_{G_y}(x)\,\mathrm{d}x \right) \mathrm{d}y = \int_{\mathbb{R}^m} m(G_y)\,\mathrm{d}y = \int_{\mathbb{R}^m} f(y)\,\mathrm{d}y. \end{aligned}$$

Set $H := \{y \in \mathbb{R}^m : f(y) \ge m^*(A)\}$. Now since

$$f(y) := m(G_y) = \int_{\mathbb{R}^{n+m}} \chi_G(x,y)\,\mathrm{d}x,$$

which gives f is measurable since χ_G is measurable and by Tonelli's theorem. Because f is measurable, the set H is measurable. Moreover, from the previous observation we have $B \subset H$. Therefore, by monotonicity of outer measure, $m(H) \geq m^*(B)$. Using the fact that $f(y) \geq m^*(A)$ on H, we obtain

$$m(G) = \int_{\mathbb{R}^m} f(y)dy \geq \int_H f(y)dy \geq \int_H m^*(A)dy = m^*(A)m(H) \geq m^*(A)m^*(B).$$

Thus every open set $G \supset A \times B$ satisfies $m(G) \geq m^*(A)m^*(B)$. Taking the infimum over all such open sets G, we conclude that $m^*(A \times B) \geq m^*(A)m^*(B)$. Combining this with the upper bound proved in Step 1, we obtain $m^*(A \times B) = m^*(A)m^*(B)$. This completes the proof. ■

Chapter 4

Some Comments and Further Directions

Lecture 19

We have now reached the end of our study of the main topics in Analysis II. The material developed in this course provides a bridge between classical real analysis and several central areas of modern analysis. In particular, the notions of metric spaces, compactness, uniform convergence, differentiation, Riemann integration, and Lebesgure measure and integral are not isolated topics. Rather, they form part of the foundation for further study in measure theory, functional analysis, partial differential equations, probability theory, Fourier analysis, and geometric analysis. May 7.

One natural next step is to study Lebesgue measure and Lebesgue integration. The Riemann integral is very useful for continuous functions and for many functions arising in elementary analysis, but it is not flexible enough for many limiting processes. The Lebesgue integral is designed to interact well with limits of functions. This is why convergence theorems such as the monotone convergence theorem, Fatou's lemma, and the dominated convergence theorem play such a central role in modern analysis.

A very suitable continuation is Terence Tao's book *An Introduction to Measure Theory*. This book develops the theory of Lebesgue measure and Lebesgue integration systematically, beginning with measure theory on Euclidean spaces and then moving toward abstract measure spaces. It also discusses measurable functions, modes of convergence, differentiation theorems, outer measures, premeasures, and product measures. Thus it provides a natural continuation from the analysis of functions on $\mathbb{R}^n$ to the more general framework needed in modern analysis. The book is available at https://terrytao.wordpress.com/wp-content/uploads/2012/12/gsm-126-tao5-measure-book.pdf.

The spaces $L^p(\Omega)$

One of the most important constructions arising from Lebesgue integration is the family of L^p-spaces. Let $\Omega \subset \mathbb{R}^n$ be a measurable set. For $0 < p < \infty$ we define

$$L^p(\Omega) := \{f : \Omega \to \mathbb{R} \text{ measurable} : \int_\Omega |f(x)|^p dx < \infty\},$$

where two functions are identified if they are equal almost everywhere on Ω. This identification is essential, because Lebesgue integration does not distinguish functions that differ only on a set of measure zero. For $1 \le p < \infty$ the quantity

$$\|f\|_{L^p(\Omega)} := \left(\int_\Omega |f(x)|^p dx\right)^{1/p}$$

defines a norm on $L^p(\Omega)$. Hence $L^p(\Omega)$ becomes a metric space with metric

$$d_p(f,g) := \|f-g\|_{L^p(\Omega)} = \left(\int_\Omega |f(x)-g(x)|^p dx\right)^{1/p}.$$

The reason we identify functions that agree almost everywhere is that $\|f-g\|_{L^p(\Omega)} = 0$ implies only that $f = g$ almost everywhere, not necessarily pointwise everywhere. With this identification, d_p is a genuine metric for $1 \le p < \infty$. In fact, $L^p(\Omega)$ is complete with respect to this metric, and hence is a Banach space.

The endpoint case $p = \infty$ is also important. We define

$$L^\infty(\Omega) := \{f : \Omega \to \mathbb{R} \text{ measurable: } f \text{ is essentially bounded}\},$$

where a function f is essentially bounded if there exists $M \in \mathbb{R}$ s.t. $|f(x)| \le M$ for almost every $x \in \Omega$. Its norm is

$$\|f\|_{L^\infty(\Omega)} := \operatorname*{ess\,sup}_{x\in\Omega} |f(x)| = \inf\{M \in \mathbb{R} : |f(x)| \le M \text{ for almost every } x \in \Omega\}.$$

With this norm, $L^\infty(\Omega)$ is also a Banach space.

For $0 < p < 1$, the formula

$$\left(\int_\Omega |f-g|^p\right)^{1/p}$$

does not define a metric because the triangle inequality fails. However,

$$d_p(f,g) := \int_\Omega |f-g|^p$$

does define a metric on $L^p(\Omega)$, after identifying functions that agree almost everywhere. Thus $L^p(\Omega)$ is a metric space for every $0 < p < \infty$, but it is a normed space only when $1 \le p < \infty$.

Hölder's inequality

One of the fundamental estimates in the theory of L^p-spaces is Hölder's inequality. Let $1 \le p, q \le \infty$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

The numbers p and q are called conjugate exponents. Thus, if $1 < p < \infty$ then $q = \frac{p}{p-1}$. If $p = 1$ then $q = \infty$, and if $p = \infty$, then $q = 1$.

Theorem 4.0.1 (Hölder's inequality)**.** Let $1 \le p, q \le \infty$ be conjugate exponents. If $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$, then $fg \in L^1(\Omega)$, and

$$\int_\Omega |f(x)g(x)|dx \le \|f\|_{L^p(\Omega)}\|g\|_{L^q(\Omega)}$$

Equivalently,

$$\|fg\|_{L^1(\Omega)} \le \|f\|_{L^p(\Omega)}\|g\|_{L^q(\Omega)}$$

Proof. We first consider the case $1 < p < \infty$, so that $1 < q < \infty$. The key ingredient is Young's inequality: for all $a, b \ge 0$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Assume first that $f \ne 0$ in $L^p(\Omega)$ and $g \ne 0$ in $L^q(\Omega)$. Define

$$F(x) := \frac{|f(x)|}{\|f\|_{L^p(\Omega)}}, \quad G(x) := \frac{|g(x)|}{\|g\|_{L^q(\Omega)}}.$$

Then $\|F\|_{L^p(\Omega)} = 1$, $\|G\|_{L^q(\Omega)} = 1$. Applying Young's inequality pointwise gives

$$F(x)G(x) \le \frac{F(x)^p}{p} + \frac{G(x)^q}{q}$$

Integrating over Ω, we obtain

$$\int_\Omega F(x)G(x)dx \le \frac{1}{p}\int_\Omega F(x)^p dx + \frac{1}{q}\int_\Omega G(x)^q dx = \frac{1}{p} + \frac{1}{q} = 1.$$

Therefore

$$\int_\Omega \frac{|f(x)g(x)|}{\|f\|_{L^p(\Omega)}\|g\|_{L^q(\Omega)}}dx \le 1,$$

which implies

$$\int_\Omega |f(x)g(x)|dx \le \|f\|_{L^p(\Omega)}\|g\|_{L^q(\Omega)}.$$

If $f = 0$ in $L^p(\Omega)$ or $g = 0$ in $L^q(\Omega)$, the inequality is immediate. It remains to consider the endpoint cases. If $p = 1$ and $q = \infty$, then

$$|f(x)g(x)| \le |f(x)|\|g\|_{L^\infty(\Omega)} \text{ for almost every } x \in \Omega.$$

Hence

$$\int_\Omega |f(x)g(x)|dx \le \|g\|_{L^\infty(\Omega)} \int_\Omega |f(x)|dx = \|f\|_{L^1(\Omega)}\|g\|_{L^\infty(\Omega)}$$

The case $p = \infty$ and $q = 1$ is identical.

Remark 4.0.1. If $f, g : \Omega \to \mathbb{R}^*$ are two Lebesgue integrable function s.t. $f \le g$ almost everwhere, then $\int_\Omega f \le \int_\Omega g$. This is because we can modify the value of f on the set $\{x \in \Omega : f(x) > g(x)\}$, which has measure zero, and thus modify value of f on this set will not change the value of integral, and thus suppose the modified function is f', then $f' \le g$ almost everywhere, and $\int_\Omega f' = \int_\Omega f$, so we have

$$\int_\Omega f = \int_\Omega f' \le \int_\Omega g.$$

■

Remark 4.0.2. Hölder's inequality generalizes the Cauchy-Schwarz inequality. Indeed, when $p = q = 2$, we obtain

$$\int_\Omega |f(x)g(x)|dx \le \left(\int_\Omega |f(x)|^2dx\right)^{1/2}\left(\int_\Omega |g(x)|^2dx\right)^{1/2}$$

Thus the Cauchy-Schwarz inequality is precisely Hölder's inequality in the special case $L^2(\Omega)$.

Remark 4.0.3. Hölder's inequality shows that multiplication is compatible with conjugate L^p-spaces. More precisely, if $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$, where p and q are conjugate exponents, then the product fg is absolutely integrable. This is one of the main reasons why $L^q(\Omega)$ is naturally related to the dual space of $L^p(\Omega)$ when $1 < p < \infty$.

Further topics

We briefly mention several directions that naturally continue the material of this course.

1. **Measure theory and Lebesgue integration.** A first direction is to study Lebesgue measure, measurable functions, and the Lebesgue integral in depth. This gives a more flexible theory of integration than the Riemann integral and allows one to handle limits of functions much more effectively. Important results include the monotone convergence theorem, Fatou's lemma, the dominated convergence theorem, and Fubini's theorem.

2. **Functional analysis.** The spaces $C(K)$, $L^p(\Omega)$, and $C^k(\Omega)$ are examples of infinite-dimensional function spaces. Functional analysis studies such spaces using the language of normed spaces, Banach spaces, Hilbert spaces, bounded linear operators, and dual spaces. In this subject, compactness is often more subtle than in finite-dimensional spaces, and the Arzelà-Ascoli theorem is a model example of a compactness theorem in an infinite-dimensional setting.

3. **Hilbert spaces and L^2-theory.** Among the L^p-spaces, the space $L^2(\Omega)$ is especially important

because it has an inner product:

$$\langle f, g\rangle_{L^2} := \int_\Omega f(x)g(x)dx.$$

This makes $L^2(\Omega)$ into a Hilbert space. The geometry of Hilbert spaces generalizes Euclidean geometry to infinite dimensions. Orthogonal projection, Fourier series, weak convergence, and spectral theory all naturally arise from this point of view.

4. **Fourier analysis.** Fourier analysis studies how functions can be decomposed into basic oscillatory functions such as sine waves, cosine waves, or complex exponentials. It is one of the most important applications of L^p-spaces and Hilbert space theory. Fourier series, the Fourier transform, Plancherel's theorem, and convolution are fundamental tools in analysis, partial differential equations, probability, and signal processing.

5. **Sobolev spaces and weak derivatives.** In many applications, especially in partial differential equations, one needs to study functions that may not be differentiable in the classical sense. Sobolev spaces provide a framework for defining and studying weak derivatives. For example, the Sobolev space $W^{1,p}(\Omega)$ consists of functions in $L^p(\Omega)$ whose first weak derivatives also belong to $L^p(\Omega)$. These spaces are indispensable in the modern theory of elliptic and parabolic partial differential equations.

6. **Partial differential equations.** Many equations in geometry, physics, and applied mathematics are partial differential equations. Examples include the Laplace equation, heat equation, wave equation, and minimal surface equation. The analytic tools developed in measure theory, L^p-spaces, Sobolev spaces, and compactness theory are essential for studying existence, uniqueness, regularity, and qualitative properties of solutions.

7. **Probability theory.** Measure theory also provides the foundation of modern probability. A probability space is a measure space with total measure one. Random variables are measurable functions, expectation is integration, and convergence theorems from measure theory become fundamental tools in probability. From this perspective, L^p-spaces measure the size of random variables through their moments:
$$\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p}.$$

8. **Geometric analysis.** Another direction is geometric analysis, where tools from analysis are used to study geometric problems. Examples include harmonic functions on manifolds, minimal surfaces, curvature equations, and geometric evolution equations. In this area, compactness theorems, integral estimates, Sobolev inequalities, and weak convergence methods play a central role.

In summary, the material of this course should be viewed as the beginning of a larger analytic framework. The study of metric spaces, compactness, continuous functions, and integration prepares us for the more advanced language of measure theory, L^p-spaces, Banach and Hilbert spaces, Fourier analysis, Sobolev spaces, and partial differential equations. These subjects form the core of modern analysis and have deep connections with geometry, probability, mathematical physics, and data science.

Chapter 5

The Arzela-Ascoli Theorem and Applications

5.1 Normed spaces and Spaces of Bounded Functions

Before studying spaces of functions, it is useful to recall how one measures size in an abstract vector space. In $\mathbb{R}^n$, the Euclidean length allows us to speak about distance, convergence, Cauchy sequences, continuity, and completeness. A norm is the corresponding abstract notion of length. Once a vector space is equipped with a norm, it becomes a metric space in a natural way.

Definition 5.1.1. Let V be a vector space over $\mathbb{R}$ and $\mathbb{C}$. A norm is a function $\|\cdot\| : V \to [0, \infty)$ satisfying the following properties for all $x, y \in V$ and all scalars λ.:

(1) $\|x\| \geq 0$, $\|x\| = 0 \Leftrightarrow x = 0$.

(2) $\|\lambda x\| = |\lambda| \cdot \|x\|$.

(3) $\|x + y\| \leq \|x\| + \|y\|$.

A vector space equippeed with a norm is called a normed (vector) space and is denoted by $(V, \|\cdot\|)$.

Remark 5.1.1. Every normed space $(V, \|\cdot\|)$ carries a natural metric $d(x, y) := \|x - y\|$.

Definition 5.1.2. A normed space $(V, \|\cdot\|)$ is called Banach space if it is complete with respect to the metric induced by the norm. Equivalently, every Cauchy sequence in V converges to an element of V, i.e.

$$\{x_n\} \text{ is Cauchy } \Leftrightarrow \|x_n - x_m\| \to 0 \text{ as } n, m \to \infty \Leftrightarrow \exists x^* \in V \text{ s.t. } \|x_n - x^*\| \to 0 \text{ as } n \to \infty.$$

Recall that suppose (X, d_X) and (Y, d_Y) are metric spaces. Let

$$B(X \to Y) := \{f \mid f : X \to Y \text{ is a bounded function}\}.$$

If X is nonempty, we define $d_\infty(f, g) = \sup_{x \in X} d_Y(f(x), g(x))$ on $B(X \to Y)$, then this is a metric on $B(X \to Y)$. We also write $d_{B(X \to Y)}$ as a synonym for d_∞. If X is empty, then we define $d_\infty(f, g) = 0$ for all f, g.

Observation. The distance $d_\infty(f, g)$ is always finite because both f, g are bounded functions on X. Therefore $(B(X \to Y), d_\infty)$ forms a well-defined metric space.

Also, we have following proposition:

Proposition 5.1.1 (Analysis I).

$$f^{(n)} \to f \text{ in } d_{B(X\to Y)} \Leftrightarrow f^{(n)} \to f \text{ uniformly on } X,$$

Now we can define

$$C(X \to Y) \coloneqq \{f : f \in B(X \to Y) \text{ and } f \text{ is continuous}\}.$$

Theorem 5.1.1. Let (X, d_X) be a metric space, and (Y, d_Y) a complete metric space. Then

$$\left(C(X \to Y), d_{B(X\to Y)}|_{C(X\to Y)\times C(X\to Y)}\right)$$

is a complete metric subspace of $(B(X \to Y), d_{B(X\to Y)})$. Equivalently, every Cauchy sequence of continuous bounded functions converges uniformly to a continuous bounded function in $C(X \to Y)$.

5.2 The Banach Space $C(K)$

Throughout this section, (K, d) denotes a non-empty compact metric space. We write

$$\mathcal{B}(K) \coloneqq B(K \to \mathbb{R}) = \{f : K \to \mathbb{R}, f \text{ is bounded}\},$$

and

$$C(K) \coloneqq C(K \to \mathbb{R}) = \{f : K \to \mathbb{R}, f \text{ is continuous}\}.$$

Note that $C(K \to \mathbb{R}) = \{f : K \to \mathbb{R}, f \text{ is continuous}\}$ since K is compact and f is continuous, so $f(K)$ is compact and thus bounded. Hence, $C(K) \subseteq \mathcal{B}(K)$.

Both $\mathcal{B}(K)$ and $C(K)$ are vector spaces over $\mathbb{R}$, with usual pointwise operations

$$(f + g)(x) \coloneqq f(x) + g(x), \quad (\lambda f)(x) \coloneqq \lambda f(x).$$

Indeed, if $f, g \in \mathcal{B}(K)$, and $a, b \in \mathbb{R}$, then $af + bg$ is bounded, because

$$\|af + bg\|_\infty \le |a|\|f\|_\infty + |b|\|g\|_\infty.$$

Similarly, if $f, g \in C(K)$, then $af + bg \in C(K)$.

On $\mathcal{B}(K)$, we can define $\|f\|_\infty \coloneqq \sup_{x\in K} |f(x)|$, where for $f \in C(K)$, the supremum is actually a maximum: $\|f\|_\infty = \max_{x\in K} |f(x)|$.

Proposition 5.2.1. The function

$$\|\cdot\|_\infty : \mathcal{B}(K) \to [0, \infty), \quad \|f\|_\infty \coloneqq \sup_{x\in K} |f(x)|$$

is a norm on $\mathcal{B}(K)$. Its restriction to $C(K)$ is a norm on $C(K)$.

Remark 5.2.1. Note that

$$d_{B(K\to\mathbb{R})}(f, g) = \sup_{x\in K} d_Y(f(x), g(x)) = \sup_{x\in K} |f(x) - g(x)|.$$

Therefore,

$$d_{B(K\to\mathbb{R})}(f, g) = \|f - g\|_\infty.$$

Thus, we can modify Proposition 5.1.1 to be

$$f_n \to f \text{ in } \|\cdot\|_\infty \Leftrightarrow f_n \to f \text{ uniformly on } K.$$

Proposition 5.2.2. The normed space $(\mathcal{B}(K), \|\cdot\|_\infty)$ is a Banach space.

Proof. In chapter 3, we proved that the space of bounded maps into complete metric space is complete with respect to the uniform metric. Applying this result to $X = K$ and $Y = \mathbb{R}$ and using the completeness of $\mathbb{R}$, we see that

$$(B(K \to \mathbb{R}), d_{B(K\to\mathbb{R})})$$

is a complete metric space. Since $B(K \to \mathbb{R}) = \mathcal{B}(K)$, and since previous remark shows that

$$d_{B(K\to\mathbb{R})}(f,g) = \|f-g\|_\infty,$$

it follows that $(\mathcal{B}(K), \|\cdot\|_\infty)$ is complete. ∎

Proposition 5.2.3. Every function $f \in C(K)$ is bounded and uniformly continuous.

Proposition 5.2.4. The space $C(K)$ is a closed vector subspace of $\mathcal{B}(K)$.

Proof. We already know that $C(K)$ is a vector subspace of $\mathcal{B}(K)$. It remains to prove that $C(K)$ is closed.

Let $\{f_n\}_{n=1}^\infty \subseteq C(K)$ and suppose that $f_n \to f$ in $\mathcal{B}(K)$ with respect to the supremum norm, for some $f \in \mathcal{B}(K)$, so $f_n \to f$ uniformly on K by previous remark. Since each f_n is continuous and $f_n \to f$ uniformly, we have f continuous, i.e. $f \in C(K)$. Therefore, $C(K)$ is closed in $\mathcal{B}(K)$.

Remark 5.2.2. "$\{f_n\} \subseteq C(K)$ and $f_n \to f$ implies $f \in C(K)$" is equivalent to $C(K)$ is closed.

∎

Proposition 5.2.5. $(C(K), \|\cdot\|_\infty)$ is a Banach space.

Proof. Since we have shown $(\mathcal{B}(K), \|\cdot\|_\infty)$ is complete and $(C(K), \|\cdot\|_\infty)$ is a closed subspace of $\mathcal{B}(K)$. A closed subspace of a complete metric space is complete. Therefore, $C(K)$, equipped with the norm $\|\cdot\|_\infty$ is complete. ∎

5.3 Equi-continuity and Arzela-Ascoli Theorem

Definition 5.3.1. Let $\mathcal{F} \subseteq C(K)$,

(i) The family $\mathcal{F}$ is called pointwise bounded if for every $x \in K$, $\exists$ constant $M_x < \infty$ s.t. $|f(x)| \leq M_x$ for every $f \in \mathcal{F}$.

(ii) The family $\mathcal{F}$ is uniformly bounded if $\exists$ a constant $M < \infty$ s.t. $\|f\|_\infty \leq M$ for every $f \in \mathcal{F}$.

(iii) The family $\mathcal{F}$ is called equi-continuous if for every $\varepsilon > 0$, $\exists \delta > 0$ s.t. $d(x,y) < \delta$ implies $|f(x) - f(y)| < \varepsilon$ for all $x, y \in K$ and all $f \in \mathcal{F}$. (δ is independent of f and x.)

Remark 5.3.1. The adjective "equi-continuous" means that the same δ works simultaneously for every function in the family.

Lemma 5.3.1 (Pointwise boundedness becomes uniform boundedness)**.** If $\mathcal{F} \subseteq C(K)$ is pointwise bounded and equi-continuous, then $\mathcal{F}$ is uniformly bounded.

Proof. Apply equicontinuity with $\varepsilon = 1$, there exists $\delta > 0$ s.t. $d(x,y) < \delta$ implies $|f(x) - f(y)| < 1$

for all $f \in \mathcal{F}$. Since K is compact, choose finitely many points $x_1, \dots, x_L \in K$ s.t.

$$K \subseteq \bigcup_{i=1}^{L} B(x_i, \delta).$$

For each i, pointwise boundedness gives a constant M_i s.t.

$$|f(x_i)| \leq M_i \quad \text{for every } f \in \mathcal{F}.$$

Let $M := 1 + \max_{1 \leq i \leq L} M_i$. For any $x \in K$, choose i with $d(x, x_i) < \delta$, then

$$f(x) \leq |f(x) - f(x_i)| + |f(x_i)| < 1 + M_i \leq M.$$

Taking the supremum over all $x \in K$, then $\|f\|_\infty \leq M$ for every $f \in \mathcal{F}$. ■

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Theorem 5.3.1 (Arzela-Ascoli's theorem: sequential form)**.** Let (K, d) be a compact metric space. If $\{f_n\}_{n=1}^{\infty} \subseteq C(K)$ is uniformly bounded and equi-continuous, then $\{f_n\}$ has a subsequence which converges uniformly on K to a function in $C(K)$.

Proof.

- **Step 1. Compact metric spaces are separable.** For each $k \in \mathbb{N}$, the collection of balls

  $$\left\{ B\left(x, \frac{1}{k}\right) : x \in K \right\}$$

  is an open cover of K. By compactness, choose finitely many points $x_{k,1}, x_{k,2}, \dots, x_{k,N_k} \in K$ s.t.

  $$K = \bigcup_{j=1}^{N_k} B\left(x_{k,j}, \frac{1}{k}\right).$$

  Set

  $$S_k := \{x_{k,1}, x_{k,2}, \dots, x_{k,N_k}\}, \quad S := \bigcup_{k=1}^{\infty} S_k.$$

  Then S is countable. We claim that S is dense in K. Indeed, given $x \in K$ and $\varepsilon > 0$, choose $k \in \mathbb{N}$ so large that $\frac{1}{k} < \varepsilon$. Since $K = \bigcup_{j=1}^{N_k} B(x_{k,j}, \frac{1}{k})$, there exists j s.t. $x \in B(x_{k,j}, \frac{1}{k})$. Hence,

  $$d(x, x_{k,j}) < \frac{1}{k} < \varepsilon,$$

  and $x_{k,j} \in S$. Therefore, S is dense in K.

- **Step 2. A diagonal subsequence converges on the dense set.** List the countable dense set as $S = \{p_1, p_2, \dots\}$. If S is finite, we repeat its elements so that it is still written as an infinite sequence.

  Since $\{f_n\}$ is uniformly bounded, there exists a constant $M > 0$ s.t.

  $$|f_n(x)| \leq M \quad \forall n \in \mathbb{N} \text{ and } \forall x \in K.$$

  In particular, for each fixed $p_j \in S$, the numerical sequence $\{f_n(p_j)\}_{n=1}^{\infty}$ is bounded in $\mathbb{R}$. We now extract subsequences one point at a time.

  First consider the point p_1. Since $\{f_n(p_1)\}$ is bounded in $\mathbb{R}$, the Bolzano-Weierstrass theorem gives a subsequence $\left\{f_n^{(1)}\right\}_{n=1}^{\infty}$ of $\{f_n\}$ s.t. $\left\{f_n^{(1)}(p_1)\right\}_{n=1}^{\infty}$ converges in $\mathbb{R}$.

Next, consider p_2. The sequence $\left\{f_n^{(1)}(p_2)\right\}$ is bounded again, so it has a convergent subsequence. Thus, we may choose a subsequence $\left\{f_n^{(2)}\right\}_{n=1}^{\infty}$ of $\left\{f_n^{(1)}\right\}_{n=1}^{\infty}$ s.t. $\left\{f_n^{(2)}(p_2)\right\}_{n=1}^{\infty}$ converges. Since $\left\{f_n^{(2)}\right\}$ is a subsequence of $\left\{f_n^{(1)}\right\}$, the sequence $\left\{f_n^{(2)}(p_1)\right\}$ also converges.

Continuing inductively, suppose that for some $r \geq 1$ we have constructed a subsequence $\left\{f_n^{(r)}\right\}_{n=1}^{\infty}$ s.t. $\left\{f_n^{(r)}(p_j)\right\}_{n=1}^{\infty}$ converges for each $j = 1, \ldots, r$. Since the sequence $\left\{f_n^{(r)}(p_{r+1})\right\}_{n=1}^{\infty}$ is bounded in $\mathbb{R}$, it has a convergent subsequence. Choose such a subsequence and call it $\left\{f_n^{(r+1)}\right\}_{n=1}^{\infty}$. Then $\left\{f_n^{(r+1)}(p_{n+1})\right\}_{n=1}^{\infty}$ converges. Moreover, because $\left\{f_n^{(r+1)}\right\}$ is a subsequence of $\left\{f_n^{(r)}\right\}$, convergence at the earlier point $p_1, \ldots, p_r$ is preserved.

In this way we obtain a nested seuqence of sequences:

$$\left\{f_n^{(1)}\right\} \supseteq \left\{f_n^{(2)}\right\} \supseteq \left\{f_n^{(3)}\right\} \supseteq \cdots,$$

where the r-th subsequence $\left\{f_n^{(r)}\right\}$ converges at the first r points $p_1, \ldots, p_r$. We now define the diagonal subsequence by

$$g_n \coloneqq f_n^{(n)}, \quad n = 1, 2, 3, \ldots.$$

This is a subsequence of the original sequence $\{f_n\}$. We claim that for every fixed $p_j \in S$, the numerical sequence $\{g_n(p_j)\}_{n=1}^{\infty}$ converges.

Indeed, fix $j \in \mathbb{N}$. For every $n \geq j$, the term $g_n = f_n^{(n)}$ belongs to the subsequence $\left\{f_m^{(j)}\right\}_{m=1}^{\infty}$, because the subsequences are nested. Hence the tail $\{g_n(p_j)\}_{n \geq j}$ is a subsequence of the convergent sequence $\left\{f_m^{(j)}(p_j)\right\}_{m=1}^{\infty}$. Therefore, $\{g_n(p_j)\}_{n \geq j}$ converges. Adding or removing finitely many terms does not affect convergence, so $\{g_n(p_j)\}_{n=1}^{\infty}$ converges. Since j was arbitrary, the diagonal subsequence $\{g_n\}$ converges pointwise on the dense set S.

- **Step 3. Equicontinuity upgrades pointwise convergence on S to uniform Cauchy convergence on K.** Let $\varepsilon > 0$. By equicontinuity, choose $\delta > 0$ such that

$$d(x, y) < \delta \Rightarrow |g_n(x) - g_n(y)| < \frac{\varepsilon}{3}$$

for every $n \in \mathbb{N}$. Choose $k \in \mathbb{N}$ so large that $\frac{1}{k} < \delta$. Since S_k is finite and $\{g_n(s)\}$ converges for each $s \in S_k$, there exists $N \in \mathbb{N}$ s.t.

$$m, n \geq N \Rightarrow |g_n(s) - g_m(s)| < \frac{\varepsilon}{3} \quad \text{for every } s \in S_k.$$

Fix any $x \in K$, choose $s \in S_k$ with $d(x, s) < \frac{1}{k} < \delta$. Then, for $m, n \geq N$,

$$\begin{aligned} |g_n(x) - g_m(x)| &\leq |g_n(x) - g_n(s)| + |g_n(s) - g_m(s)| + |g_m(s) - g_m(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence, $\{g_n\}$ is Cauchy in $(C(K), \|\cdot\|_\infty)$. Since $C(K)$ is complete, there exists $g \in C(K)$ such that

$$\|g_n - g\|_\infty \to 0.$$

Thus, $g_n \to g$ uniformly on K, as desired.

■

Corollary 5.3.1 (Compactness criterion in $C(K)$)**.** A subset $\mathcal{F} \subseteq C(K)$ is compact if and only if it is closed, pointwise bounded, and equicontinuous. Equivalently, since pointwise boundedness plus equicontinuity implies uniform boundedness, compact subsets of $C(K)$ are precisely the closed, uniformly bounded, equicontinuous subsets.

Proof. First suppose $\mathcal{F}$ is compact. Since $C(K)$ is a metric space, $\mathcal{F}$ is closed. The evaluation map

$$E_x : C(K) \to \mathbb{R}, \quad E_x(f) := f(x),$$

is continuous for each $x \in K$ because

$$|E_x(f) - E_x(g)| = |f(x) - g(x)| \le \sup_{t\in K} |f(t) - g(t)| = \|f - g\|_\infty.$$

Hence, $E_x(\mathcal{F})$ is compact in $\mathbb{R}$, and therefore bounded. Thus, $\mathcal{F}$ is pointwise bounded.

It remains to prove equicontinuity. Let $\varepsilon > 0$. Since $\mathcal{F}$ is compact in the metric induced by $\|\cdot\|_\infty$, it is totally bounded. Thus, there exist $f_1, \dots, f_L \in \mathcal{F}$ such that for every $f \in \mathcal{F}$,

$$\|f - f_i\|_\infty < \frac{\varepsilon}{3}$$

for some i. Since each $f_i \in \mathcal{F} \subseteq C(K)$ is uniformly continuous and there are only finitely many of them, there exists $\delta > 0$ s.t.

$$d(x, y) < \delta \Rightarrow |f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$$

for every $i = 1, 2, \dots, L$. If $f \in \mathcal{F}$ and i is chosen with $\|f - f_i\|_\infty < \frac{\varepsilon}{3}$, then $d(x, y) < \delta$ implies

$$|f(x) - f(y)| \le |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < \varepsilon.$$

Thus $\mathcal{F}$ is equicontinuous.

Conversely, suppose $\mathcal{F}$ is closed, pointwise bounded, and equicontinuous. Let $\{f_n\}$ be any sequence in $\mathcal{F}$. By the preceding lemma, $\mathcal{F}$ is uniformly bounded. The sequential Arzela-Ascoli theorem therefore gives a subsequence $\{f_{n_j}\}$ converging uniformly to some $f \in C(K)$. Since $\mathcal{F}$ is closed, $f \in \mathcal{F}$. Hence every sequence in $\mathcal{F}$ has a convergent subsequence with limit in $\mathcal{F}$. Because $C(K)$ is a metric space, sequential compactness is equivalent to compactness. Thus, $\mathcal{F}$ is compact. ∎

Corollary 5.3.2. Let $\{f_n\} \subseteq C(K)$ be pointwise bounded and equicontinuous. Then $\{f_n\}$ has a subsequence which converges uniformly on K. In particular, the same conclusion holds if the sequence is uniformly bounded and equicontinuous.

Proof. By Lemma 5.3.1 and Arzela-Ascoli Theorem, this is true. ∎

Example 5.3.1. Let K be a finite set with n points, equipped with the discrete metric

$$d(x, y) = \begin{cases} 0, & \text{if } x = y; \\ 1, & \text{if } x \ne y. \end{cases}$$

Show that $C(K)$ may be identified with $\mathbb{R}^n$, and that the supremum norm on $C(K)$ is equivalent to the usual Euclidean norm on $\mathbb{R}^n$. Conclude that the Bolzano-Weierstrass theorem is a special case of Arzela-Ascoli theorem.

Proof. Let

$$K = \{x_1, x_2, \dots, x_n\}$$

be a finite set equipped with the discrete metric

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \ne y. \end{cases}$$

Since K is finite and discrete, every subset of K is open. Hence every function $f : K \to \mathbb{R}$ is continuous, so

$$C(K) = \{f : K \to \mathbb{R}\}.$$

Define

$$T : C(K) \to \mathbb{R}^n$$

by

$$T(f) = \big(f(x_1), f(x_2), \ldots, f(x_n)\big).$$

The map T is linear and bijective, since a function on K is completely determined by its values at the points $x_1, \ldots, x_n$. Thus $C(K)$ may be identified with $\mathbb{R}^n$.

Now let $f \in C(K)$. The supremum norm on $C(K)$ is

$$\|f\|_\infty = \sup_{x \in K} |f(x)|.$$

Because K has finitely many points,

$$\|f\|_\infty = \max_{1 \le i \le n} |f(x_i)|.$$

Under the identification with $\mathbb{R}^n$, this is exactly the max norm

$$\|(a_1, \ldots, a_n)\|_{\max} = \max_{1 \le i \le n} |a_i|.$$

We now compare this norm with the Euclidean norm

$$\|(a_1, \ldots, a_n)\|_2 = \left(\sum_{i=1}^n a_i^2\right)^{1/2}.$$

For every $a = (a_1, \ldots, a_n) \in \mathbb{R}^n$,

$$\|a\|_{\max} \le \|a\|_2$$

since

$$|a_i|^2 \le \sum_{j=1}^n a_j^2$$

for each i. Also,

$$\|a\|_2^2 = \sum_{i=1}^n a_i^2 \le \sum_{i=1}^n \|a\|_{\max}^2 = n\|a\|_{\max}^2,$$

so

$$\|a\|_2 \le \sqrt{n}\,\|a\|_{\max}.$$

Hence the supremum norm and the Euclidean norm are equivalent on $\mathbb{R}^n$.

Finally, observe that every family of functions on K is equicontinuous. Indeed, if $\varepsilon > 0$, choose $\delta = \frac{1}{2}$. Then

$$d(x, y) < \delta \Rightarrow x = y,$$

and therefore

$$|f(x) - f(y)| = 0 < \varepsilon$$

for every function $f \in C(K)$.

Thus, in this setting, the Arzelà–Ascoli theorem states that every uniformly bounded sequence in $C(K) \cong \mathbb{R}^n$ has a uniformly convergent subsequence. Since uniform convergence corresponds to convergence in the max norm, and this norm is equivalent to the Euclidean norm, we conclude that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

This is precisely the Bolzano–Weierstrass theorem. ⊛

5.4 Application: Compactness of Integral Operators

We now give an important application of the Arzela-Ascoli theorem. Let $K : [a, b] \times [a, b] \to \mathbb{R}$ be continuous. For each $f \in C([a, b])$, define

$$(Tf)(x) := \int_a^b K(x, y)f(y)\,\mathrm{d}y, \quad x \in [a, b].$$

We shall first check that Tf is continuous, so that T is indeed a map from $C([a, b])$ into itself. Then we shall prove that T is compact with respect to the supremum norm.

Proposition 5.4.1. Let $K : [a, b] \times [a, b] \to \mathbb{R}$ be continuous, and define

$$(Tf)(x) := \int_a^b K(x, y)f(y)\,dy, \qquad x \in [a, b].$$

Then for every $f \in C([a, b])$, the function Tf is continuous on $[a, b]$.

Proof. Fix $f \in C([a, b])$. Since f is continuous on the compact interval $[a, b]$, it is bounded. Hence there exists $M > 0$ such that

$$|f(y)| \leq M \qquad \forall y \in [a, b].$$

Let $\{x_n\}_{n=1}^{\infty} \subseteq [a, b]$ satisfy $x_n \to x \in [a, b]$. We shall show that

$$(Tf)(x_n) \to (Tf)(x).$$

For each n,

$$\begin{aligned}
|(Tf)(x_n) - (Tf)(x)| &= \left| \int_a^b K(x_n, y)f(y)\,dy - \int_a^b K(x, y)f(y)\,dy \right| \\
&= \left| \int_a^b \big(K(x_n, y) - K(x, y)\big) f(y)\,dy \right| \\
&\leq \int_a^b |K(x_n, y) - K(x, y)|\,|f(y)|\,dy \\
&\leq M \int_a^b |K(x_n, y) - K(x, y)|\,dy.
\end{aligned}$$

Since K is continuous on the compact set $[a, b] \times [a, b]$, it is uniformly continuous. Therefore,

$$\sup_{y \in [a,b]} |K(x_n, y) - K(x, y)| \to 0 \qquad \text{as } n \to \infty.$$

Consequently,

$$\begin{aligned}
\int_a^b |K(x_n, y) - K(x, y)|\,dy &\leq (b - a) \sup_{y \in [a,b]} |K(x_n, y) - K(x, y)| \\
&\to 0.
\end{aligned}$$

It follows that

$$|(Tf)(x_n) - (Tf)(x)| \to 0,$$

and hence

$$(Tf)(x_n) \to (Tf)(x).$$

Therefore Tf is continuous on $[a, b]$, i.e.

$$Tf \in C([a, b]).$$

■

Theorem 5.4.1 (Compactness of integral operators). Let $K : [a, b] \times [a, b] \to \mathbb{R}$ be continuous. Define $T : C([a, b]) \to C([a, b])$ by

$$(Tf)(x) := \int_a^b K(x, y) f(y) \, dy.$$

Then T is compact. More precisely, if $\{f_n\} \subseteq C([a, b])$ is bounded in the supremum norm, then the sequence $\{Tf_n\}$ has a uniformly convergent subsequence in $C([a, b])$.

Proof. We divide the proof into two steps.

- **Step 1. The operator is well-defined.** Let $f \in C([a, b])$. By previous proposition, we khow that $Tf \in C([a, b])$.
- **Step 2. The image of every bounded sequence is relatively compact.** Let $\{f_n\} \subseteq C([a, b])$ be bounded in the supremum norm. Then there exists a constant $M > 0$ s.t.

$$\|f_n\|_\infty \le M \quad \text{for all } n \in \mathbb{N}.$$

We shall prove the family $\{Tf_n : n \in \mathbb{N}\}$ is uniformly bounded and equicontinuous. Then Arzela-Ascoli theorem will then imply that $\{Tf_n\}$ has a uniformly convergent subsequence in $C([a, b])$.

First, since K is continuous on the compact set $[a, b] \times [a, b]$, it is bounded. Hence, there exists $C > 0$ s.t.

$$|K(x, y)| \le C \quad \text{for all } (x, y) \in [a, b] \times [a, b].$$

Therefore, for every $x \in [a, b]$,

$$|(Tf_n)(x)| = \left| \int_a^b K(x, y) f_n(y) \, dy \right| \le \int_a^b |K(x, y)||f_n(y)| \, dy \le \int_a^b CM \, dy = CM(b - a).$$

Thus, $\|Tf_n\|_\infty \le CM(b - a)$ for all $n \in \mathbb{N}$. So the family $\{Tf_n\}$ is uniformly bounded. Next, we prove equicontinuity. Since K is uniformly continuous on $[a, b] \times [a, b]$, given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$|x - x'| < \delta \Rightarrow |K(x, y) - K(x', y)| < \frac{\varepsilon}{M(b - a)}$$

for every $y \in [a, b]$. If $M = 0$, then $f_n = 0$ for every n, and the claim is trivial. Hence we may assume $M > 0$.

Thus, if $|x - x'| < \delta$, then

$$\begin{aligned}
|(Tf_n)(x) - (Tf_n)(x')| &= \left| \int_a^b \left(K(x, y) - K(x', y)\right) f_n(y) \, dy \right| \\
&\le \int_a^b |K(x, y) - K(x', y)||f_n(y)| \, dy \\
&< \int_a^b \frac{\varepsilon}{M(b - a)} M \, dy = \varepsilon.
\end{aligned}$$

The estimate is independent of n. Hence the family $\{Tf_n\}$ is equicontinuous, and we're done.

∎

Chapter 6

The Stone-Weierstrass Theorem

In this chapter, we discuss an general approximation theorem that has many applications to approximation problems. The theorem gives a simple and useful criterion for deciding when all continuous functions on a compact metric space can be uniformly approximated by elements of a given algebra of functions.

As consequences, we shall recover approximation results for polynomials in several variables and for trigonometric polynomials.

6.1 Algebras and vector lattices

Definition 6.1.1 (Algebra of continuous functions). Let X be a compact metric space. A subset $A \subseteq C(X)$, where $C(X)$ denotes the space of real-valued continuous functions on X, is called an algebra if A is a vector subspace of $C(X)$ and is closed under pointwise multiplication. That is, if $f, g \in A$, then $fg \in A$.

For $f, g \in C(X)$, we define

$$(f \vee g)(x) := \max\{f(x), g(x)\}, \quad (f \wedge g)(x) := \min\{f(x), g(x)\}.$$

Definition 6.1.2 (Vector lattice). A subset $L \subseteq C(X)$ is called a vector lattice if L is a vector subspace of $C(X)$ and is closed under the two operations

$$f, g \mapsto f \vee g, \quad f, g \mapsto f \wedge g.$$

Equivalently, wheanever $f, g \in L$, one has

$$f \vee g \in L, \quad f \wedge g \in L.$$

The operations $f \vee g$ and $f \wedge g$ can be expressed using the absolute value:

$$f \vee g = \frac{1}{2}(f+g) + \frac{1}{2}|f-g|, \quad f \wedge g = \frac{1}{2}(f+g) - \frac{1}{2}|f-g|.$$

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Conversely, May 14.

$$|f| = f \vee (-f).$$

Therefore, we can give a corollary:

Corollary 6.1.1. An algebra $A \subseteq C(X)$ is a vector lattice if and only if

$$|f| \in A \quad \text{for every } f \in A.$$

Proof. If an algebra $A \subseteq C(X)$ is a vector lattice, then for $f \in A$, we have $-f \in A$ since A is a vector space, and $|f| = f \vee (-f) \in A$ since A is a vector lattice.

Now if $|f| \in A$ for every $f \in A$, where A is an algebra, then for all $f, g \in A$, we have $f-g, f+g \in A$ since A is a vector space, and thus $|f-g| \in A$. Hence, we have

$$f \vee g = \frac{1}{2}(f+g) + \frac{1}{2}|f-g| \in A, \quad f \wedge g = \frac{1}{2}(f+g) - \frac{1}{2}|f-g| \in A.$$

Thus, A is a vector lattice. ∎

Definition 6.1.3 (Separating points and vanishing at a point)**.** Let S be a set of functions on X. We say S separates points if for every pair of distinct points $x, y \in X$, there exists $f \in S$ such that

$$f(x) \neq f(y).$$

We say that S vanishes at a point $x_0 \in X$ if

$$f(x_0) = 0 \quad \text{for every } f \in S.$$

In order for an algebra $A \subseteq C(X)$ to approximate arbitrary continuous functions on X, it is necessary that A separates points. Moreover, A cannot vanish at any point, since otherwise it could not approximate the constant function 1.

The Stone-Weierstrass theorem states that these two natural conditions are also sufficient.

6.2 The Stone-Weierstrass theorem

Theorem 6.2.1 (Stone-Weierstrass theorem)**.** Let X be a compact metric space, and let $A \subseteq C(X)$ be an algebra of real-valued continuous functions. Suppose that A separates points and does not vanish at any point of X. Then A is dense in $C(X)$ with respect to the uniform norm. In other words, for every $f \in C(X)$ and every $\varepsilon > 0$, there exists $g \in A$ s.t.

$$\|f - g\|_X < \varepsilon.$$

Here

$$\|f\|_X \coloneqq \sup_{x \in X} |f(x)|$$

denotes the uniform norm on X.

We break the proof into several lemmas.

Lemma 6.2.1. Let $n \in \mathbb{N}$ and $0 \leq x \leq 1$. Then

$$\sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n}.$$

Proof. We first record three elementary identities which will be used in the proof. They are consequences of the binomial theorem. For two variables x and y, we have

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

First, set $y = 1 - x$. Then $x + y = 1$, and hence

$$\sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} = (x + (1-x))^n = 1.$$

Next, differentiate the binomial identity with respect to x. Since y is treated as an independent variable, we obtain

$$n(x+y)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k} x^{k-1} y^{n-k}.$$

Multiplying both sides by x, we get

$$nx(x+y)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k} x^{k} y^{n-k}.$$

The term $k=0$ is zero, so we may write the right-hand side as a sum from $k=0$ to n. Setting $y = 1-x$, we obtain

$$\sum_{k=0}^{n} k\binom{n}{k} x^k (1-x)^{n-k} = nx(x+(1-x))^{n-1} = nx.$$

Finally, differentiate the binomial identity twice with respect to x. This gives

$$n(n-1)(x+y)^{n-2} = \sum_{k=2}^{n} k(k-1)\binom{n}{k} x^{k-2} y^{n-k}.$$

Multiplying both sides by x^2, we obtain

$$n(n-1)x^2(x+y)^{n-2} = \sum_{k=2}^{n} k(k-1)\binom{n}{k} x^{k} y^{n-k}.$$

The terms $k=0$ and $k=1$ are zero, so again we may extend the sum to start from $k=0$. Setting $y=1-x$, we get

$$\sum_{k=0}^{n} k(k-1)\binom{n}{k} x^k (1-x)^{n-k} = n(n-1)x^2(x+(1-x))^{n-2} = n(n-1)x^2.$$

Now expand the square:

$$\begin{aligned}
&\sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} \\
&= \frac{1}{n^2}\sum_{k=0}^{n} k^2 \binom{n}{k} x^k (1-x)^{n-k} - \frac{2x}{n}\sum_{k=0}^{n} k\binom{n}{k} x^k (1-x)^{n-k} + x^2 \sum_{k=0}^{n}\binom{n}{k} x^k (1-x)^{n-k}.
\end{aligned}$$

Since $k^2 = k(k-1)+k$, we have

$$\begin{aligned}
\sum_{k=0}^{n} k^2 \binom{n}{k} x^k (1-x)^{n-k} &= \sum_{k=0}^{n} k(k-1)\binom{n}{k} x^k (1-x)^{n-k} + \sum_{k=0}^{n} k\binom{n}{k} x^k (1-x)^{n-k} \\
&= n(n-1)x^2 + nx.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} &= \frac{n(n-1)x^2 + nx}{n^2} - \frac{2x}{n}(nx) + x^2 \\
&= \frac{n-1}{n}x^2 + \frac{x}{n} - 2x^2 + x^2 = \frac{x}{n} - \frac{x^2}{n} = \frac{x(1-x)}{n}.
\end{aligned}$$

■

Theorem 6.2.2 (Weierstrass approximation theorem)**.** Let $h : [a,b] \to \mathbb{R}$ be continuous. Then there exists a sequence of polynomials $\{p_n\}_{n=1}^{\infty}$ such that $p_n \to h$ uniformly on $[a, b]$. Equivalently,

$$\lim_{n\to\infty} \sup_{x\in[a,b]} |p_n(x) - h(x)| = 0.$$

Proof. We first prove the theorem on the interval $[0, 1]$. Let $h : [0, 1] \to \mathbb{R}$ be continuous. For each $n \in \mathbb{N}$, define Bernstein polynomial of degree n by

$$B_n h(x) := \sum_{k=0}^{n} h\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}, \quad 0 \le x \le 1.$$

This is indeed a polynomial in x. We will prove that $B_n h \to h$ uniformly on $[0, 1]$.

Fix $x \in [0, 1]$. Since

$$\sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} = (x + (1-x))^n = 1,$$

we have

$$h(x) = \sum_{k=0}^{n} h(x) \binom{n}{k} x^k (1-x)^{n-k},$$

so we may write

$$B_n h(x) - h(x) = \sum_{k=0}^{n} \left[h\left(\frac{k}{n}\right) - h(x) \right] \binom{n}{k} x^k (1-x)^{n-k}.$$

Therefore,

$$|B_n h(x) - h(x)| \le \sum_{k=0}^{n} \left| h\left(\frac{k}{n}\right) - h(x) \right| \binom{n}{k} x^k (1-x)^{n-k}.$$

Since h is continuous on the compact interval $[0, 1]$, it is uniformly continuous. Thus, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|u - v| < \delta \Rightarrow |h(u) - h(v)| < \frac{\varepsilon}{2}$$

for all $u, v \in [0, 1]$.

Also, since h is continuous on $[0, 1]$, it is bounded. Hence, there exists $M > 0$ s.t.

$$|h(x)| \le M \quad \text{for every } x \in [0, 1].$$

We split the sum into two parts:

$$A_n(x) := \left\{ k \in [n] : \left| \frac{k}{n} - x \right| < \delta \right\}, \quad A_n(x)^c := [n] \setminus A_n(x).$$

For $k \in A_n(x)$, uniform continuity gives

$$\left| h\left(\frac{k}{n}\right) - h(x) \right| < \frac{\varepsilon}{2}.$$

For all k, the boundedness of h gives

$$\left| h\left(\frac{k}{n}\right) - h(x) \right| \le 2M.$$

Hence,

$$|B_n h(x) - h(x)| \le \frac{\varepsilon}{2} \sum_{k\in A_n(x)} \binom{n}{k} x^k (1-x)^{n-k} + 2M \sum_{k\in A_n(x)^c} \binom{n}{k} x^k (1-x)^{n-k}.$$

Since
$$\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}=1,$$
we have
$$|B_nh(x)-h(x)|\le\frac{\varepsilon}{2}+2M\sum_{\left|\frac{k}{n}-x\right|\ge\delta}\binom{n}{k}x^k(1-x)^{n-k}.$$
It remains to estimate the last sum. We shall use the elementary identity established in Lemma 6.2.1:
$$\sum_{k=0}^{n}\left(\frac{k}{n}-x\right)^2\binom{n}{k}x^k(1-x)^{n-k}=\frac{x(1-x)}{n}.$$
Since $0\le x(1-x)\le\frac{1}{4}$, we have
$$\sum_{k=0}^{n}\left(\frac{k}{n}-x\right)^2\binom{n}{k}x^k(1-x)^{n-k}\le\frac{1}{4n}.$$
On the set where $\left|\frac{k}{n}-x\right|\ge\delta$, we have
$$1\le\frac{1}{\delta^2}\left(\frac{k}{n}-x\right)^2.$$
Therefore,
$$\sum_{\left|\frac{k}{n}-x\right|\ge\delta}\binom{n}{k}x^k(1-x)^{n-k}\le\frac{1}{\delta^2}\sum_{k=0}^{n}\left(\frac{k}{n}-x\right)^2\binom{n}{k}x^k(1-x)^{n-k}\le\frac{1}{4n\delta^2}.$$
Consequently,
$$|B_nh(x)-h(x)|\le\frac{\varepsilon}{2}+\frac{M}{2n\delta^2}.$$
Choose $N\in\mathbb{N}$ such that
$$\frac{M}{2N\delta^2}<\frac{\varepsilon}{2}.$$
Then, for every $n\ge N$ and every $x\in[0,1]$, we have
$$|B_nh(x)-h(x)|<\varepsilon.$$
Thus,
$$\lim_{n\to\infty}\sup_{x\in[0,1]}|B_nh(x)-h(x)|=0.$$

Now let $h:[a,b]\to\mathbb{R}$ be continuous. Define
$$\phi(t):=h(a+(b-a)t),\quad 0\le t\le 1.$$
Then, ϕ is continuous on $[0,1]$. By the result just proved, there exists a seuqence of polynomials $q_n(t)$ such that $q_n\to\phi$ uniformly on $[0,1]$.

For $x\in[a,b]$, set
$$t=\frac{x-a}{b-a}.$$
Define
$$p_n(x):=q_n\left(\frac{x-a}{b-a}\right),$$
then since q_n is polynomial in t, the function p_n is a polynomial in x. Moreover,
$$p_n(x)-h(x)=q_n\left(\frac{x-a}{b-a}\right)-\phi\left(\frac{x-a}{b-a}\right).$$

Taking the supremum over $x \in [a, b]$, we obtain

$$\sup_{x\in[a,b]} |p_n(x) - h(x)| = \sup_{t\in[0,1]} |q_n(t) - \phi(t)|.$$

Since the right-hand side tends to 0 as $n \to \infty$, we conclude that $p_n \to h$ uniformly on $[a, b]$.
This proves the Weierstrass approximation theorem. ■

Lemma 6.2.2. Let $A \subseteq C(X)$ be an algebra of real-valued continuous functions. Then its uniform closure $\overline{A}$ is a closed algebra and a vector lattice.

Proof. It is straightforward to check that the closure of a vector subspace is again a vector subspace. Thus, $\overline{A}$ is a vector subspace of $C(X)$.

We next show that $\overline{A}$ is closed under multiplication. Let $f, g \in \overline{A}$. Choose seuqences $\{f_n\}$ and $\{g_n\}$ in A s.t. $f_n \to f$ and $g_n \to g$ uniformly on X. Since A is an algebra, $f_n g_n \in A$ for every n. Also, because uniform convergence preserves products of bounded continuous functions, we have $f_n g_n \to fg$ uniformly on X. Hence, $fg \in \overline{A}$. Therefore $\overline{A}$ is an algebra.

It remains to show that $\overline{A}$ is a vector lattice. By Corollary 6.1.1, it suffices to prove that $|f| \in \overline{A}$ for every $f \in \overline{A}$.

Fix $f \in \overline{A}$. Consider the function

$$h(t) = |t|, \quad t \in [-\|f\|_X, \|f\|_X].$$

By the Weierstrass approximation theorem, there exists a sequence of polynomials p_n s.t. $p_n \to h$ uniformly on $[-\|f\|_X, \|f\|_X]$.

We may arrange that the approximating polynomials vanish at 0. Indeed, since $h(0) = 0$ and $p_n(0) \to 0$, define

$$q_n(t) := p_n(t) - p_n(0).$$

Then $q_n(0) = 0$ and

$$\|h - q_n\|_\infty \le \|h - p_n\|_\infty + |p_n(0)| \to 0.$$

Thus, $q_n \to h$ uniformly on $[-\|f\|_X, \|f\|_X]$, and each q_n has the form

$$q_n(t) = a_1 t + a_2 t^2 + \cdots + a_k t^k.$$

Since $\overline{A}$ is an algebra and $f \in \overline{A}$, every power $f^j \in \overline{A}$. Hence,

$$q_n(f) = a_1 f + a_2 f^2 + \cdots + a_k f^k \in \overline{A}.$$

Moreover, for any two such polynomials p, q,

$$\|p(f) - q(f)\|_X = \sup_{x\in X} |p(f(x)) - q(f(x))| \le \sup_{|t|\le\|f\|_X} |p(t) - q(t)|.$$

Since $\{q_n\}$ is a uniformly Cauchy sequence on $[-\|f\|_X, \|f\|_X]$, it follows that $\{q_n(f)\}$ is Cauchy in $C(X)$. Its uniform limit is

$$\lim_{n\to\infty} q_n(f(x)) = h(f(x)) = |f(x)|.$$

Becuase $\overline{A}$ is closed, this limit belongs to $\overline{A}$. Therefore, $|f| \in \overline{A}$, and thus $\overline{A}$ is a vector lattice. ■

Lecture 22

May 19.

Lemma 6.2.3. Let $A \subseteq C(X)$ be an algebra that separates points and does not vanish at any point. Then. for any two distinct points $x, y \in X$ and any $\alpha, \beta \in \mathbb{R}$, there exists function $h \in A$ s.t.

$$h(x) = \alpha, \quad h(y) = \beta.$$

Proof. Since A separates points, there exists a function $f \in A$ s.t. $f(x) \ne f(y)$. We shall construct h by considering the possible values of $f(x)$ and $f(y)$.

First suppose that $f(x) \neq 0$ and $f(y) \neq 0$. In this case we look for h in the form

$$h = af + bf^2,$$

where $a, b \in \mathbb{R}$ are constants to be chosen. Since A is an algebra, both f and f^2 belong to A, and hence $h \in A$.

We want $h(x) = \alpha$ and $h(y) = \beta$. Equivalently,

$$af(x) + bf(x)^2 = \alpha, \quad af(y) + bf(y)^2 = \beta.$$

This is a linear system for the two unknowns a and b:

$$\begin{pmatrix} f(x) & f(x)^2 \\ f(y) & f(y)^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Its determinant is

$$f(x)f(y)^2 - f(y)f(x)^2 = f(x)f(y)(f(y) - f(x)).$$

This determinant is nonzero because $f(x), f(y) \neq 0$, and $f(x) \neq f(y)$. Therefore the system has a unique solution (a, b) to the system, and thus we can pick $h = af + bf^2$ with this solution (a, b).

It remains to treat the case where one of $f(x)$ and $f(y)$ is zero. Since $f(x) \neq f(y)$, they cannot be both zero.

We claim that there exists $g \in A$ s.t. the matrix

$$\begin{pmatrix} g(x) & f(x) \\ g(y) & f(y) \end{pmatrix}$$

has non-zero determinant. Indeed, if $f(x) = 0$, then $f(y) \neq 0$. Since A does not vanish at x, there exists $g \in A$ s.t. $g(x) \neq 0$. Hence,

$$\det \begin{pmatrix} g(x) & f(x) \\ g(y) & f(y) \end{pmatrix} = g(x)f(y) - g(y)f(x) = g(x)f(y) \neq 0.$$

Similarly, if $f(y) = 0$, then $f(x) \neq 0$. Since A does not vanish at y, there exists $g \in A$ s.t. $g(y) \neq 0$. Therefore,

$$\det \begin{pmatrix} g(x) & f(x) \\ g(y) & f(y) \end{pmatrix} = g(x)f(y) - g(y)f(x) = -g(y)f(x) \neq 0.$$

Now look for h in the form

$$h = ag + bf,$$

where $a, b \in \mathbb{R}$ are to be chosen. Since A is a vector space and $f, g \in A$, we have $h \in A$. The interpolation conditions

$$h(x) = \alpha, \quad h(y) = \beta$$

are equivalent to the linear system

$$\begin{pmatrix} g(x) & f(x) \\ g(y) & f(y) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

By the determinant computation above, the coefficient matrix is invertible, and we're done. ■

Now we can prove the Stone-Weierstrass theorem: Let X be a compact metric space, and let $A \subseteq C(X)$ be an algebra of real-valued continuous functions. Suppose that A separates points and does not vanish at any point of X. Then A is dense in $C(X)$ with respect to the uniform norm. In other words, for every $f \in C(X)$ and every $\varepsilon > 0$, there exists $g \in A$ s.t.

$$\|f - g\|_X < \varepsilon.$$

Here

$$\|f\|_X := \sup_{x \in X} |f(x)|$$

denotes the uniform norm on X.

Proof. Let $f \in C(X)$ and let $\varepsilon > 0$. We shall prove that $f \in \overline{A}$ where $\overline{A}$ denotes the closure of A in the uniform norm. Since f is arbitrary, this will imply

$$\overline{A} = C(X),$$

and hence A is dense in $C(X)$ ($f \in \overline{A}$ implies there is a sequence of A converging to f).

By Lemma 6.2.2, $\overline{A}$ is a closed vector lattice. In particular, if $u_1, \ldots, u_m \in \overline{A}$, then

$$u_1 \vee \cdots \vee u_m \in \overline{A}, \quad u_1 \wedge \cdots \wedge u_m \in \overline{A}.$$

We shall use this lattice property to combine finitely many local approximations.

First, fix a point $y \in X$. For each point $x \in X$ with $x \neq y$, Lemma 6.2.3 gives a function $h_{x,y} \in A$ s.t.

$$h_{x,y}(x) = f(x), \quad h_{x,y}(y) = f(y).$$

Since $h_{x,y}$ agrees with f at x, by continuity it remains close to f in a neighborhood of x. Since it also agrees with f at y, it remains close to f in a neighborhood of y. More precisely, define

$$U_{x,y} \coloneqq \{z \in X : h_{x,y}(z) > f(z) - \varepsilon\}.$$

Then, $U_{x,y}$ is open, because $h_{x,y} - f$ is continuous, and

$$U_{x,y} = (h_{x,y} - f)^{-1}((-\varepsilon, \infty)),$$

and $(-\varepsilon, \infty)$ is open. Also, note that $x, y \in U_{x,y}$.

We claim that, for this fixed y, the family

$$\{U_{x,y} : x \in X, x \neq y\}$$

covers X. Indeed, if $z \neq y$, then by taking $x = z$ we have $z \in U_{z,y}$. On the other hands, $y \in U_{x,y}$ for all $x \in X$. Hence the family covers X.

Since X is compact, there exist finitely many points

$$x_1, \ldots, x_k \in X \setminus \{y\}$$

s.t.

$$X = U_{x_1,y} \cup \cdots \cup U_{x_k,y}.$$

Define

$$H_y \coloneqq h_{x_1,y} \vee h_{x_2,y} \vee \cdots \vee h_{x_k,y}.$$

Each $h_{x_j,y} \in A$ and hence to $\overline{A}$. Since $\overline{A}$ is a vector lattice, it follows that $H_y \in \overline{A}$.

We record two important properties of H_y. First, since every $h_{x_j,y}$ agrees with f at y, we have

$$h_{x_j,y}(y) = f(y) \quad \text{for } j = 1, \ldots, k.$$

Therefore, $H_y(y) = f(y)$. Second, for every $z \in X$, since the sets $\{U_{x_j,y}\}$ covers X, there exists j s.t. $z \in U_{x_j,y}$, i.e.

$$h_{x_j,y}(z) > f(z) - \varepsilon.$$

Since $H_y \geq h_{x_j,y}$, we obtain

$$H_y(z) > f(z) - \varepsilon \quad \text{for every } z \in X.$$

Hence, for each fixed $y \in X$, we have constructed a function $H_y \in \overline{A}$ s.t. $H_y(y) = f(y)$ and $H_y > f - \varepsilon$ on X. We now use these functions H_y to obtain an upper bound as well. For each $y \in X$, define

$$V_y \coloneqq \{z \in X : H_y(z) < f(z) + \varepsilon\}.$$

Since $H_y - f$ is continuous, V_y is open. Moreovr, because $H_y(y) = f(y)$, we have $y \in V_y$. Thus, the family

$$\{V_y : y \in X\}$$

is an open cover of X. By compactness, there exist finitely many points $y_1, \dots, y_\ell \in X$ s.t.

$$X = V_{y_1} \cup \cdots \cup V_{y_\ell}.$$

Define

$$g := H_{y_1} \wedge H_{y_2} \cdots \wedge H_{y_\ell}.$$

Since each $H_{y_i} \in \overline{A}$ and $\overline{A}$ is a vector lattice, we have $g \in \overline{A}$.

We now vertify that g approximates f uniformly. First, for each $i = 1, \dots, \ell$, the construction of H_{y_i} gives

$$H_{y_i}(z) > f(z) - \varepsilon \quad \text{for every } z \in X.$$

Taking the minimum over $i = 1, \dots, \ell$, we still have

$$g(z) > f(z) - \varepsilon \quad \text{for every } z \in X.$$

On the other hand, let $z \in X$. Since the sets V_{y_i} cover X, there exists an index i such that $z \in V_{y_i}$, i.e.

$$H_{y_i}(z) < f(z) + \varepsilon.$$

Becuase

$$g(z) = \min_{1 \le j \le \ell} H_{y_j}(z) \le H_{y_i}(z),$$

we obtain $g(z) < f(z) + \varepsilon$. Combining two estimates, we het

$$f(z) - \varepsilon < g(z) < f(z) + \varepsilon \quad \text{for every } z \in X.$$

Equivalently,

$$|g(z) - f(z)| < \varepsilon \quad \text{for every } z \in X.$$

Therefore,

$$\|g - f\|_X < \varepsilon.$$

We have shown that for every $\varepsilon > 0$, there exists $g \in \overline{A}$ s.t. $\|g - f\|_X < \varepsilon$. This shows $f \in \overline{\overline{A}}$. Since $\overline{A}$ is closed, we have $f \in \overline{A}$, and we're done. ∎

6.3 Polynomial approximation on compact subsets of $\mathbb{R}^n$

Corollary 6.3.1. Let $X \subseteq \mathbb{R}^n$ be compact. Then the algebra of all polynomials

$$p(x_1, x_2, \dots, x_n)$$

in the n coordinate functions is dense in $C(X)$.

Proof. Let $\mathcal{P}$ denote the set of all polynomial functions on X. It is clear that $\mathcal{P}$ is an algebra.

The constant function 1 belongs to $\mathcal{P}$, so $\mathcal{P}$ does not vanish at any point of X.

Finally, $\mathcal{P}$ separates points. Indeed, if $x, y \in X$ are distinct, then $x_i \neq y_i$ for at least one coordinate i. The coordinate function $p(z) = z_i$ belongs to $\mathcal{P}$ and satisfies $p(x) \neq p(y)$. Thus, $\mathcal{P}$ satisfies the hypotheses of the Stone-Weierstrass theorem. Hence $\mathcal{P}$ is dense in $C(X)$. ∎

6.4 Trigonometric polynomial approximation

Recall that a trigonometric polynomial is a function of the form

$$f(t) = a_0 + \sum_{k=1}^{n} (a_k \cos kt + b_k \sin kt),$$

where $a_0, a_k, b_k \in \mathbb{R}$.

Corollary 6.4.1. The set of all trigonometric polynomials is dense in

$$C_*[-\pi,\pi] := \{f \in C[-\pi,\pi] : f(-\pi) = f(\pi)\}.$$

Equivalently, trigonometric polynomials are dense in the space of continuous 2π-periodic functions.

Proof. Let $\mathcal{TP}$ denote the set of trigonometric polynomial. It is clear that $\mathcal{TP}$ is a vector space. We first check that it is an algebra.

Using the product-to-sum identities,

$$\begin{aligned}
\sin kt \sin lt &= \frac{1}{2}\cos((k-l)t) - \frac{1}{2}\cos((k+l)t),\\
\sin kt \cos lt &= \frac{1}{2}\sin((k+l)t) + \frac{1}{2}\sin((k-l)t),\\
\cos kt \cos lt &= \frac{1}{2}\cos((k-l)t) + \frac{1}{2}\cos((k+l)t),
\end{aligned}$$

we see that the product of two trigonometric polynomials is again a trigonometric polynomial. Therefore $\mathcal{TP}$ is an algebra.

To apply the Stone-Weierstrass theorem, it is convenient to identify $C_*[-\pi,\pi]$ with the space of continuous functions on the unit circle. Let

$$T := \left\{(x_1,x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\right\}$$

be the unit circle, and define

$$P : \mathbb{R} \to T, \quad P(t) = (\cos t, \sin t).$$

Then $P(s) = P(t)$ if and only if $s - t \in 2\pi\mathbb{Z}$. In particular, the interval $[-\pi,\pi]$ parametrizes the unit circle, with the two endpoints $-\pi$ and π identified.

Every function $g \in C(T)$ gives a function $f \in C_*[-\pi,\pi]$ by

$$f(t) = g(P(t)).$$

Conversely, every $f \in C_*[-\pi,\pi]$ arises this way from a unique function $g \in C(T)$. This correspondence is linear, isometric, and preserves multiplication.

Under this identification, the coordinate functions

$$x_1, x_2 : T \to \mathbb{R}$$

correspond respectively to $\cos t, \sin t$. Therefore polynomial functions in x_1 and x_2 on T correspond exactly to trigonometric polynomials in t.

By Corollary 6.3.1, polynomial functions in x_1, x_2 are dense in $C(T)$. Hence, trigonometric polynomials are dense in $C_*[-\pi,\pi]$. ■

Chapter 7

Foundations of Functional Analysis

This chapter introduces several fundamental ideas that play a central role in functional analysis. Broadly speaking, functional analysis studies normed vector spaces and bounded linear operators between them. Thus it combines linear algebra, which studies vector spaces and linear maps, with topology, which studies continuity, convergence, and compactness.

Most of the topologies appearing in this chapter come from norms or metrics. Consequently, the metric space results developed earlier will be used throughout. Among these, compactness is especially important. We shall use the characterization of compactness in metric spaces by sequential compactness and open covers, as well as the Arzela-Ascoli theorem, which gives a powerful compactness criterion for families of continuous functions on compact metric spaces.

The chapter is organized as follows. In Section 12.1, we study finite-dimensional normed vector spaces. We prove that any two norms on a finite-dimensional vector space are equivalent and derive several important consequences: finite-dimensional normed spaces are complete, their linear subspaces are closed, linear maps defined on them are continuous, and their closed and bounded subsets are compact. We also show that compactness of the closed unit ball characterizes finite-dimensionality. Along the way, we introduce bounded linear operators, as well as quotient and product spaces.

The next section introduces the dual space of a normed vector space. We discuss basic examples and develop some elementary Hilbert space theory, including the Cauchy-Schwarz inequality and the Riesz representation theorem. We then study power series in Banach algebras. As an application of the geometric series, we show that the set of invertible operators on a Banach space is open and that the inverse map is continuous. Finally, we discuss the Baire category theorem, one of the basic structural results in the theory of complete metric spaces.

7.1 Finite-Dimensional Banach Spaces

The purpose of this section is to examine finite-dimensional normed vector spaces and to highlight the properties that distinguish them from infinite-dimensional normed vector spaces. In finite dimensions, the analytic structure is especially rigid. Every finite-dimensional normed vector space is complete, every finite-dimensional linear subspace of a normed vector space is closed, every linear map defined on a finite-dimensional normed vector space is continuous, and every closed and bounded subset is compact. These facts are familiar from Euclidean spaces, but they remain true for arbitrary norms because all norms on a finite dimensional vector space are equivalent.

We shall also see that compactness of the closed unit ball is not merely a consequence of finite-dimensionality; it is a characterization of it. More precisely, a normed vector space is finite-dimensional if and only if its closed unit ball is compact. This fact explains one of the central di!erences between finite-dimensional linear algebra and infinite-dimensional functional analysis.

Before proving these results, we first introduce bounded linear operators.

The basic morphisms in functional analysis are bounded linear operators. These are the linear maps that are compatible with the norm structures on the domain and target spaces. In functional analysis one often says linear operator rather than linear map. The meaning is the same: a map between vector spaces that preserves addition and scalar multiplication. The terminology reflects the fact that many important vector spaces in analysis are spaces of functions, and the linear maps acting on them are often viewed as operators.

If the domain and target are normed vector spaces, it is natural to require a linear operator to be continuous with respect to the norm topologies. For linear maps, this continuity condition is equivalent to a simple global estimate.

Definition 7.1.1 (Bounded linear operator)**.** Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be real normed vector spaces. A linear operator $A : X \to Y$ is called bounded if there exists a constant $c > 0$ s.t.

$$\|Ax\|_Y \leq c\|x\|_X \quad \text{for all } x \in X. \tag{7.1}$$

Remark 7.1.1. The smallest constant $c \geq 0$ for which Equation 7.1 holds is called the operator norm of A. It is denoted by

$$\|A\| := \|A\|_{\mathcal{L}(X,Y)} := \sup_{x \in X \setminus \{0\}} \frac{\|Ax\|_Y}{\|x\|_X}.$$

Equivalently,

$$\|A\|_{\mathcal{L}(X,Y)} = \sup_{\|x\|_X \leq 1} \|Ax\|_Y.$$

Remark 7.1.2. A bounded linear operator with values in $Y = \mathbb{R}$ is called a bounded linear functional on X.

The collection of all bounded linear operators from X to Y is denoted by

$$\mathcal{L}(X, Y) := \{A : X \to Y \mid A \text{ is linear and bounded}\}.$$

With the operator norm, this becomes a normed vector space:

$$\left(\mathcal{L}(X, Y), \|\cdot\|_{\mathcal{L}(X,Y)}\right).$$

The topology induced by this norm is called the *uniform operator topology.*

Convergence in this topology means convergence uniformly on the unit ball. More precisely, a sequence $A_n \in \mathcal{L}(X, Y)$ converges to $A \in \mathcal{L}(X, Y)$ if and only if

$$\|A_n - A\|_{\mathcal{L}(X,Y)} \to 0,$$

or equivalently,

$$\sup_{\|x\|_X \leq 1} \|(A_n - A)x\|_Y \to 0.$$

Remark 7.1.3. Many authors use the notation $\mathcal{B}(X, Y)$ instead of $\mathcal{L}(X, Y)$ for the space of bounded linear operators.

Theorem 7.1.1 (Boundedness and continuity of linear maps)**.** Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be real normed vector spaces, and let $A : X \to Y$ be linear operator. Then the following statements are equivalent:

(i) A is bounded.

(ii) A is continuous on X.

(iii) A is continuous at 0.

Proof. We prove the implications

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)} \Rightarrow \text{(i)}.$$

First assume A is bounded. Then there exists a constant $C \geq 0$ such that

$$\|Ax\|_Y \leq C\|x\|_X \quad \text{for every } x \in X.$$

Hence, for any $x, x' \in X$, linearity gives

$$Ax - Ax' = A(x - x').$$

Therefore,

$$\|Ax - Ax'\|_Y = \|A(x - x')\|_Y \leq C\|x - x'\|_X.$$

Thus, A is Lipschitz continuous. In particular, A is continuous on X. Hence, (i) implies (ii).

The implication (ii) $\Rightarrow$ (iii) is immediate.

It remains to prove that continuity at 0 implies boundedness. Assume that A is continuous at 0. Taking $\varepsilon = 1$, there exists $\delta > 0$ s.t.

$$\|u\|_X < \delta \Rightarrow \|Au\|_Y < 1.$$

We shall show that this local estimate near 0 implies a global linear estimate for A.

Let $x \in X$. If $x = 0$, then $\|Ax\|_Y = 0$, so there is nothing to prove. Suppose $x \neq 0$. Define

$$u := \frac{\delta}{2\|x\|_X} x.$$

Then

$$\|u\|_X = \frac{\delta}{2\|x\|_X} \|x\|_X = \frac{\delta}{2} < \delta.$$

By the choice of δ, we have $\|Au\|_Y < 1$. Using linearity of A, we also have

$$Au = A\left(\frac{\delta}{2\|x\|_X} x\right) = \frac{\delta}{2\|x\|_X} Ax.$$

Hence

$$\frac{\delta}{2\|x\|_X} \|Ax\|_Y = \|Au\|_Y < 1.$$

Therefore

$$\|Ax\|_Y < \frac{2}{\delta} \|x\|_X.$$

Thus, A is bounded, and we're done. ■

Recall that the *kernel* and *image* of a linear operator $A : X \to Y$ are defined by

$$\ker(A) := \{x \in X : Ax = 0\} \subseteq X,$$

and

$$\mathrm{Im}(A) := \{Ax : x \in X\} \subseteq Y.$$

If X, Y are normed vector spaces and $A : X \to Y$ is a bounded linear operator, then $\ker(A)$ is a closed linear subspace of X since

$$\ker(A) = A^{-1}(\{0\})$$

while A is bounded implies A is continuous and $\{0\}$ is closed. However, the image of a bounded linear operator need not be closed in Y.

Since boundedness, continuity, closedness, convergence, and completeness are all defined using the norm, it is natural to ask when two different norms on the same vector space give essentially the same analytic and topological structure. This leads to the following definition.

Definition 7.1.2 (Equivalent norms)**.** Let X be a real vector space. Two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are called *equivalent* if there exists a constant $c \geq 1$ s.t.

$$\frac{1}{c}\|x\| \leq \|x\|' \leq c\|x\| \quad \text{for all } x \in X.$$

Remark 7.1.4. Equivalence of norms means that the two norms measure in size in the same way up to fixed multiplicative constants. In particular, equivalent norms define the same convergent sequences, the same Cauchy sequences, and the same open sets.

Exercise 7.1.1. Let X be a real vector space.

(i) Show that equivalence of norms defines an equivalence relation on the set of all norms on X.

(ii) Show that two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are equivalent if and only if the identity maps

$$\mathrm{id} : (X, \|\cdot\|) \to (X, \|\cdot\|')$$

and

$$\mathrm{id} : (X, \|\cdot\|') \to (X, \|\cdot\|)$$

are bounded linear operators.

(iii) Show that two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are equivalent if and only if they induce the same topology on X:

$$\mathcal{U}(X, \|\cdot\|) = \mathcal{U}(X, \|\cdot\|').$$

(iv) If $\|\cdot\|$ and $\|\cdot\|'$ are equivalent norms on X, show that $(X, \|\cdot\|)$ is complete if and only if $(X, \|\cdot\|')$ is complete.

The fundamental structural fact about finite-dimensional normed vector spaces is that the choice of norm is essentially irrelevant: any two norms define the same notion of convergence, the same Cauchy sequences, and the same topology. This is one of the main reasons finite-dimensional analysis is much simpler than infinite-dimensional functional analysis.

Theorem 7.1.2 (Equivalence of norms in finite dimensions). Let X be a finite-dimensional real vector space. Then for any two norms on X are equivalent.

Proof. It is enough to prove that every norm on X is equivalent to one fixed norm. We shall compare an arbitrary norm on X with the Euclidean norm coming from a choice of basis.

Let $e_1, \dots, e_n$ be a basis of X. Then every $x \in X$ can be written uniquely as

$$x = \sum_{i=1}^{n} x_i e_i, \quad x_i \in \mathbb{R}.$$

Using these coordinates, define

$$\|x\|_2 := \left(\sum_{i=1}^{n} |x_i|^2 \right)^{\frac{1}{2}}.$$

This is the Euclidean coordinate norm on X.

Let $\|\cdot\|$ be any norm on X. We will show that there exist constants $\delta > 0$ and $C > 0$ s.t.

$$\delta \|x\|_x \le \|x\| \le C\|x\|_2 \quad \text{for all } x \in X.$$

This will prove that $\|\cdot\|$ and $\|\cdot\|_2$ are equivalent.

Step 1. The norm $\|\cdot\|$ is bounded above by a multiple of $\|\cdot\|_2$. Define

$$C := \left(\sum_{i=1}^{n} \|e_i\|^2 \right)^{\frac{1}{2}}.$$

For $x = \sum_{i=1}^{n} x_i e_i$, the triangle inequality gives

$$\|x\| = \left\| \sum_{i=1}^{n} x_i e_i \right\| \le \sum_{i=1}^{n} |x_i| \|e_i\|.$$

By the Cauchy-Schwarz inequality,

$$\sum_{i=1}^{n} |x_i| \|e_i\| \leq \left(\sum_{i=1}^{n} |x_i|^2 \right)^{\frac{1}{2}} \left(\sum_{i=1}^{n} \|e_i\|^2 \right)^{\frac{1}{2}}.$$

Therefore

$$\|x\| \leq C\|x\|_2 \quad \text{for all } x \in X.$$

Step 2. The norm $\|\cdot\|$ is bounded below by a positive multiple of $\|\cdot\|_2$. Consider the Euclidean unit sphere

$$S := \{x \in X : \|x\|_2 = 1\}.$$

Under the coordinate identification

$$x = \sum_{i=1}^{n} x_i e_i \leftrightarrow (x_1, \ldots, x_n) \in \mathbb{R}^n,$$

the set S corresponds to the usual unit sphere in $\mathbb{R}^n$. Therefore S is compact by the Heine-Borel theorem w.r.t. the Euclidean coordinate norm.

Now define a map $f : (X, \|\cdot\|_2) \to \mathbb{R}$ by $f(x) = \|x\|$. By Step 1, the function f is continuous with respect to the Euclidean coordiante norm $\|\cdot\|_2$. Indeed, for $x, y \in X$,

$$|f(x) - f(y)| = |\|x\| - \|y\|| \leq \|x - y\| \leq C\|x - y\|_2.$$

Hence f is continuous on the compact set S. It therefore attains its minimum on S. Thus there exists $x_0 \in S$ s.t. $\|x_0\| = \min_{x \in S} \|x\|$. Since $x_0 \in S$, we have $x_0 \neq 0$. Hence, because $\|\cdot\|$ is a norm, $\|x_0\| > 0$. Set $\delta := \|x_0\| > 0$. Then $\|x\| \geq \delta$ for every $x \in S$. Now let $x \in X$ be arbitrary. If $x = 0$, then

$$\delta\|x\|_2 \leq \|x\|.$$

If $x \neq 0$, then

$$\frac{x}{\|x\|_2} \in S.$$

Therefore

$$\left\| \frac{x}{\|x\|_2} \right\| \geq \delta.$$

Multiplying by $\|x\|_2$, we obtain $\|x\| \geq \delta\|x\|_2$.

Now we have

$$\delta\|x\|_2 \leq \|x\| \leq C\|x\|_2,$$

and we're done. ■

Remark 7.1.5. The essential finite-dimensional ingredient in the proof is the compactness of the Euclidean unit sphere. This compactness allows a continuous norm to attain a positive minimum on the unit sphere, which gives the lower bound needed for equivalence of norms. In infinite-dimensional normed spaces, the unit sphere is not compact in general, and norms on the same vector space need not be equivalent.

Corollary 7.1.1. Every finite-dimensional normed vector space is complete.

Proof. Let $(X, \|\cdot\|)$ be a finite-dimensional normed vector space. Choose a basis $e_1, \ldots, e_n$ of X. Then every $x \in X$ can be written uniquely as $x = \sum_{i=1}^{n} x_i e_i$, where $x_i \in \mathbb{R}$. Define the Euclidean coordiante norm by

$$\|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{\frac{1}{2}}.$$

With this norm, X is naturally identified with $\mathbb{R}^n$, and hence $(X, \|\cdot\|_2)$ is complete.

By the equivalence of norms in finite dimensions, the given norm $\|\cdot\|$ is equivalent to $\|\cdot\|_2$. Thus there exists a constant $C \geq 1$ such that

$$\frac{1}{C}\|x\|_2 \leq \|x\| \leq C\|x\|_2 \quad \text{for all } x \in X.$$

It follows that a seuqnece is Cauchy with respect to $\|\cdot\|$ if and only if it is Cauchy with respect to $\|\cdot\|_2$. Since $(X, \|\cdot\|_2)$ is complete, every $\|\cdot\|$-Cauchy sequence converges in X. Therefore $(X, \|\cdot\|)$ is complete. ■

Corollary 7.1.2. Let X be a normed vector space. Then every finite-dimensional linear subspace of X is closed in X.

Proof. Let $Y \subseteq X$ be a finite-dimensional linear subspace. We equip Y with the norm inherited from X, namely $\|y\|_Y := \|y\|_X$ for all $y \in Y$. By the previous corollary, the finite-dimensional normed vector space $(Y, \|\cdot\|_Y)$ is complete.

We now show that Y is closed in X. Let (y_k) be a sequence in Y such that $y_k \to x$ in X. Then (y_k) is a Cauchy sequence in X. Since the norm on Y is the restriction of the norm on X, it is also a Cauchy sequence in Y. Because Y is complete, there exists $y \in Y$ s.t. $y_k \to y$ in Y. Equivalently, $\|y_k - y\|_X \to 0$, so $y_k \to y$ in X. But limits in a normed vector space are unique. Since $y_n \to x$ in X and $y_n \to y$ in X, we must have $x = y$.

Now since $x = y \in Y$, so we know Y is closed in X. ■

Corollary 7.1.3 (Heine-Borel theorem in finite dimensions)**.** Let X be a finite-dimensional normed vector space, and let $K \subseteq X$. Then K is compact if and only if K is closed and bounded.

Proof. Choose a basis $e_1, \ldots, e_n$ of X. Then every $x \in X$ can be written uniquely as $x = \sum_{i=1}^n x_i e_i$, where $x_i \in \mathbb{R}$. Now we can define the Euclidean coordinate norm:

$$\|x\|_2 = \left(\sum_{i=1}^n |x_i|^2\right)^{\frac{1}{2}}.$$

By the equivalence of norms in finite dimensions, the given norm on X is equivalent to $\|\cdot\|_2$. Hence, the two norms induce the same topology on X. In particuler, a subset of X is closed for one norm if and only if it is closed for the other, and a subset is compact for one norm if and only if it is compact for the other. Also, boundedness is the same for equivalent norms.

Therefore, the statement reduces to the classical Heine-Borel theorem in $\mathbb{R}^n$, which says that a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. Hence, $K \subseteq X$ is compact if and only if K is closed and bounded. ■

Corollary 7.1.4. Let $(X, \|X\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed vector spaces. If $\dim X < \infty$, then every linear operator $A : X \to Y$ is bounded. Equivalently, every linear operator defined on a finite-dimensional normed vector space is continuous.

Proof. Let $A : X \to Y$ be linear. We define a new norm on X by

$$\|x\|_A := \|x\|_X + \|Ax\|_Y, \quad x \in X.$$

Claim 7.1.1. $\|\cdot\|_A$ is a norm.

Since X is finite dimensional, all norms on X are equivalent. Therefore $\|\cdot\|_A$ is equivalent to the given norm $\|\cdot\|_X$. In particular, there exists a constant $C \geq 1$ s.t.

$$\|x\|_A \leq C\|x\|_X \quad \text{for all } x \in X.$$

Hence,

$$\|x\|_X + \|Ax\|_Y = \|x\|_A \leq C\|x\|_X,$$

and thus

$$\|Ax\|_Y \le (C-1)\|x\|_X.$$

This shows A is bounded. ■

The preceding corollaries depend essentially on finite-dimensionality. The following examples illustrate what can fail in infinite-dimensional normed spaces.

Example 7.1.1 (Two natural norms need not be equivalent). Let $X := C([0,1])$ be the vector space of real-valued continuous functions on $[0,1]$. Define two norms on X by

$$\|f\|_\infty := \sup_{0\le t\le 1} |f(t)|$$

and

$$\|f\|_2 := \left(\int_0^1 |f(t)|^2 \,\mathrm{d}t\right)^{\frac{1}{2}}.$$

Both are norms on $C([0,1])$. However, they are not equivalent.

Proof. Indeed, the space $C([0,1])$ is complete with respect to $\|\cdot\|_\infty$, but it is not complete with respect to $\|\cdot\|_2$. For example, one may approximate the discontinuous function $\chi_{[\frac{1}{2},1]}$ in the L^2-norm by continuous functions $\{f_n\}$:

$$f_n(t) := \begin{cases} 0, & \text{if } t \le \frac{1}{2} - \frac{1}{n}; \\ n\left(t - (\frac{1}{2} - \frac{1}{n})\right), & \text{if } \frac{1}{2} - \frac{1}{n} \le t \le \frac{1}{2}; \\ 1, & \text{if } t \ge \frac{1}{2}. \end{cases}$$

Such a sequence is Cauchy in $\|\cdot\|_2$, but it cannot converge in $C([0,1])$ with respect to $\|\cdot\|_2$ since $\chi_{[\frac{1}{2},1]}$ is discontinuous, because its natural L^2-limit is not continuous.

If the two norms $\|\cdot\|_\infty$ and $\|\cdot\|_2$ were equivalent, then completeness with respect to one would imply completeness with respect to the other. Therefore the two norms are not equivalent. ⊛

Lecture 23

May 21.

Example 7.1.2 (A dense proper subspace need not be closed). The space $C^1([0,1])$ of continuously differentiable functions on $[0,1]$ is a linear subspace of $C([0,1])$. With respect to the supremum norm, $C^1([0,1])$ is dense in $C([0,1])$. For instance, every continuous function on $[0,1]$ can be uniformly approximated by polynomials, and polynomials belong to $C^1([0,1])$.

However, $C^1([0,1]) \ne C([0,1])$. For example, the function

$$f(t) := \left|t - \frac{1}{2}\right|$$

is continuous on $[0,1]$, but it is not differentiable at $t = \frac{1}{2}$. Hence, $C^1([0,1])$ is a dense proper subspace of $C([0,1])$. Now since we know a sequence of $C^1([0,1])$ does not converge in $C^1([0,1])$, $C^1([0,1])$ is not closed in

$$(C([0,1]), \|\cdot\|_\infty).$$

This shows that, in infitnite-dimensional normed spaces, linear subspaces need not be closed.

Example 7.1.3 (Closed and bounded need not imply compact). Consider the Banach space

$$(C([0,1]), \|\cdot\|_\infty),$$

and let

$$B := \{f \in C([0,1]) : \|f\|_\infty \le 1\}.$$

Then B is closed and bounded. However, B is not compact.

To see this, define

$$f_n(t) := \sin(2^n \pi t), \quad 0 \le t \le 1.$$

Then $\|f_n\|_\infty \le 1$, so $f_n \in B$ for every n.

We claim that the family $\{f_n : n \in \mathbb{N}\}$ is not equicontinuous. Indeed, take $\varepsilon_0 = \frac{1}{2}$. Let $\delta > 0$ be arbitrary. Since $2^{-(n+1)} \to 0$, we may choose n sufficiently large so that

$$\frac{1}{2^{n+1}} < \delta.$$

Set

$$s_n = 0, \quad t_n = \frac{1}{2^{n+1}}.$$

Then

$$|s_n - t_n| = \frac{1}{2^{n+1}} < \delta.$$

However,

$$f_n(s_n) = f_n(0) = 0,$$

whereas

$$f_n(t_n) = \sin\left(2^n \pi \cdot \frac{1}{2^{n+1}}\right) = \sin\left(\frac{\pi}{2}\right) = 1.$$

Therefore

$$|f_n(t_n) - f_n(s_n)| = 1 > \frac{1}{2} = \varepsilon_0.$$

Thus no single $\delta > 0$ can make all the functions f_n uniformly continuous with the same modules of continuity. Hence $\{f_n\}$ is not equicontinuous.

Since $\{f_n : n \in \mathbb{N}\} \subseteq B$, it follows that B itself is not equicontinuous. By Corollary 5.3.1, B is not compact.

Example 7.1.4 (A linear functional need not be continuous)**.** Let $(X, \|\cdot\|)$ be an infinite-dimensional normed vector space over $\mathbb{R}$. We will construct a linear functional on X which is not continuous.

Choose a basis E of X. By rescaling each basis vector if necessary, we may assume that $\|e\| = 1$ for every $e \in E$. This means that every non-zero vector $x \in X$ can be written uniquely in the form

$$x = \sum_{i=1}^{\ell} x_i e_i,$$

where $e_1, \dots, e_\ell \in E$ are distinct basis vectors and $x_i \in \mathbb{R} \setminus \{0\}$.

Since X is infinite-dimensional, the basis E must be infinite.

Because E is infinite, we may choose a sequence of distinct basis vectors $e_1, e_2, e_3, \dots \in E$. Define a function $\lambda : E \to \mathbb{R}$ by setting

$$\lambda(e_n) = n \quad \text{for } n = 1, 2, 3, \dots,$$

and, for all other basis vectors $e \in E \setminus \{e_1, e_2, e_3, \dots\}$, set $\lambda(e) = 0$. Then λ is unbounded on E. This means that there is no constant $M \ge 0$ s.t.

$$|\lambda(e)| \le M \quad \text{for every } e \in E.$$

Indeed, if such an M existed, then for every positive integer n we would have

$$n = |\lambda(e_n)| \le M,$$

which is impossible for all n. Equivalently, for every $M > 0$, one can choose $n > M$ and then $|\lambda(e_n)| = n > M$. Thus, the values of λ on the basis vectors become arbitrarily large.

We now use λ to define a linear functional on all of X. If

$$x = \sum_{i=1}^{\ell} x_i e_i$$

is the unique expansion of x with respect to the Hamel basis E, define

$$\Phi_\lambda(x) := \sum_{i=1}^{\ell} \lambda(e_i) x_i.$$

This definition is well-defined because the representation of x as a finite linear combination of basis vectors is unique.

Moreover, Φ_λ is linear. Indeed, the map is defined by prescribing its value on each basis vector,

$$\Phi_\lambda(e) = \lambda(e), \quad e \in E,$$

and then extending linearly to finite linear combinations of basis vectors. Therefore, for all $x, y \in X$ and all $a, b \in \mathbb{R}$,

$$\Phi_\lambda(ax + by) = a\Phi_\lambda(x) + b\Phi_\lambda(y).$$

Now we claim that Φ_λ is not bounded. Recall that a linear functional $\Phi : X \to \mathbb{R}$ is bounded if there exists a constant $C \geq 0$ s.t.

$$|\Phi(x)| \leq C\|x\| \quad \text{for every } x \in X.$$

In particular, if Φ_λ were bounded, then for every basis vector $e \in E$ we would have

$$|\lambda(e)| = |\Phi_\lambda(e)| \leq C\|e\| = C$$

because $\|e\| = 1$, which is impossible.

Finally, for linear maps between normed vector spaces, boundedness is equivalent to continuity. Hence Φ_λ is a linear functional on X which is not continuous.

We now return to the compactness characterization of finite-dimensional normed spaces. Let $(X, \|\cdot\|)$ be a normed vector space, and denote its closed unit ball and unit sphere by

$$B := \{x \in X : \|x\| \leq 1\}, \quad S := \{x \in X : \|x\| = 1\}.$$

In finite dimensions, the Heine-Borel theorem tells us B is compact. The result below shows that this property is not merely a consequence of finite-dimensionality, but actually characterizes it.

Lemma 7.1.1 (Riesz's lemma)**.** Let $(X, \|\cdot\|)$ be a normed vector space, and let $Y \subseteq X$ be a proper closed linear subspace. Then for every $0 < \delta < 1$, there exists $x \in X$ s.t. $\|x\| = 1$ and

$$\inf_{y \in Y} \|x - y\| \geq 1 - \delta.$$

Equivalently,

$$\operatorname{dis}(x, Y) \geq 1 - \delta.$$

Proof. Since $Y \neq X$, we may choose a point $x_0 \in X \setminus Y$. Geometrically, x_0 is a point outside the closed subspace Y.

Because Y is closed and $x_0 \notin Y$, the distance from x_0 to Y is strictly positive. Set

$$d := \operatorname{dis}(x_0, Y) = \inf_{y \in Y} \|x_0 - y\|.$$

Then $d > 0$. The number d meausres how far x_0 is from the subspace Y.

Although there may not be a point of Y which realizes this distance, we can choose a point

$y_0 \in Y$ such that
$$\|x_0 - y_0\| \le \frac{d}{1-\delta},$$
which is because $0 < \delta < 1$.

Now consider the vector from $y_0 \in Y$ to x_0: $x_0 - y_0$. We normalize this vector and define
$$x := \frac{x_0 - y_0}{\|x_0 - y_0\|}.$$
Then immediately $\|x\| = 1$. Thus x is the unit vector pointing in the direction from the subspace Y toward the exterior point x_0.

We claim that this unit vector remains at distance at least $1 - \delta$ fro the whole subspace Y. To prove this, let $y \in Y$ be arbitrary. We want to estimate $\|x - y\|$ from below.

Using the definition of x, we have
$$x - y = \frac{x_0 - y_0}{\|x_0 - y_0\|} - y.$$
Hence,
$$\begin{aligned}\|x - y\| &= \frac{1}{\|x_0 - y_0\|}\|x_0 - y_0 - \|x_0 - y_0\|y\| \\ &= \frac{1}{\|x_0 - y_0\|}\left\|x_0 - (y_0 + \|x_0 - y_0\|y)\right\|.\end{aligned}$$
Now since $y_0 + \|x_0 - y_0\|y \in Y$, we have
$$\|x_0 - (y_0 + \|x_0 - y_0\|y)\| \ge \inf_{y \in Y}\|x_0 - y\| = d.$$
Hence,
$$\|x - y\| \ge \frac{d}{\|x_0 - y_0\|}.$$
Now since
$$\|x_0 - y_0\| \le \frac{d}{1-\delta},$$
we have
$$\|x - y\| \ge \frac{d}{\|x_0 - y_0\|} \ge 1 - \delta.$$
Since y is arbitrarily chosen, we know
$$\inf_{y \in Y}\|x - y\| \ge 1 - \delta.$$
∎

We now apply Riesz's lemma to the compactness problem. If X is infinite-dimensional, then after choosing finitely many unit vectors $x_1, \ldots, x_k$, their span is still a proper finite-dimensional subspace of X. Since finite-dimensional subspaces are closed, Riesz's lemma allows us to choose another unit vector x_{k+1} that stays a fixed positive distance away from this span. Iterating this construction produces infinitely many unit vectors separated from one another by a uniform distance. Such a sequence cannot have a convergent subsequence, and therefore the unit sphere cannot be compact.

This gives the desired converse to the finite-dimensional Heine-Borel theorem.

Theorem 7.1.3 (Compactness of the unit ball)**.** Let $(X, \|\cdot\|)$ be a normed vector space. Then the following are equivalent:

(i) $\dim X < \infty$.

(ii) The closed unit ball B is compact.

(iii) The unit sphere S is comapct.

Proof. We prove the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

First assumte that $\dim X < \infty$. By the finite-dimensional Heine-Borel theorem, a subset of X is compact if and only if it is closed and bounded. The closed unit ball

$$B = \{x \in X : \|x\| \leq 1\}$$

is closed and bounded. Therefore B is comapct. This proves (i) $\Rightarrow$ (ii).

Now assume that B is compact. The unit sphere

$$S = \{x \in X : \|x\| = 1\}$$

is closed in X. Indeed, the norm map

$$\|\cdot\| : X \to \mathbb{R}, \quad x \mapsto \|x\|$$

is continuous, and $S = \|\cdot\|^{-1}(\{1\})$. Since $\{1\}$ is closed in $\mathbb{R}$, it follows that S is closed in X. Since $S \subseteq B$, the sphere S is a closed subset of the compact set B. Hence, S is compact. Thus, (ii) $\Rightarrow$ (iii).

It remains to prove (iii) $\Rightarrow$ (i). We prove the contrapositive. Suppose X is infinite-dimensional. We will show that S is not compact.

The idea is to construct a sequence of unit vectors that stay uniformly separated from one another. More precisely, we shall construct a sequence $(x_k)_{k=1}^{\infty} \subseteq S$ such that $\|x_i - x_j\| \geq \frac{1}{2}$ whenever $i \neq j$. Such a sequence cannot have a convergent subsequence, because every convergent sequence is Cauchy. Therefore, the existence of such a sequence implies that S is not compact.

Choose $x_1 \in S$. Suppose that $x_1, \ldots, x_k \in S$ have already been chosen so that $\|x_i - x_j\| \geq \frac{1}{2}$ for $i \neq j$. Let $Y_k := \operatorname{span}\{x_1, \ldots, x_k\}$. Then Y_k is finite-dimensional, hence closed in X. Since X is infinite-dimensional, $Y_k \neq X$. Thus Y_k is a proper closed linear subspace of X. By Riesz's lemma, applied to Y_k, there exists $x_{k+1} \in X$ such that $\|x_{k+1}\| = 1$ and $\inf_{y \in Y_k} \|x_{k+1} - y\| \geq \frac{1}{2}$. Since $x_1, \ldots, x_k \in Y_k$, we have

$$\|x_{k+1} - x_i\| \geq \frac{1}{2} \quad \text{for } i = 1, \ldots, k.$$

Thus the induction continues.

Consequently, we obtain a sequence $(x_k) \subseteq S$ such that

$$\|x_i - x_j\| \geq \frac{1}{2} \quad \forall i \neq j,$$

and we're done. ■

7.2 The Dual Space

Let X and Y be normed vector spaces. Recall that $\mathcal{L}(X, Y)$ denotes the vector space of all bounded linear operators $A : X \to Y$, equipped with the operator norm

$$\|A\|_{\mathcal{L}(X,Y)} := \sup_{\|x\|_X \leq 1} \|Ax\|_Y.$$

Equivalently,

$$\|Ax\|_Y \leq \|A\|_{\mathcal{L}(X,Y)} \|x\|_X \quad \text{for all } x \in X.$$

The next theorem shows that $\mathcal{L}(X, Y)$ is complete whenever the target space Y is complete. Notice that no completeness assumption is needed on X.

Theorem 7.2.1. Let X be a normed vector space and let Y be a Banach space. Then $\mathcal{L}(X, Y)$ is a Banach space with respect to the operator norm.

Proof. Let $(A_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{L}(X, Y)$. Thus, for every $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ s.t.

$$m, n \geq n_0 \Rightarrow \|A_m - A_n\|_{\mathcal{L}(X,Y)} < \varepsilon.$$

For each fixed $x \in X$, we have

$$\|A_n x - A_m x\|_Y = \|(A_n - A_m)x\|_Y \le \|A_n - A_m\|_{\mathcal{L}(X,Y)} \|x\|_X.$$

Hence $(A_n x)$ is a Cauchy sequence in Y. Since Y is complete, the limit

$$Ax := \lim_{n\to\infty} A_n x$$

exists for every $x \in X$. This defines a map $A : X \to Y$.

We first show that A is linear. Let $x, x' \in X$ and $\alpha, \beta \in \mathbb{R}$. Then

$$A(\alpha x+\beta x') = \lim_{n\to\infty} A_n(\alpha x+\beta x') = \lim_{n\to\infty} (\alpha A_n x + \beta A_n x') = \alpha \lim_{n\to\infty} A_n x+\beta \lim_{n\to\infty} A_n x' = \alpha Ax+\beta Ay.$$

Thus, A is linear.

It remains to show that A is bounded and that $A_n \to A$ in operator norm. Fix $\varepsilon > 0$. Since (A_n) is Cauchy in operator norm, choose $n_0 \in \mathbb{N}$ s.t.

$$m, n \ge n_0 \Rightarrow \|A_m - A_n\|_{\mathcal{L}(X,Y)} < \varepsilon.$$

Fix $n \ge n_0$. For every $x \in X$, we have

$$\|Ax - A_n x\|_Y = \lim_{m\to\infty} \|A_m x - A_n x\|_Y \le \limsup_{m\to\infty} \|A_m - A_n\|_{\mathcal{L}(X,Y)} \|x\|_X \le \varepsilon \|x\|_X.$$

Therefore

$$\|(A - A_n)x\|_Y \le \varepsilon \|x\|_X \quad \text{for all } x \in X.$$

This implies $A - A_n \in \mathcal{L}(X, Y)$ and

$$\|A - A_n\|_{\mathcal{L}(X,Y)} \le \varepsilon \quad \text{for every } n \ge n_0.$$

In particular, taking $n = n_0$, we get

$$\|Ax\|_Y \le \|Ax - A_{n_0} x\|_Y + \|A_{n_0} x\|_Y \le \left(\varepsilon + \|A_{n_0}\|_{\mathcal{L}(X,Y)}\right) \|x\|_X.$$

Hence A is bounded.

Moreover,

$$\|A - A_n\|_{\mathcal{L}(X,Y)} \le \varepsilon \quad \text{for all } n \ge n_0.$$

Thus $A_n \to A$ in $\mathcal{L}(X, Y)$.

Since every Cauchy sequence in $\mathcal{L}(X, Y)$ converges in $\mathcal{L}(X, Y)$, $\mathcal{L}(X, Y)$ is a Banach space. ■

A particularly important case is when the target space is $\mathbb{R}$. The dual space of a normed vector space X is defined by

$$X^* := \mathcal{L}(X, \mathbb{R}).$$

Thus X^* is the space of all bounded linear functionals $\Lambda : X \to \mathbb{R}$. By the preceding theorem, X^* is always a Banach space, whether or not X itself is complete.

Definition 7.2.1 (Dual space)**.** Let X be a normed vector space. The dual space of X is $X^* := \mathcal{L}(X, \mathbb{R})$. Elements of X^* are called bounded linear functionals or continuous linear functionals on X.

Example 7.2.1 (Dual space of $L^p(\mu)$)**.** Let $(M, \mathcal{A}, \mu)$ be a measure space, and let $1 < p < \infty$. Let q be the conjugate exponent of p, that is,

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Equivalently,

$$q = \frac{p}{p-1}, \quad q - 1 = \frac{1}{p-1}, \quad (q-1)p = q.$$

Holder's inequality says that if $f \in L^p(\mu)$ and $g \in L^q(\mu)$, then $fg \in L^1(\mu)$ and

$$\left| \int_M fg \, \mathrm{d}\mu \right| \le \int_M |f||g| \, \mathrm{d}\mu \le \|f\|_p \|g\|_q. \tag{7.2}$$

Consequently, every function $g \in L^q(\mu)$ defines a linear functional

$$\Lambda_g : L^p(\mu) \to \mathbb{R}$$

by

$$\Lambda_g(f) := \int_M fg \, \mathrm{d}\mu, \quad f \in L^p(\mu).$$

The functional Λ_g is well-defined because $fg \in L^1(\mu)$. It is linear by the linearity of the integral. Moreover, by Equation 7.2,

$$|\Lambda_g(f)| \le \|f\|_p \|g\|_q \quad \text{for every } f \in L^p(\mu).$$

Therefore, Λ_g is bounded and

$$\|\Lambda_g\|_{\mathcal{L}(L^p(\mu),\mathbb{R})} \le \|g\|_q. \tag{7.3}$$

In fact, the estimate is sharp. More precisely, one has

$$\|\Lambda_g\|_{\mathcal{L}(L^p(\mu),\mathbb{R})} = \|g\|_q.$$

We now prove the reverse inequality. If $g = 0$, then $\Lambda_g = 0$, and hence the equality is immediate. Thus assume $g \ne 0$. Define

$$f := \frac{\operatorname{sgn}(g)|g|^{q-1}}{\|g\|_q^{q-1}},$$

where, in the real-valued case,

$$\operatorname{sgn}(g)(x) = \begin{cases} 1, & \text{if } g(x) > 0; \\ -1, & \text{if } g(x) < 0; \\ 0, & \text{if } g(x) = 0. \end{cases}$$

Thus

$$\operatorname{sgn}(g)g = |g|, \quad \text{and} \quad |f| = \frac{|g|^{q-1}}{\|g\|_q^{q-1}}.$$

Since $(q-1)p = q$, we have

$$|f|^p = \frac{|g|^{(q-1)p}}{\|g\|_q^{(q-1)p}} = \frac{|g|^q}{\|g\|_q^q}.$$

Because $g \in L^q(\mu)$, this shows that $f \in L^p(\mu)$. Moreover,

$$\|f\|_p^p = \int_M |f|^p \, \mathrm{d}\mu = \frac{1}{\|g\|_q^q} \int_M |g|^q \, \mathrm{d}\mu = 1.$$

Hence, $\|f\|_p = 1$. Now we evaluate Λ_g at this particular function f. Using $\operatorname{sgn}(g)g = |g|$, we get

$$\Lambda_g(f) = \int_M \frac{\operatorname{sgn}(g)|g|^{q-1}}{\|g\|_q^{q-1}} g \, \mathrm{d}\mu = \frac{1}{\|g\|_q^{q-1}} \int_M |g|^q \, \mathrm{d}\mu = \frac{\|g\|_q^q}{\|g\|_q^{q-1}} = \|g\|_q.$$

Since $\|f\|_p = 1$, it follows from the definition of the operator norm that

$$\|\Lambda_g\|_{\mathcal{L}(L^p(\mu),\mathbb{R})} \ge |\Lambda_g(f)| = \|g\|_q.$$

Combining this lower bound with Equation 7.3, we obtain

$$\|\Lambda_g\|_{\mathcal{L}(L^p(\mu),\mathbb{R})} = \|g\|_q.$$

7.3 Hilbert Space

In the following, we introduce some elementary Hilbert space theory. The main result is the Riesz representation theorem, which shows that every Hilbert space is naturally isometrically isomorphic to its dual space.

Definition 7.3.1 (Innder Product)**.** Let H be a real vector space. A bilinear map $H \times H \to \mathbb{R}$, $(x, y) \mapsto \langle x, y\rangle$, is called an inner product if it satisfies the following two properties:

1. **Symmetry:** $\langle x, y\rangle = \langle y, x\rangle$ for all $x, y \in H$.
2. **Positive definiteness:** $\langle x, x\rangle \geq 0$ for all $x \in H$, and $\langle x, x\rangle = 0$ iff $x = 0$.

Remark 7.3.1. The inner product induces a norm on H by

$$\|x\| := \sqrt{\langle x, x\rangle}, \quad x \in H.$$

This norm is called the norm associated with the inner product.

Lemma 7.3.1 (Cauchy-Schwarz inequality)**.** Let H be a real vector space equipped with an inner product and the norm. Then the inner product and norm satisfy the Cauchy-Schwarz inequality

$$|\langle x, y\rangle| \leq \|x\|\|y\|$$

and the triangle inequality

$$\|x + y\| \leq \|x\| + \|y\|$$

for all $x, y \in H$. Thus, the norm induced by the inner product is actually a norm.

Proof. The Cauchy-Schwarz inequality is obvious when $x = 0$ or $y = 0$. Hence assume that $x, y \neq 0$. Define

$$\xi := \|x\|^{-1}x, \quad \eta := \|y\|^{-1}y.$$

Then $\|\xi\| = \|\eta\| = 1$. Hence

$$0 \leq \|\eta - \langle \xi, \eta\rangle\xi\|^2 = \langle \eta, \eta\rangle - \langle \xi, \eta\rangle^2 = 1 - \langle \xi, \eta\rangle^2.$$

This implies

$$|\langle \xi, \eta\rangle| \leq 1.$$

Therefore

$$|\langle x, y\rangle| \leq \|x\|\|y\|.$$

It follows that

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Hence, we have the triangle equality

$$\|x + y\| \leq \|x\| + \|y\|.$$

■

Definition 7.3.2 (Hilbert space)**.** Let H be a real vector space equipped with an inner product

$$\langle \cdot, \cdot\rangle : H \times H \to \mathbb{R}.$$

The pair $(H, \langle \cdot, \cdot\rangle)$ is called a Hilbert space if H is complete with respect to the norm induced by the inner product, namely

$$\|x\| = \sqrt{\langle x, x\rangle}.$$

Equivalently, H is a Hilbert space if every Cauchy sequence $\{x_n\}_{n=1}^{\infty}$ in H, with respect to the metric

$$d(x, y) := \|x - y\|,$$

converges to an element of H.

Example 7.3.1 (Dual space of a Hilbert space). Let H be a real Hilbert space. Thus H is a Banach space whose norm is induced by an inner product

$$\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}, \quad \|x\| = \sqrt{\langle x, x \rangle}.$$

Every element $y \in H$ determines a linear functional $\Lambda_y : H \to \mathbb{R}$ by

$$\Lambda_y(x) := \langle x, y \rangle, \quad x \in H.$$

This functional is bounded by the Cauchy-Schwarz inequality:

$$|\Lambda_y(x)| = |\langle x, y \rangle| \leq \|x\|\|y\|.$$

Hence,

$$\|\Lambda_y\|_{H^*} \leq \|y\|.$$

In fact, taking $x = y$ when $y \neq 0$, or $x = \frac{y}{\|y\|}$, shows that

$$\|\Lambda_y\|_{H^*} = \|y\|.$$

The Riesz representation theorem states that every bounded linear functional on H is of this form. Therefore the map $H \to H^*$, $y \mapsto \Lambda_y$, is an isometric isomorphism.

Example 7.3.2. Let $(M, \mathcal{A}, \mu)$ be a measure space. Then $H := L^2(\mu)$ is a Hilbert space. The inner product is induced by the bilinear map

$$\mathcal{L}^2(\mu) \times \mathcal{L}^2(\mu) \to \mathbb{R}, \quad (f, g) \mapsto \langle f, g \rangle := \int_M f g \, \mathrm{d}\mu.$$

It is well-defined because the product of two L^2-functions $f, g : M \to \mathbb{R}$ is integrable by the Cauchy-Schwarz inequality. That it is bilinear and symmetric follows directly from the properties of Lebesgue integral.

Theorem 7.3.1 (Riesz representation theorem). Let H be a Hilbert space and let $\Lambda : H \to \mathbb{R}$ be a bounded linear functional. Then there exists a unique element $y \in H$ such that

$$\Lambda(x) = \langle y, x \rangle \quad \text{for all } x \in H.$$

This element $y \in H$ satisfies

$$\|y\| = \sup_{0 \neq x \in H} \frac{|\langle y, x \rangle|}{\|x\|} = \|\Lambda\|.$$

Thus the map

$$H \to H^*, \quad y \mapsto \langle y, \cdot \rangle$$

is an isometry of normed vector spaces.

Remark 7.3.2. Riesz's representation theorem shows $H \simeq H^*$ for any Hilbert space H.

Definition 7.3.3 (Convex set). Let V be a real vector space. A subset $C \subseteq V$ is called convex if for every $x, y \in C$ and every $t \in [0,1]$, one has

$$tx + (1-t)y \in C.$$

Equivalently, C contains the line segment joining any two of its points.

Theorem 7.3.2 (Projection theorem)**.** Let H be a Hilbert space and let $E \subseteq H$ be a nonempty closed convex subset. Then there exists a unique elements $x_0 \in E$ s.t.

$$\|x_0\| \le \|x\| \quad \text{for all } x \in E.$$

Proof. Define

$$\delta := \inf\{\|x\| : x \in E\}.$$

We first prove uniqueness. Suppose that $x_0, x_1 \in E$ both satisfy $\|x_0\| = \|x_1\| = \delta$. Since E is convex,

$$\frac{x_0 + x_1}{2} \in E.$$

Therefore,

$$\left\|\frac{x_0 + x_1}{2}\right\| \ge \delta.$$

Using the parallelogram identity, we get

$$\|x_0 - x_1\|^2 = 2\|x_0\|^2 + 2\|x_1\|^2 - \|x_0 + x_1\|^2 \le 4\delta^2 - 4\delta^2 = 0.$$

Hence, $x_0 = x_1$.

Now we prove the existence. Choose a sequence $(x_i) \subseteq E$ s.t. $\lim_{i\to\infty}\|x_i\| = \delta$. We claim that (x_i) is a Cauchy sequence.

Fix $\varepsilon > 0$, $\exists i_0 \in \mathbb{N}$ s.t.

$$i \ge i_0 \Rightarrow \|x_i\|^2 < \delta^2 + \frac{\varepsilon^2}{4}.$$

Let $i, j \in \mathbb{N}$ be such that $i \ge i_0, j \ge i_0$. Since E convex, $\frac{x_i + x_j}{2} \in E$. Hence,

$$\left\|\frac{x_i + x_j}{2}\right\| \ge \delta.$$

Using parallelogram identity again,

$$\|x_i - x_j\|^2 = 2\|x_i\|^2 + 2\|x_j\|^2 - \|x_i + x_j\|^2 < 4\left(\delta^2 + \frac{\varepsilon^2}{4}\right) - 4\delta^2 = \varepsilon^2.$$

Thus, $\|x_i - x_j\| < \varepsilon$ for all $i, j \ge i_0$ and thus (x_i) is a Cauchy sequence.

Now since H is complete, the limit $x_0 := \lim_{i\to\infty} x_i$ exists in H. Moreover, $x_0 \in E$ because E is closed. Finally, by continuity of the norm, $\|x_0\| = \delta$. Hence,

$$\|x_0\| \le \|x\| \quad \text{for all } x \in E.$$

■

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Now we proved the Riesz's representation theorem: Let H be a Hilbert space and let $\Lambda : H \to \mathbb{R}$ be a bounded linear functional. Then there exists a unique element $y \in H$ such that May 26.

$$\Lambda(x) = \langle y, x\rangle \quad \text{for all } x \in H.$$

This element $y \in H$ satisfies

$$\|y\| = \sup_{0 \ne x \in H} \frac{|\langle y, x\rangle|}{\|x\|} = \|\Lambda\|.$$

Thus the map

$$H \to H^*, \quad y \mapsto \langle y, \cdot \rangle$$

is an isometry of normed vector spaces.

Proof. We prove the existence first. If $\Lambda = 0$, then we can pick $y = 0$, and this is true.

Hence, we assume that $\Lambda \neq 0$ and define

$$E \coloneqq \{x \in H : \Lambda(x) = 1\}.$$

Then $E \neq \varnothing$. Indeed, since Λ is not the zero functional, there exists $\xi \in H$ such that $\Lambda(\xi) \neq 0$. Hence, if we set

$$x \coloneqq \Lambda(\xi)^{-1}\xi,$$

then by linearity of Λ,

$$\Lambda(x) = \Lambda\left(\Lambda(\xi)^{-1}\xi\right) = \Lambda(\xi)^{-1}\Lambda(\xi) = 1.$$

Thus, $x \in E$, and thus $E \neq \varnothing$.

The set E is closed because it is the inverse image of the closed set $\{1\} \subseteq \mathbb{R}$ under the continuous map $\Lambda : H \to \mathbb{R}$.

Finally, E is convex. To see this, let $x, y \in E$ and let $t \in [0,1]$. Since $x, y \in E$, we have $\Lambda(x) = 1$ and $\Lambda(y) = 1$, so by the linearity of Λ,

$$\Lambda\left(tx + (1-t)y\right) = t\Lambda(x) + (1-t)\Lambda(y) = 1.$$

This shows Λ is convex.

Now by Theorem 7.3.2, we know there exists an element $x_0 \in E$ s.t.

$$\|x_0\| \leq \|x\| \quad \text{for all } x \in E.$$

We prove that $x \in H$ and $\Lambda(x) = 0$ implies $\langle x_0, x \rangle = 0$.

To see this, fix an element $x \in H$ s.t. $\Lambda(x) = 0$. Then $x_0 + tx \in E$ for all $t \in \mathbb{R}$. This implies

$$\|x_0\|^2 \leq \|x_0 + tx\|^2 = \|x_0\|^2 + 2t\langle x_0, x \rangle + t^2\|x\|^2 \quad \text{for all } t \in \mathbb{R}.$$

Thus the differentiable function

$$t \mapsto \|x_0 + tx\|^2$$

attains its minimum at $t = 0$. Therefore its derivative vanishes at $t = 0$, and hence

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \|x_0 + tx\|^2 = 2\langle x_0, x \rangle.$$

This shows $\langle x_0, x \rangle = 0$.

Now define

$$y \coloneqq \frac{x_0}{\|x_0\|^2}.$$

Fix an element $x \in H$ and set $\lambda \coloneqq \Lambda(x)$. Then

$$\Lambda(x - \lambda x_0) = \Lambda(x) - \lambda = 0.$$

Hence, it follows that

$$0 = \langle x_0, x - \lambda x_0 \rangle = \langle x_0, x \rangle - \lambda\|x_0\|^2.$$

Therefore

$$\langle y, x \rangle = \frac{\langle x_0, x \rangle}{\|x_0\|^2} = \lambda = \Lambda(x).$$

Now we show

$$\|y\| = \sup_{0 \neq x \in H} \frac{|\langle y, x \rangle|}{\|x\|} = \|\Lambda\|.$$

If $y = 0$, then $\Lambda = 0$, and so

$$\|y\| = 0 = \|\Lambda\|.$$

Hence assume that $y \neq 0$. Then

$$\|y\| = \frac{\|y\|^2}{\|y\|} = \frac{\Lambda(y)}{\|y\|} \leq \sup_{0 \neq x \in H} \frac{|\Lambda(x)|}{\|x\|} = \|\Lambda\|.$$

Conversely, by the Cauchy-Schwarz inequality,

$$|\Lambda(x)| = |\langle y, x\rangle| \leq \|y\|\|x\| \quad \text{for all } x \in H.$$

Hence, $\|\Lambda\| \leq \|y\|$, and thus we have

$$\|y\| = \|\Lambda\|.$$

Now we prove the uniqueness. Assume that $y, z \in H$ both satisfy

$$\langle y, x\rangle = \langle z, x\rangle = \Lambda(x) \quad \text{for all } x \in H.$$

Then

$$\langle y - z, x\rangle = 0 \quad \text{for all } x \in H.$$

Taking $x := y - z$, we obtain

$$\|y - z\|^2 = \langle y - z, y - z\rangle = 0.$$

Hence, $y = z$, and we're done. ■

7.4 Banach Algebras

We begin the discussion with a result about convergent series in a Banach space. It extends the basic assertion in first-year analysis that every absolutely convergent series of real numbers converges. We will use the following lemma to study power series in a Banach algebra.

Lemma 7.4.1 (Convergent series). Let $(X, \|\cdot\|)$ be a Banach space and let $(x_i)_{i\in\mathbb{N}}$ be a sequence in X s.t.

$$\sum_{i=1}^{\infty} \|x_i\| < \infty.$$

Then the sequence

$$\xi_n := \sum_{i=1}^{n} x_i$$

converges in X. Its limit is denoted by

$$\sum_{i=1}^{\infty} x_i := \lim_{n\to\infty} \sum_{i=1}^{n} x_i.$$

Proof. Define

$$s_n := \sum_{i=1}^{n} \|x_i\|, \quad n \in \mathbb{N}.$$

This sequence is nondecreasing and converges by assumption. Moreover, for every pair of integers $n > m \geq 1$, we have

$$\|\xi_n - \xi_m\| = \left\| \sum_{i=m+1}^{n} x_i \right\| \leq \sum_{i=m+1}^{n} \|x_i\| = s_n - s_m.$$

Hence, (ξ_n) is a Cauchy sequence in X. Since X is complete, this sequence converges. ■

Definition 7.4.1 (Banach algebra). A real, respectively complex, Banach algebra is a pair consisting of a real, respectively complex, Banach space $(\mathcal{A}, \|\cdot\|)$ and a bilinear map

$$\mathcal{A} \times \mathcal{A} \to \mathcal{A}, \quad (a, b) \mapsto ab,$$

called the product, such that the product is associative, that is,

$$(ab)c = a(bc) \quad \text{for all } a, b, c \in \mathcal{A},$$

and satisfies

$$\|ab\| \leq \|a\|\|b\| \quad \text{for all } a, b \in \mathcal{A}.$$

Remark 7.4.1. A Banach algebra $\mathcal{A}$ is called commutative if

$$ab = ba \quad \text{for all } a, b \in \mathcal{A}.$$

It is called unital if there exists an element

$$1 \in \mathcal{A} \setminus \{0\}$$

such that

$$1a = a1 = a \quad \text{for all } a \in \mathcal{A}.$$

The unit 1, if it exists, is uniquely determined by the product.

An element $a \in \mathcal{A}$ of a unital Banach algebra $\mathcal{A}$ is called invertible if there exists an element $b \in \mathcal{A}$ such that

$$ab = ba = 1.$$

The element b, if it exists, is uniquely determined by a. It is called the inverse of a, and is denoted by $a^{-1} := b$. The invertible elements form a group $\mathcal{G} \subseteq \mathcal{A}$.

Example 7.4.1. The following are standard examples of Banach algebras.

1. Let X be a Banach space. The archetypal example of a Banach algebra is the space

$$\mathcal{L}(X) := \mathcal{L}(X, X)$$

of bounded linear operators from X to itself, equipped with the operator norm

$$\|T\|_{\text{op}} := \sup_{\|x\|_X \leq 1} \|Tx\|_X.$$

The product is given by composition:

$$ST := S \circ T, \quad S, T \in \mathcal{L}(X).$$

We now explain why this is a Banach algebra. First, $\mathcal{L}(X)$ is a Banach space. Indeed, since X is complete, the space of bounded linear maps from X to X, equipped with the operator norm, is complete. More explicitly, if (T_n) is a Cauchy sequence in $\mathcal{L}(X)$, then for each $x \in X$,

$$\|T_n x - T_m x\|_X \leq \|T_n - T_m\|_{\text{op}} \|x\|_X.$$

Thus, $(T_n x)$ is a Cauchy sequence in X. Since X is complete, the limit

$$Tx := \lim_{n \to \infty} T_n x$$

exists for each $x \in X$. One checks from the linearity of the T_n's that T is linear. Moreover, since (T_n) is Cauchy in operator norm, the sequence $(\|T_n\|_{\text{op}})$ is bounded; say

$$\|T_n\|_{\text{op}} \leq M \quad \text{for all } n.$$

Therefore

$$\|Tx\|_X = \lim_{n \to \infty} \|T_n x\|_X \leq M\|x\|_X,$$

so $T \in \mathcal{L}(X)$. Finally, one shows that $T_n \to T$ in operator norm. Given $\varepsilon > 0$, choose N s.t.

$$\|T_n - T_m\|_{\text{op}} < \varepsilon \quad \text{whenever } n, m \geq N.$$

Then for every $x \in X$ with $\|x\|_X \leq 1$,

$$\|T_n x - Tx\|_X = \lim_{m \to \infty} \|T_n x - T_m x\|_X \leq \varepsilon.$$

Taking the supremum over all $\|x\|_X \leq 1$, we obtain

$$\|T_n - T\|_{\text{op}} \leq \varepsilon \quad \text{for all } n \geq N.$$

Hence, $\mathcal{L}(X)$ is complete. Next, we verify the algebra structure. If $S, T \in \mathcal{L}(X)$, then $ST = S \circ T$ is again a bounded linear operator. Indeed, ST is linear, and for every $x \in X$,

$$\|STx\|_X = \|S(Tx)\|_X \leq \|S\|_{\text{op}} \|Tx\|_X \leq \|S\|_{\text{op}} \|T\|_{\text{op}} \|x\|_X.$$

Hence $ST \in \mathcal{L}(X)$, and

$$\|ST\|_{\text{op}} \leq \|S\|_{\text{op}} \|T\|_{\text{op}}.$$

This is precisely the submultiplicative estimate required in the definition of a Banach algebra. Note that the product is associative, which is trivial. The product is also bilinear, which is also trivial. Therefore $\mathcal{L}(X)$, equipped with the operator norm and composition, is a Banach algebra.

If $X \neq \{0\}$, then $\mathcal{L}(X)$ is unital. Its unit is the identity operator

$$I_X : X \to X, \quad I_X x = x.$$

This Banach algebra is usually noncommutative. In general,

$$ST \neq TS.$$

For example, on $X = \mathbb{R}^2$, let

$$S = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$ST = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad TS = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $ST \neq TS$. Finally, the invertible elements of $\mathcal{L}(X)$ are precisely the bijective bounded linear operators from X to itself. Also, for $T \in \mathcal{L}(X)$, the inverse of T, say T^{-1}, is also linear and bounded, but the boundedness is untrivial: since X is a Banach space and $T : X \to X$ is a bijective bounded linear operator, the Open Mapping Theorem implies that T^{-1} is also bounded.

Therefore, the group of invertible elements of $\mathcal{L}(X)$ is exactly the group of bounded linear isomorphisms of X.

2. An example of commutative unital Banach algebra is the space of real-valued bounded continuous functions on a nonempty topological space, equipped with the supremum norm and pointwise multiplication.

Let $\mathcal{A}$ be a real Banach algebra and let

$$f(z) = \sum_{n=0}^{\infty} c_n z^n$$

be a power series with real coefficients $c_n \in \mathbb{R}$ and convergence radius

$$\rho := \frac{1}{\limsup_{n \to \infty} |c_n|^{\frac{1}{n}}} > 0.$$

Choose an element $a \in \mathcal{A}$ with $\|a\| < \rho$. Then the sequence $(c_n a^n)_{n\in\mathbb{N}}$ satisfies

$$\sum_{n=0}^{\infty} \|c_n a^n\| \leq |c_0|\|1\| + \sum_{n=1}^{\infty} |c_n|\|a\|^n < \infty.$$

Therefore, by Lemma 7.4.1, the sequence

$$\xi_n \coloneqq \sum_{i=0}^{n} c_i a^i$$

converges. Denote the limit by

$$f(a) \coloneqq \sum_{n=0}^{\infty} c_n a^n, \quad a \in \mathcal{A}, \|a\| < \rho. \tag{7.4}$$

When the power series f has complex coefficients, the definition extends to complex Banach algebras.

Exercise 7.4.1. The map

$$f : \{a \in \mathcal{A} : \|a\| < \rho\} \to \mathcal{A}$$

defined by Equation 7.4 is continuous.

Hint. For $n \in \mathbb{N}$, define

$$f_n : \mathcal{A} \to \mathcal{A}, \quad f_n(a) \coloneqq \sum_{i=0}^{n} c_i a^i.$$

Prove that f_n is continuous. Then prove that the sequence (f_n) converges uniformly to f on the set

$$\{a \in \mathcal{A} : \|a\| \leq r\}$$

for every $r < \rho$.

Theorem 7.4.1 (Inverse)**.** Let $\mathcal{A}$ be a real unital Banach algebra

1. For every $a \in \mathcal{A}$, the limit

$$r_a \coloneqq \lim_{n\to\infty} \|a^n\|^{\frac{1}{n}} = \inf_{n\in\mathbb{N}} \|a^n\|^{\frac{1}{n}} \leq \|a\|$$

exists. It is called the spectral radius of a.

2. If $a \in \mathcal{A}$ satisfies $r_a < 1$, then the element $1 - a$ is invertible and

$$(1-a)^{-1} = \sum_{n=0}^{\infty} a^n.$$

3. The group $\mathcal{G} \subseteq \mathcal{A}$ of invertible elements is an open subset of $\mathcal{A}$, and the map

$$\mathcal{G} \to \mathcal{G}, \quad a \mapsto a^{-1},$$

is continuous. More precisely, if $a \in \mathcal{G}$ and $b \in \mathcal{A}$ satisfy

$$\|a - b\|\|a^{-1}\| < 1,$$

then $b \in \mathcal{G}$,

$$b^{-1} = \sum_{n=0}^{\infty} \left(1 - a^{-1}b\right)^n a^{-1},$$

and

$$\|b^{-1} - a^{-1}\| \leq \frac{\|a-b\|\|a^{-1}\|^2}{1 - \|a-b\|\|a^{-1}\|}, \quad \|b^{-1}\| \leq \frac{\|a^{-1}\|}{1 - \|a-b\|\|a^{-1}\|}.$$

Proof. We prove the three assertions in order.

1. Fix $a \in \mathcal{A}$. Since the norm in a Banach algebra is submultiplicative, we have

$$\|a^{m+n}\| \leq \|a^m\|\|a^n\| \quad \text{for all } m, n \in \mathbb{N}.$$

Thus the sequence $\alpha_n := \|a^n\|$ is submultiplicative. We shall show that

$$\lim_{n\to\infty} \alpha_n^{\frac{1}{n}} = \inf_{n\in\mathbb{N}} \alpha_n^{\frac{1}{n}}.$$

It is immediate that $0 \leq r \leq \|a\|$ since the term $n = 1$ gives $r \leq \|a\|$.
We claim that

$$\lim_{n\to\infty} \|a^n\|^{\frac{1}{n}} = r.$$

Let $\varepsilon > 0$. By the definition of the infimum, we may choose $m \in \mathbb{N}$ s.t.

$$\|a^m\|^{\frac{1}{m}} < r + \varepsilon.$$

Every sufficiently large integer n can be written uniquely in the form

$$n = qm + \ell, \quad q \in \mathbb{N} \cup \{0\}, \quad 0 \leq \ell \leq m - 1.$$

Using submultiplicativity, we have

$$\|a^n\| = \|a^{qm+\ell}\| \leq \|a^m\|^q\|a^\ell\| \leq (r + \varepsilon)^{mq}\|a^\ell\|.$$

Taking the n-th root gvies

$$\|a^n\|^{\frac{1}{n}} \leq (r + \varepsilon)^{\frac{mq}{n}}\|a^\ell\|^{\frac{1}{n}} = (r + \varepsilon)^{1-\frac{\ell}{n}}\|a^\ell\|^{\frac{1}{n}}.$$

Because ℓ can take only finitely many values $0, 1, \dots, m - 1$, the numbers $\|a^\ell\|$ are bounded. More precisely, set

$$C := \max_{0\leq\ell\leq m-1}\|a^\ell\|.$$

Since $0 \leq \ell \leq m - 1$, we have

$$1 - \frac{m}{n} \leq 1 - \frac{\ell}{n} \leq 1.$$

Therefore

$$(r + \varepsilon)^{1-\frac{\ell}{n}} \leq \max\left\{r + \varepsilon, (r + \varepsilon)^{1-\frac{m}{n}}\right\}.$$

Thus

$$\|a^n\|^{\frac{1}{n}} \leq \max\left\{(r + \varepsilon)C^{\frac{1}{n}}, (r + \varepsilon)^{1-\frac{m}{n}}C^{\frac{1}{n}}\right\}.$$

Both terms on the right-hand side converge to $r + \varepsilon$. Hence,

$$\limsup_{n\to\infty}\|a^n\|^{\frac{1}{n}} \leq r + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that

$$\limsup_{n\to\infty}\|a^n\|^{\frac{1}{n}} \leq r.$$

On the other hand, by the definition of r,

$$r \leq \|a^n\|^{\frac{1}{n}} \quad \text{for every } n \in \mathbb{N}.$$

Hence

$$r \leq \liminf_{n\to\infty}\|a^n\|^{\frac{1}{n}}.$$

Combining the two inequalities, we obtain

$$\lim_{n\to\infty} \|a^n\|^{\frac{1}{n}} = r = \inf_{n\in\mathbb{N}} \|a^n\|^{\frac{1}{n}}.$$

This proves the existence of r_a nad the formula

$$r_a = \inf_{n\in\mathbb{N}} \|a^n\|^{\frac{1}{n}} \leq \|a\|.$$

2. Assume now that $r_a < 1$. Choose a number α such that $r_a < \alpha < 1$. By the definition of r_a, there exists $N \in \mathbb{N}$ s.t.

$$\|a^n\|^{\frac{1}{n}} \leq \alpha \quad \text{for all } n \geq N.$$

Equivalently,

$$\|a^n\| \leq \alpha^n \quad \text{for all } n \geq N.$$

Therefore

$$\sum_{n=0}^{\infty} \|a^n\| < \infty.$$

Since $\mathcal{A}$ is a Banach space, by Lemma 7.4.1, $\sum_{n=0}^{\infty} a^n$ converges in $\mathcal{A}$. Denote its sum by

$$b \coloneqq \sum_{n=0}^{\infty} a^n.$$

We claim that $b = (1-a)^{-1}$. Let $b_N \coloneqq \sum_{n=0}^{N} a^n$. Then the usual finite geometric identity gives

$$(1-a)b_N = b_N(1-a) = 1 - a^{N+1}.$$

Since $\|a^{N+1}\| \to 0$, we may pass to the limit and obtain

$$(1-a)b = b(1-a) = 1.$$

This shows $b = (1-a)^{-1}$, and we're done.

3. Let $a \in \mathcal{G}$ and $b \in \mathcal{A}$ satisfy

$$\|a-b\|\|a^{-1}\| < 1.$$

We write

$$b = a - a\left(a^{-1}(a-b)\right) = a\left(1 - a^{-1}(a-b)\right).$$

Set $h \coloneqq a^{-1}(a-b)$. Then

$$\|h\| \leq \|a^{-1}\|\|a-b\| < 1.$$

By a. and 2., we know $1-h$ is invertible, and

$$(1-h)^{-1} = \sum_{n=0}^{\infty} h^n.$$

Since $b = a(1-h)$, it follows that b is invertible, with

$$b^{-1} = (1-h)^{-1}a^{-1}.$$

Therefore,

$$b^{-1} = \sum_{n=0}^{\infty} \left(a^{-1}(a-b)\right)^n a^{-1}. \tag{7.5}$$

Equivalently, since

$$a^{-1}(a-b) = 1 - a^{-1}b,$$

we may write

$$b^{-1} = \sum_{n=0}^{\infty} \left(1 - a^{-1}b\right)^n a^{-1}.$$

This already shows that every b satisfying

$$\|a - b\| < \frac{1}{\|a^{-1}\|}$$

is invertible. Hence

$$B\left(a, \frac{1}{\|a^{-1}\|}\right) \subseteq \mathcal{G},$$

and therefore $\mathcal{G}$ is open in $\mathcal{A}$.

It remains to estimate b^{-1} and $b^{-1} - a^{-1}$. From Equation 7.5, we have

$$\|b^{-1}\| \leq \sum_{n=0}^{\infty} \|h\|^n \|a^{-1}\| = \frac{\|a^{-1}\|}{1 - \|h\|}.$$

Since

$$\|h\| \leq \|a^{-1}\| \|a - b\|,$$

we obtain

$$\|b^{-1}\| \leq \frac{\|a^{-1}\|}{1 - \|a - b\| \|a^{-1}\|}.$$

Similarly,

$$b^{-1} - a^{-1} = \left(\sum_{n=0}^{\infty} h^n - a\right) a^{-1} = \sum_{n=1}^{\infty} h^n a^{-1}.$$

Hence,

$$\|b^{-1} - a^{-1}\| \leq \sum_{n=1}^{\infty} \|h\|^n \|a^{-1}\| = \frac{\|h\| \|a^{-1}\|}{1 - \|h\|}.$$

Using

$$\|h\| \leq \|a^{-1}\| \|a - b\|,$$

we have

$$\|b^{-1} - a^{-1}\| \leq \frac{\|a - b\| \|a^{-1}\|^2}{1 - \|a - b\| \|a^{-1}\|}.$$

This shows given any $a \in \mathcal{A}$, for any $b \in \mathcal{A}$ close enough to $a \in \mathcal{G}$, $b \in \mathcal{G}$ and $\|b^{-1} - a^{-1}\|$ can be arbitrarily small, so the map $\mathcal{G} \to \mathcal{G}$, $a \mapsto a^{-1}$ is continuous.

∎

Definition 7.4.2 (Invertible Operator)**.** Let X and Y be Banach spaces. A bounded linear operator $A : X \to Y$ is called invertible if there exists a bounded linear operator $B : Y \to X$ such that

$$BA = 1_X, \quad AB = 1_Y.$$

The operator B is uniquely determined by A and is denoted by $B =: A^{-1}$. It is called the inverse of A.

Remark 7.4.2. When $X = Y$, the space of invertible bounded linear operators in $\mathcal{L}(X)$ is denoted by

$$\mathrm{Aut}(X) := \{A \in \mathcal{L}(X) : \text{there exists } B \in \mathcal{L}(X) \text{ such that } AB = BA = 1\}.$$

The spectral radius of a bounded linear operator $A \in \mathcal{L}(X)$ is the real number $r_A \geq 0$ defined by

$$r_A := \lim_{n\to\infty} \|A^n\|^{\frac{1}{n}} = \inf_{n\in\mathbb{N}} \|A^n\|^{\frac{1}{n}} \leq \|A\|.$$

Corollary 7.4.1 (Spectral radius)**.** Let X and Y be Banach spaces. Then the following hold.

1. If $A \in \mathcal{L}(X)$ has spectral radius $r_A < 1$, then

$$1_X - A \in \mathrm{Aut}(X), \quad (1_X - A)^{-1} = \sum_{n=0}^{\infty} A^n.$$

2. $\mathrm{Aut}(X)$ is an open subset of $\mathcal{L}(X)$ with respect to the norm topology, and the map

$$\mathrm{Aut}(X) \to \mathrm{Aut}(X), \quad A \mapsto A^{-1},$$

is continuous.

3. Let $A, P \in \mathcal{L}(X, Y)$ be bounded linear operators. Assume A is invertible and

$$\|P\|\|A^{-1}\| < 1.$$

Then $A - P$ is invertible,

$$(A - P)^{-1} = \sum_{n=0}^{\infty} (A^{-1}P)^n A^{-1},$$

and

$$\left\|(A-P)^{-1} - A^{-1}\right\| \leq \frac{\|P\|\|A^{-1}\|^2}{1 - \|P\|\|A^{-1}\|}.$$

Proof. Assertion 1. and 2. follow from Theorem 7.4.1 with $\mathcal{A} = \mathcal{L}(X)$.
To prove part 3., observe that

$$\|A^{-1}P\| \leq \|A^{-1}\|\|P\| < 1.$$

Hence it follows from part 1. that the operator $1_X - A^{-1}P$ is invertible and that its inverse is given by

$$\left(1_X - A^{-1}P\right)^{-1} = \sum_{k=0}^{\infty} (A^{-1}P)^k.$$

Multiplying this identity by A^{-1} on the right gives

$$\left(1_X - A^{-1}P\right)^{-1} A^{-1} = \sum_{k=0}^{\infty} (A^{-1}P)^k A^{-1},$$

and note that

$$\left(1_X - A^{-1}P\right)^{-1} A^{-1} = \left(A(1_X - A^{-1}P)\right)^{-1} = (A - P)^{-1}.$$

Hence,

$$(A - P)^{-1} = \sum_{k=0}^{\infty} (A^{-1}P)^k A^{-1}.$$

Finally,

$$\begin{aligned}\|(A-P)^{-1} - A^{-1}\| &= \left\|\left(\sum_{k=0}^{\infty} (A^{-1}P)^k - 1_X\right) A^{-1}\right\| \leq \|A^{-1}\| \left\|\left(\sum_{k=0}^{\infty} (A^{-1}P)^k - 1_X\right)\right\| \\ &= \|A^{-1}\| \left\|\sum_{k=1}^{\infty} (A^{-1}P)^k\right\| \leq \frac{\|P\|\|A^{-1}\|^2}{1 - \|P\|\|A^{-1}\|}.\end{aligned}$$

■

Chapter 8

Principles of Functional Analysis

Lecture 25

This chapter presents the most important general principles in functional analysis. These results explain why completeness is such a powerful assumption in the study of normed vector spaces. The first principle is the Uniform Boundedness Principle, also known as the Banach–Steinhaus Theorem. It says that a pointwise bounded family of bounded linear operators on a Banach space is automatically uniformly bounded in operator norm. The second principle is the Open Mapping Theorem. It says that a surjective bounded linear operator between Banach spaces maps open sets to open sets. Its most important consequence is the Inverse Operator Theorem: a bijective bounded linear operator between Banach spaces has a bounded inverse. An equivalent formulation is the Closed Graph Theorem, which says that a linear operator between Banach spaces is bounded if and only if its graph is closed. The third principle is the Hahn–Banach Theorem. It says that bounded linear functionals defined on a subspace of a normed vector space can be extended to the whole space without increasing their norm. This theorem is the main tool for producing linear functionals and separating points from closed subspaces. May 28.

8.1 The Baire Category Theorem

The Baire Category Theorem is the topological foundation behind the Uniform Boundedness Principle and the Open Mapping Theorem.

Definition 8.1.1 (Nowhere dense)**.** Let X be a metric space. A subset $E \subseteq X$ is called nowhere dense if

$$\operatorname{int}(\overline{E}) = \varnothing.$$

Theorem 8.1.1 (Baire Category Theorem)**.** Let X be a nonempty complete metric space. If

$$X = \bigcup_{n=1}^{\infty} F_n,$$

where each $F_n \subseteq X$ is closed, then at least one of the sets F_n has nonempty interior. Equivalently, a nonempty complete metric space cannot be written as a countable union of closed nowhere dense subsets.

Proof. We prove the contrapositive. Suppose that each F_n is closed and has empty interior. We shall prove that

$$X \neq \bigcup_{n=1}^{\infty} F_n.$$

In other words, we will construct a point $x \in X$ such that $x \notin F_n$ for every $n \in \mathbb{N}$.

The proof is based on the following idea. Since F_1 has empty interior, it cannot contain any open ball. Hence we can find a small closed ball avoiding F_1. Inside that ball, since F_2 also has empty interior, we can find a smaller closed ball avoiding F_2. Continuing in this way, we construct a nested sequence of closed balls whose radii tend to zero. Completeness will then guarantee that these balls contain a limiting point. By construction, that limiting point avoids every F_n.

We begin the construction. Since F_1 has empty interior, $\operatorname{int}(F_1) = \varnothing$. Thus F_1 contains no nonempty open ball. Since X is nonempty and X itself is open in the metric topology, it follows that $F_1 \neq X$. Hence there exists a point $y_1 \in X \setminus F_1$. Because F_1 is closed, its complement $X \setminus F_1$ is open. Therefore there exists $\omega_1 > 0$ such that $B_{\omega_1}(y_1) \subseteq X \setminus F_1$. Choose $r_1 > 0$ so small that $0 < r_1 < 1$ and $r_1 < \omega_1$. Then

$$\overline{B_{r_1}}(y_1) \subseteq B_{\omega_1}(y_1) \subseteq X \setminus F_1.$$

Renaming y_1 as x_1, we have found a closed ball

$$\overline{B_{r_1}}(x_1) \subseteq X \setminus F_1, \quad 0 < r_1 < 1.$$

We now proceed inductively. Suppose that, for some $n \geq 1$, we have already chosen a closed ball $\overline{B_{r_n}}(x_n)$. Since F_{n+1} has empty interior, it cannot contain the nonempty open ball $B_{r_n}(x_n)$. Therefore there exists a point $y_{n+1} \in B_{r_n}(x_n) \setminus F_{n+1}$. Because F_{n+1} is closed, the complement $X \setminus F_{n+1}$ is open. Hence there exists $\omega_{n+1} > 0$ such that $B_{\omega_{n+1}}(y_{n+1}) \subseteq X \setminus F_{n+1}$. Also, since $y_{n+1} \in B_{r_n}(x_n)$, the open ball $B_{r_n}(x_n)$ is a neighborhood of y_{n+1}. Thus, by choosing the radius sufficiently small, we may ensure that the next closed ball lies inside both $B_{r_n}(x_n)$ and $X \setminus F_{n+1}$. More precisely, choose $r_{n+1} > 0$ such that

$$0 < r_{n+1} < \frac{1}{n+1}$$

and $\overline{B_{r_{n+1}}}(y_{n+1}) \subseteq B_{r_n}(x_n) \cap (X \setminus F_{n+1})$. Renaming y_{n+1} as x_{n+1}, we obtain

$$\overline{B_{r_{n+1}}}(x_{n+1}) \subseteq B_{r_n}(x_n) \setminus F_{n+1},$$

with

$$0 < r_{n+1} < \frac{1}{n+1}. \tag{8.1}$$

By induction, we obtain a sequence of closed balls $\overline{B_{r_1}}(x_1), \overline{B_{r_2}}(x_2), \overline{B_{r_3}}(x_3), \ldots$ such that

$$\overline{B_{r_{n+1}}}(x_{n+1}) \subseteq \overline{B_{r_n}}(x_n) \quad \text{for every } n \in \mathbb{N}, \tag{8.2}$$

and

$$\overline{B_{r_n}}(x_n) \cap F_n = \varnothing \quad \text{for every } n \in \mathbb{N}. \tag{8.3}$$

Moreover, by Equation 8.1, $r_n \to 0$.

We claim that (x_n) is a Cauchy sequence. Let $m \geq n$. Since the balls are nested, Equation 8.2 implies that $x_m \in \overline{B_{r_n}}(x_n)$. Therefore $d(x_m, x_n) \leq r_n$. Since $r_n \to 0$, it follows that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $r_N < \varepsilon$. Hence, whenever $m \geq n \geq N$,

$$d(x_m, x_n) \leq r_n \leq r_N < \varepsilon.$$

Thus (x_n) is Cauchy. Since X is complete, there exists $x \in X$ such that $x_n \to x$.

We now show that this limit point avoids every F_n. Fix $n \in \mathbb{N}$. For every $m \geq n$, the nestedness condition gives $x_m \in \overline{B_{r_n}}(x_n)$. Since $\overline{B_{r_n}}(x_n)$ is closed and $x_m \to x$, we may pass to the limit and obtain $x \in \overline{B_{r_n}}(x_n)$. But by Equation 8.3, the ball $\overline{B_{r_n}}(x_n)$ is disjoint from F_n. Hence $x \notin F_n$. Since $n \in \mathbb{N}$ was arbitrary, we conclude that

$$x \notin \bigcup_{n=1}^{\infty} F_n.$$

Therefore $X \neq \bigcup_{n=1}^{\infty} F_n$. This proves the contrapositive. Consequently, if a nonempty complete metric space X is written as a countable union of closed sets, $X = \bigcup_{n=1}^{\infty} F_n$, then at least one of the sets F_n must have nonempty interior. ■

8.2 The Uniform Boundedness Principle

Let X be a set. A family $\{f_i\}_{i\in I}$ of functions $f_i : X \to Y_i$, where each Y_i is a normed vector space, is called **pointwise bounded** if

$$\sup_{i\in I}\|f_i(x)\|_{Y_i} < \infty \quad \text{for every } x \in X.$$

Theorem 8.2.1 (Uniform Boundedness Principle)**.** Let X be a Banach space, let I be an index set, and, for each $i \in I$, let Y_i be a normed vector space. Suppose $A_i : X \to Y_i$ is a bounded linear operator for every $i \in I$. If the family $\{A_i\}_{i\in I}$ is pointwise bounded, that is,

$$\sup_{i\in I}\|A_i x\|_{Y_i} < \infty \quad \text{for every } x \in X,$$

then

$$\sup_{i\in I}\|A_i\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$F_n := \left\{ x \in X : \sup_{i\in I}\|A_i x\|_{Y_i} \le n \right\}.$$

Each F_n is closed. Indeed, if $x_m \in F_n$ and $x_m \to x$, then for each $i \in I$,

$$\|A_i x\|_{Y_i} = \lim_{m\to\infty}\|A_i x_m\|_{Y_i} \le n,$$

and hence $x \in F_n$. By pointwise boundedness,

$$X = \bigcup_{n=1}^{\infty} F_n.$$

Since X is complete, the Baire Category Theorem implies that some F_N has nonempty interior. Hence there exist $x_0 \in X$ and $\varepsilon > 0$ such that $B_\varepsilon(x_0) \subseteq F_N$. Thus

$$\|A_i x\|_{Y_i} \le N \quad \text{for every } i \in I \text{ and every } x \in B_\varepsilon(x_0).$$

We now show that this local uniform bound implies a uniform bound on operator norms. Let $x \in X$ with $x \neq 0$. Then both points

$$x_0 + \frac{\varepsilon}{2}\frac{x}{\|x\|} \quad \text{and} \quad x_0 - \frac{\varepsilon}{2}\frac{x}{\|x\|}$$

belong to $B_\varepsilon(x_0)$. Hence, by the local uniform bound, for every $i \in I$,

$$\left\| A_i\left(x_0 + \frac{\varepsilon}{2}\frac{x}{\|x\|}\right)\right\|_{Y_i} \le N, \qquad \left\| A_i\left(x_0 - \frac{\varepsilon}{2}\frac{x}{\|x\|}\right)\right\|_{Y_i} \le N.$$

Using the linearity of A_i, we have

$$A_i\left(x_0 + \frac{\varepsilon}{2}\frac{x}{\|x\|}\right) - A_i\left(x_0 - \frac{\varepsilon}{2}\frac{x}{\|x\|}\right) = A_i\left(\frac{\varepsilon x}{\|x\|}\right) = \frac{\varepsilon}{\|x\|}A_i x.$$

Therefore, by the triangle inequality,

$$\frac{\varepsilon}{\|x\|}\|A_i x\|_{Y_i} \le 2N.$$

Equivalently, $\|A_i x\|_{Y_i} \le \frac{2N}{\varepsilon}\|x\|$. Thus $\|A_i\| \le \frac{2N}{\varepsilon}$ for every $i \in I$. Consequently, $\sup_{i\in I}\|A_i\| < \infty$. ∎

Corollary 8.2.1 (Pointwise limits of bounded operators). Let X be a Banach space and let Y be a normed vector space. Suppose $A_n : X \to Y$ is a sequence of bounded linear operators such that, for every $x \in X$, the limit

$$Ax := \lim_{n\to\infty} A_n x$$

exists in Y. Then $A : X \to Y$ is a bounded linear operator, and

$$\|A\| \leq \liminf_{n\to\infty} \|A_n\|.$$

Proof. For each $x \in X$, the sequence $(A_n x)$ converges, and hence is bounded. Therefore

$$\sup_{n\in\mathbb{N}} \|A_n x\|_Y < \infty$$

for every $x \in X$. By the Uniform Boundedness Principle, $\sup_{n\in\mathbb{N}} \|A_n\| < \infty$. The map A is linear because it is defined as a pointwise limit of linear maps. Moreover, for every $x \in X$,

$$\|Ax\|_Y = \lim_{n\to\infty} \|A_n x\|_Y = \liminf_{n\to\infty} \|A_n x\|_Y \leq \left(\liminf_{n\to\infty} \|A_n\|\right) \|x\|_X.$$

Taking the supremum over $\|x\|_X \leq 1$, we obtain $\|A\| \leq \liminf_{n\to\infty} \|A_n\|$. ∎

8.3 The Open Mapping Theorem

A map $T : X \to Y$ between topological spaces is called **open** if $T(U)$ is open in Y whenever U is open in X.

Theorem 8.3.1 (Open Mapping Theorem). Let X and Y be Banach spaces. If $T : X \to Y$ is a surjective bounded linear operator, then T is open.

We prove the theorem in two steps.

Lemma 8.3.1. Let X and Y be Banach spaces, and let $T : X \to Y$ be a surjective bounded linear operator. Let $B_X := \{x \in X : \|x\| < 1\}$. Then there exists $\delta > 0$ such that

$$B_Y(0, \delta) \subseteq \overline{T(B_X)}.$$

Proof. Since T is surjective, we have $Y = T(X)$. Also,

$$X = \bigcup_{n=1}^{\infty} nB_X,$$

because for every $x \in X$, one can choose $n > \|x\|_X$, and then $x \in nB_X$. Hence

$$Y = T(X) = \bigcup_{n=1}^{\infty} T(nB_X) = \bigcup_{n=1}^{\infty} nT(B_X),$$

where the last equality follows from the linearity of T. Therefore

$$Y = \bigcup_{n=1}^{\infty} \overline{nT(B_X)}.$$

Indeed, since $nT(B_X) \subseteq \overline{nT(B_X)}$, the inclusion $Y \subseteq \bigcup_{n=1}^{\infty} \overline{nT(B_X)}$ follows, while the reverse inclusion holds because each $\overline{nT(B_X)}$ is a subset of Y. Each set $\overline{nT(B_X)}$ is closed in Y. Since Y is a Banach space, it is a complete metric space. Hence, by the Baire Category Theorem, there exists $n_0 \in \mathbb{N}$ such that

$$\operatorname{int}\left(\overline{n_0 T(B_X)}\right) \neq \varnothing.$$

Since scalar multiplication by n_0 is a homeomorphism of Y, we have $\overline{n_0 T(B_X)} = n_0 \overline{T(B_X)}$. Thus $\overline{T(B_X)}$ also has nonempty interior.

Consequently, there exist $y_0 \in Y$ and $r > 0$ such that $B_Y(y_0, r) \subseteq \overline{T(B_X)}$. We now show that this implies that $\overline{T(B_X)}$ contains an open ball centered at the origin. Since B_X is convex and symmetric, the set $T(B_X)$ is also convex and symmetric. Therefore its closure $\overline{T(B_X)}$ is convex and symmetric.

Let $y \in Y$ satisfy $\|y\|_Y < r$. Then

$$y_0 + y \in B_Y(y_0, r) \quad \text{and} \quad y_0 - y \in B_Y(y_0, r).$$

Hence both $y_0 + y$ and $y_0 - y$ lie in $\overline{T(B_X)}$. By symmetry of $\overline{T(B_X)}$, $-(y_0 - y) = -y_0 + y \in \overline{T(B_X)}$. Since $\overline{T(B_X)}$ is convex,

$$y = \frac{1}{2}(y_0 + y) + \frac{1}{2}(-y_0 + y) \in \overline{T(B_X)}.$$

Thus $B_Y(0, r) \subseteq \overline{T(B_X)}$. Taking $\delta = r$ proves the lemma. ■

Lemma 8.3.2. Let X and Y be Banach spaces, and let $T : X \to Y$ be a bounded linear operator. Suppose that for some $\delta > 0$,

$$B_Y(0, \delta) \subseteq \overline{T(B_X)}.$$

Then

$$B_Y(0, \delta) \subseteq T(B_X).$$

Proof. Let $y \in Y$ satisfy $\|y\|_Y < \delta$. Choose a number σ such that $\|y\|_Y < \sigma < \delta$. We shall construct a sequence $(x_n)_{n=1}^{\infty}$ in X such that

$$\sum_{n=1}^{\infty} \|x_n\|_X < 1 \quad \text{and} \quad y = \sum_{n=1}^{\infty} T x_n.$$

Then $x := \sum_{n=1}^{\infty} x_n$ will belong to B_X, and by continuity of T, we will have $Tx = y$.

We use the assumption $B_Y(0, \delta) \subseteq \overline{T(B_X)}$ in the following scaled form. If $z \in Y$ and $a > 0$ satisfy $\|z\|_Y < a\delta$, then $\|z/a\|_Y < \delta$, so $z/a \in \overline{T(B_X)}$. Therefore, for every $\varepsilon > 0$, there exists $u \in B_X$ such that $\|z/a - Tu\|_Y < \varepsilon/a$. Setting $x := au$, we get $\|x\|_X < a$ and

$$\|z - Tx\|_Y = a\|z/a - Tu\|_Y < \varepsilon.$$

Thus we have proved the following useful fact:

$$\|z\|_Y < a\delta \Rightarrow \text{for every } \varepsilon > 0,\ \exists x \in X \text{ s.t. } \|x\|_X < a \text{ and } \|z - Tx\|_Y < \varepsilon.$$

We now begin the construction. Since $\|y\|_Y < \sigma = \delta \cdot \frac{\sigma}{\delta}$, we apply the above with $z = y$, $a = \frac{\sigma}{\delta}$, and $\varepsilon = \frac{\delta - \sigma}{2}$. Thus there exists $x_1 \in X$ such that

$$\|x_1\|_X < \frac{\sigma}{\delta} \quad \text{and} \quad \|y - Tx_1\|_Y < \frac{\delta - \sigma}{2}.$$

Define $r_1 := y - Tx_1$. Then $\|r_1\|_Y < \frac{\delta - \sigma}{2}$.

We now carry out the induction. Suppose that, for some $n \geq 1$, we have already chosen $x_1, \ldots, x_n \in X$ such that the remainder

$$r_n := y - \sum_{k=1}^{n} T x_k$$

satisfies $\|r_n\|_Y < \frac{\delta - \sigma}{2^n}$. Set $a_n := \frac{\delta - \sigma}{\delta \cdot 2^n}$. Then $a_n \delta = \frac{\delta - \sigma}{2^n}$, and by the induction hypothesis, $\|r_n\|_Y < a_n \delta$. Therefore we may apply the above with $z = r_n$, $a = a_n$, and $\varepsilon = \frac{\delta - \sigma}{2^{n+1}}$. Hence there exists $x_{n+1} \in X$ such that

$$\|x_{n+1}\|_X < \frac{\delta - \sigma}{\delta \cdot 2^n} \quad \text{and} \quad \left\| y - \sum_{k=1}^{n+1} Tx_k \right\|_Y < \frac{\delta - \sigma}{2^{n+1}}.$$

This completes the induction.

We now estimate the series $\sum x_n$. Using the above bounds,

$$\sum_{n=1}^{\infty} \|x_n\|_X = \|x_1\|_X + \sum_{n=1}^{\infty} \|x_{n+1}\|_X < \frac{\sigma}{\delta} + \sum_{n=1}^{\infty} \frac{\delta - \sigma}{\delta \cdot 2^n} = \frac{\sigma}{\delta} + \frac{\delta - \sigma}{\delta} = 1.$$

Thus the series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent in X. Since X is Banach, it converges to some element $x := \sum_{n=1}^{\infty} x_n \in X$. Moreover, $\|x\|_X \leq \sum_{n=1}^{\infty} \|x_n\|_X < 1$, hence $x \in B_X$. Since T is bounded, it is continuous. Therefore

$$Tx = T\left(\sum_{n=1}^{\infty} x_n\right) = \sum_{n=1}^{\infty} Tx_n.$$

On the other hand, the remainder estimate gives $\|y - \sum_{k=1}^{n} Tx_k\|_Y < \frac{\delta - \sigma}{2^n} \to 0$, hence $\sum_{n=1}^{\infty} Tx_n = y$. Consequently, $Tx = y$, and since $x \in B_X$, we have $y \in T(B_X)$. This proves $B_Y(0, \delta) \subseteq T(B_X)$. ∎

Now we can prove open mapping theorem.

Proof of Theorem 8.3.1. By Lemma 8.3.1, there exists $\delta > 0$ such that $B_Y(0, \delta) \subseteq \overline{T(B_X)}$. By Lemma 8.3.2, $B_Y(0, \delta) \subseteq T(B_X)$.

Let $U \subseteq X$ be open. We prove that $T(U)$ is open. Choose $y_0 \in T(U)$, say $y_0 = Tx_0$ with $x_0 \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_X(x_0, \varepsilon) \subseteq U$. We claim that $B_Y(y_0, \delta\varepsilon) \subseteq T(U)$. Indeed, if $\|y - y_0\|_Y < \delta\varepsilon$, then $\|\varepsilon^{-1}(y - y_0)\|_Y < \delta$. Hence there exists $u \in X$ such that $\|u\|_X < 1$ and $Tu = \varepsilon^{-1}(y - y_0)$. Therefore $y = y_0 + \varepsilon Tu = T(x_0 + \varepsilon u)$. Since $x_0 + \varepsilon u \in B_X(x_0, \varepsilon) \subseteq U$, we get $y \in T(U)$. Hence $T(U)$ is open. ∎

8.4 The Inverse Operator and Closed Graph Theorems

Theorem 8.4.1 (Inverse Operator Theorem)**.** Let X and Y be Banach spaces. If $T : X \to Y$ is a bijective bounded linear operator, then $T^{-1} : Y \to X$ is bounded.

Proof. Since T is surjective and bounded, the Open Mapping Theorem implies that T is open. Therefore, for every open set $U \subseteq X$,

$$(T^{-1})^{-1}(U) = T(U)$$

is open in Y. Hence T^{-1} is continuous. Since T^{-1} is linear, it is bounded. ∎

Definition 8.4.1 (Graph of a linear operator)**.** Let X and Y be normed vector spaces, and let $T : X \to Y$ be linear. The graph of T is

$$\operatorname{graph}(T) := \{(x, Tx) : x \in X\} \subseteq X \times Y.$$

We equip $X \times Y$ with the norm

$$\|(x, y)\|_{X \times Y} := \|x\|_X + \|y\|_Y.$$

If X and Y are Banach spaces, then $X \times Y$ is Banach.

Theorem 8.4.2 (Closed Graph Theorem)**.** Let X and Y be Banach spaces. Let $T : X \to Y$ be a linear operator. Then T is bounded if and only if $\operatorname{graph}(T)$ is closed in $X \times Y$.

Proof. First suppose that T is bounded. Let $(x_n, Tx_n) \to (x, y)$ in $X \times Y$. Then $x_n \to x$ in X, and $Tx_n \to y$ in Y. Since T is continuous, $Tx_n \to Tx$. Hence $y = Tx$, and so $(x, y) \in \mathrm{graph}(T)$. Thus the graph is closed.

Conversely, suppose that $\mathrm{graph}(T)$ is closed. Since $X \times Y$ is Banach, the closed subspace $\mathrm{graph}(T)$ is also Banach. Consider the projection

$$\pi_X : \mathrm{graph}(T) \to X, \quad \pi_X(x, Tx) = x.$$

This map is linear, bijective, and bounded because

$$\|\pi_X(x, Tx)\|_X = \|x\|_X \le \|x\|_X + \|Tx\|_Y.$$

By the Inverse Operator Theorem, π_X^{-1} is bounded. Hence there exists $C > 0$ such that

$$\|(x, Tx)\|_{X\times Y} = \|\pi_X^{-1} x\|_{X\times Y} \le C\|x\|_X.$$

Therefore $\|Tx\|_Y \le C\|x\|_X$ for every $x \in X$. Thus T is bounded. ■

Remark 8.4.1. The Closed Graph Theorem is often used in the following form: to prove that a linear map $T : X \to Y$ between Banach spaces is bounded, it suffices to prove that

$$x_n \to x, \quad Tx_n \to y \Rightarrow Tx = y.$$

8.5 The Hahn–Banach Theorem

We now turn to the extension theorem for bounded linear functionals.

Definition 8.5.1 (Sublinear function). Let X be a real vector space. A function $p : X \to \mathbb{R}$ is called sublinear if

$$p(x + y) \le p(x) + p(y) \quad \text{for all } x, y \in X,$$

and

$$p(\lambda x) = \lambda p(x) \quad \text{for all } x \in X,\ \lambda \ge 0.$$

Theorem 8.5.1 (Hahn–Banach theorem, real form). Let X be a real vector space, let $M \subseteq X$ be a linear subspace, and let $p : X \to \mathbb{R}$ be sublinear. Suppose $f : M \to \mathbb{R}$ is linear and satisfies

$$f(x) \le p(x) \quad \text{for every } x \in M.$$

Then there exists a linear functional $F : X \to \mathbb{R}$ such that

$$F|_M = f \quad \text{and} \quad F(x) \le p(x) \quad \text{for every } x \in X.$$

Proof. We first prove the one-dimensional extension step. Let $z \in X \setminus M$ and set $M_1 := M + \mathbb{R}z$. We seek an extension of the form

$$f_1(x + tz) = f(x) + t\alpha, \quad x \in M,\ t \in \mathbb{R}.$$

We need $f(x)+t\alpha \le p(x+tz)$ for all $x \in M$, $t \in \mathbb{R}$. For $t > 0$, this is equivalent to $\alpha \le p(u+z)-f(u)$ for all $u \in M$. For $t < 0$, it is equivalent to $\alpha \ge f(u) - p(u - z)$ for all $u \in M$. Thus it suffices to choose α such that

$$f(u) - p(u - z) \le \alpha \le p(v + z) - f(v) \quad \text{for all } u, v \in M.$$

Such an α exists because

$$f(u) + f(v) = f(u + v) \le p(u + v) \le p(u - z) + p(v + z),$$

and hence $f(u) - p(u - z) \leq p(v + z) - f(v)$. Therefore we may choose

$$\alpha \in \left[\sup_{u \in M} \left(f(u) - p(u - z) \right), \ \inf_{v \in M} \left(p(v + z) - f(v) \right) \right].$$

With this choice of α, we have, for every $x \in M$,

$$f(x) + \alpha \leq p(x + z), \tag{8.4}$$

and

$$f(x) - \alpha \leq p(x - z). \tag{8.5}$$

We now explain why these two inequalities, which correspond to the coefficients 1 and -1 of the new direction z, imply the desired estimate for all real coefficients t. The important point is that Equation 8.4 and Equation 8.5 hold for every element of M, not just for one fixed element.

Let $x \in M$ and $t \in \mathbb{R}$. If $t > 0$, then $x + tz = t(x/t + z)$ and $x/t \in M$. Applying Equation 8.4 to x/t and multiplying by $t > 0$, using the linearity of f and the positive homogeneity of p, gives $f(x) + t\alpha \leq p(x + tz)$. If $t < 0$, write $t = -s$ where $s > 0$. Then $x + tz = s(x/s - z)$ and $x/s \in M$. Applying Equation 8.5 to x/s and multiplying by $s > 0$, using the linearity of f and the positive homogeneity of p, gives $f(x) - s\alpha \leq p(x - sz)$. Since $t = -s$, this is exactly $f(x) + t\alpha \leq p(x + tz)$. If $t = 0$, the desired inequality reduces to $f(x) \leq p(x)$, which holds by assumption on M. Thus

$$f_1(x + tz) = f(x) + t\alpha \leq p(x + tz) \quad \text{for all } x \in M, \ t \in \mathbb{R}.$$

Therefore f_1 extends f and satisfies $f_1 \leq p$ on M_1.

Now let $\mathcal{E}$ be the collection of all pairs (N, g), where $M \subseteq N \subseteq X$ is a linear subspace and $g : N \to \mathbb{R}$ is a linear functional such that $g|_M = f$ and $g(x) \leq p(x)$ for every $x \in N$. Thus $\mathcal{E}$ consists of all possible extensions of f to larger subspaces, still dominated by p. We define a partial order on $\mathcal{E}$ by extension: for $(N_1, g_1), (N_2, g_2) \in \mathcal{E}$, we write $(N_1, g_1) \leq (N_2, g_2)$ if $N_1 \subseteq N_2$ and $g_2|_{N_1} = g_1$.

Notice first that $\mathcal{E}$ is nonempty, because $(M, f) \in \mathcal{E}$.

We claim that every chain in $\mathcal{E}$ has an upper bound. Let $\mathcal{C} = \{(N_i, g_i)\}_{i \in I}$ be a chain in $\mathcal{E}$. This means that for any $i, j \in I$, either $(N_i, g_i) \leq (N_j, g_j)$ or $(N_j, g_j) \leq (N_i, g_i)$. Define $N_{\mathcal{C}} := \bigcup_{i \in I} N_i$. We first check that $N_{\mathcal{C}}$ is a linear subspace of X. Let $x, y \in N_{\mathcal{C}}$ and $a, b \in \mathbb{R}$. Then there exist $i, j \in I$ such that $x \in N_i$ and $y \in N_j$. Since $\mathcal{C}$ is a chain, the two subspaces N_i and N_j are comparable. For example, suppose $N_i \subseteq N_j$. Then $x, y \in N_j$, and since N_j is a linear subspace, $ax + by \in N_j \subseteq N_{\mathcal{C}}$. The other case $N_j \subseteq N_i$ is identical. Hence $N_{\mathcal{C}}$ is a linear subspace.

We now define a functional $g_{\mathcal{C}} : N_{\mathcal{C}} \to \mathbb{R}$ as follows. Given $x \in N_{\mathcal{C}}$, choose $i \in I$ such that $x \in N_i$, and set $g_{\mathcal{C}}(x) := g_i(x)$. This definition is well-defined. Indeed, if $x \in N_i \cap N_j$, then because $\mathcal{C}$ is a chain, the pairs (N_i, g_i) and (N_j, g_j) are comparable. Suppose, for instance, that $(N_i, g_i) \leq (N_j, g_j)$. Then $g_j|_{N_i} = g_i$, and therefore $g_j(x) = g_i(x)$. The other case is the same. Hence $g_{\mathcal{C}}(x)$ is independent of the choice of i.

Next we verify that $g_{\mathcal{C}}$ is linear. Let $x, y \in N_{\mathcal{C}}$ and $a, b \in \mathbb{R}$. Choose $i, j \in I$ such that $x \in N_i$ and $y \in N_j$. Since $\mathcal{C}$ is a chain, after possibly interchanging i and j, we may assume that $(N_i, g_i) \leq (N_j, g_j)$. Then $x, y \in N_j$, and hence $ax + by \in N_j$. Therefore

$$g_{\mathcal{C}}(ax + by) = g_j(ax + by) = a g_j(x) + b g_j(y) = a g_{\mathcal{C}}(x) + b g_{\mathcal{C}}(y).$$

Thus $g_{\mathcal{C}}$ is linear. Moreover, $g_{\mathcal{C}}$ extends f, since every $(N_i, g_i) \in \mathcal{E}$ satisfies $M \subseteq N_i$ and $g_i|_M = f$. Finally, $g_{\mathcal{C}}(x) = g_i(x) \leq p(x)$ for any $x \in N_i$. Therefore $(N_{\mathcal{C}}, g_{\mathcal{C}}) \in \mathcal{E}$. It is also an upper bound for $\mathcal{C}$, because for every $i \in I$, $N_i \subseteq N_{\mathcal{C}}$ and $g_{\mathcal{C}}|_{N_i} = g_i$.

Thus every chain in $\mathcal{E}$ has an upper bound. By Zorn's lemma, there exists a maximal element $(N_0, g_0) \in \mathcal{E}$. We claim that $N_0 = X$. Suppose, for contradiction, that $N_0 \neq X$. Then there exists $z \in X \setminus N_0$. Since $z \notin N_0$, the subspace $N_0 + \mathbb{R}z$ is strictly larger than N_0. By the one-dimensional extension step, there exists a linear functional $\widetilde{g} : N_0 + \mathbb{R}z \to \mathbb{R}$ such that $\widetilde{g}|_{N_0} = g_0$ and $\widetilde{g}(x) \leq p(x)$ for every $x \in N_0 + \mathbb{R}z$. Therefore $(N_0 + \mathbb{R}z, \widetilde{g}) \in \mathcal{E}$. Moreover, $(N_0, g_0) < (N_0 + \mathbb{R}z, \widetilde{g})$, and this extension is strict because $z \notin N_0$. This contradicts the maximality of (N_0, g_0).

Hence $N_0 = X$. Consequently, $g_0 : X \to \mathbb{R}$ is a linear extension of f to all of X, and it satisfies $g_0(x) \leq p(x)$ for every $x \in X$. This is the desired Hahn–Banach extension. ∎

Theorem 8.5.2 (Hahn–Banach Extension Theorem). Let X be a real normed vector space, and let $M \subseteq X$ be a linear subspace. If $f : M \to \mathbb{R}$ is a bounded linear functional, then there exists $F \in X^*$ such that

$$F|_M = f \quad \text{and} \quad \|F\|_{X^*} = \|f\|_{M^*}.$$

Proof. Define $p(x) := \|f\|_{M^*}\|x\|$ for $x \in X$. Then p is sublinear. For $x \in M$,

$$f(x) \le |f(x)| \le \|f\|_{M^*}\|x\| = p(x).$$

By the Hahn–Banach theorem, there exists a linear functional $F : X \to \mathbb{R}$ such that $F|_M = f$ and $F(x) \le p(x)$ for every $x \in X$. Applying this inequality to $-x$, we also get $-F(x) \le p(x)$. Therefore

$$|F(x)| \le \|f\|_{M^*}\|x\| \quad \text{for every } x \in X.$$

Thus F is bounded and $\|F\|_{X^*} \le \|f\|_{M^*}$. The reverse inequality follows from $F|_M = f$. Hence $\|F\|_{X^*} = \|f\|_{M^*}$. ∎

Remark 8.5.1 (Complex Hahn–Banach theorem). The complex version is also true. If X is a complex normed vector space and $M \subseteq X$ is a complex linear subspace, then every bounded complex-linear functional $f : M \to \mathbb{C}$ extends to a bounded complex-linear functional $F : X \to \mathbb{C}$ with

$$F|_M = f, \quad \|F\| = \|f\|.$$

One proves this by applying the real Hahn–Banach theorem to $\operatorname{Re} f$ and then reconstructing the complex-linear functional.

8.6 Basic Consequences of Hahn–Banach

Corollary 8.6.1 (Existence of norming functionals). Let X be a normed vector space and let $x_0 \in X$, $x_0 \neq 0$. Then there exists $F \in X^*$ such that

$$\|F\| = 1 \quad \text{and} \quad F(x_0) = \|x_0\|.$$

Proof. Let $M = \operatorname{span}\{x_0\}$, and define $f : M \to \mathbb{R}$ by $f(tx_0) := t\|x_0\|$. Then

$$|f(tx_0)| = |t|\|x_0\| = \|tx_0\|,$$

so $\|f\| = 1$. By the Hahn–Banach Extension Theorem, f extends to some $F \in X^*$ with $\|F\| = 1$. Since $F(x_0) = f(x_0) = \|x_0\|$, the result follows. ∎

Corollary 8.6.2 (Dual norm formula). Let X be a normed vector space. Then for every $x \in X$,

$$\|x\| = \sup\{|F(x)| : F \in X^*,\ \|F\| \le 1\}.$$

Proof. For every $F \in X^*$ with $\|F\| \le 1$, $|F(x)| \le \|x\|$. Thus

$$\sup\{|F(x)| : \|F\| \le 1\} \le \|x\|.$$

If $x = 0$, equality is clear. If $x \neq 0$, Corollary 8.6.1 gives $F \in X^*$ such that $\|F\| = 1$ and $F(x) = \|x\|$. Hence equality holds. ∎

Corollary 8.6.3 (Separation from a closed subspace). Let X be a normed vector space, let $M \subseteq X$ be a closed linear subspace, and let $x_0 \notin M$. Then there exists $F \in X^*$ such that

$$F|_M = 0 \quad \text{and} \quad F(x_0) \neq 0.$$

More precisely, one can choose $F \in X^*$ such that

$$\|F\| = 1, \quad F(x_0) = \operatorname{dist}(x_0, M).$$

Proof. Let $d := \operatorname{dist}(x_0, M) = \inf_{m \in M} \|x_0 - m\|$. Since M is closed and $x_0 \notin M$, we have $d > 0$. Let $N := M + \mathbb{R}x_0$. Define $f : N \to \mathbb{R}$ by

$$f(m + tx_0) := td, \quad m \in M,\ t \in \mathbb{R}.$$

This is well-defined and linear. Also, $f|_M = 0$ and $f(x_0) = d$. We claim that $\|f\| \leq 1$. If $t \neq 0$, then

$$\|m + tx_0\| = |t| \left\| x_0 + \frac{m}{t} \right\| \geq |t|d = |f(m + tx_0)|.$$

The case $t = 0$ is immediate. Thus $\|f\| \leq 1$. By the Hahn–Banach Extension Theorem, extend f to $F \in X^*$ with $\|F\| = \|f\| \leq 1$. Since $F(x_0) = d$ and F vanishes on M, we have for every $m \in M$,

$$d = |F(x_0 - m)| \leq \|F\| \|x_0 - m\|.$$

Taking the infimum over $m \in M$, we get $d \leq \|F\| \cdot d$. Since $d > 0$, this implies $\|F\| \geq 1$. Therefore $\|F\| = 1$. ■

Summary. The essential principles of this chapter are:

- **The Baire Category Theorem:** completeness prevents a space from being too small in a countable-union sense.
- **The Uniform Boundedness Principle:** pointwise bounded families of bounded linear operators on a Banach space are uniformly bounded in operator norm.
- **The Open Mapping Theorem:** surjective bounded linear operators between Banach spaces are open.
- **The Inverse Operator Theorem:** bijective bounded linear operators between Banach spaces have bounded inverses.
- **The Closed Graph Theorem:** a linear operator between Banach spaces is bounded if and only if its graph is closed.
- **The Hahn–Banach Theorem:** bounded linear functionals extend from subspaces to the whole space without increasing their norm.

Together, these results form the basic toolkit for working with Banach spaces. They show that completeness converts algebraic hypotheses into strong continuity and extension properties.

Appendix

Appendix A

Midterm Practice

A.1 HW6

Problem A.1.1. Let A be an open interval in $\mathbb{R}$. Show that

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

Proof. Let $A = (\alpha, \beta)$, then

$$A \cap (0, \infty) = \begin{cases} \varnothing, & \text{if } \beta \leq 0; \\ (\alpha, \beta), & \text{if } \alpha, \beta > 0; \\ (0, \beta), & \text{if } \alpha \leq 0, \beta > 0, \end{cases}$$

and

$$A \setminus (0, \infty) = \begin{cases} (\alpha, \beta), & \text{if } \beta \leq 0; \\ \varnothing, & \text{if } \alpha, \beta > 0; \\ (\alpha, 0], & \text{if } \alpha \leq 0, \beta > 0. \end{cases}$$

Thus, in each case

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

■

Problem A.1.2. Let A be an open box in $\mathbb{R}^n$, and let $E = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_n > 0\}$. Show that

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

Proof. Let $A = \prod_{i=1}^{\infty} (a_i, b_i)$, then

$$A \cap E = \left(\prod_{i=1}^{n-1} (a_i, b_i) \right) \times ((a_n \cap b_n) \cap (0, \infty)).$$

Now since $(a_n, b_n) \cap (0, \infty)$ is an interval,

$$m^*(A \cap E) = \operatorname{vol}(A_{n-1}) \cdot m^*(A_n \cap (0, \infty)) \text{ where } A_{n-1} = \prod_{i=1}^{n-1} (a_i, b_i),\ A_n = (a_n, b_n).$$

Similarly,

$$A \setminus E = A_{n-1} \times (A_n \setminus (0, \infty)),$$

so

$$m^*(A \setminus E) = \operatorname{vol}(A_{n-1}) \cdot m^*(A_n \setminus (0, \infty)).$$

Hence,

$$\begin{aligned}m^*(A) = \mathrm{vol}(A) = \mathrm{vol}(A_{n-1})(m^*(A_n)) &= \mathrm{vol}(A_{n-1})(m^*(A_n \cap (0,\infty)) + m^*(A_n \setminus (0,\infty)))\\ &= \mathrm{vol}(A_{n-1}) \cdot m^*(A_n \cap (0,\infty)) + \mathrm{vol}(A_{n-1}) \cdot m^*(A_n \setminus (0,\infty))\\ &= m^*(A \cap (0,\infty)) + m^*(A \setminus (0,\infty)).\end{aligned}$$

■

A.2 HW1

Problem A.2.1. Let

$$D = \left\{(x,y) \in \mathbb{R}^n, x \geq 0, 0 \leq y \leq x^3\right\},$$

and $x_0 = (0,0)$. Define $f : D \to \mathbb{R}$ by $f(x,y) = 2x + \sqrt{y}$.

(a) Prove that f is differentiable at $(0,0)$ on D.

(b) Describe the set of all derivative $L(x,y)$ for f on $(0,0)$ on D.

A.3 HW2

Problem A.3.1. Define

$$\begin{aligned}\mathrm{GL}(n,\mathbb{R}) &= \{A \text{ is a } n \times n \text{ matrix and } A \text{ is invertible}\}\\ &= \{A \mid \det(A) \neq 0\},\end{aligned}$$

which is open in $\mathbb{R}^{n+m}$. Consider the inverse map $g : \mathrm{GL}(n,r) \to \mathrm{GL}(n,r)$, i.e. $g(A) = A^{-1}$. Show that $g'(A) : \mathbb{R}^{n\times n} \to \mathrm{R}^{n\times n}$ is $g'(A)(H) = A^{-1}HA^{-1}$ for every $H \in \mathbb{R}^{n\times n}$.

Remark A.3.1. We can use the fact that if $M \in \mathbb{R}^{n+m}$ has $\|M\| < 1$, then

$$(I+M)^{-1} = I - M + M^2 - M^3 + \dots$$

converges in $\mathbb{R}^{n+m}$.

A.4 HW3

Problem A.4.1. Let $f : S \to S$ be a map between a complete metric space S. Assume there is a real sequence $\{\alpha_n\}$ which converges to 0 s.t.

$$d(f^n(x), f^n(y)) \leq \alpha_n d(x,y)$$

for all $n \geq 1$ and all $x, y \in S$. Prove that f has a unique fixed point. Hint. Apply the fixed-point theorem to f^m for a suitable m.

A.5 HW4

Problem A.5.1. Let $n \geq 2$, and let

$$f : \mathbf{R}^n \to \mathbf{R}$$

be a smooth function. Define

$$S := \{x = (x_1, \dots, x_n) \in \mathbf{R}^n : f(x) = 0\}.$$

Assume that

$$\nabla f(x) \neq 0 \qquad \text{for every } x \in S.$$

Then for every point $p \in S$, there exists an index $j \in \{1, \dots, n\}$, an open neighborhood $W \subset \mathbf{R}^n$ of p, an open set $U \subset \mathbf{R}^{n-1}$, and a differentiable function

$$g : U \to \mathbf{R}$$

such that, after possibly reordering the coordinates, the set $S \cap W$ can be written in the form

$$S \cap W = \Big\{(x_1, \dots, x_{j-1},\, g(x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n),\, x_{j+1}, \dots, x_n) : (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \in U\Big\}.$$

In other words, near every point of S, the hypersurface S is locally the graph of a differentiable function of $n-1$ variables.

A.6 HW5

Problem A.6.1. Recall that a function $f : [a, b] \to \mathbf{R}$ is *Lipschitz* if there exists $L \geq 0$ such that $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in [a, b]$. Define the graph of f as:

$$\Gamma_f := \{(x, f(x)) : x \in [a, b]\} \subset \mathbf{R}^2.$$

Prove that $m_2^*(\Gamma_f) = 0$. *Hint: Partition $[a, b]$ into n sub-intervals and cover the corresponding pieces of the graph with rectangles. Consider the limit as $n \to \infty$.*

A.7 Random Stuff

Theorem A.7.1. If f is continuoius and I is an open interval, then $f(I)$ is also an interval.

Theorem A.7.2. Any interval in $\mathbb{R}$ is measurable (open, closed, or half-open, half-closed)

Theorem A.7.3. A set E is Lebesgue measurable iff $m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$ for all $A \subseteq \mathbb{R}^n$. To show E is measurable, it suffices to show

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

Appendix B

Additional proof

Theorem B.0.1. If $f = \liminf_{n\to\infty} f_n = \limsup_{n\to\infty} f_n$, then $f_n \to f$ pointwise.

Proof. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of real-valued functions on a set Ω, and suppose that for every $x \in \Omega$,

$$f(x) = \liminf_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x).$$

We will show that $f_n(x) \to f(x)$ for every $x \in \Omega$.
Fix $x \in \Omega$. Then $\{f_n(x)\}$ is a sequence of real numbers satisfying

$$\liminf_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x) = f(x).$$

We prove that $f_n(x) \to f(x)$.

Let $\varepsilon > 0$.
By the definition of $\limsup$, we have

$$\limsup_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sup_{k\geq n} f_k(x) = f(x).$$

Hence there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$\sup_{k\geq n} f_k(x) < f(x) + \varepsilon.$$

In particular, for all $k \geq n \geq N_1$,

$$f_k(x) \leq \sup_{j\geq n} f_j(x) < f(x) + \varepsilon.$$

Thus for all $k \geq N_1$,

$$f_k(x) < f(x) + \varepsilon.$$

Similarly, by the definition of $\liminf$, we have

$$\liminf_{n\to\infty} f_n(x) = \lim_{n\to\infty} \inf_{k\geq n} f_k(x) = f(x).$$

Hence there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$\inf_{k\geq n} f_k(x) > f(x) - \varepsilon.$$

In particular, for all $k \geq n \geq N_2$,

$$f_k(x) \geq \inf_{j\geq n} f_j(x) > f(x) - \varepsilon.$$

Thus for all $k \geq N_2$,

$$f_k(x) > f(x) - \varepsilon.$$

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$f(x) - \varepsilon < f_n(x) < f(x) + \varepsilon,$$

which implies

$$|f_n(x) - f(x)| < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that

$$f_n(x) \to f(x).$$

Since $x \in \Omega$ was arbitrary, the convergence is pointwise on Ω. ∎

Theorem B.0.2. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a Lebesgue measurable function, and let $g : \mathbb{R}^n \to \mathbb{R}$ be a function such that $f = g$ almost everywhere. Then g is also Lebesgue measurable.

Proof. Since $f = g$ almost everywhere, there exists a measurable set $N \subset \mathbb{R}^n$ such that $m(N) = 0$ and

$$f(x) = g(x) \quad \text{for all } x \in \mathbb{R}^n \setminus N.$$

To prove that g is measurable, it suffices to show that for every $a \in \mathbb{R}$, the set

$$E := \{x \in \mathbb{R}^n : g(x) > a\}$$

is measurable.

Fix $a \in \mathbb{R}$ and define

$$F := \{x \in \mathbb{R}^n : f(x) > a\}.$$

Since f is measurable, the set F is measurable.

We claim that

$$E \triangle F \subseteq N,$$

where $E \triangle F = (E \setminus F) \cup (F \setminus E)$ denotes the symmetric difference.

Indeed, let $x \notin N$. Then $f(x) = g(x)$, and hence

$$g(x) > a \Leftrightarrow f(x) > a.$$

Thus $x \in E$ if and only if $x \in F$, which implies $x \notin E \triangle F$. Therefore $E \triangle F \subseteq N$.

Since N has measure zero, it is measurable. It follows that $E \triangle F$ is measurable. As F is measurable and the collection of measurable sets is closed under symmetric difference, we conclude that

$$E = F \triangle (E \triangle F)$$

is measurable.

Since $a \in \mathbb{R}$ was arbitrary, this shows that $\{x : g(x) > a\}$ is measurable for all a, and hence g is measurable. ∎